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(54) **HAPTIC PLATFORM AND ECOSYSTEM FOR IMMERSIVE COMPUTER MEDIATED ENVIRONMENTS**

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(21) Appl. No.: **18/858,721**

(22) PCT Filed: **Aug. 4, 2023**

(86) PCT No.: **PCT/US23/29559**

§ 371 (c)(1),
(2) Date: **Oct. 21, 2024**

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Publication Classification

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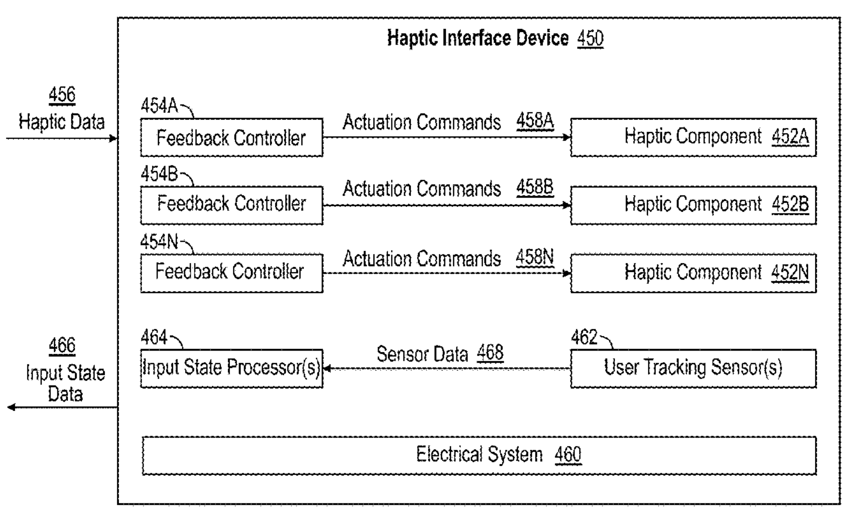
G06F 3/01 (2006.01)
G06N 3/008 (2023.01)

(52) **U.S. Cl.**
CPC ***G06F 3/016*** (2013.01); ***G06F 3/011*** (2013.01); ***G06F 3/017*** (2013.01); ***G06N 3/008*** (2013.01)

(57) **ABSTRACT**

A device may receive, by a haptic interface module that is configured to interact with one or more haptic interface devices and an application that generates a computer-mediated environment comprising an avatar corresponding to a user wearing the one or more haptic interface devices, respective sensor data for each respective haptic interface device, wherein the respective sensor data for a respective haptic interface device indicates respective positioning of respective sensors of the respective wearable haptic interface. A device may process, by the haptic interface module, the respective sensor data to generate respective relative location data for each respective haptic interface device, wherein the relative location data is relative to a reference location defined with respect to the corresponding wearable haptic interface. A device may receive, by the haptic interface module, tracked location data from one or more motion tracking sensors, wherein the tracked location data indicates respective locations of the one or more haptic interface devices relative to a spatial environment of the user. A device may generate, by the haptic interface module, a series of motion capture frames based on the tracked location data and the respective relative location data for each respective haptic interface device, wherein each respective motion capture frame indicates a set of locations and orientations for each respective haptic interface device at a given time. A device may generate, by the haptic interface module, a series of kinematic frames based on the series of motion capture frames and one or more mediation processes that collectively convert, for each of the motion capture frames, the set of locations and orientations of the one or more respective haptic interface devices into a set of intended locations and intended orientations for configuring the avatar in the computer-mediated environment. A device may output the series of kinematic frames to the application, wherein the kinematic frames are provided to the application as user input.



Related U.S. Application Data

on May 4, 2023, provisional application No. 63/467,560, filed on May 18, 2023, provisional application No. 63/395,747, filed on Aug. 5, 2022, provisional application No. 63/338,345, filed on May 4, 2022.

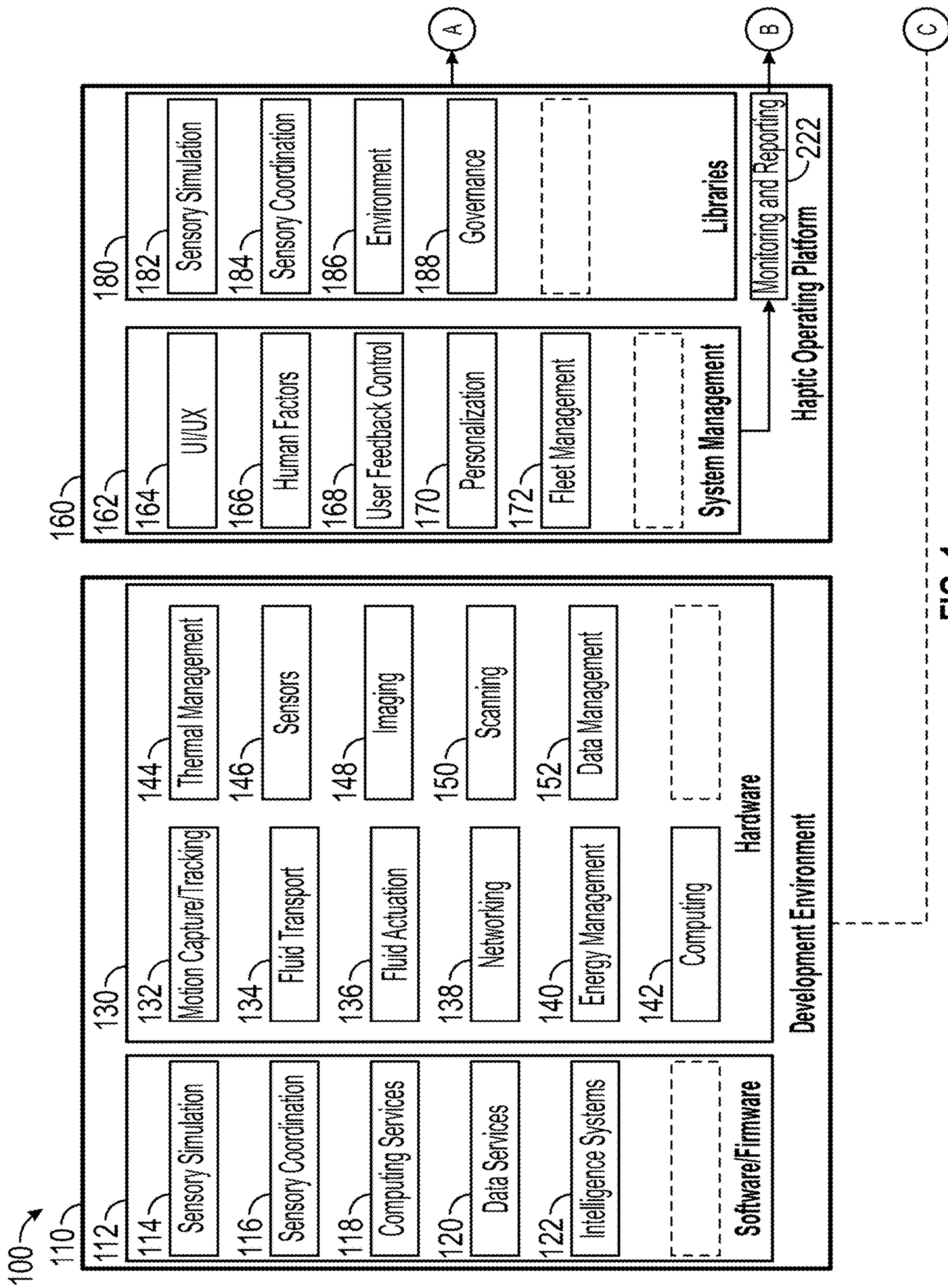

100
110
112
114 Sensory Simulation
116 Sensory Coordination
118 Computing Services
120 Data Services
122 Intelligence Systems
Software/Firmware
130
132 Motion Capture/Tracking
134 Fluid Transport
136 Fluid Actuation
138 Networking
140 Energy Management
142 Computing
144 Thermal Management
146 Sensors
148 Imaging
150 Scanning
152 Data Management
Hardware
Development Environment
160
162
164 UI/UX
166 Human Factors
168 User Feedback Control
170 Personalization
172 Fleet Management
System Management
180
182 Sensory Simulation
184 Sensory Coordination
186 Environment
188 Governance
Libraries
Monitoring and Reporting
222
Haptic Operating Platform
A
B
C

FIG. 1

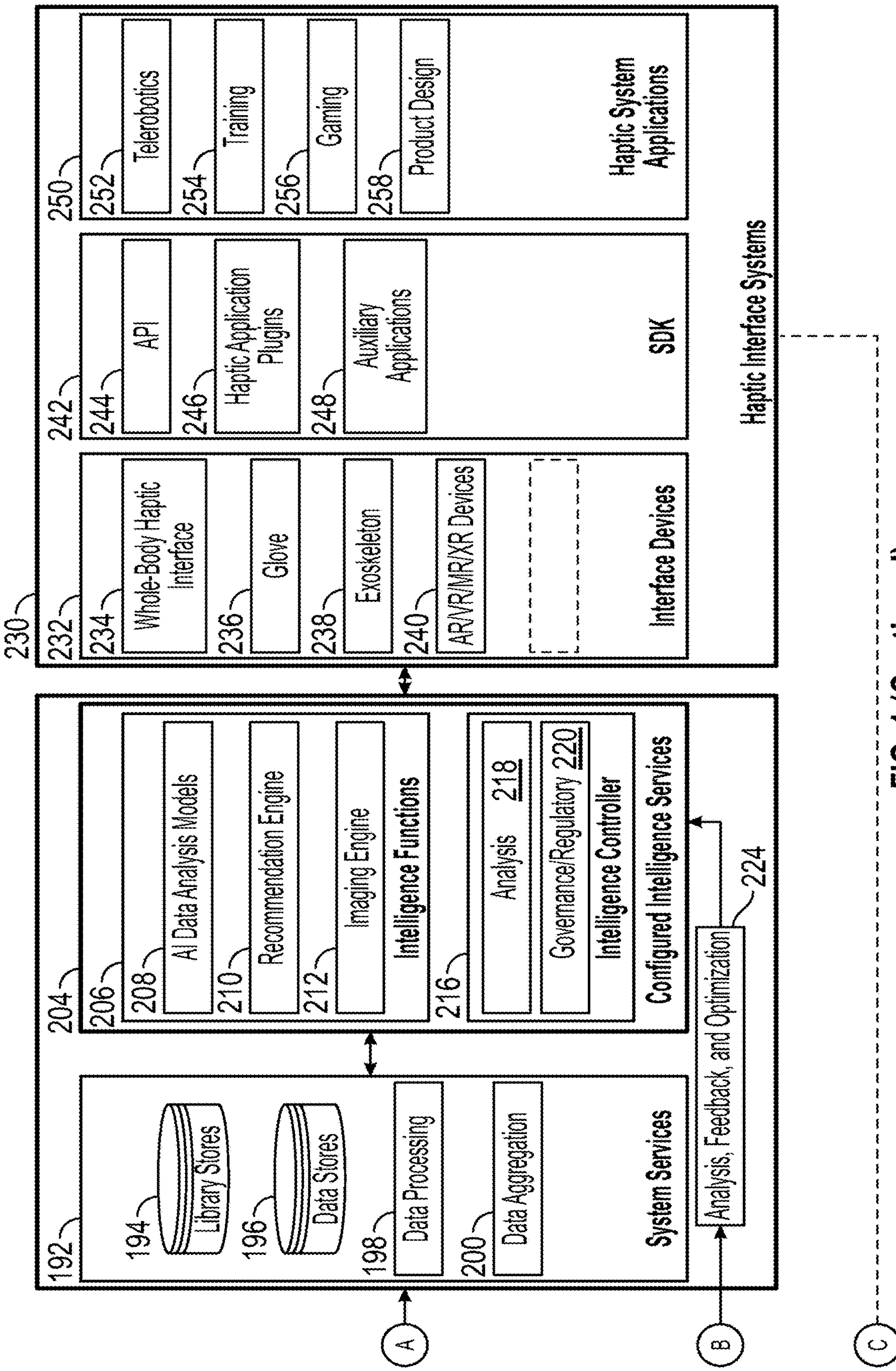


FIG. 1 (Continued)

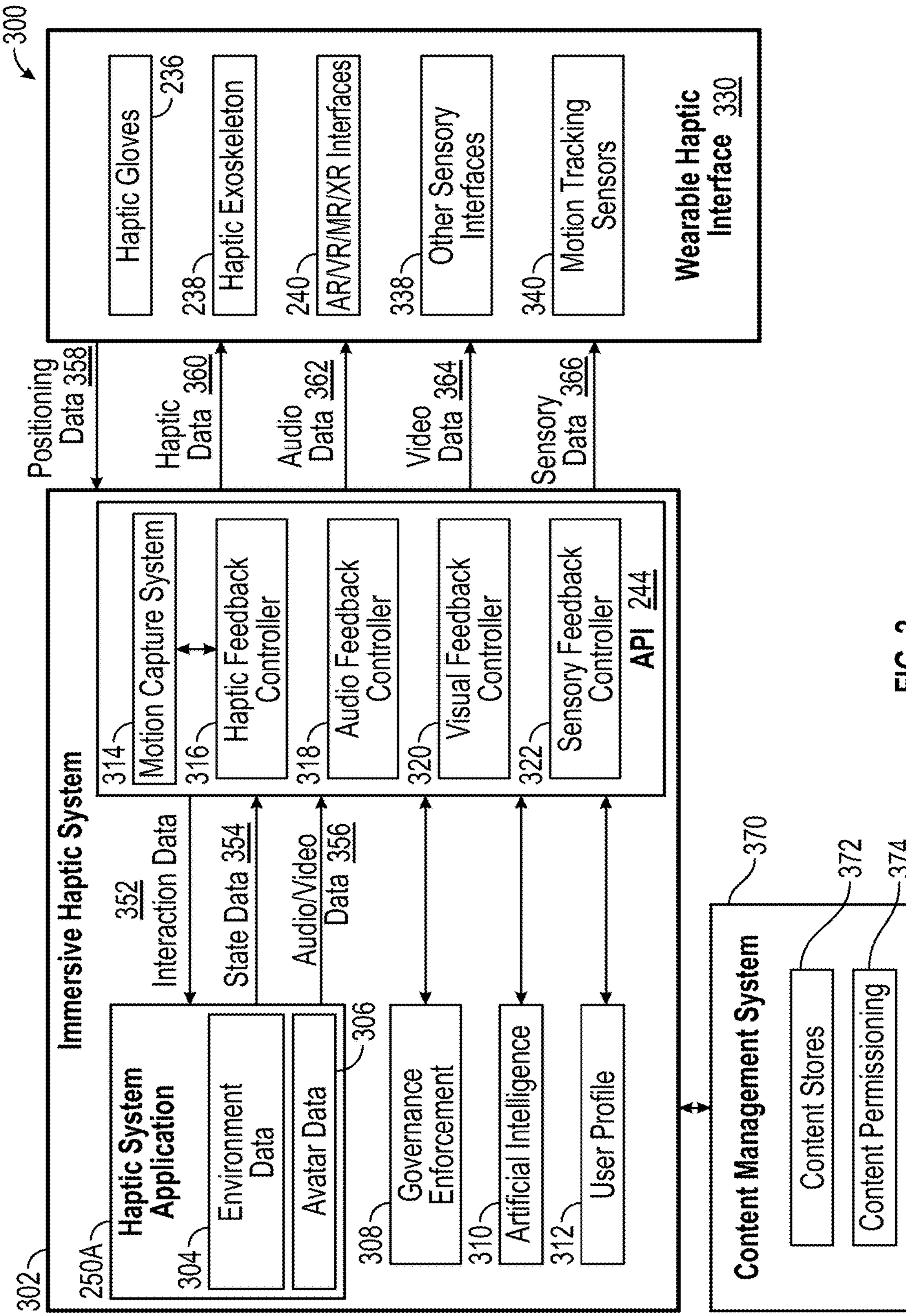

300
Wearable Haptic Interface 330
Haptic Gloves
236
238
Haptic Exoskeleton
240
AR/VR/MR/XR Interfaces
338
Other Sensory Interfaces
340
Motion Tracking Sensors
Positioning Data 358
Haptic Data 360
Audio Data 362
Video Data 364
Sensory Data 366
302
Immersive Haptic System
250A
Haptic System Application
304
Environment Data
306
Avatar Data
352
Interaction Data
State Data 354
Audio/Video Data 356
API 244
314
Motion Capture System
316
Haptic Feedback Controller
318
Audio Feedback Controller
320
Visual Feedback Controller
322
Sensory Feedback Controller
308
Governance Enforcement
310
Artificial Intelligence
312
User Profile
370
Content Management System
372
Content Stores
374
Content Permissioning

FIG. 2

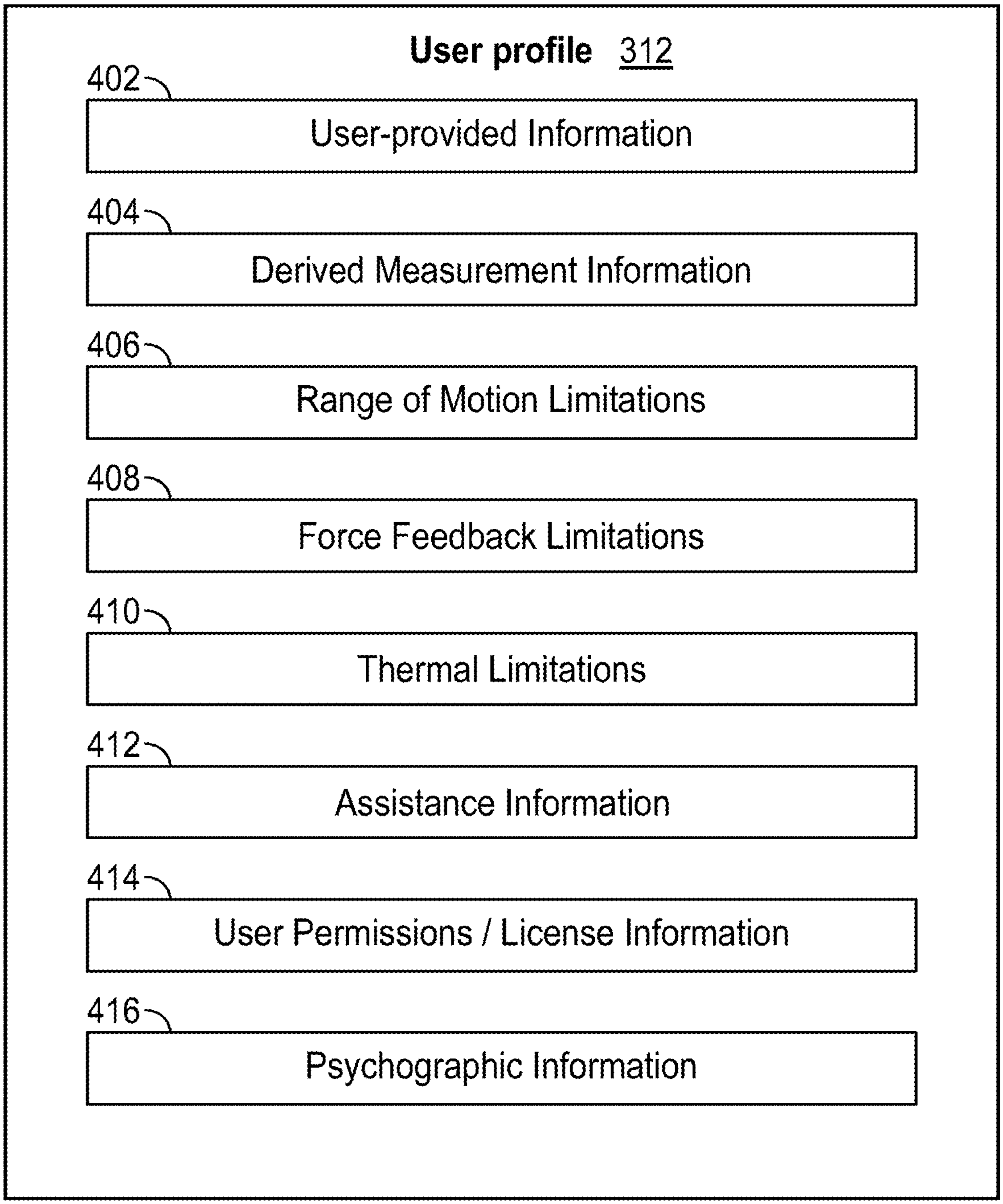

User profile 312
402
User-provided Information
404
Derived Measurement Information
406
Range of Motion Limitations
408
Force Feedback Limitations
410
Thermal Limitations
412
Assistance Information
414
User Permissions / License Information
416
Psychographic Information

FIG. 3

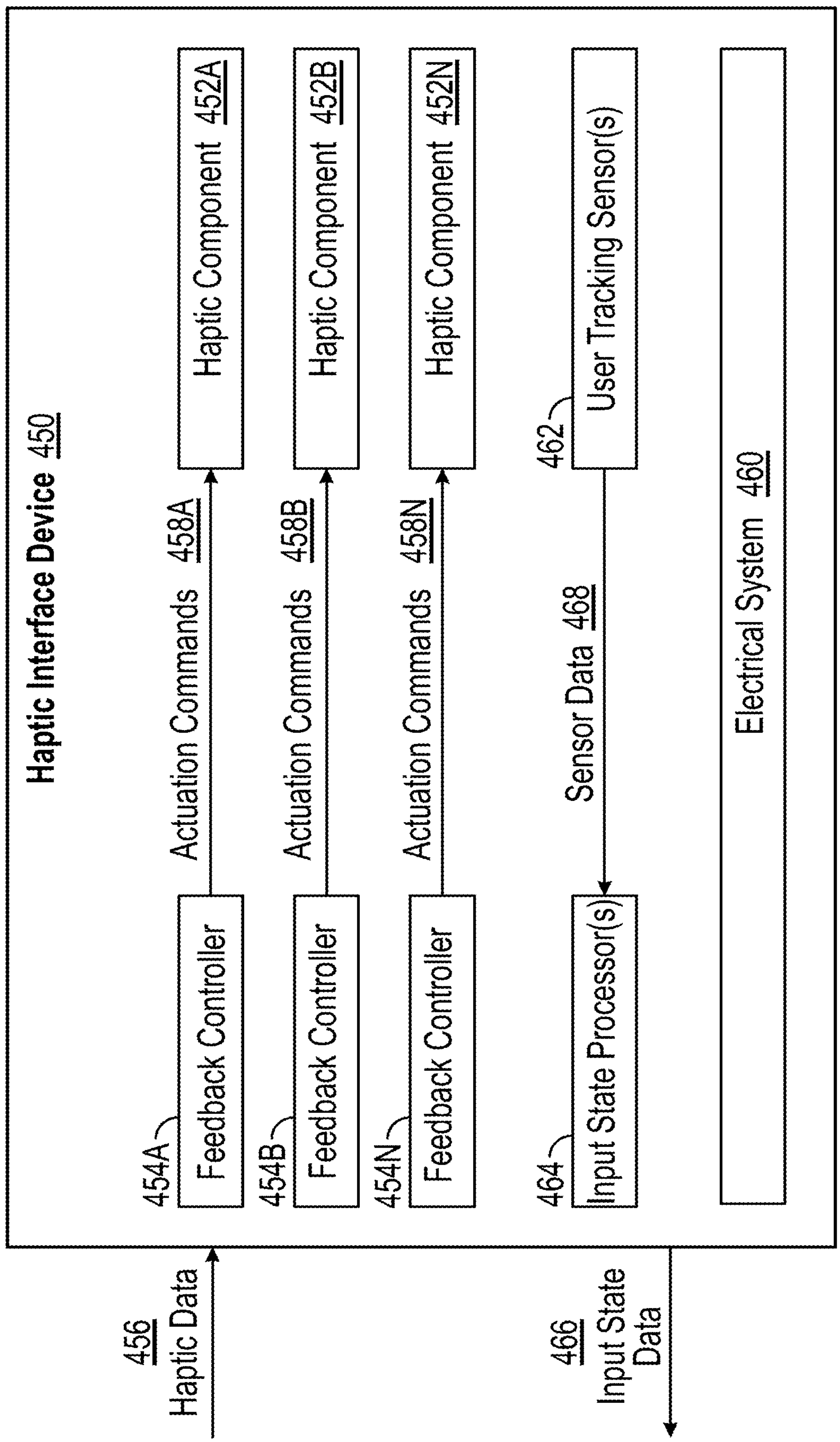
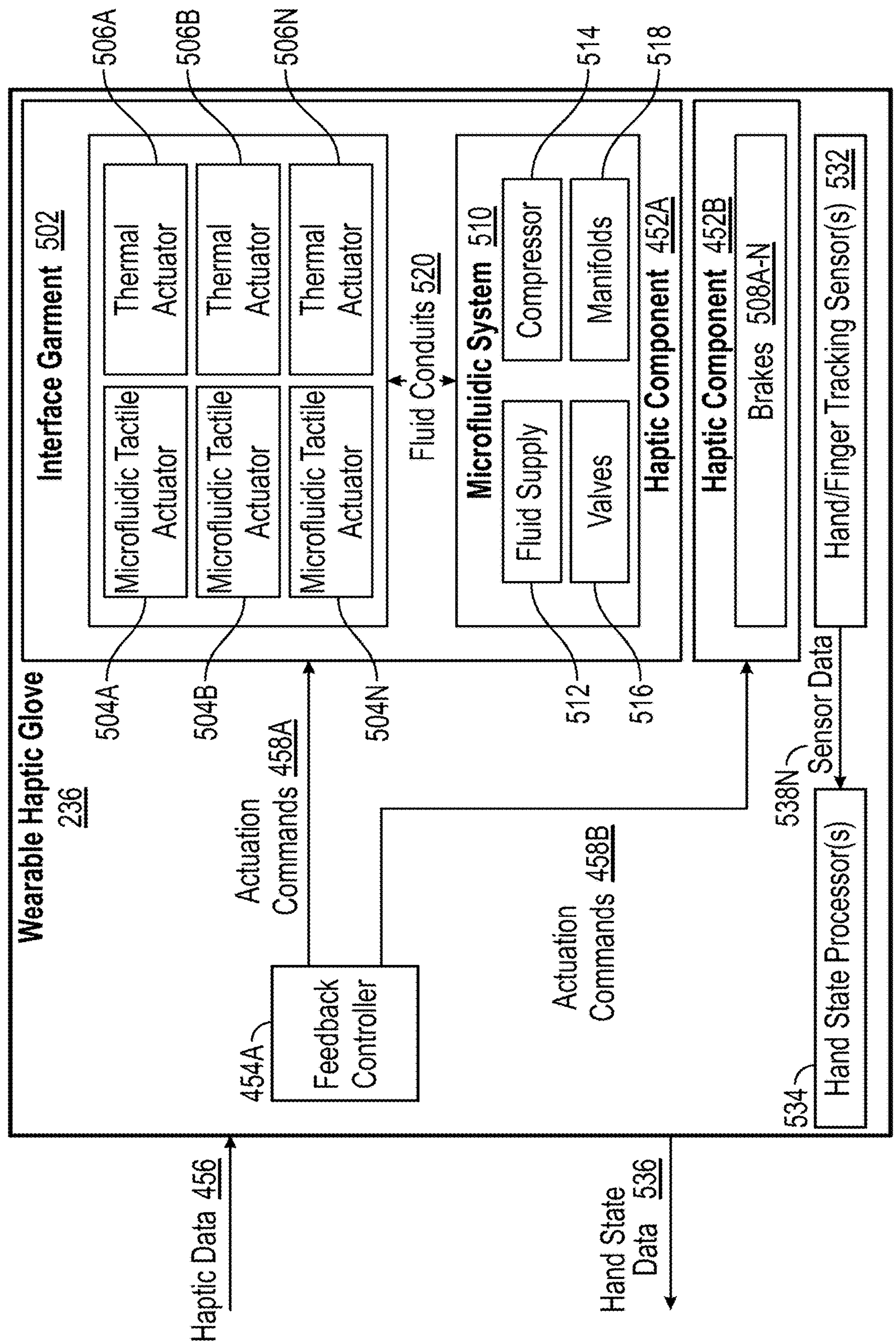

Haptic Interface Device 450
454A
Feedback Controller
Actuation Commands 458A
Haptic Component 452A
454B
Feedback Controller
Actuation Commands 458B
Haptic Component 452B
454N
Feedback Controller
Actuation Commands 458N
Haptic Component 452N
464
Input State Processor(s)
Sensor Data 468
462
User Tracking Sensor(s)
Electrical System 460
456 Haptic Data
466 Input State Data

FIG. 4



Wearable Haptic Glove
236
Interface Garment 502
Microfluidic Tactile Actuator
Thermal Actuator
Microfluidic Tactile Actuator
Thermal Actuator
Microfluidic Tactile Actuator
Thermal Actuator
506A
506B
506N
504A
504B
504N
Fluid Conduits 520
Microfluidic System 510
Fluid Supply
Compressor
Valves
Manifolds
514
518
512
516
Haptic Component 452A
Haptic Component 452B
Brakes 508A-N
Hand/Finger Tracking Sensor(s) 532
Sensor Data
538N
Hand State Processor(s)
534
Actuation Commands 458A
Actuation Commands 458B
Feedback Controller
454A
Haptic Data 456
Hand State Data 536

FIG. 5

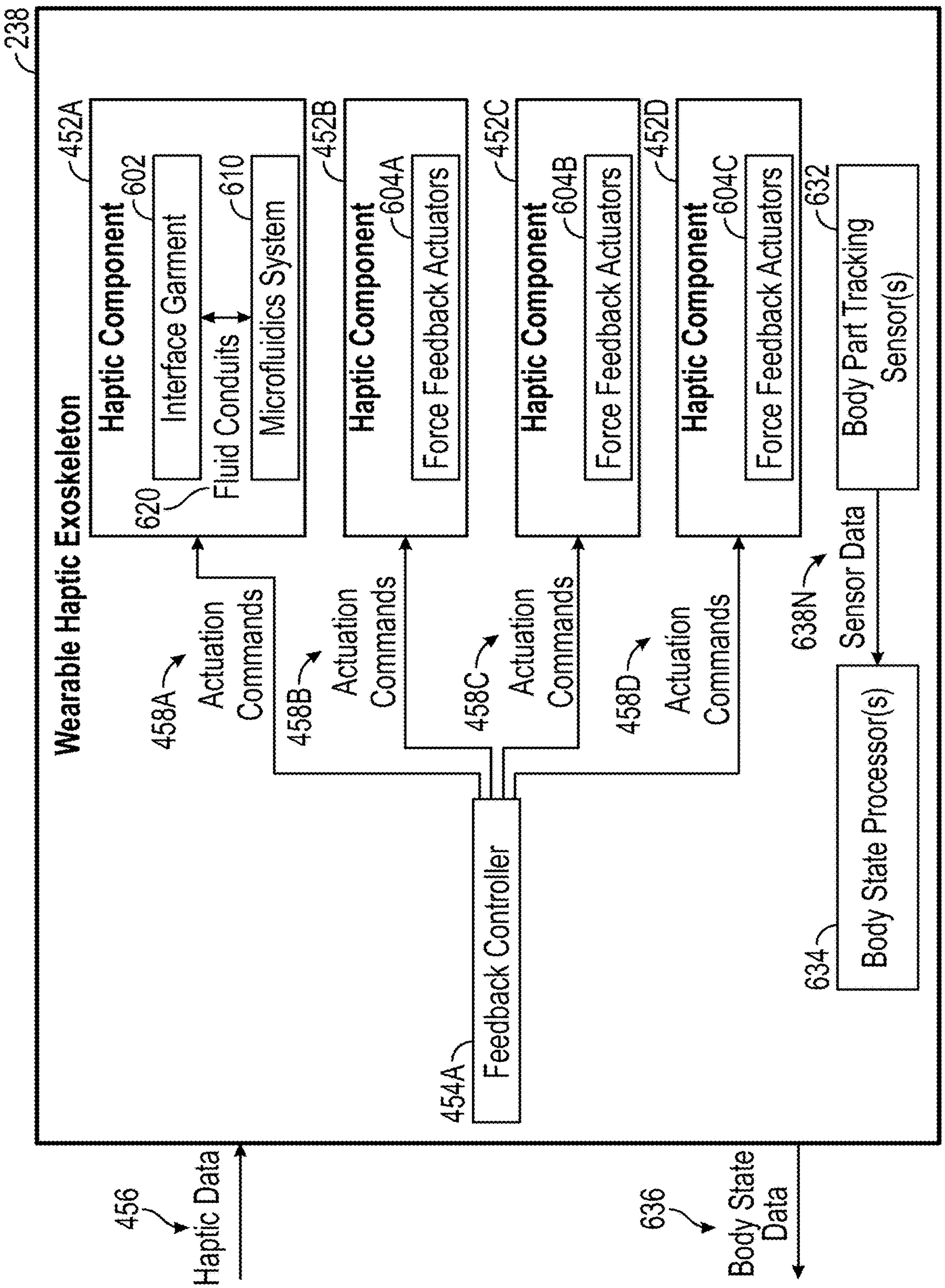

238
Wearable Haptic Exoskeleton
452A
Haptic Component
602
Interface Garment
620
Fluid Conduits
610
Microfluidics System
452B
Haptic Component
604A
Force Feedback Actuators
452C
Haptic Component
604B
Force Feedback Actuators
452D
Haptic Component
604C
Force Feedback Actuators
632
Body Part Tracking Sensor(s)
458A
Actuation Commands
458B
Actuation Commands
458C
Actuation Commands
458D
Actuation Commands
638N
Sensor Data
454A
Feedback Controller
634
Body State Processor(s)
456
Haptic Data
636
Body State Data

FIG. 6

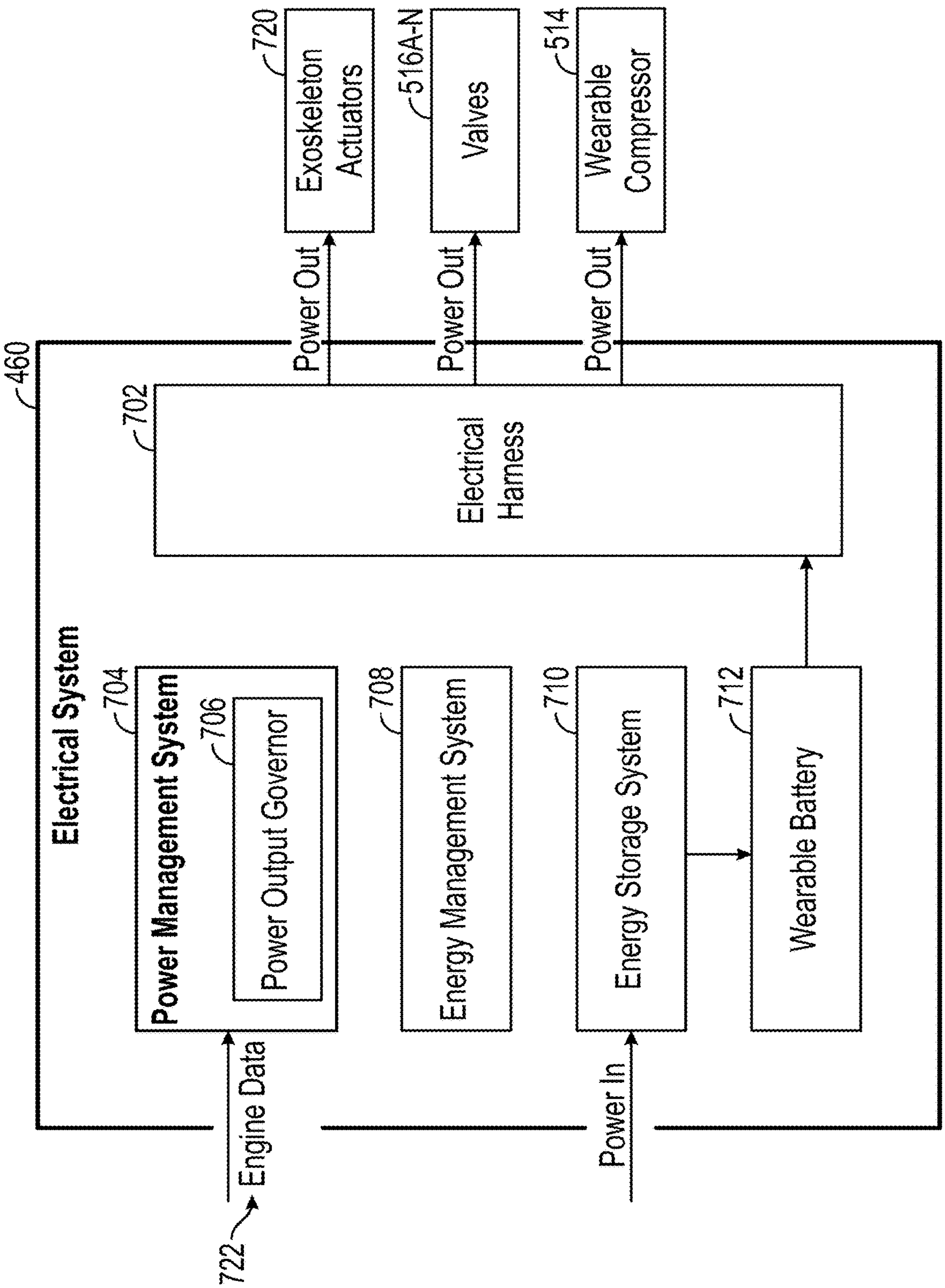

460
Electrical System
704
Power Management System
706
Power Output Governor
708
Energy Management System
710
Energy Storage System
712
Wearable Battery
702
Electrical Harness
Power Out
Power Out
Power Out
720
Exoskeleton Actuators
516A-N
Valves
514
Wearable Compressor
722
Engine Data
Power In

FIG. 7

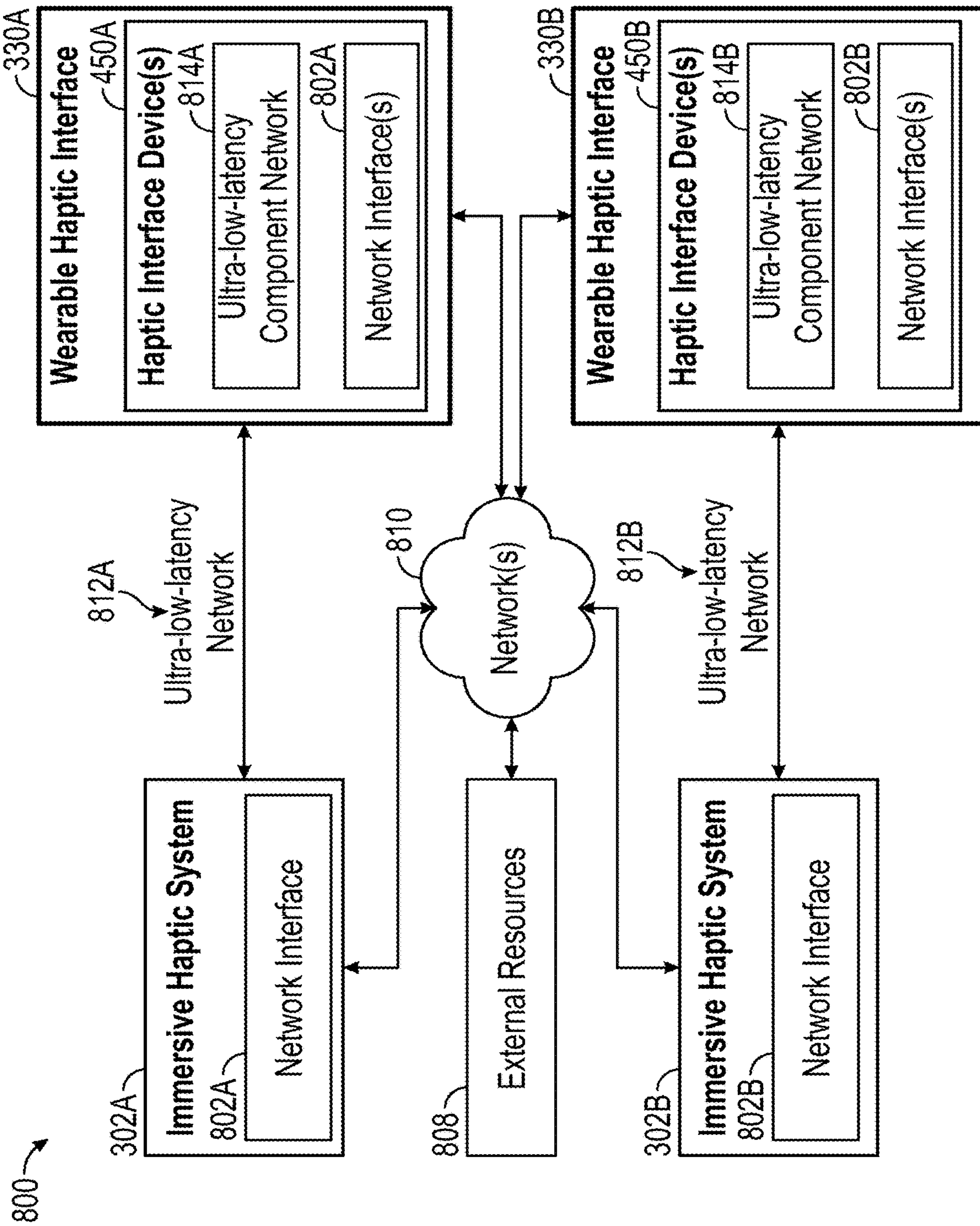

800
330A
Wearable Haptic Interface
450A
Haptic Interface Device(s)
814A
Ultra-low-latency Component Network
802A
Network Interface(s)
330B
Wearable Haptic Interface
450B
Haptic Interface Device(s)
814B
Ultra-low-latency Component Network
802B
Network Interface(s)
812A
Ultra-low-latency Network
810
Network(s)
812B
Ultra-low-latency Network
302A
Immersive Haptic System
802A
Network Interface
808
External Resources
302B
Immersive Haptic System
802B
Network Interface

FIG. 8

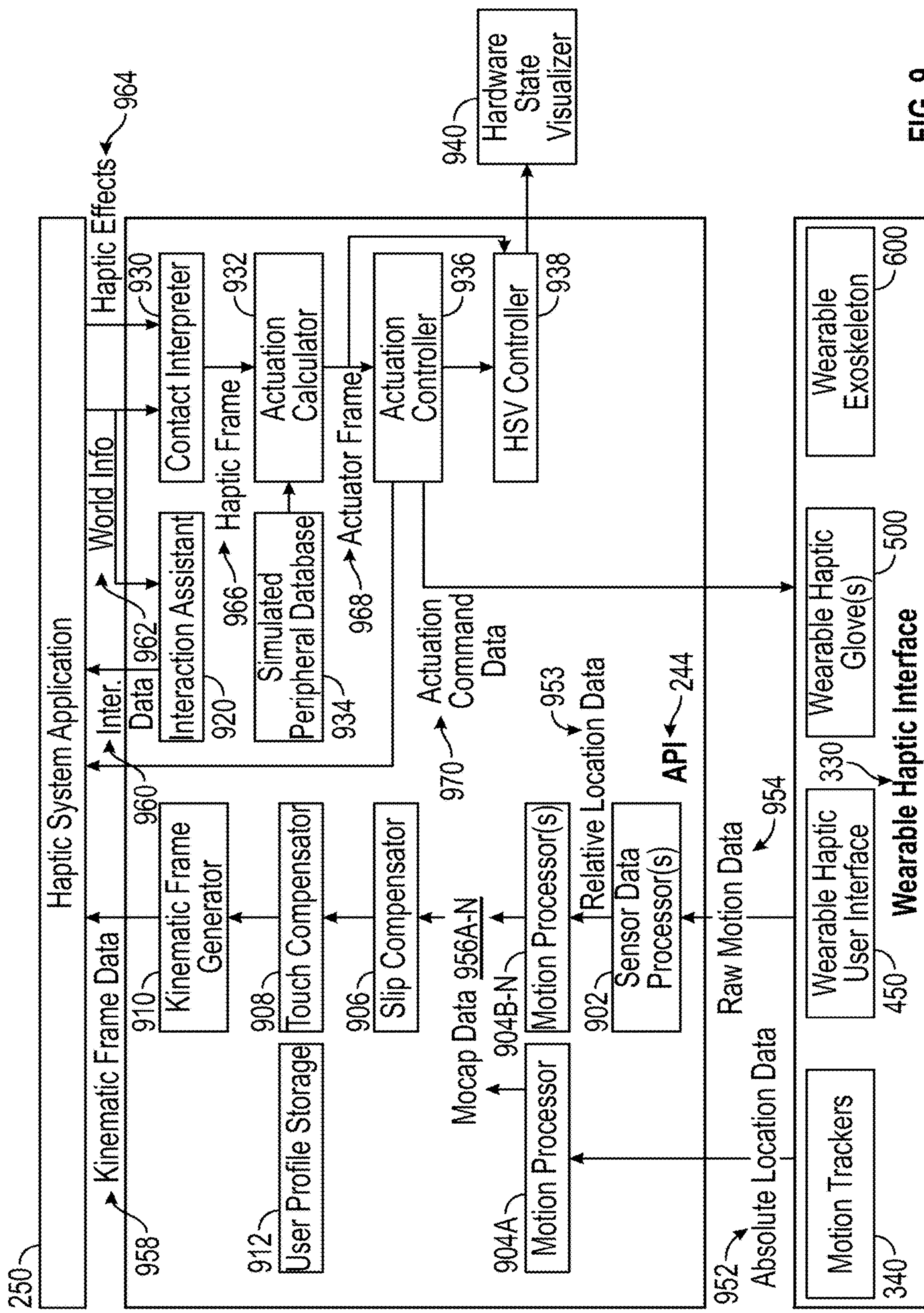

250
Haptic System Application
958
Kinematic Frame Data
960
Inter. Data
962
World Info
Haptic Effects
964
910
Kinematic Frame Generator
920
Interaction Assistant
930
Contact Interpreter
966
Haptic Frame
912
User Profile Storage
908
Touch Compensator
934
Simulated Peripheral Database
932
Actuation Calculator
968
Actuator Frame
906
Slip Compensator
936
Actuation Controller
938
HSV Controller
940
Hardware State Visualizer
970
Actuation Command Data
Mocap Data 956A-N
904B-N
Motion Processor(s)
904A
Motion Processor
953
Relative Location Data
902
Sensor Data Processor(s)
API
244
952
Absolute Location Data
954
Raw Motion Data
340
Motion Trackers
450
Wearable Haptic User Interface
330
Wearable Haptic Glove(s)
500
Wearable Exoskeleton
600
Wearable Haptic Interface

FIG. 9

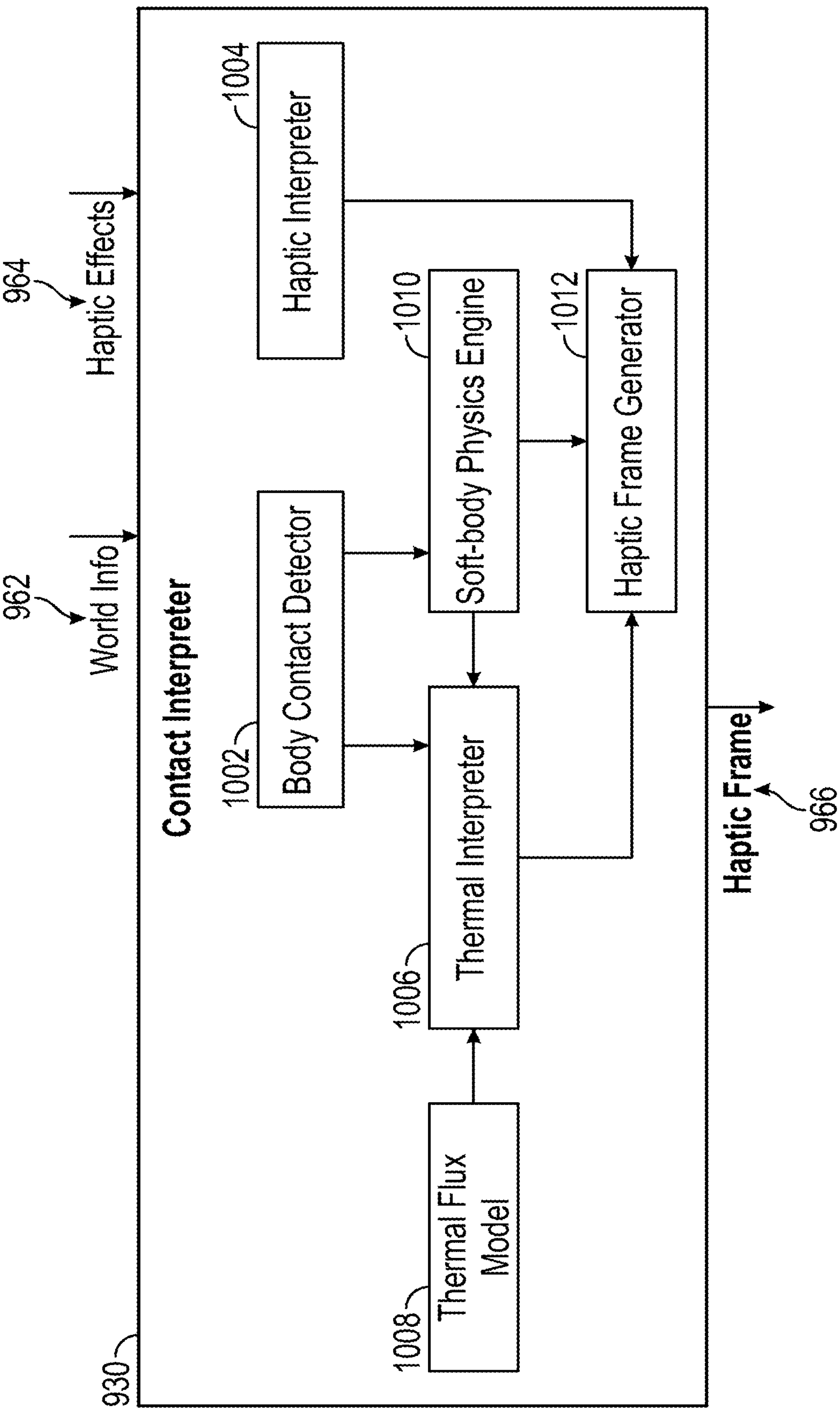

930
Contact Interpreter
962
World Info
964
Haptic Effects
1002
Body Contact Detector
1004
Haptic Interpreter
1006
Thermal Interpreter
1008
Thermal Flux Model
1010
Soft-body Physics Engine
1012
Haptic Frame Generator
Haptic Frame
966

FIG. 10

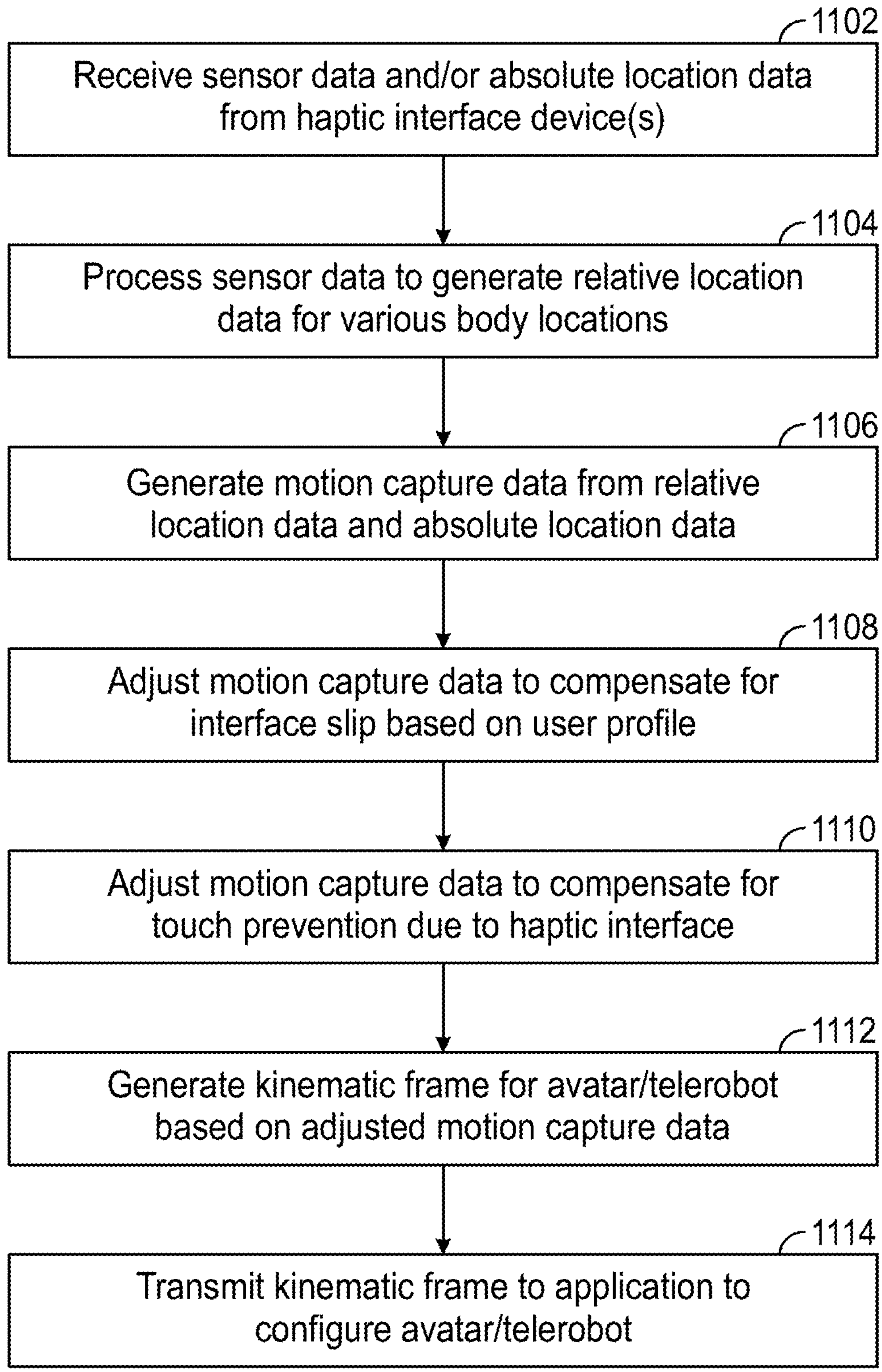

1102
Receive sensor data and/or absolute location data from haptic interface device(s)
1104
Process sensor data to generate relative location data for various body locations
1106
Generate motion capture data from relative location data and absolute location data
1108
Adjust motion capture data to compensate for interface slip based on user profile
1110
Adjust motion capture data to compensate for touch prevention due to haptic interface
1112
Generate kinematic frame for avatar/telerobot based on adjusted motion capture data
1114
Transmit kinematic frame to application to configure avatar/telerobot

FIG. 11

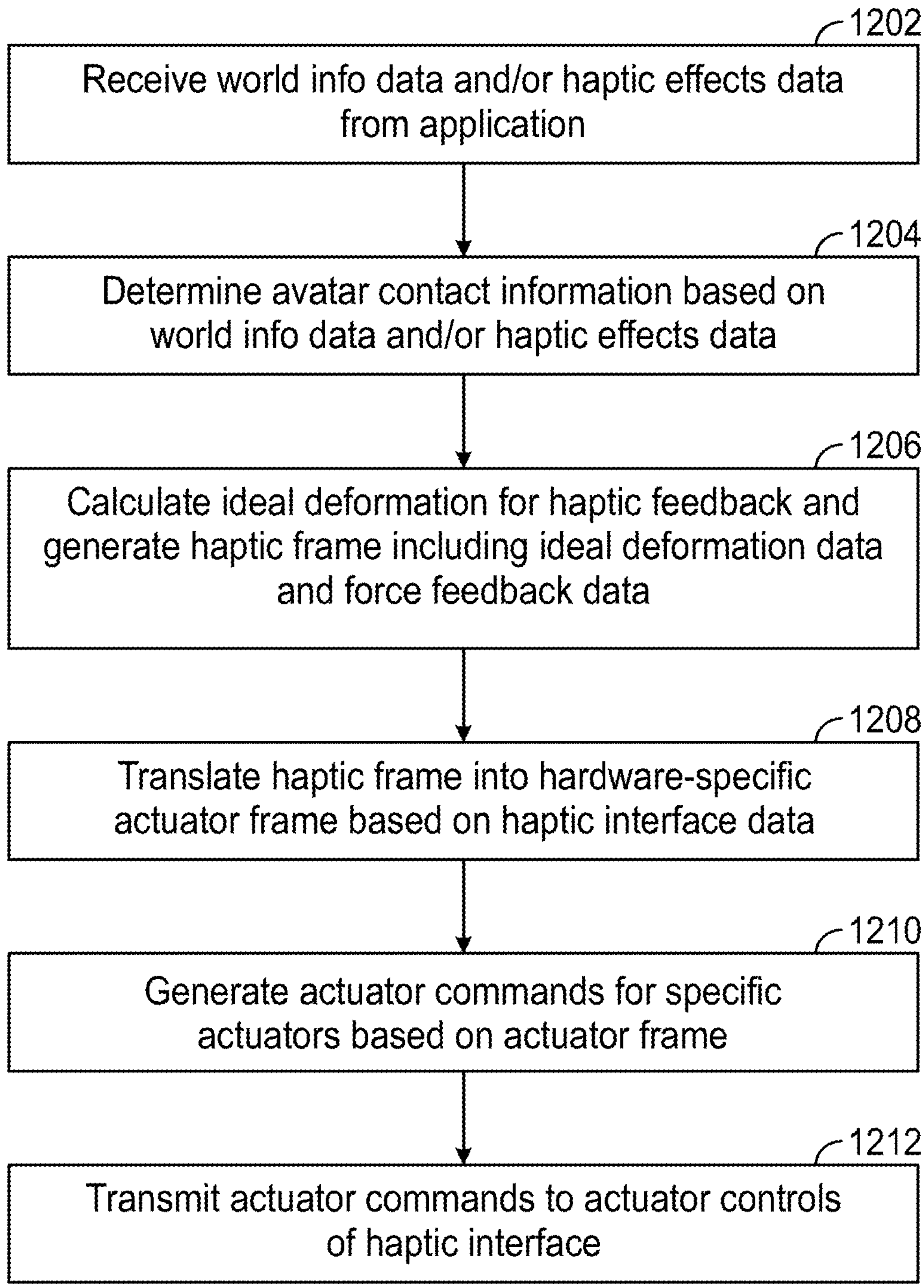

1202
Receive world info data and/or haptic effects data from application
1204
Determine avatar contact information based on world info data and/or haptic effects data
1206
Calculate ideal deformation for haptic feedback and generate haptic frame including ideal deformation data and force feedback data
1208
Translate haptic frame into hardware-specific actuator frame based on haptic interface data
1210
Generate actuator commands for specific actuators based on actuator frame
1212
Transmit actuator commands to actuator controls of haptic interface

FIG. 12

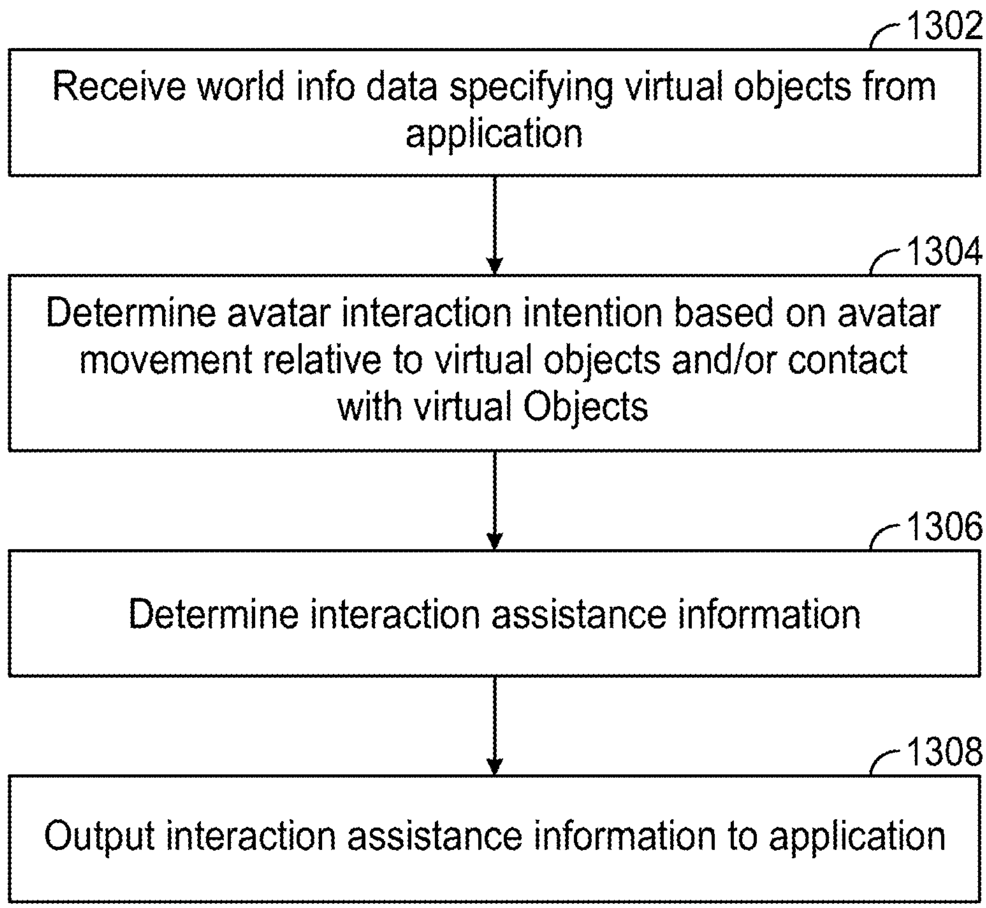

1302
Receive world info data specifying virtual objects from application
1304
Determine avatar interaction intention based on avatar movement relative to virtual objects and/or contact with virtual Objects
1306
Determine interaction assistance information
1308
Output interaction assistance information to application

FIG. 13

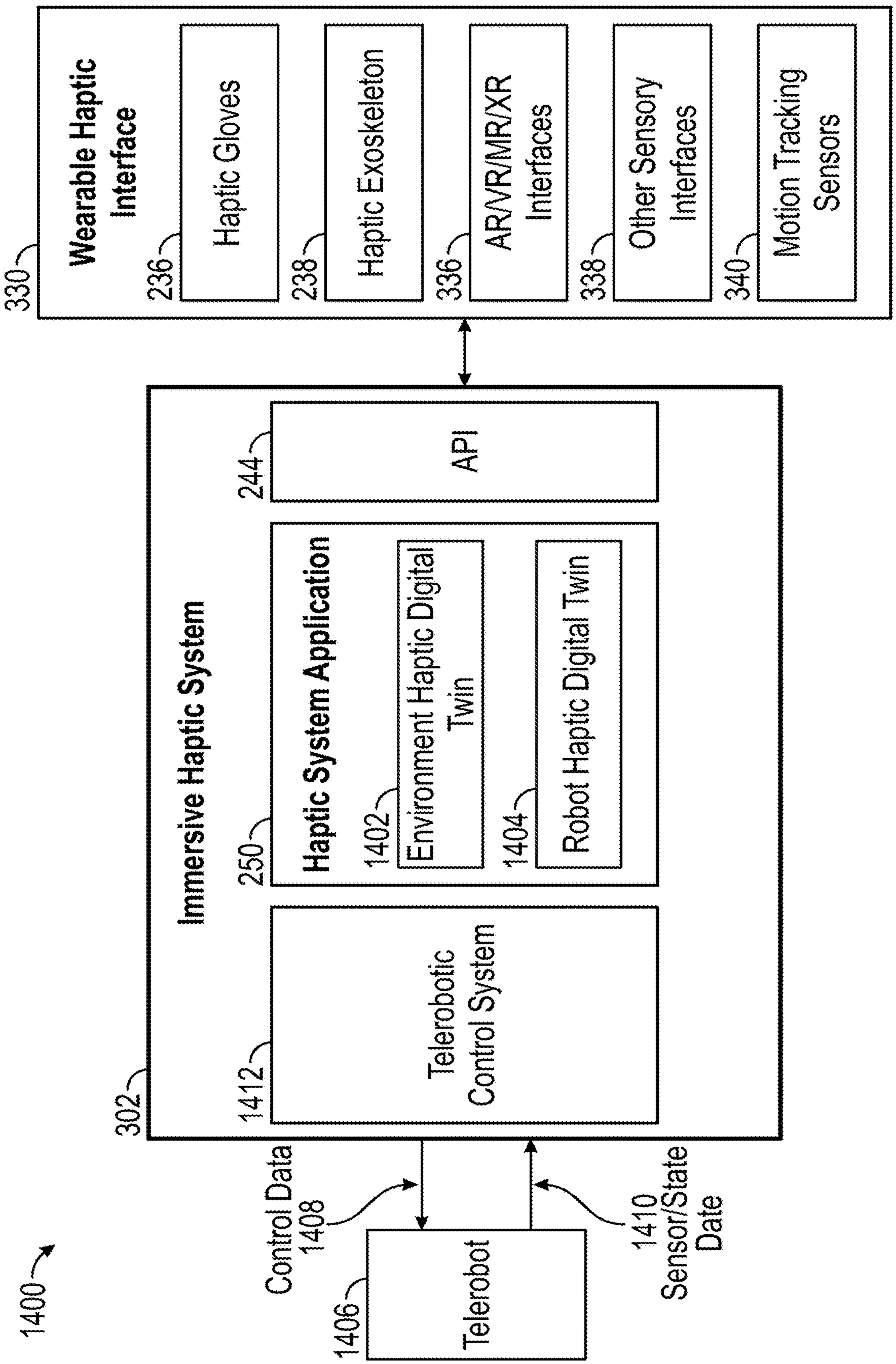

1400
330
Wearable Haptic Interface
236
Haptic Gloves
238
Haptic Exoskeleton
336
AR/VR/MR/XR Interfaces
338
Other Sensory Interfaces
340
Motion Tracking Sensors
302
Immersive Haptic System
244
API
250
Haptic System Application
1402
Environment Haptic Digital Twin
1404
Robot Haptic Digital Twin
1412
Telerobotic Control System
1406
Telerobot
Control Data
1408
1410
Sensor/State Date

FIG. 14

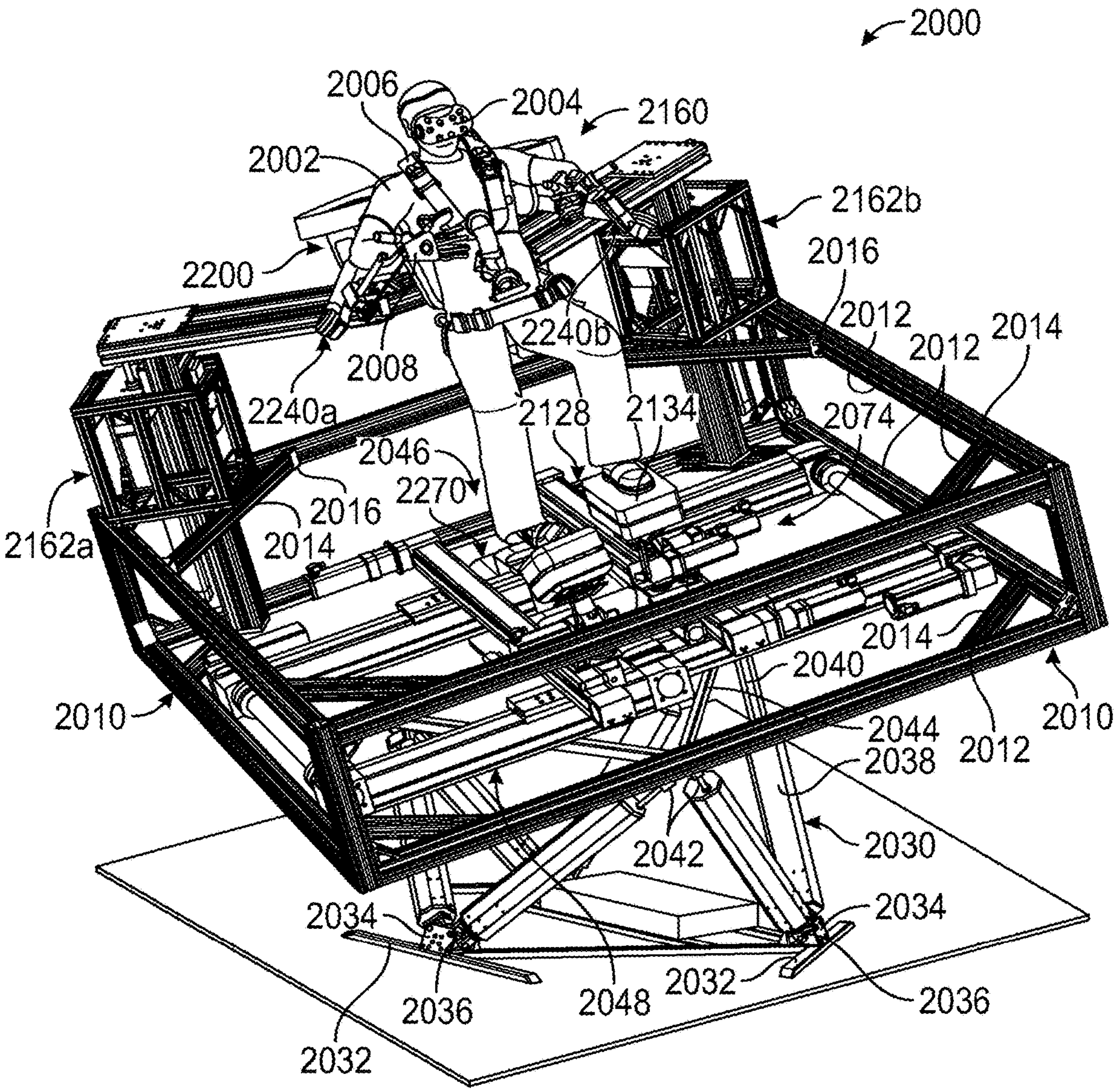

2000
2006
2004
2160
2002
2162b
2016
2200
2012
2012
2014
2240b
2008
2240a
2128
2134
2074
2046
2270
2016
2162a
2014
2014
2040
2010
2010
2044
2012
2038
2030
2042
2034
2034
2032
2036
2048
2036
2032

FIG. 15

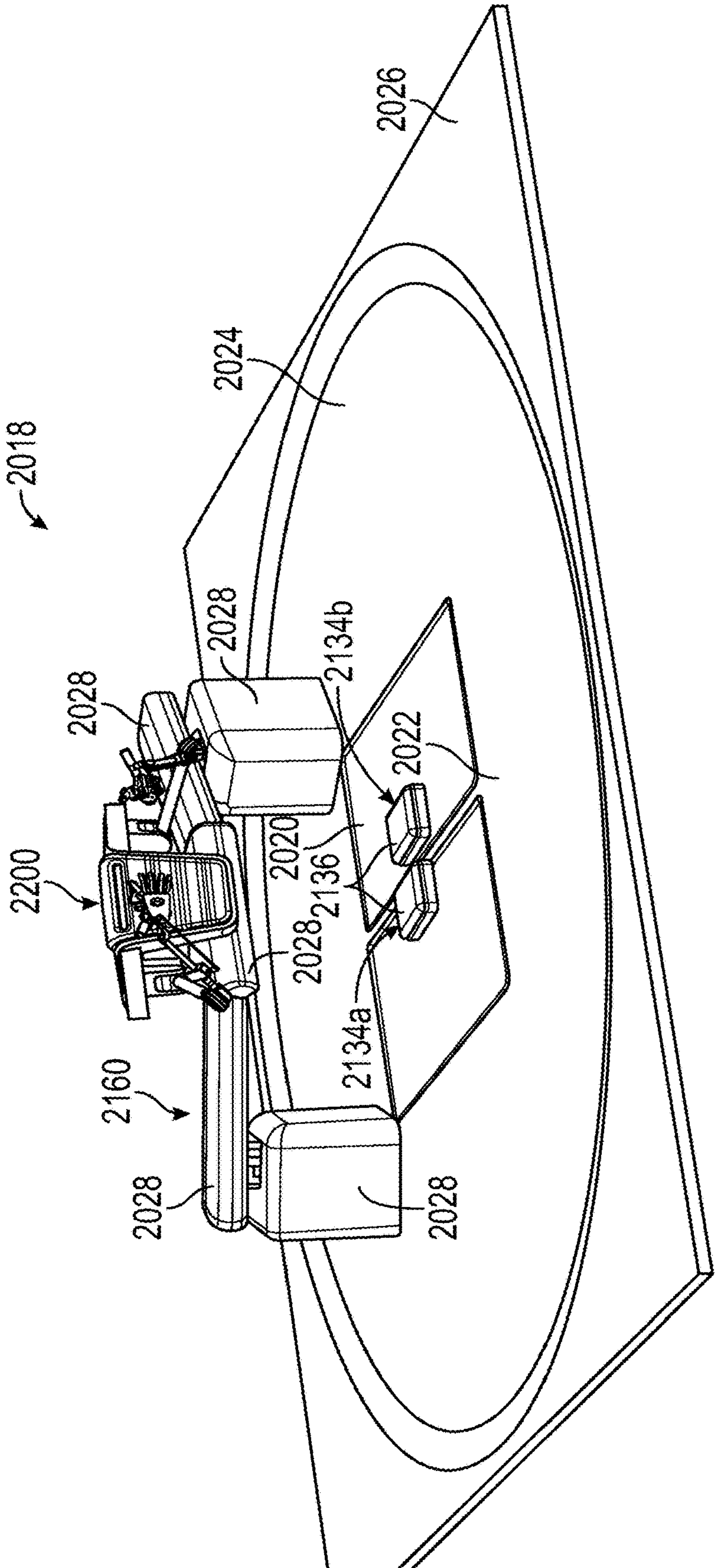

2018
2026
2024
2028
2028
2134b
2022
2200
2020
2136
2028
2134a
2160
2028
2028

FIG. 16

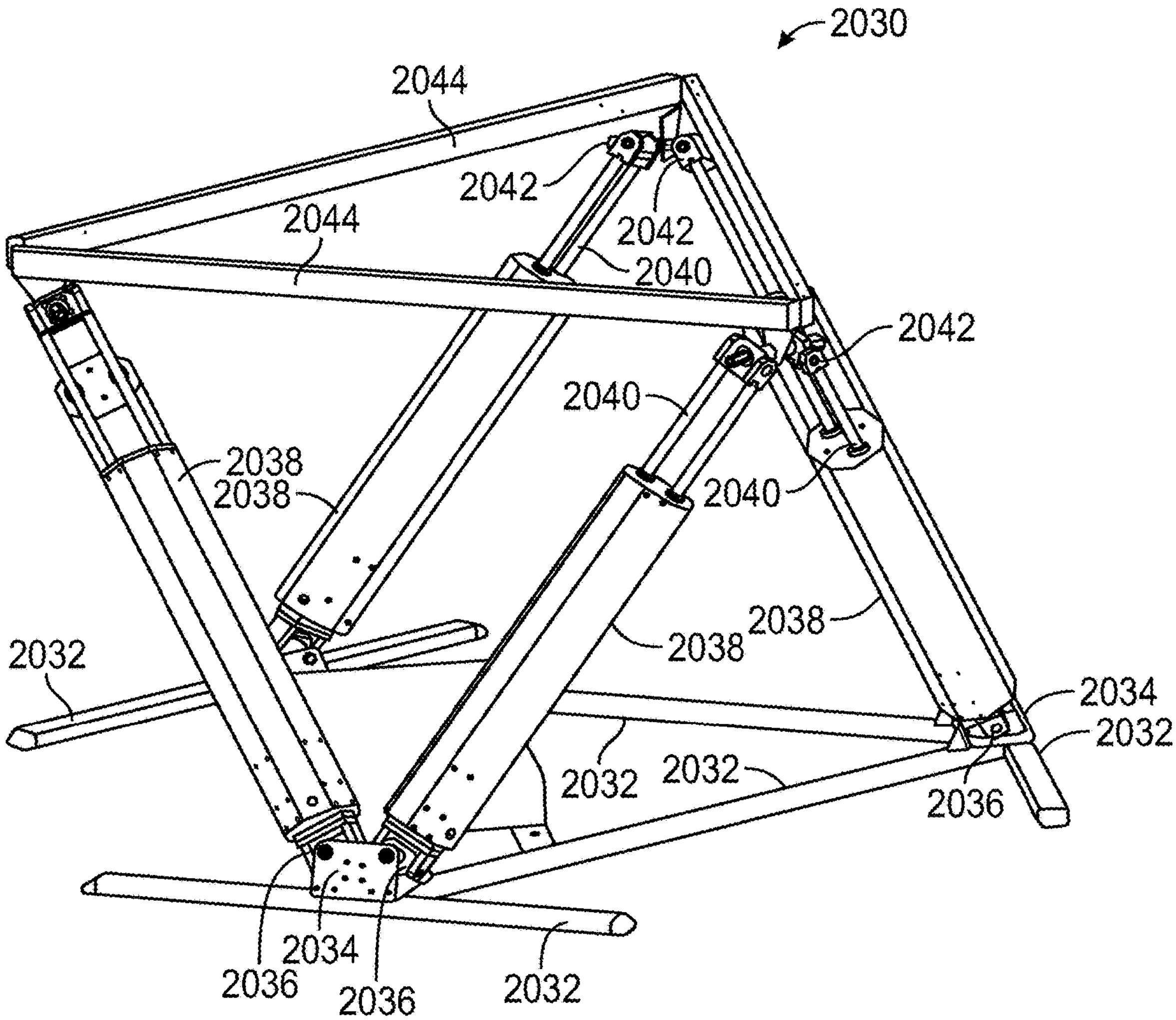

2030
2044
2044
2042
2042
2040
2042
2040
2040
2038
2038
2038
2038
2038
2032
2034
2032
2032
2032
2036
2034
2036
2036
2032

FIG. 17

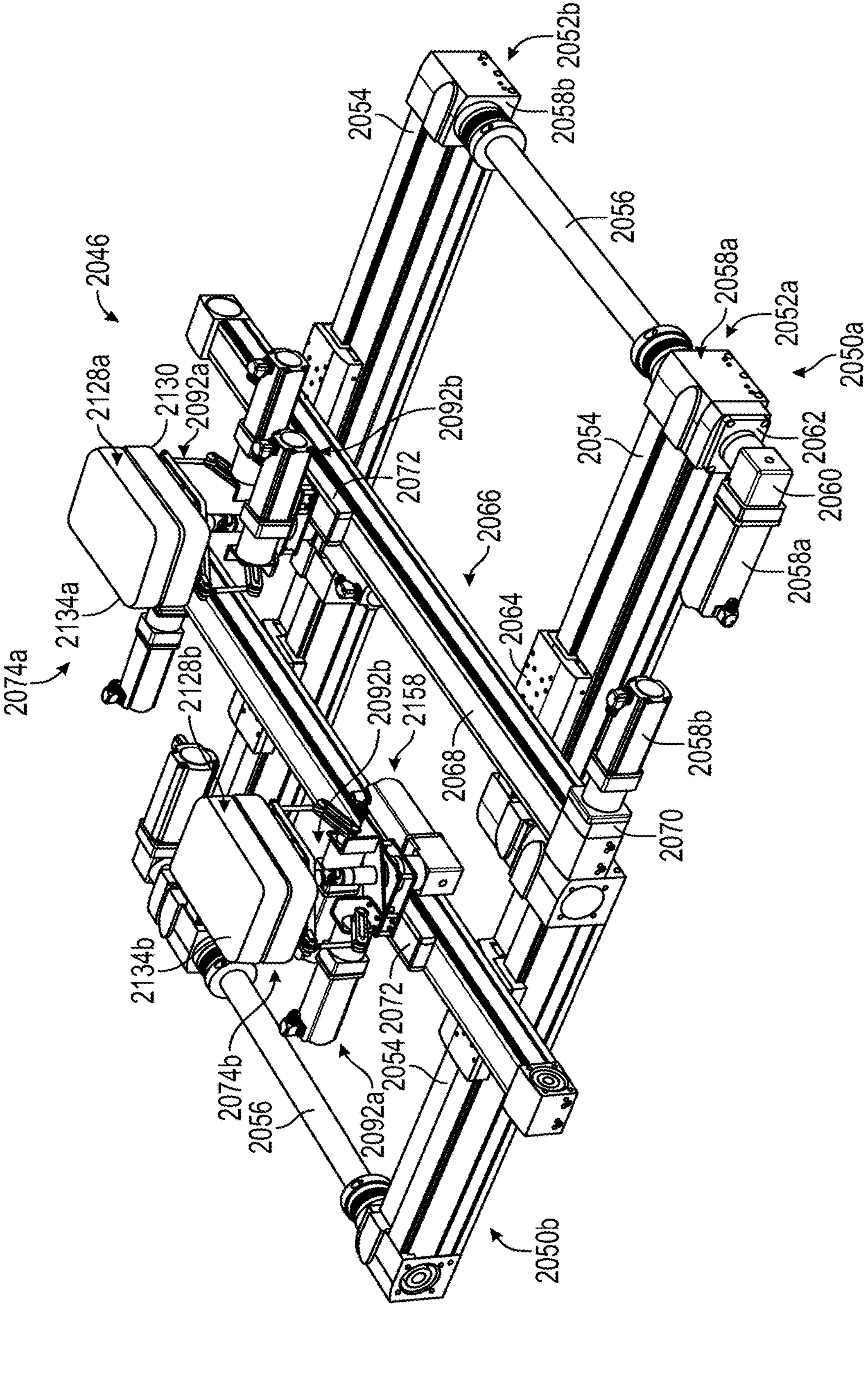

2046
2052b
2058b
2054
2056
2058a
2052a
2050a
2128a
2130
2092a
2092b
2062
2072
2060
2066
2134a
2064
2074a
2128b
2092b
2158
2068
2058b
2070
2134b
2074b
2054
2072
2092a
2056
2050b

FIG. 18

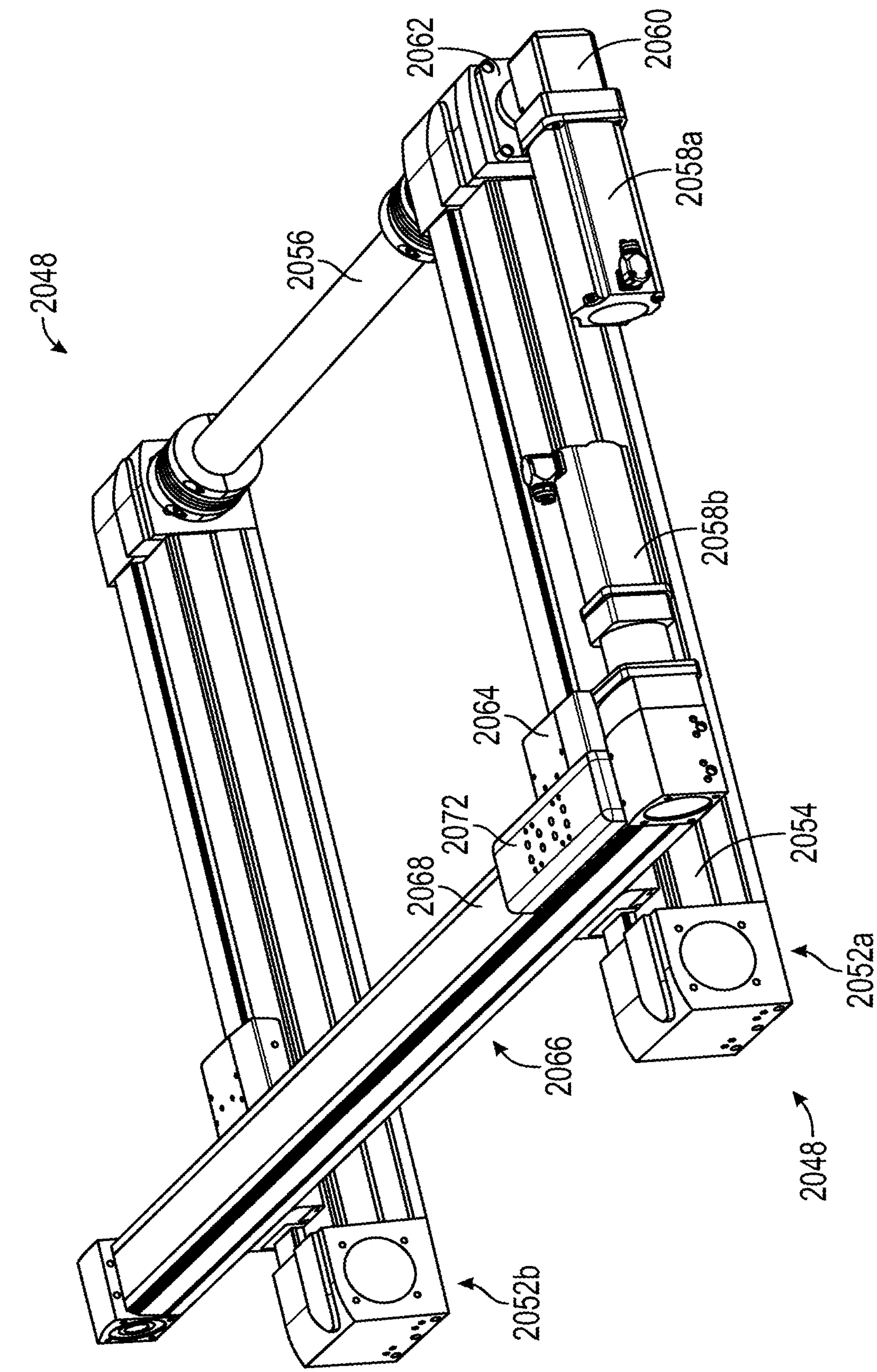

2048
2056
2062
2060
2058a
2058b
2064
2072
2068
2054
2052a
2066
2048
2052b

FIG. 19

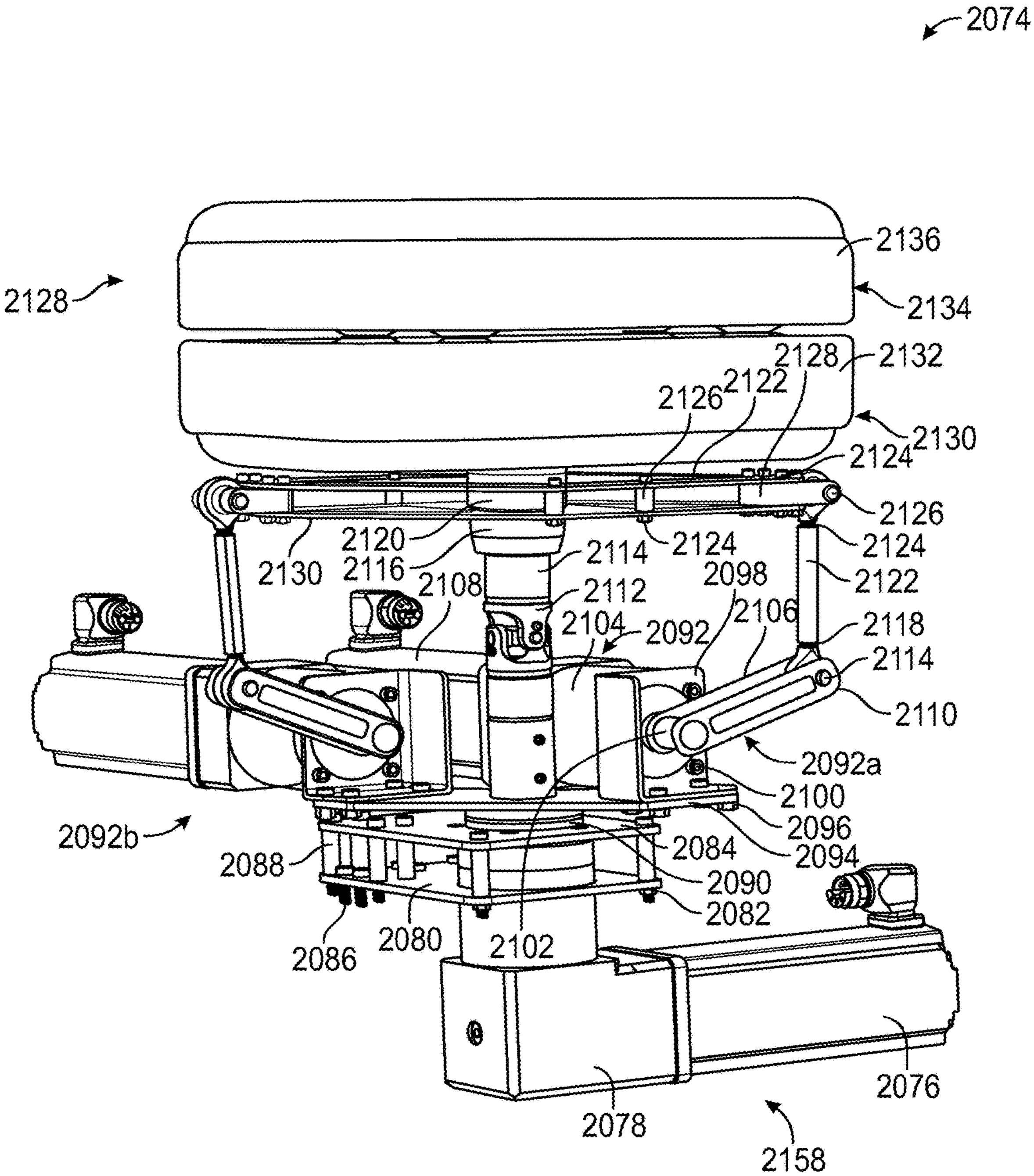

2074
2128
2136
2134
2128
2132
2126
2122
2130
2124
2126
2120
2124
2124
2116
2130
2114
2122
2108
2112
2098
2104
2092
2106
2118
2114
2110
2092a
2100
2092b
2096
2084
2094
2088
2090
2082
2080
2102
2086
2076
2078
2158

FIG. 20A

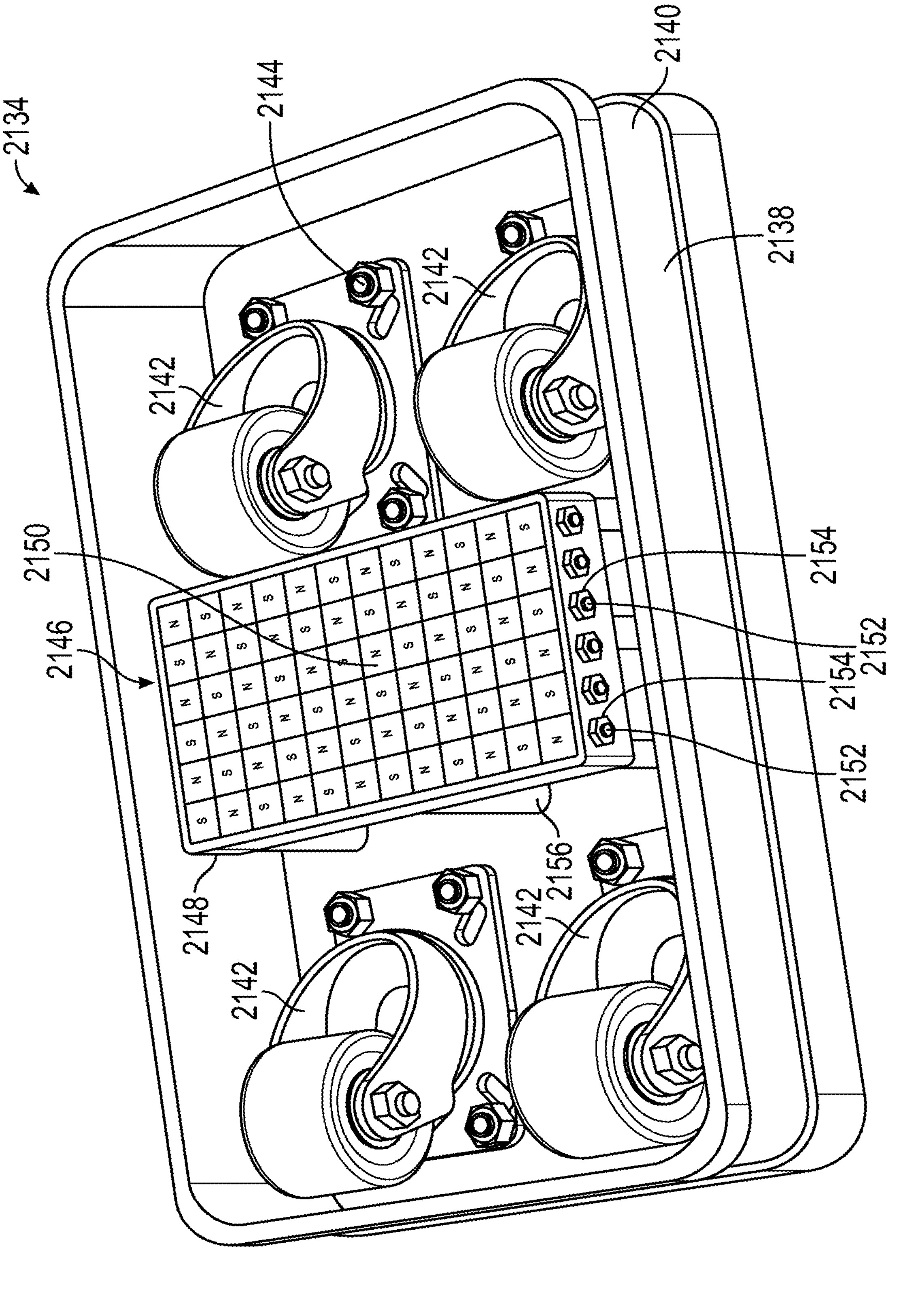

2134
2144
2140
2138
2142
2142
2150
2146
2154
2152
2154
2152
2156
2142
2148
2142

FIG. 20B

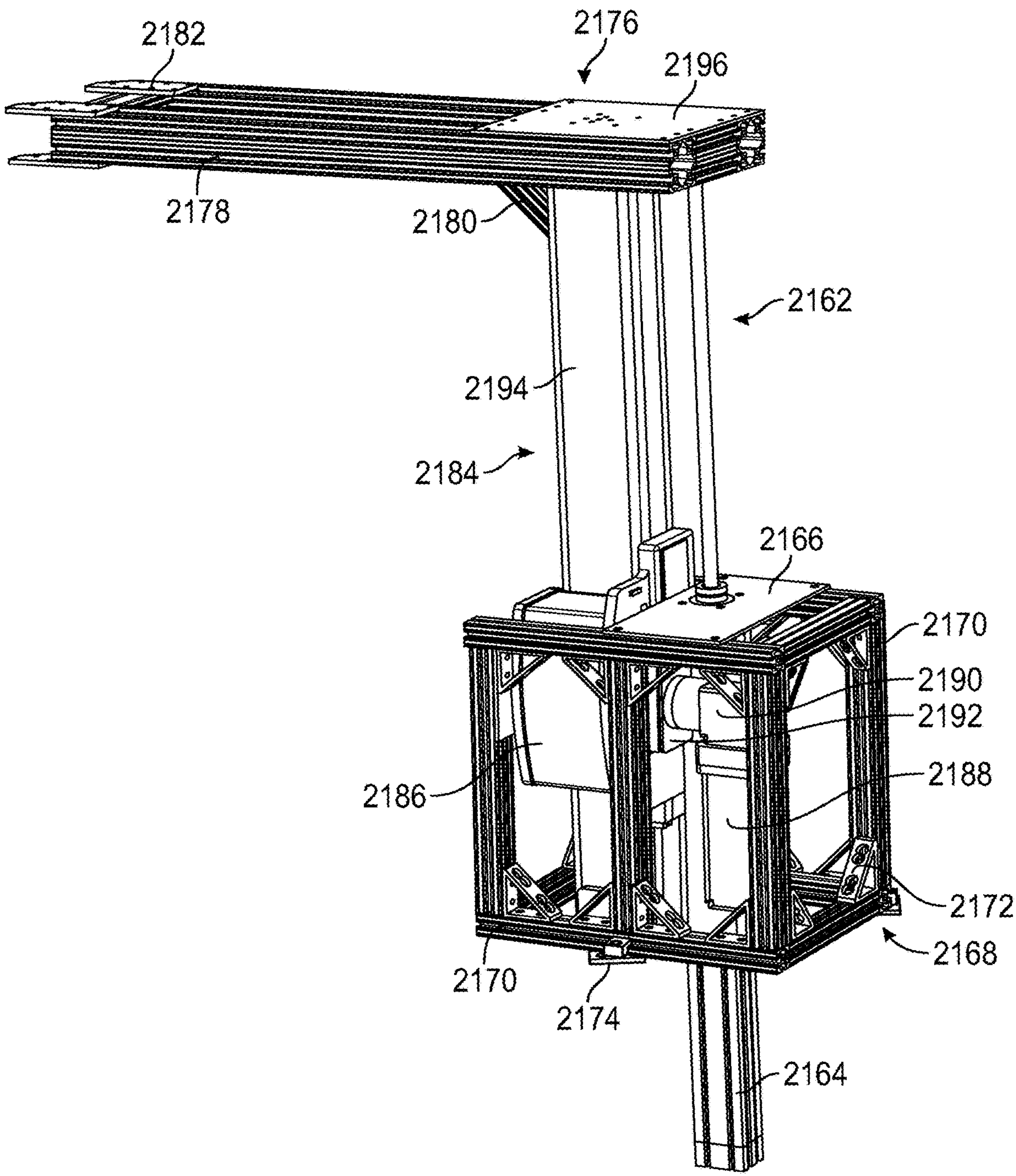

2182
2176
2196
2178
2180
2162
2194
2184
2166
2170
2190
2192
2186
2188
2172
2170
2168
2174
2164

FIG. 21A

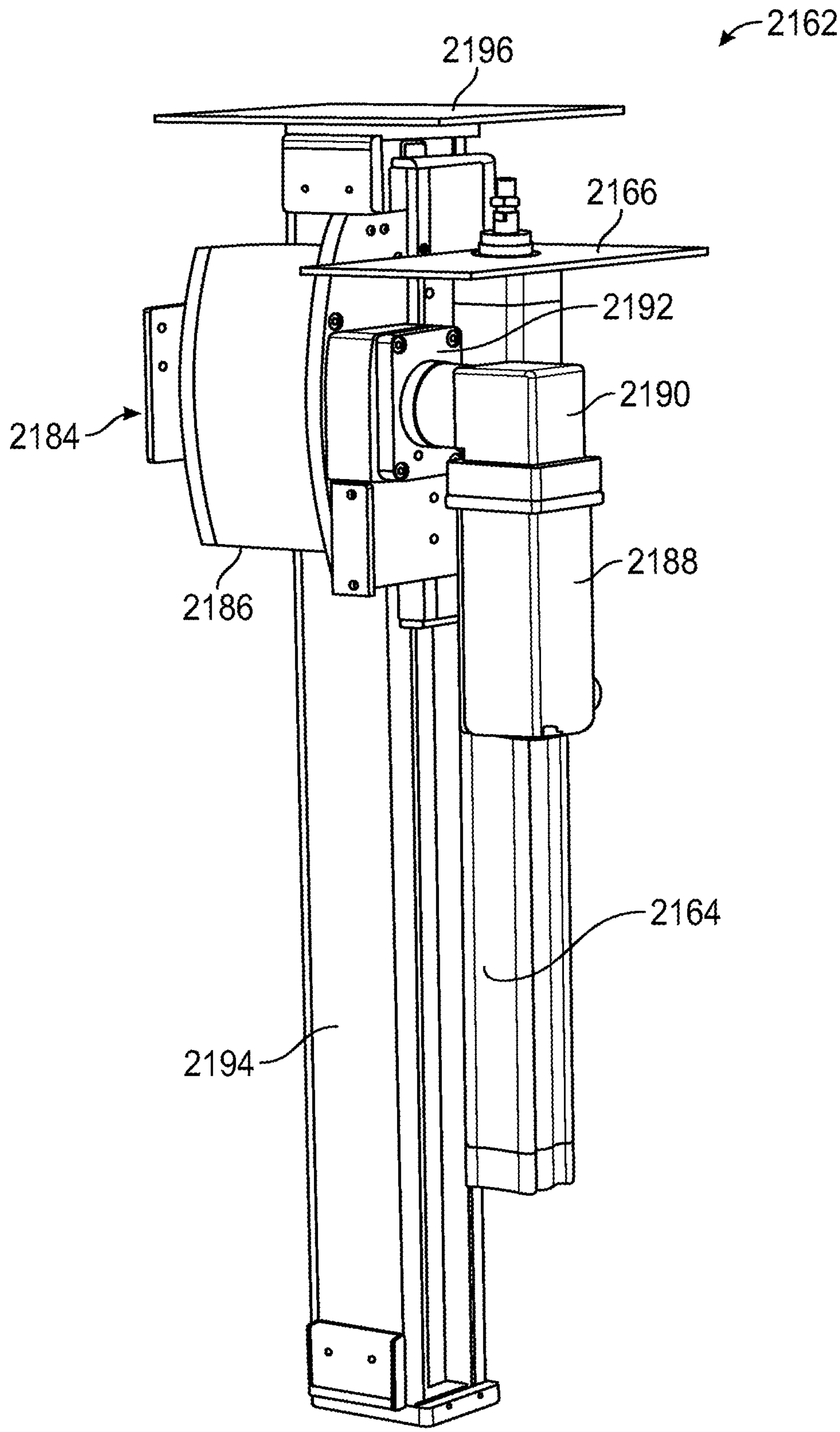


FIG. 21B

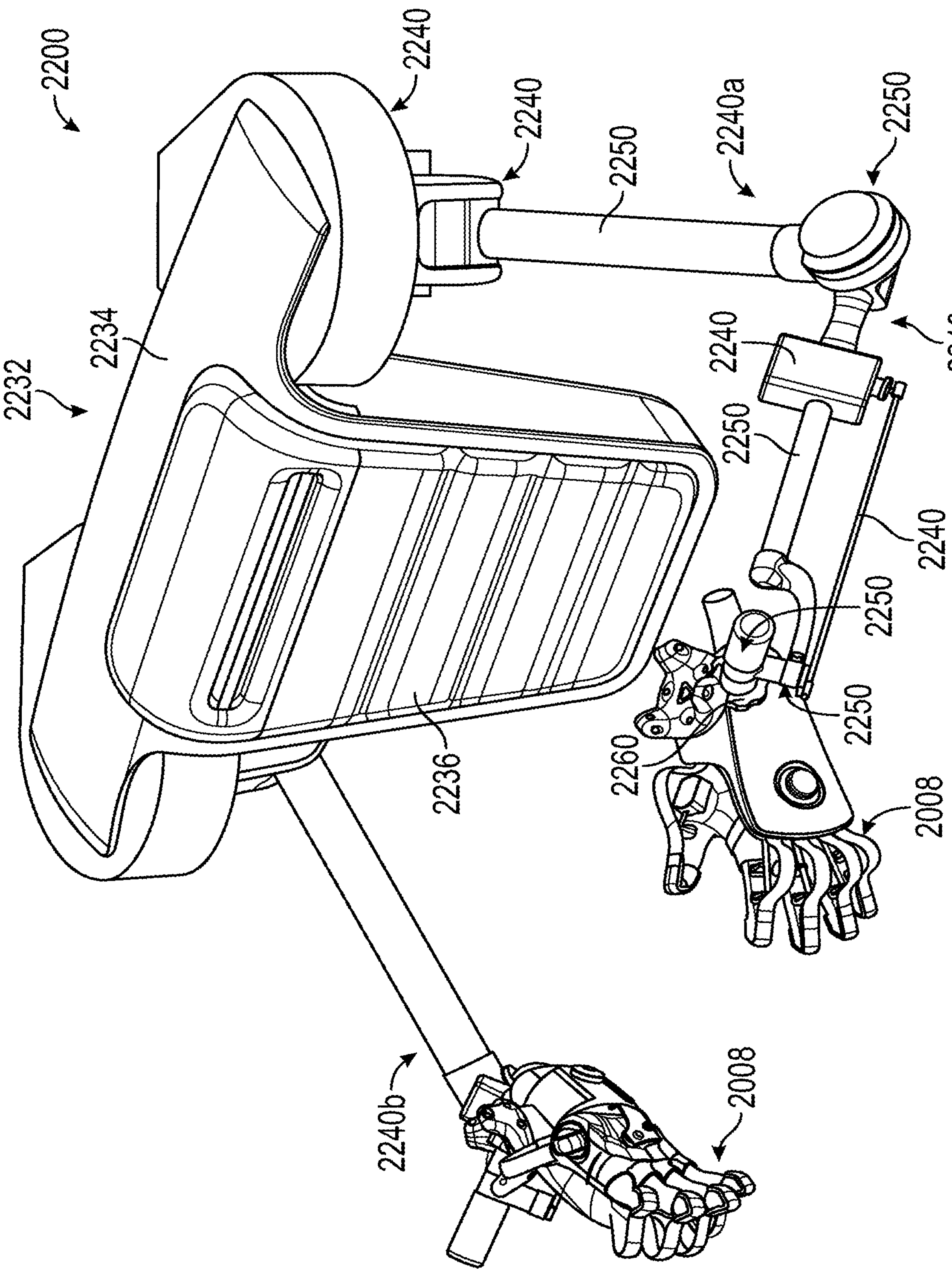

2200
2240
2240
2250
2240a
2250
2232
2234
2240
2240
2250
2240
2250
2250
2236
2260
2008
2240b
2008

FIG. 22A

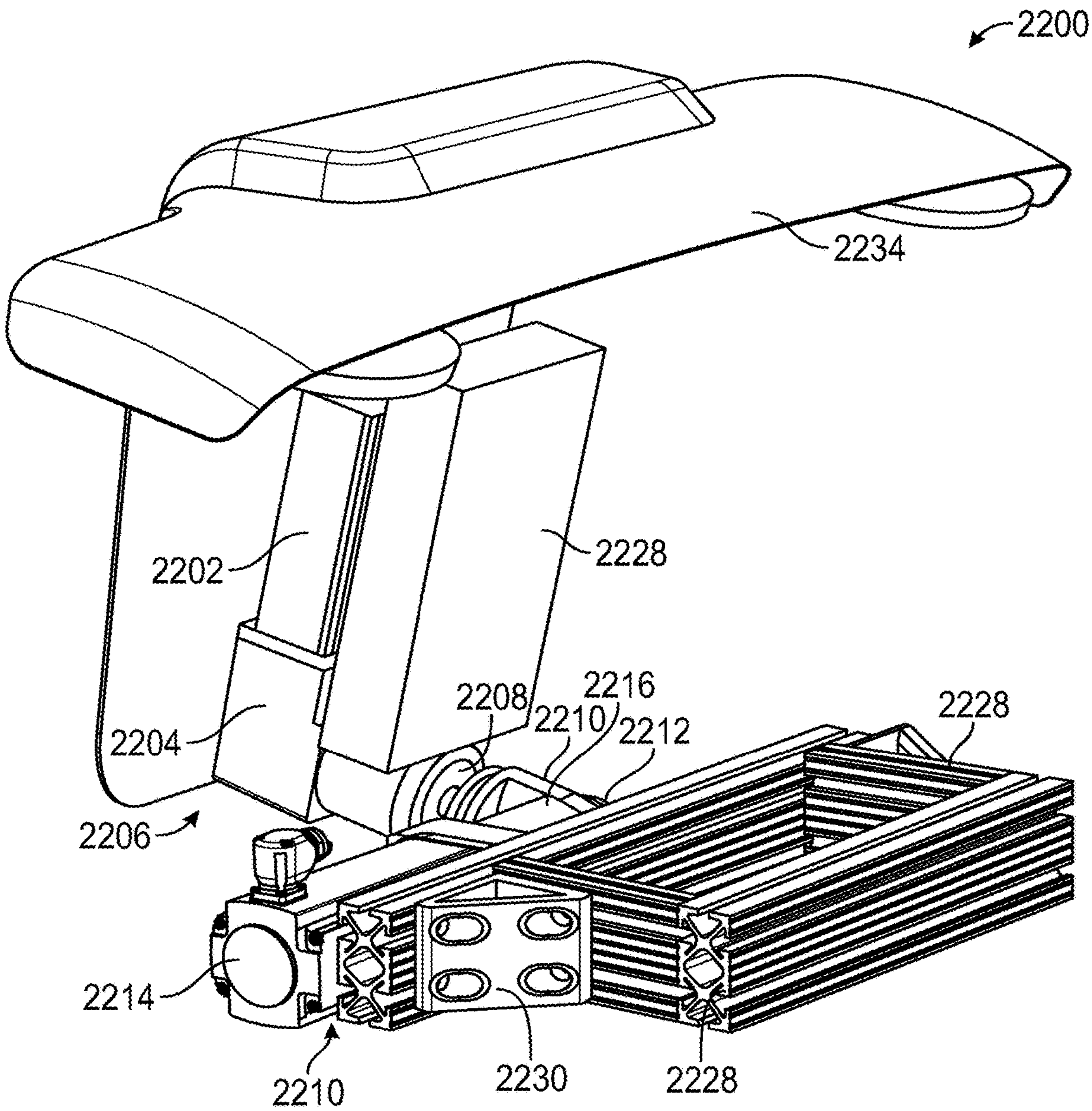

2200
2234
2202
2228
2204
2208
2216
2210
2212
2228
2206
2214
2210
2230
2228

FIG. 22B

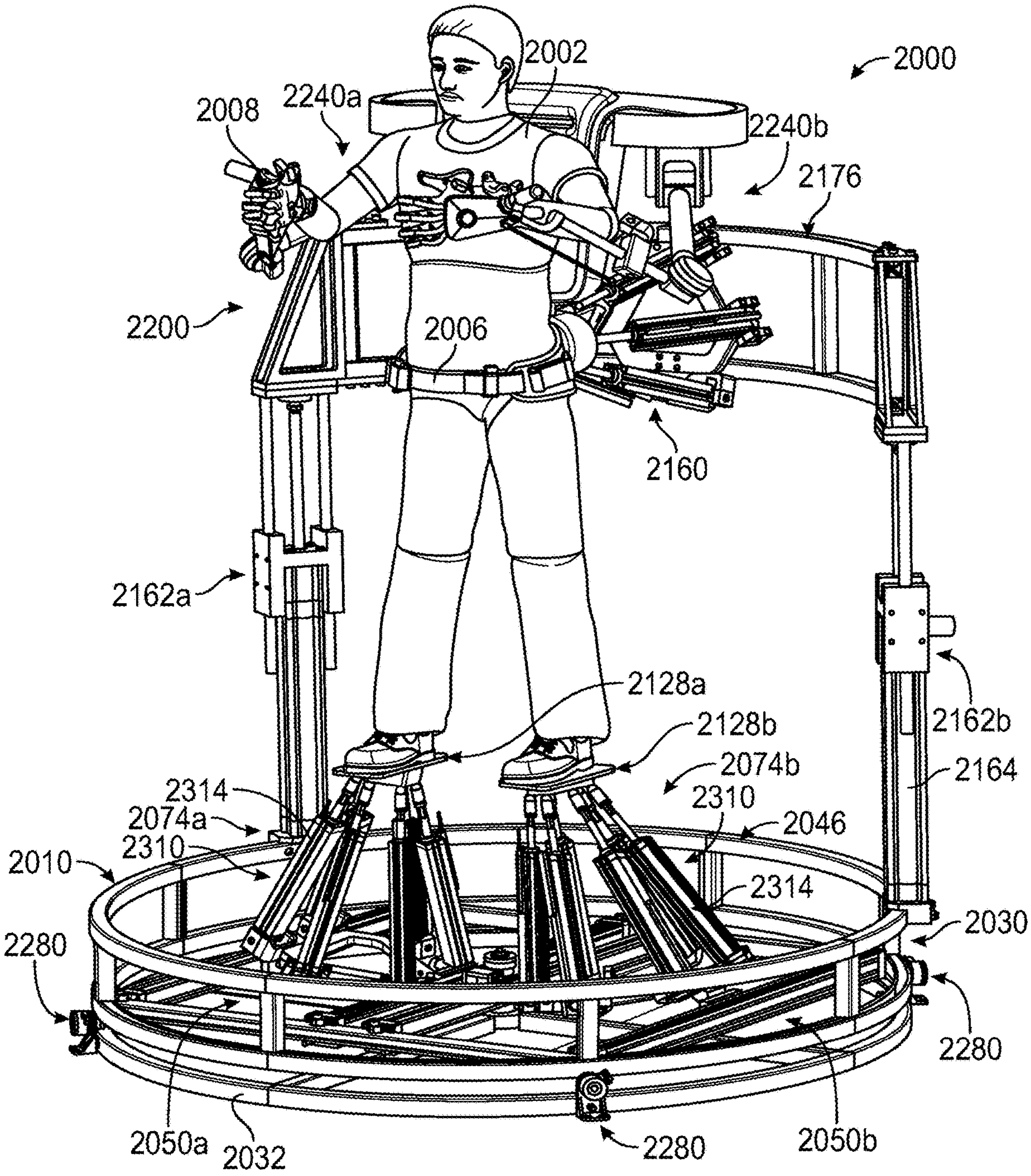

2000
2002
2008
2240a
2240b
2176
2200
2006
2160
2162a
2162b
2128a
2128b
2074b
2164
2314
2074a
2310
2046
2010
2310
2314
2030
2280
2280
2050a
2032
2280
2050b

FIG. 23

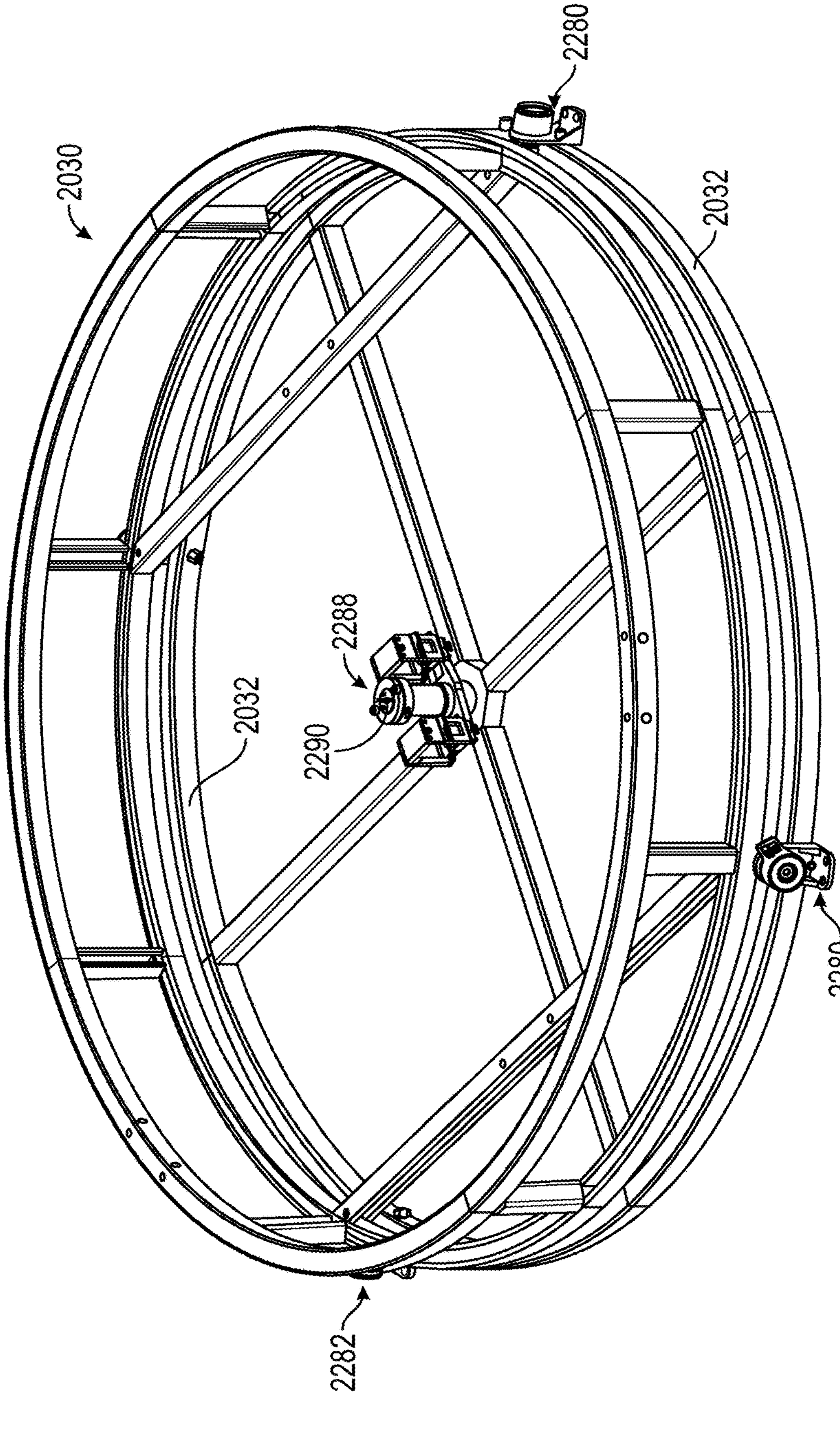


FIG. 24

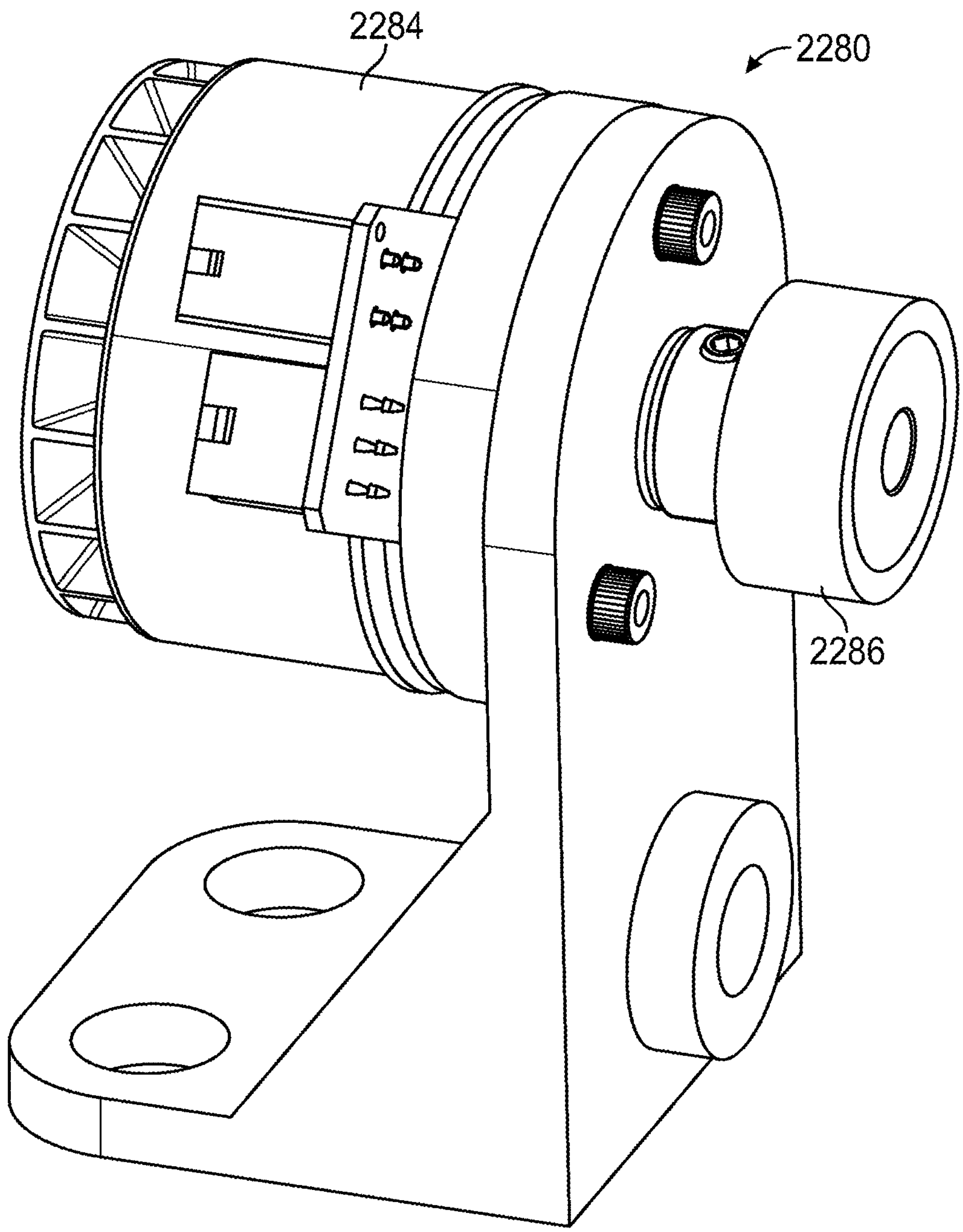


FIG. 25

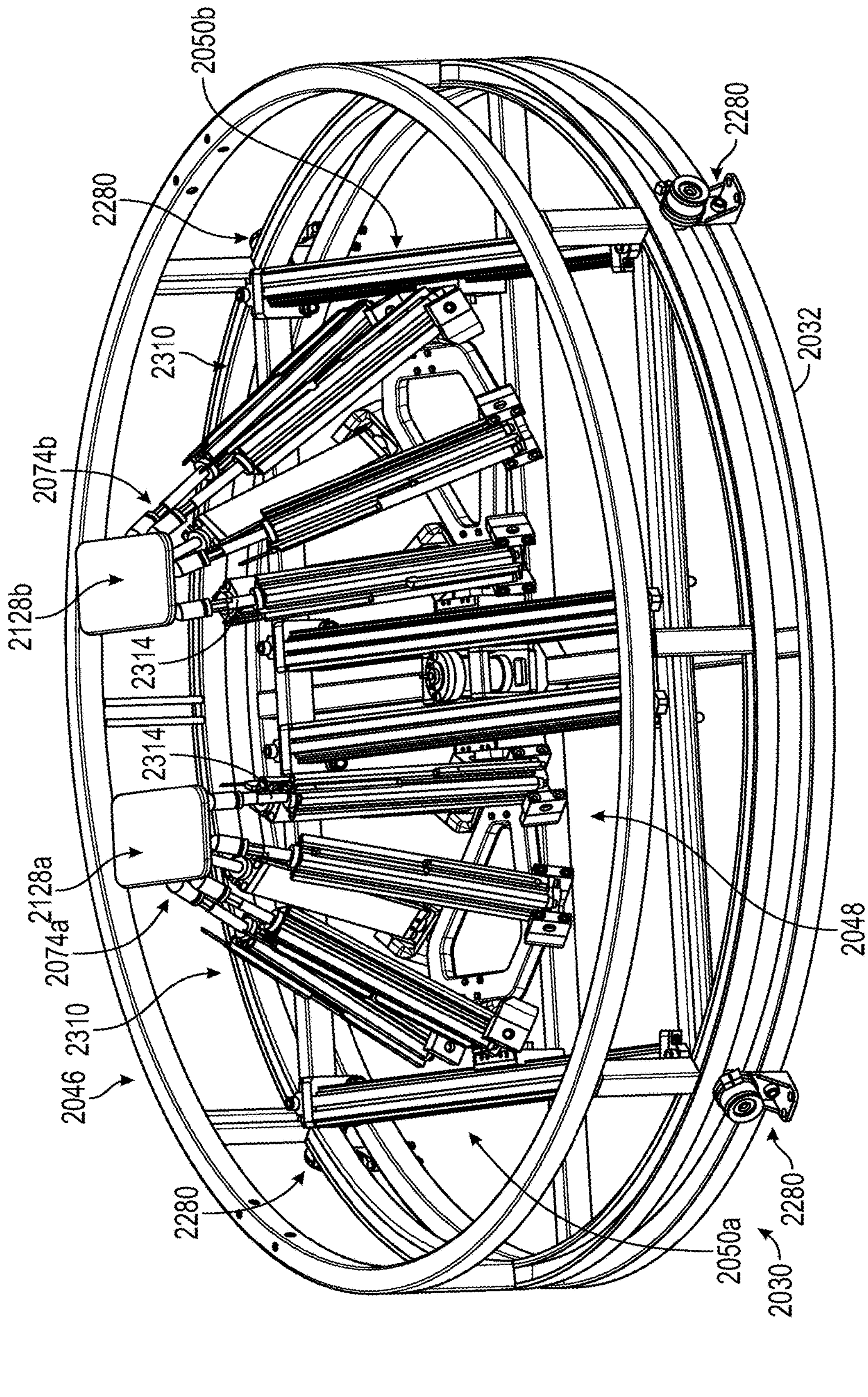

2050b
2280
2280
2310
2074b
2128b
2314
2314
2128a
2074a
2310
2046
2280
2050a
2030
2280
2048
2032
2280

FIG. 26

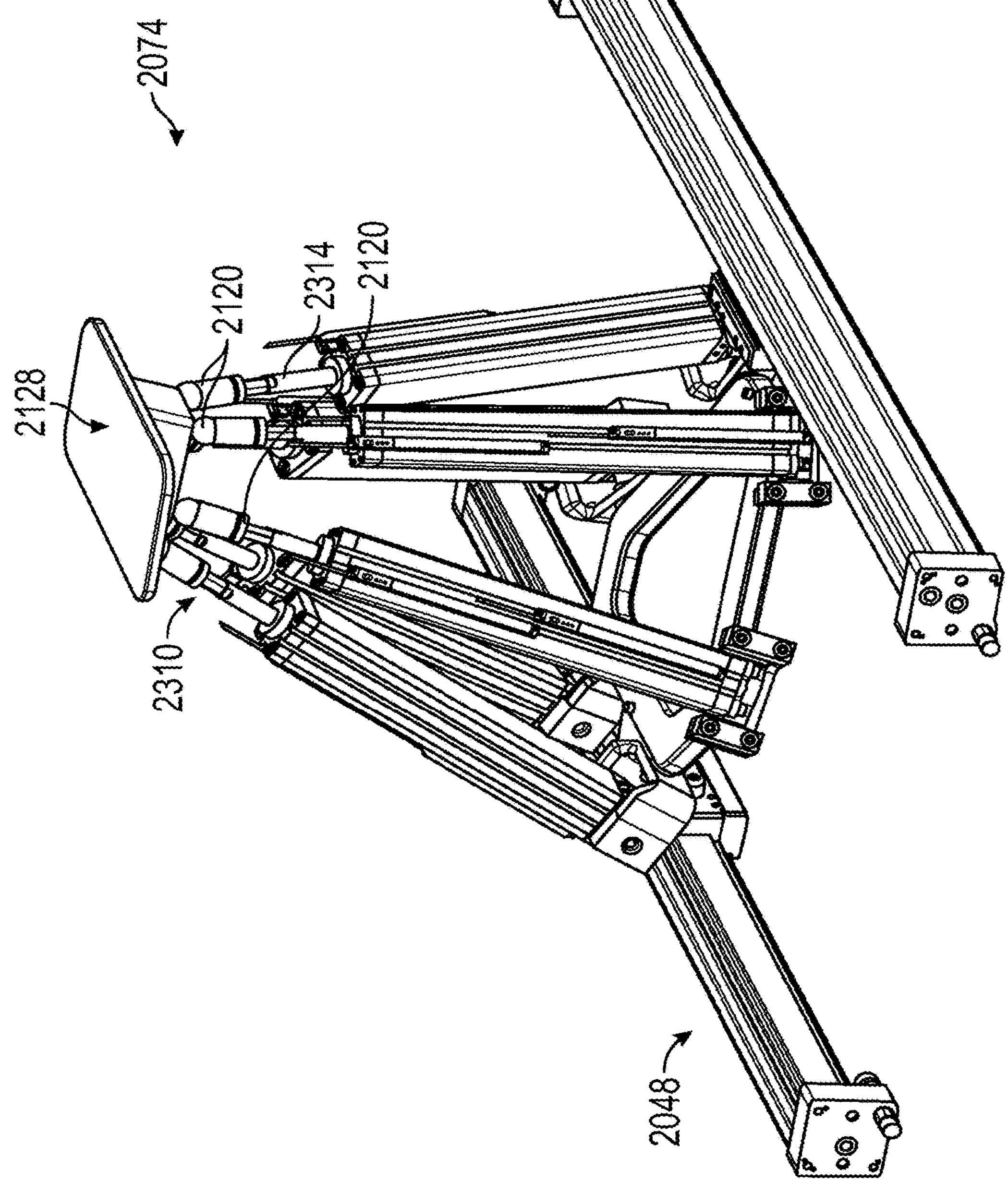


FIG. 27

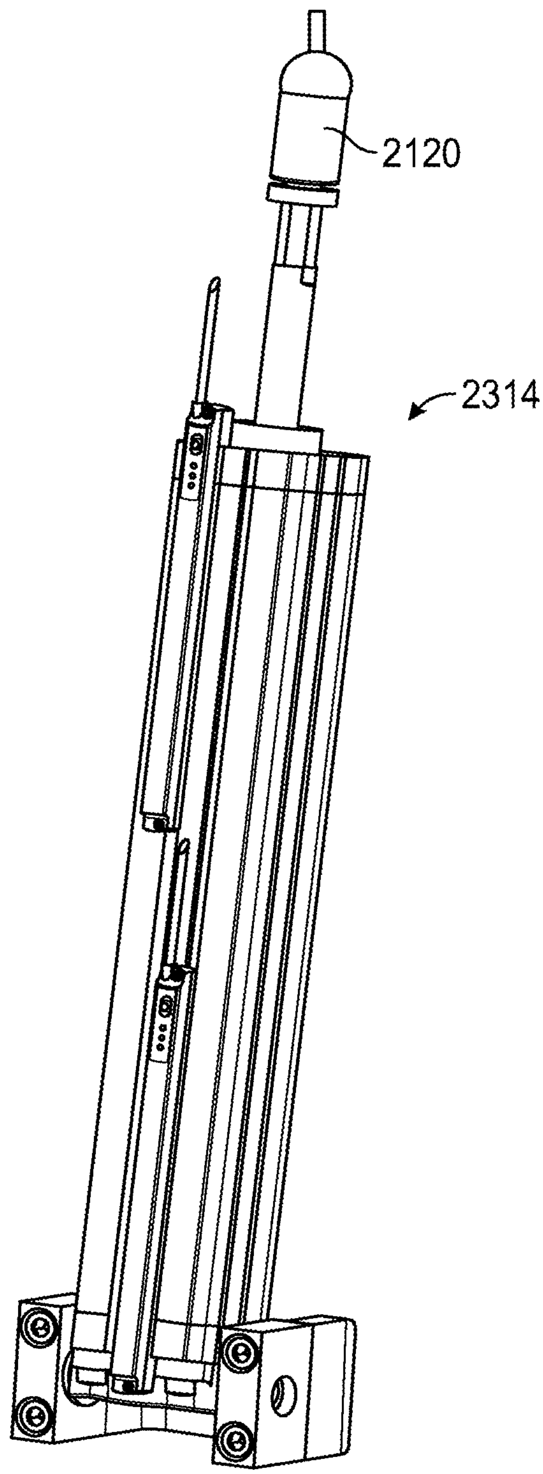


FIG. 28

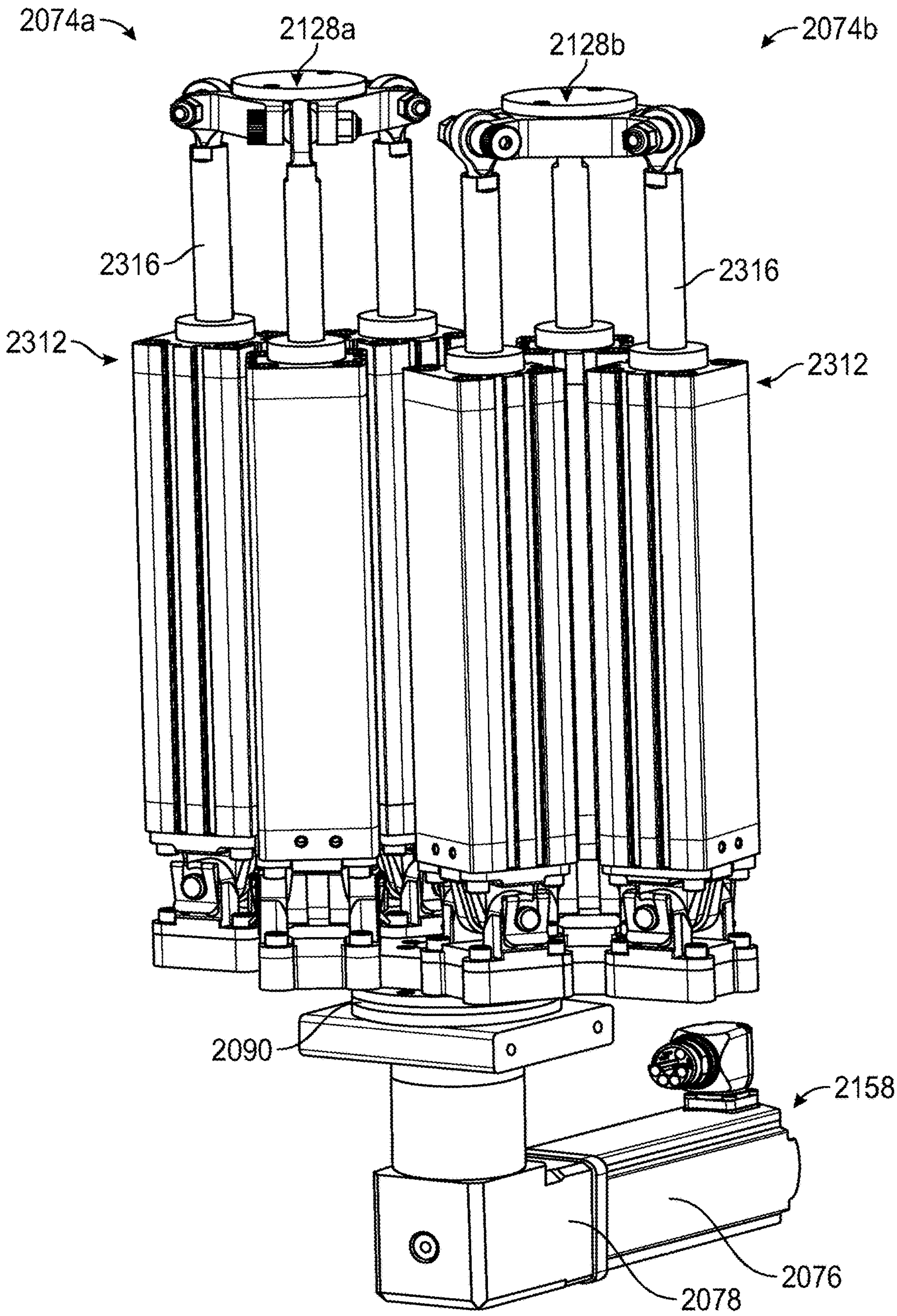

2074a
2128a
2128b
2074b
2316
2316
2312
2312
2090
2158
2076
2078

FIG. 29

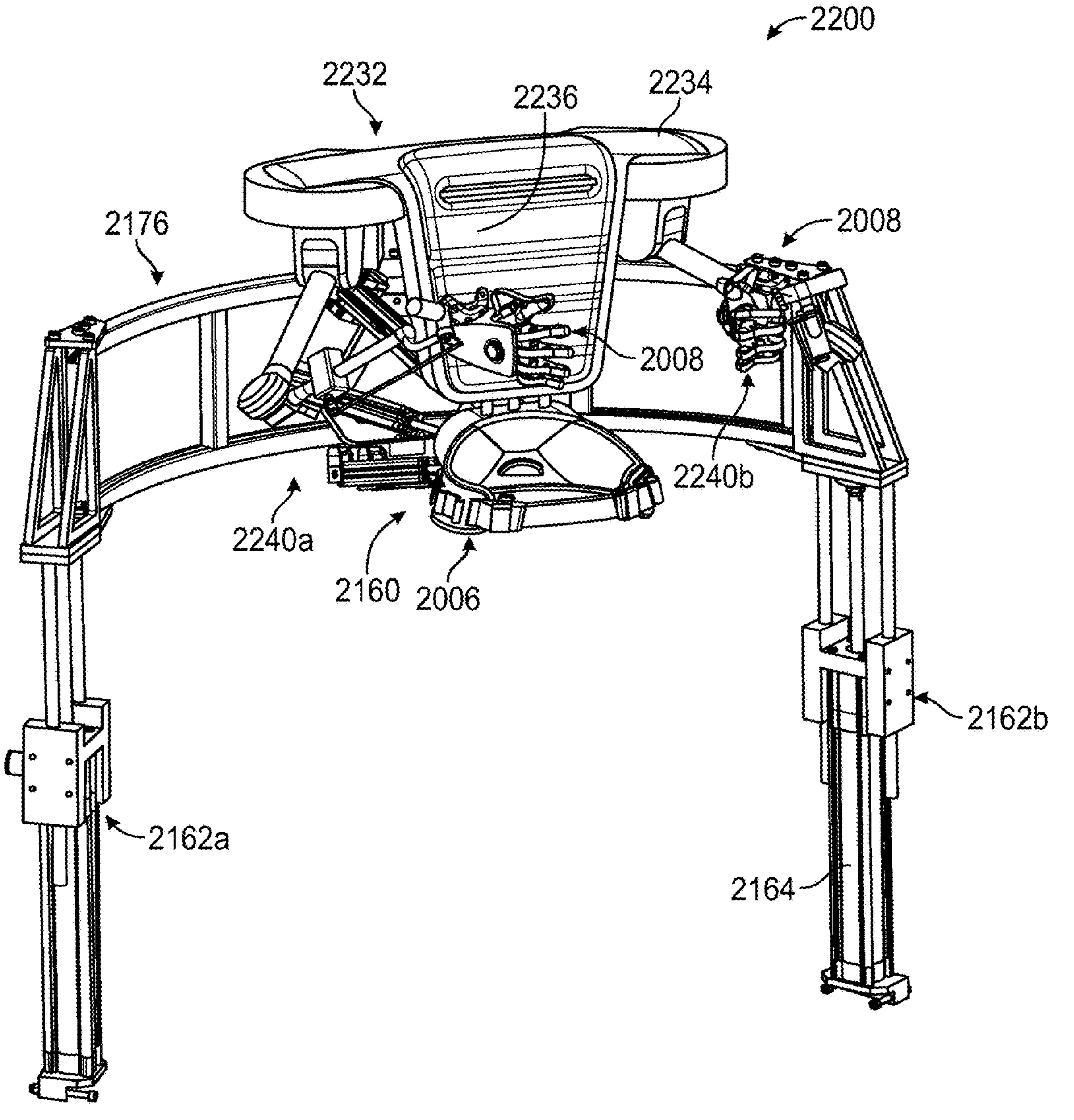

2200
2232
2236
2234
2176
2008
2008
2240b
2240a
2160
2006
2162b
2162a
2164

FIG. 30

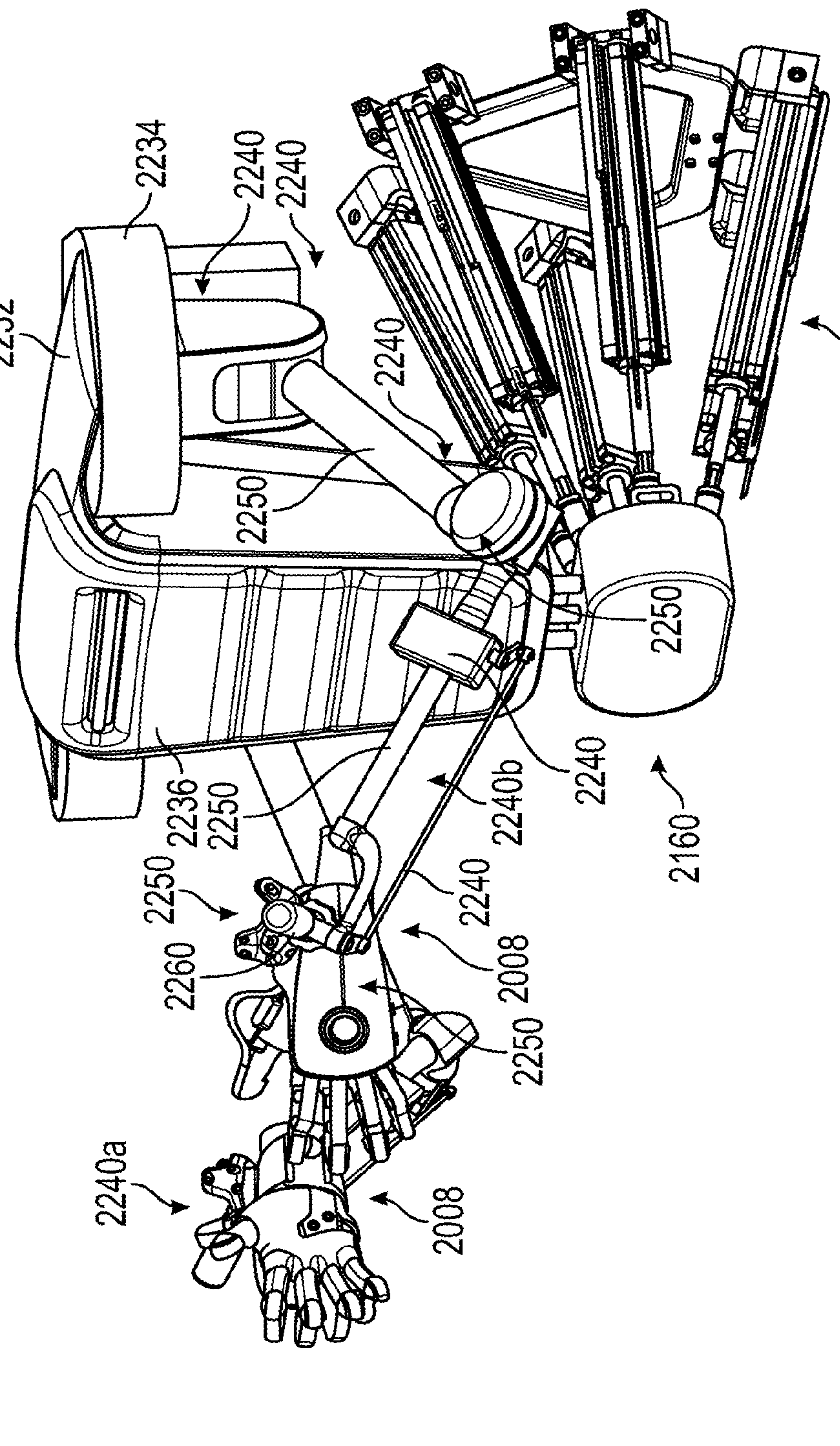

2200
2232
2234
2240
2240
2240
2250
2250
2300
2250
2240
2160
2236
2250
2240b
2240
2250
2260
2008
2250
2240a
2008

FIG. 31

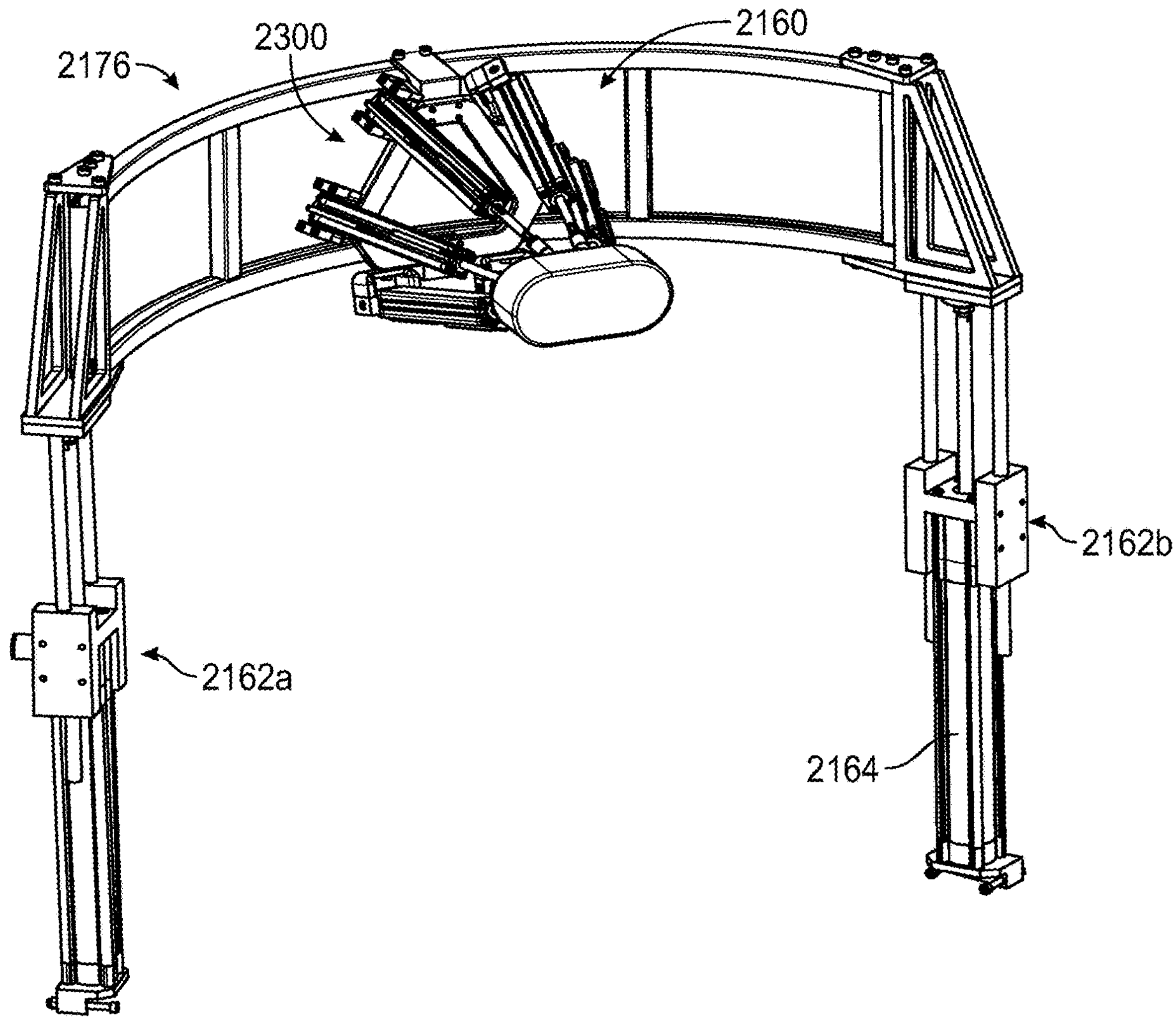


FIG. 32

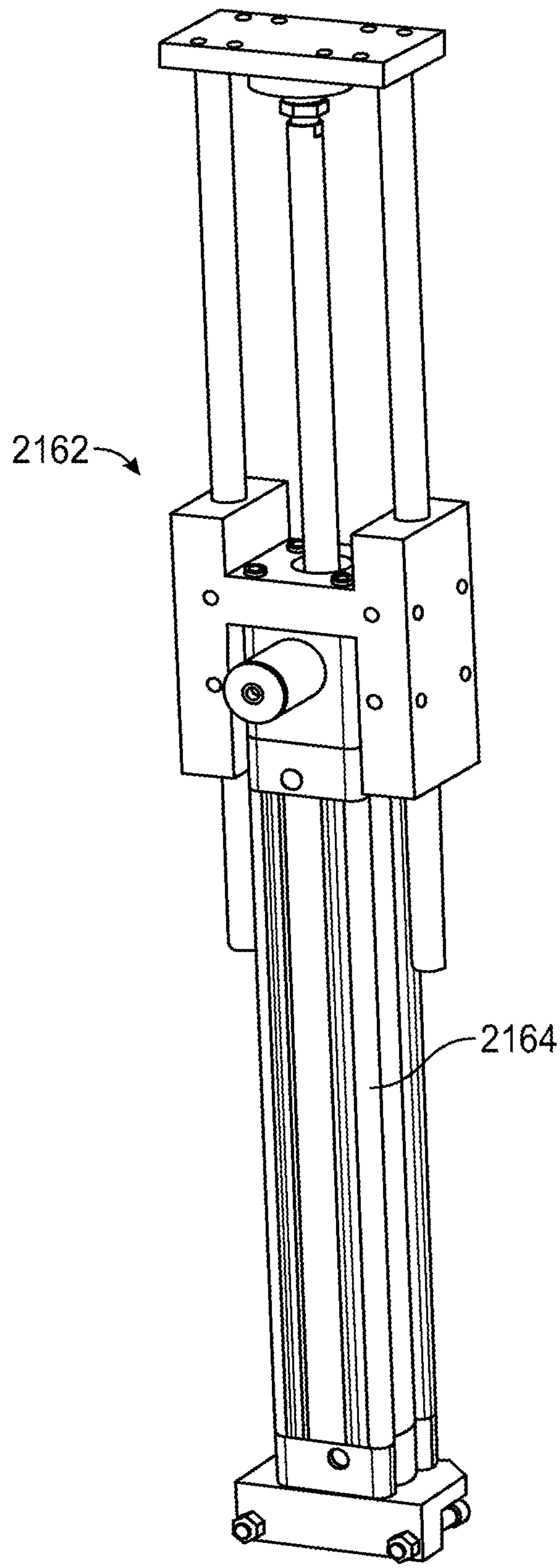


FIG. 33

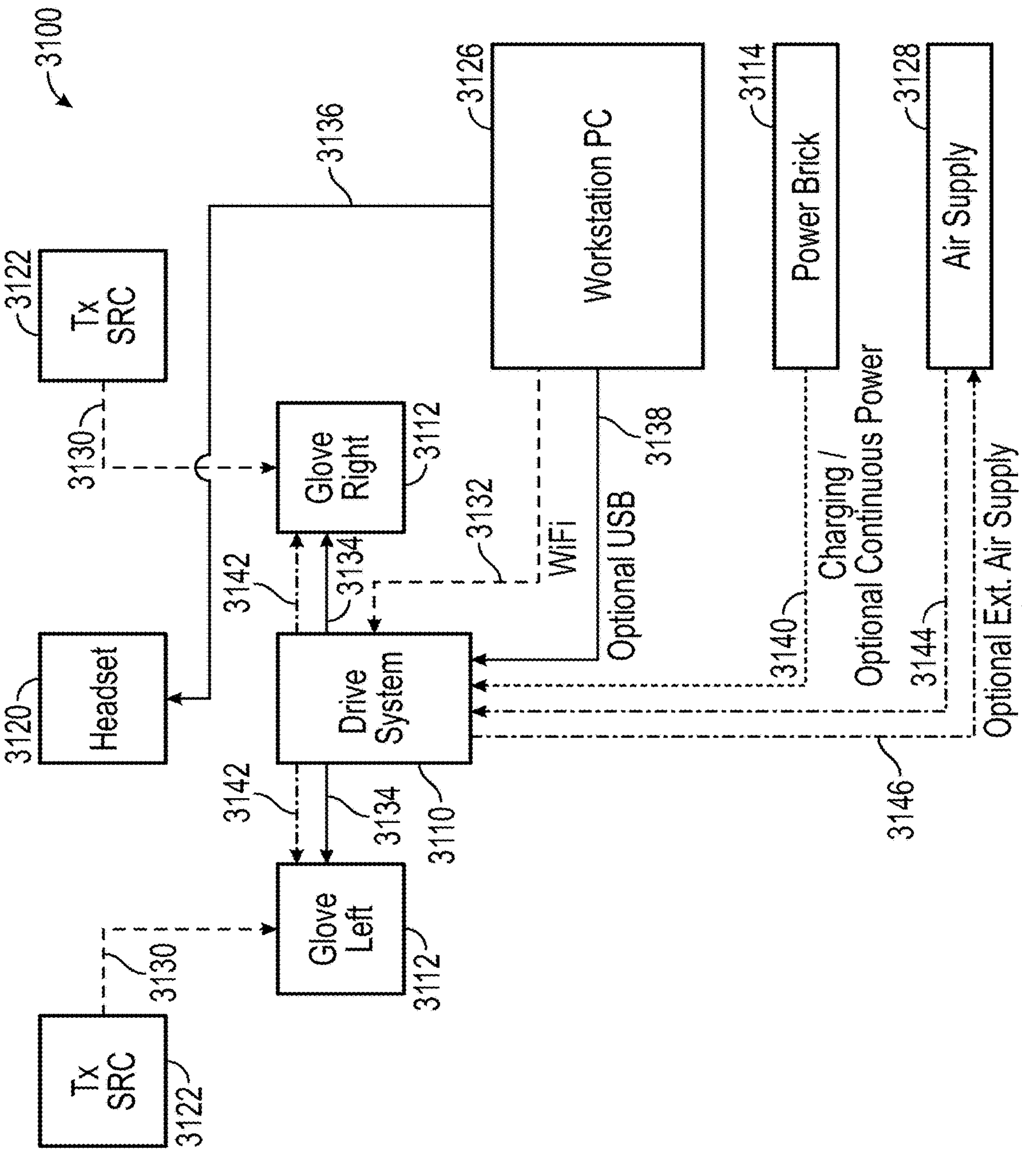

3100
Tx SRC
3122
Headset
3120
Glove Right
3112
Drive System
3110
Glove Left
3112
Tx SRC
3122
3130
3130
3142
3134
3142
3134
3136
3132
WiFi
Optional USB
3138
Workstation PC
3126
Power Brick
3114
Charging / Optional Continuous Power
3140
Air Supply
3128
3144
Optional Ext. Air Supply
3146

FIG. 34

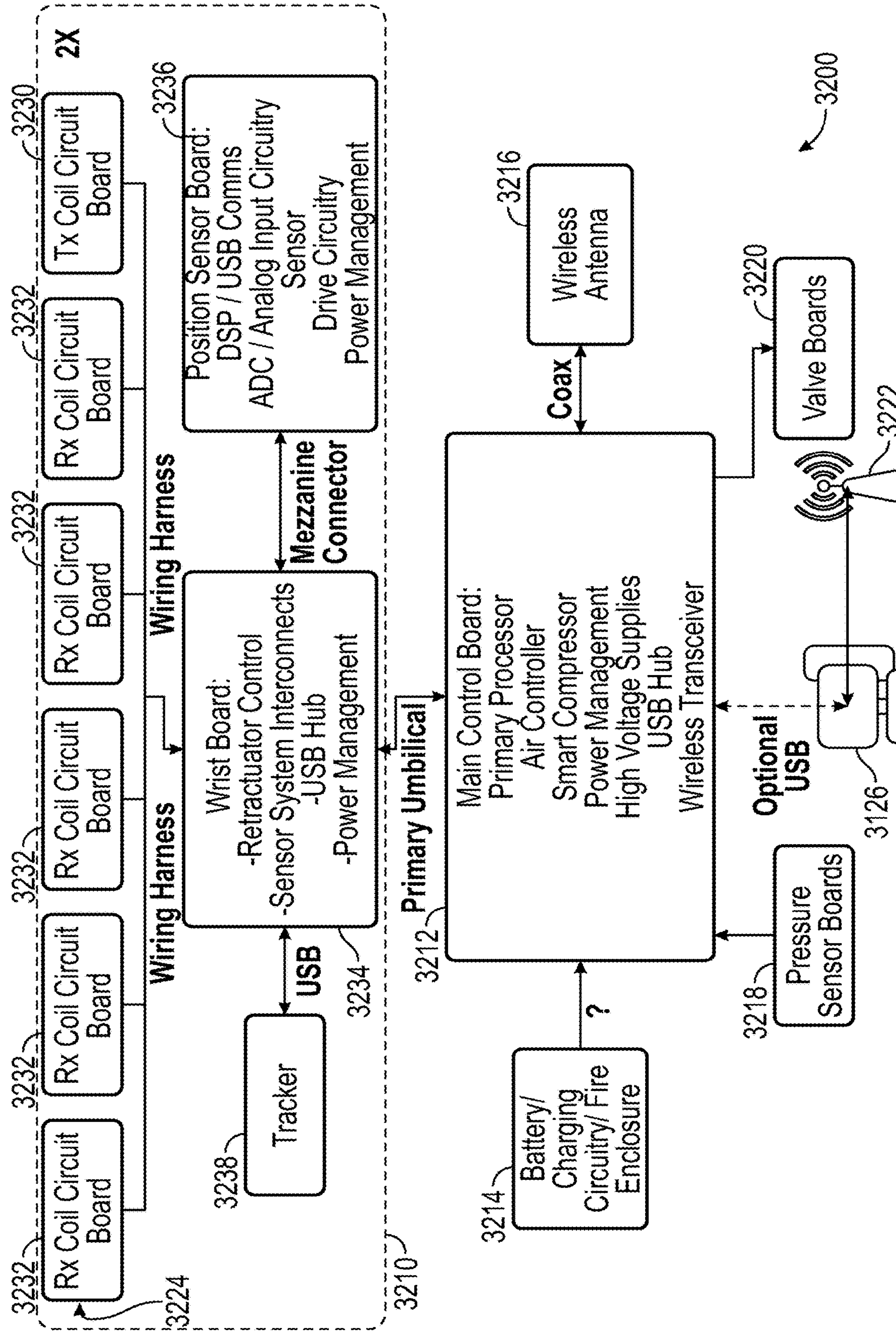

2X
3230
Tx Coil Circuit Board
3232
Rx Coil Circuit Board
3232
Rx Coil Circuit Board
3232
Rx Coil Circuit Board
3232
Rx Coil Circuit Board
3232
Rx Coil Circuit Board
3224
Wiring Harness
Wiring Harness
3236
Position Sensor Board:
DSP / USB Comms
ADC / Analog Input Circuitry
Sensor
Drive Circuitry
Power Management
Mezzanine Connector
Wrist Board:
-Retractuator Control
-Sensor System Interconnects
-USB Hub
-Power Management
3234
USB
3238
Tracker
3210
Primary Umbilical
3212
Main Control Board:
Primary Processor
Air Controller
Smart Compressor
Power Management
High Voltage Supplies
USB Hub
Wireless Transceiver
Coax
3216
Wireless Antenna
3220
Valve Boards
3222
Wireless Hub
Optional USB
3126
End PC
3218
Pressure Sensor Boards
?
3214
Battery/ Charging Circuitry/ Fire Enclosure
3200

FIG. 35

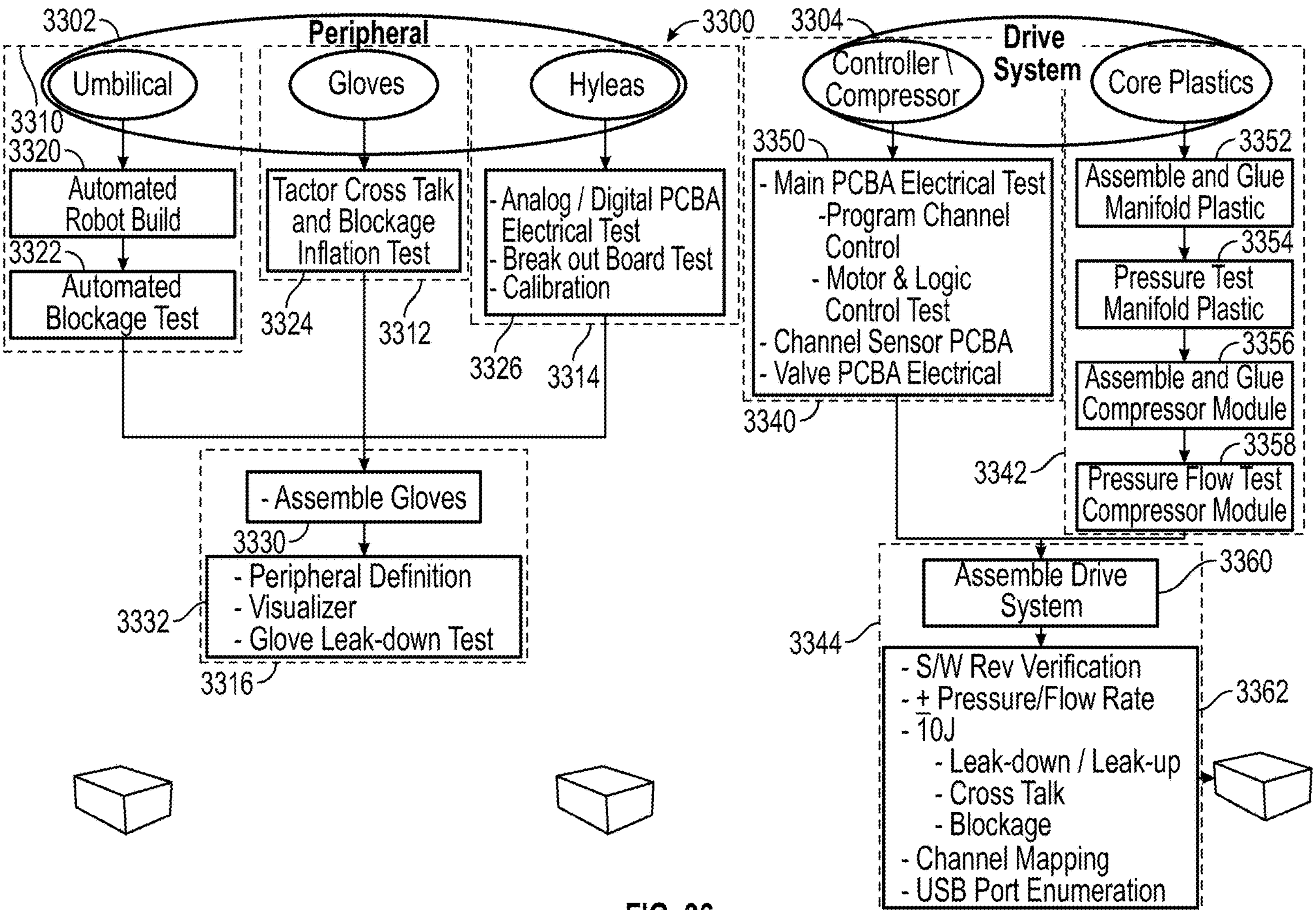

3300
3302
Peripheral
Umbilical
Gloves
Hyleas
3310
3320
Automated Robot Build
3322
Automated Blockage Test
Tactor Cross Talk and Blockage Inflation Test
3324
3312
- Analog / Digital PCBA Electrical Test
- Break out Board Test
- Calibration
3326
3314
- Assemble Gloves
3330
- Peripheral Definition
- Visualizer
- Glove Leak-down Test
3332
3316
3304
Drive System
Controller \ Compressor
Core Plastics
3350
- Main PCBA Electrical Test
-Program Channel Control
- Motor & Logic Control Test
- Channel Sensor PCBA
- Valve PCBA Electrical
3340
3352
Assemble and Glue Manifold Plastic
3354
Pressure Test Manifold Plastic
3356
Assemble and Glue Compressor Module
3358
Pressure Flow Test Compressor Module
3342
Assemble Drive System
3360
3344
- S/W Rev Verification
- ± Pressure/Flow Rate
- 10J
- Leak-down / Leak-up
- Cross Talk
- Blockage
- Channel Mapping
- USB Port Enumeration
3362

FIG. 36

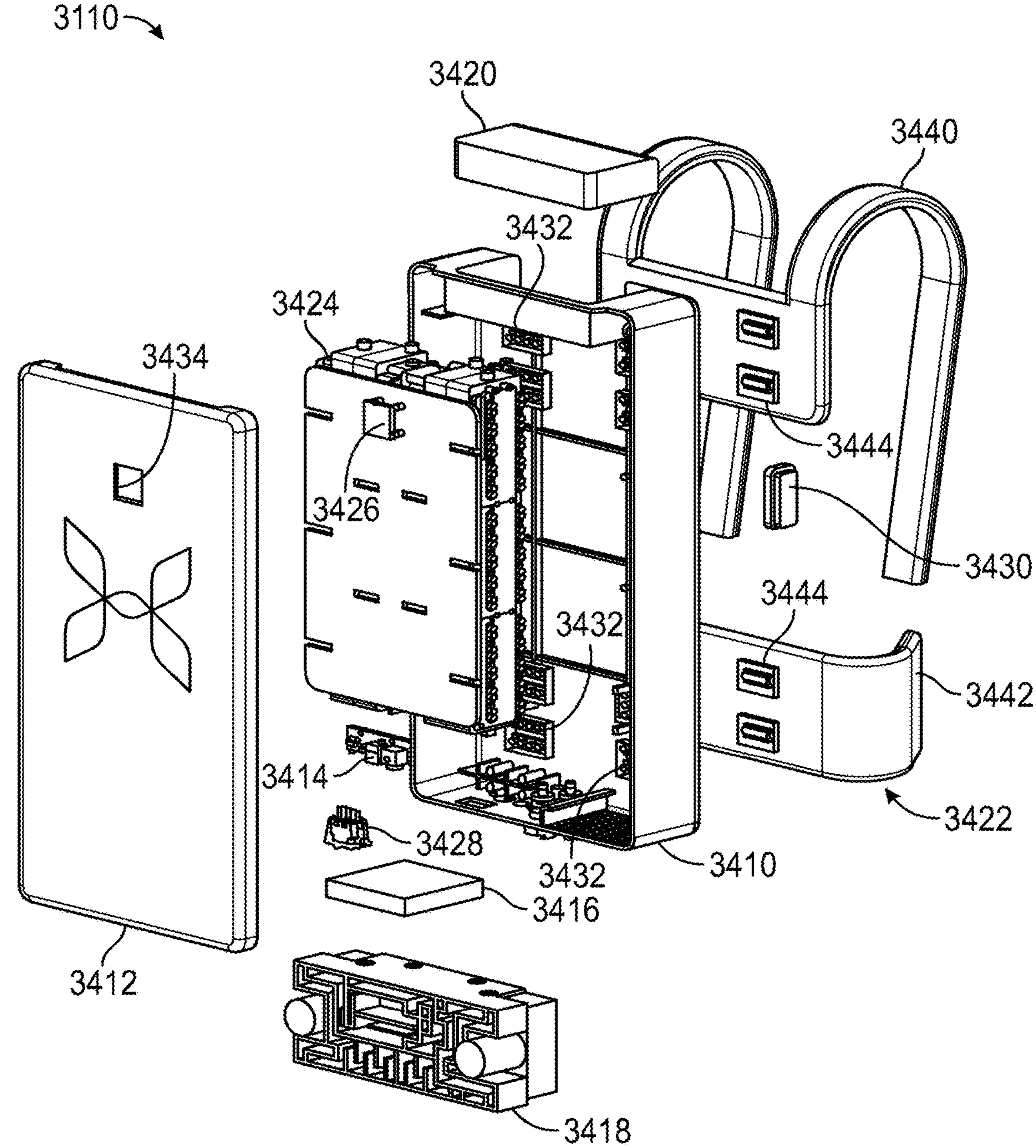

3110
3420
3440
3432
3424
3434
3444
3426
3430
3444
3432
3442
3414
3422
3428
3410
3432
3416
3412
3418

FIG. 37

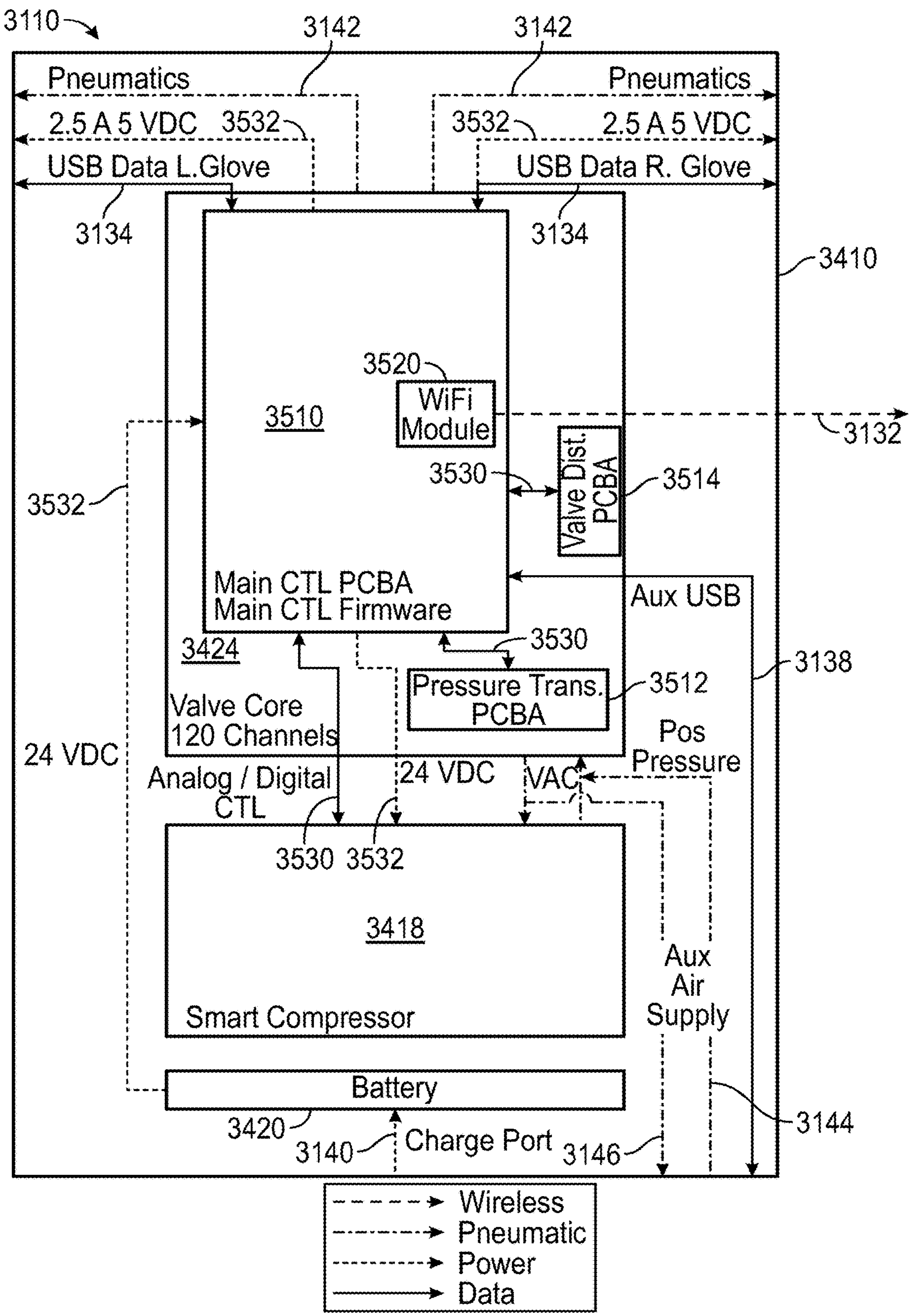

3110
3142
3142
Pneumatics
Pneumatics
2.5 A 5 VDC
3532
3532
2.5 A 5 VDC
USB Data L.Glove
USB Data R. Glove
3134
3134
3410
3520
WiFi Module
3510
3132
3530
Valve Dist. PCBA
3514
3532
Main CTL PCBA
Main CTL Firmware
Aux USB
3424
3530
3138
Pressure Trans. PCBA
3512
Valve Core
120 Channels
Pos Pressure
24 VDC
Analog / Digital CTL
24 VDC
VAC
3530
3532
3418
Aux Air Supply
Smart Compressor
Battery
3420
3140
Charge Port
3146
3144
Wireless
Pneumatic
Power
Data

FIG. 38

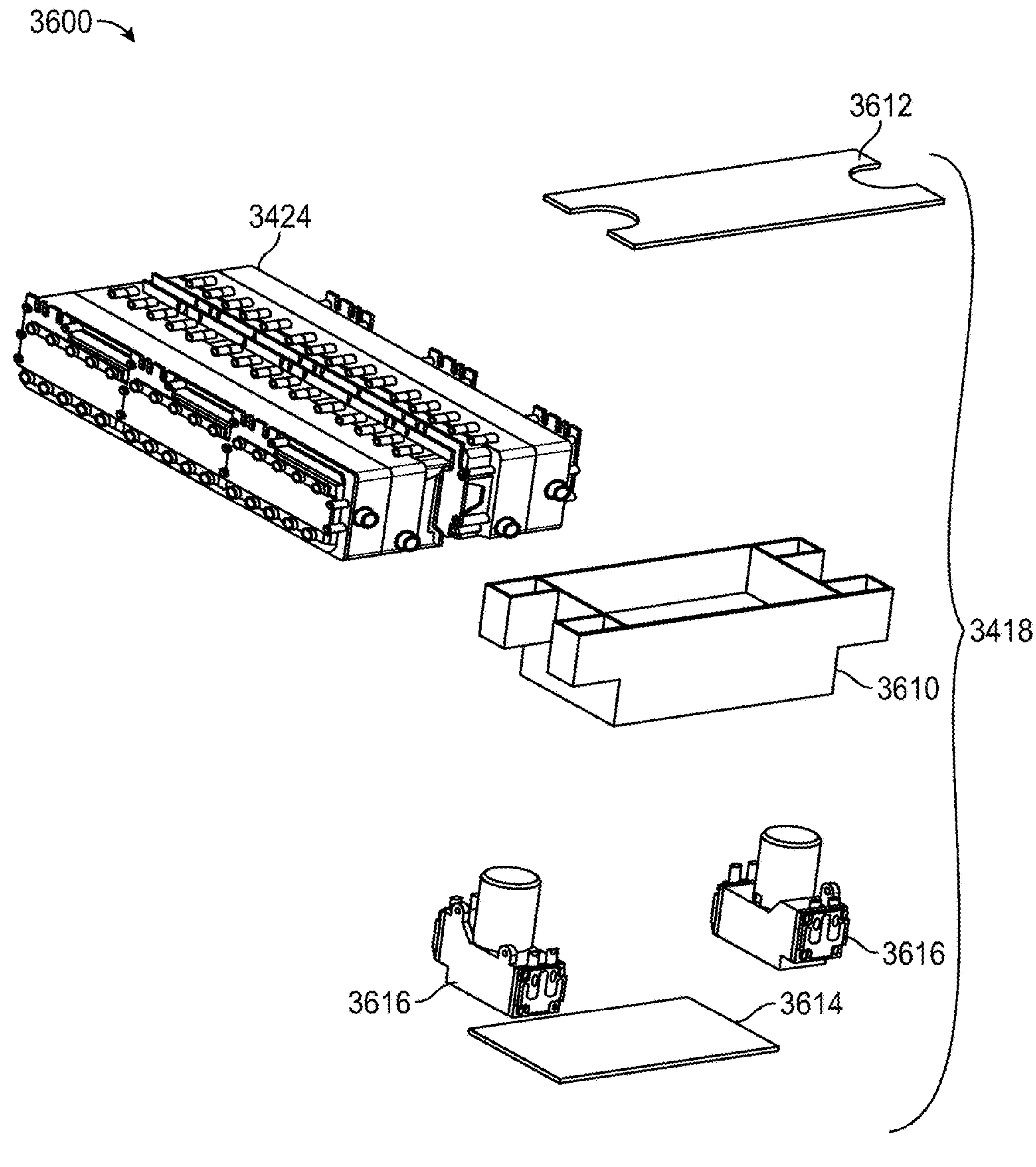


FIG. 39

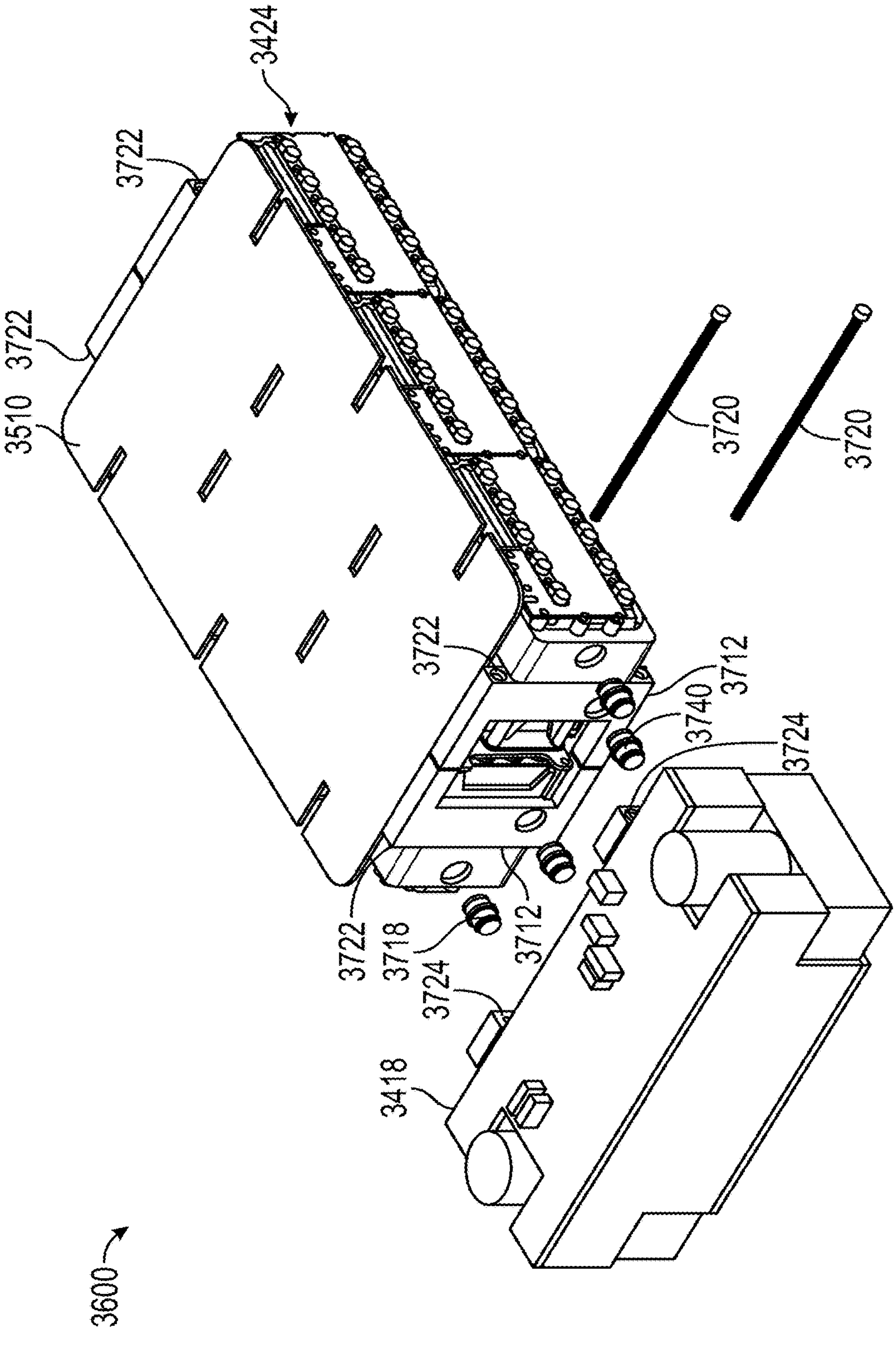

3424
3722
3722
3510
3720
3720
3722
3712
3740
3724
3722
3718
3712
3724
3418
3600

FIG. 40

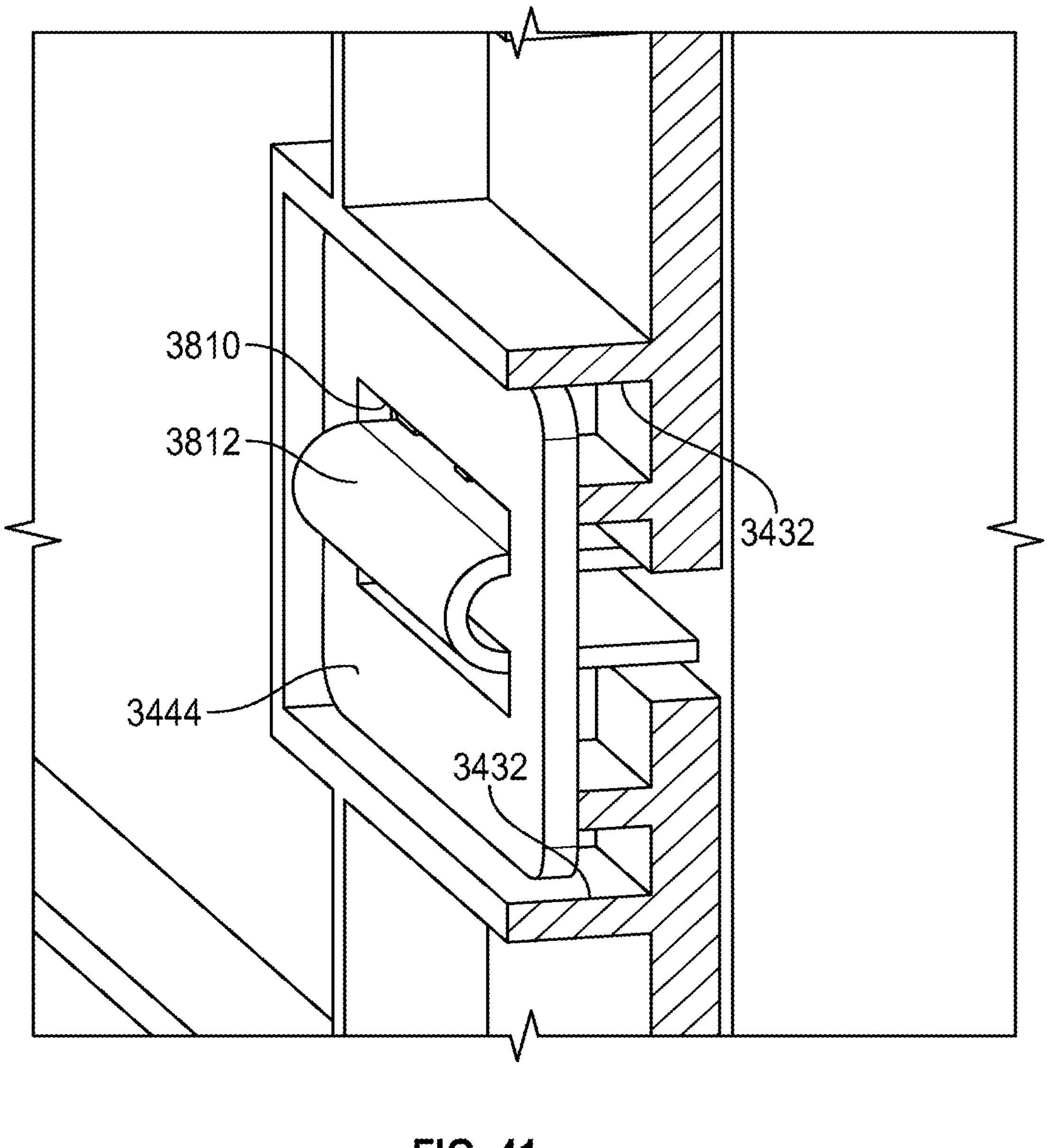


FIG. 41

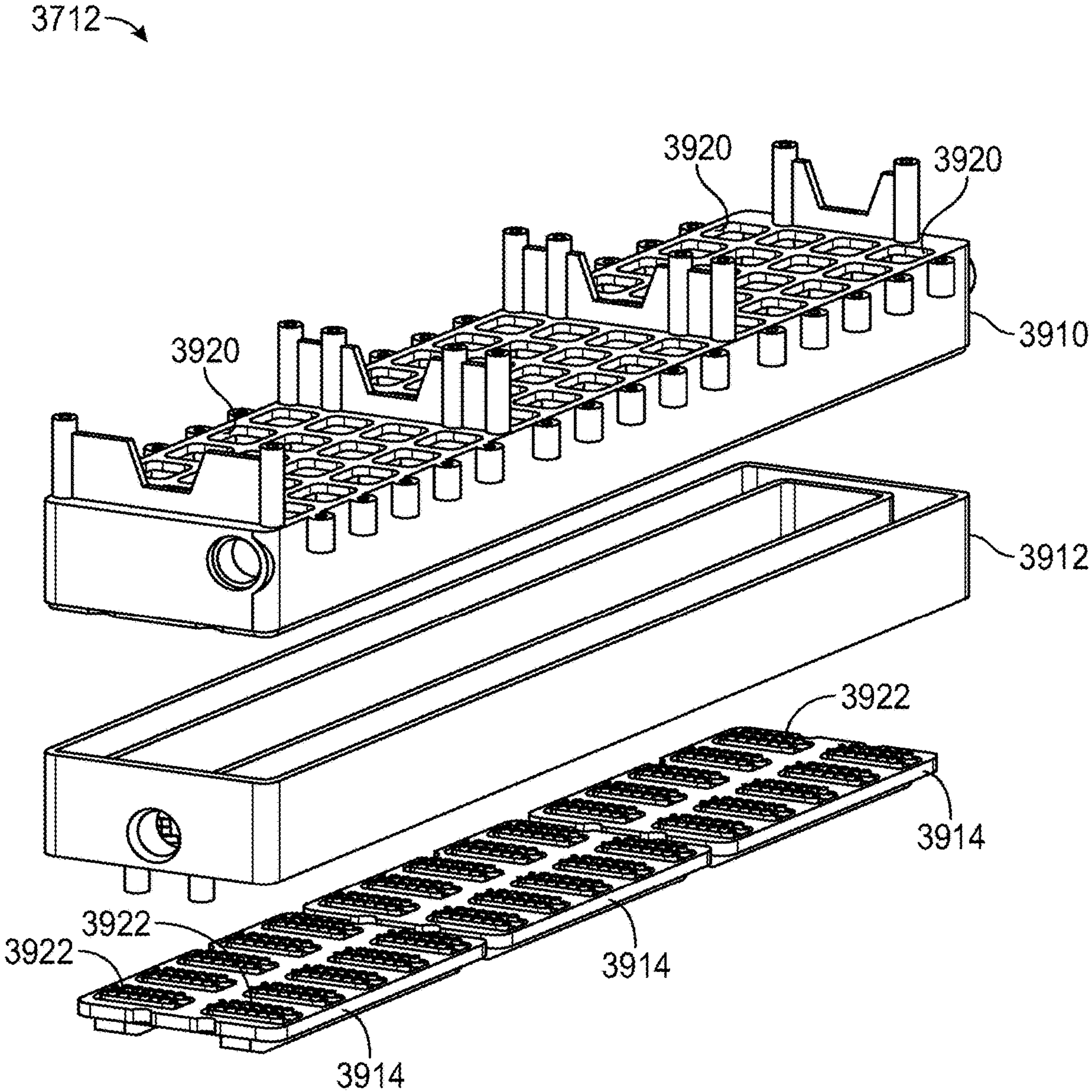


FIG. 42

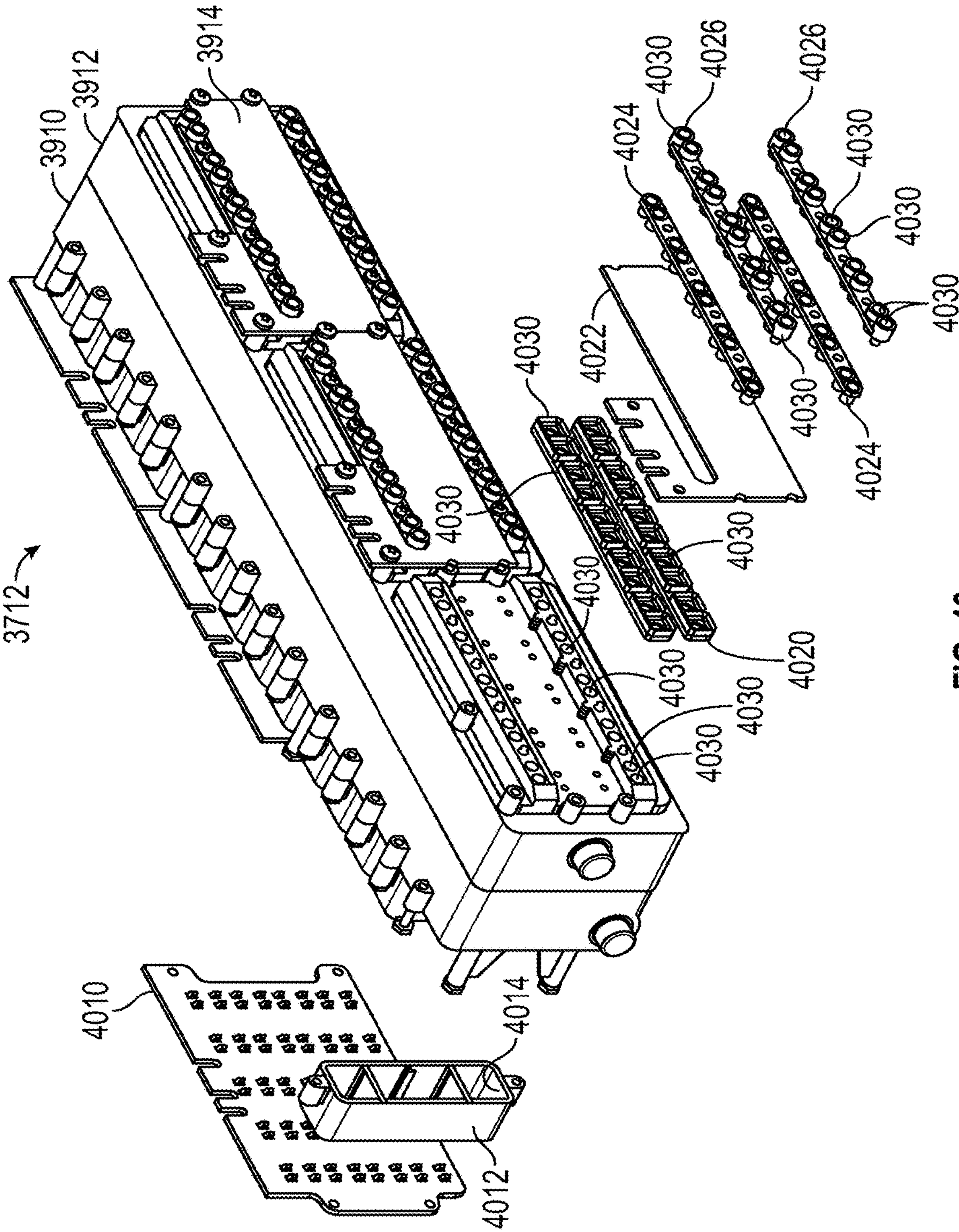


FIG. 43

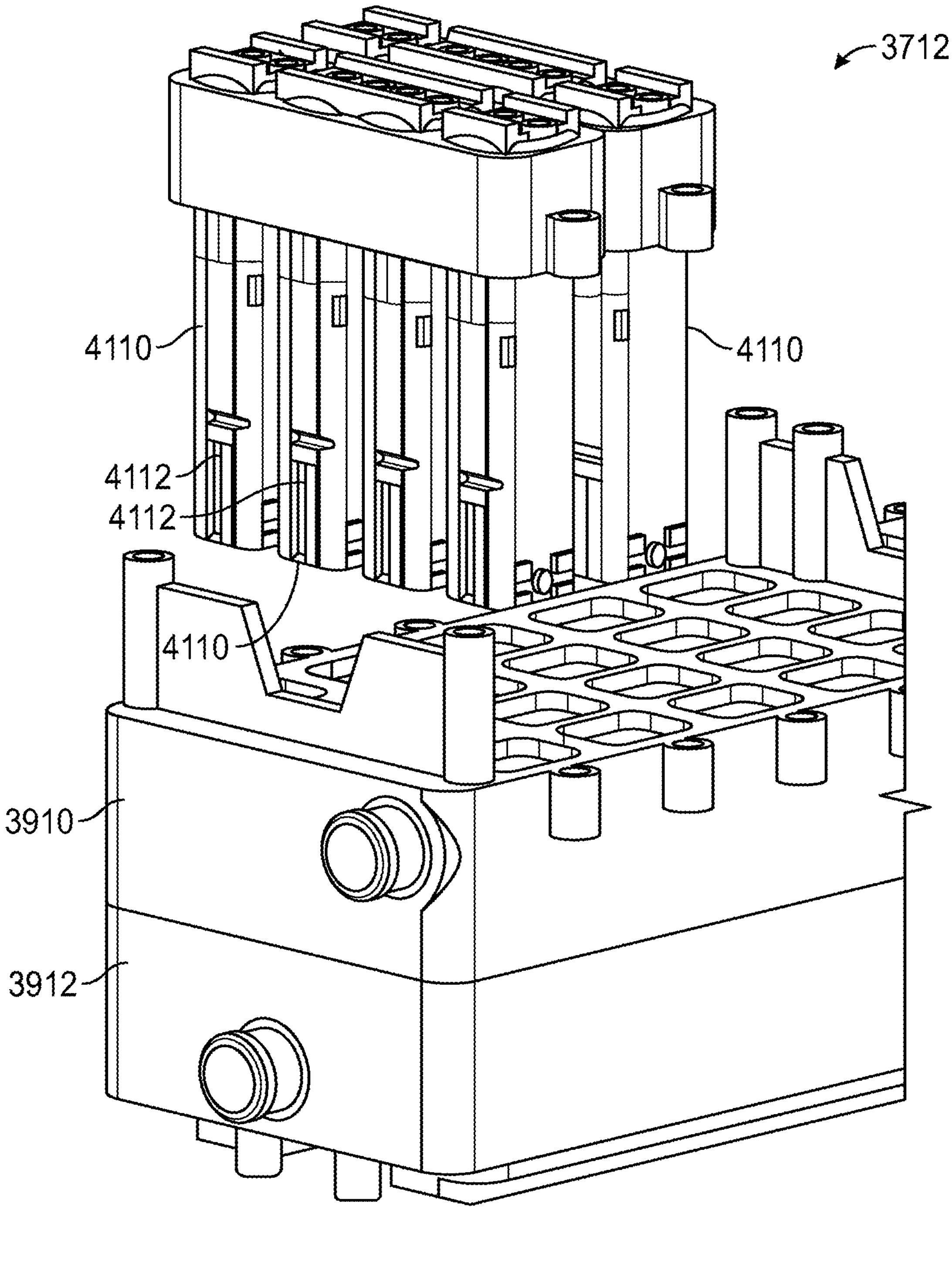


FIG. 44

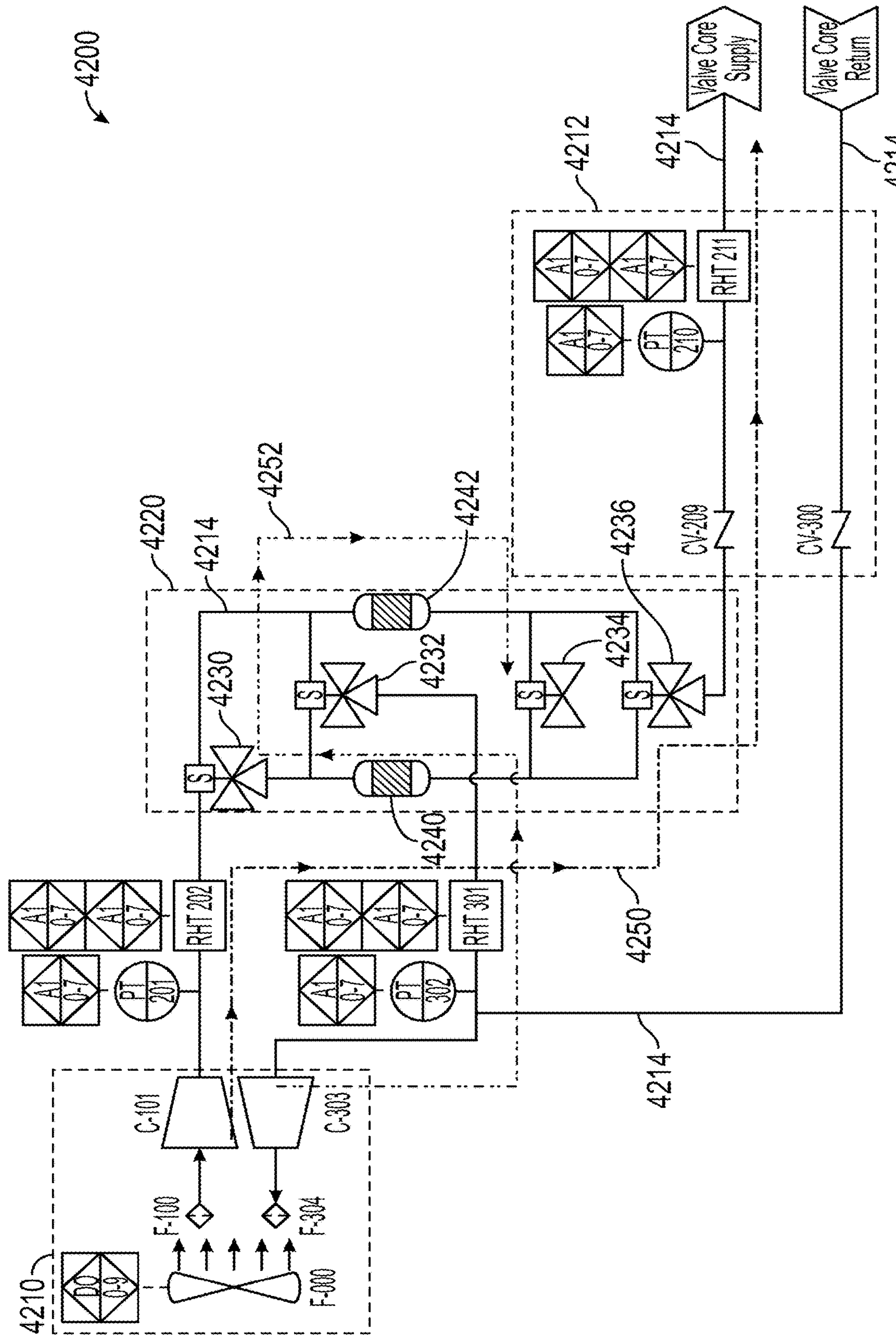
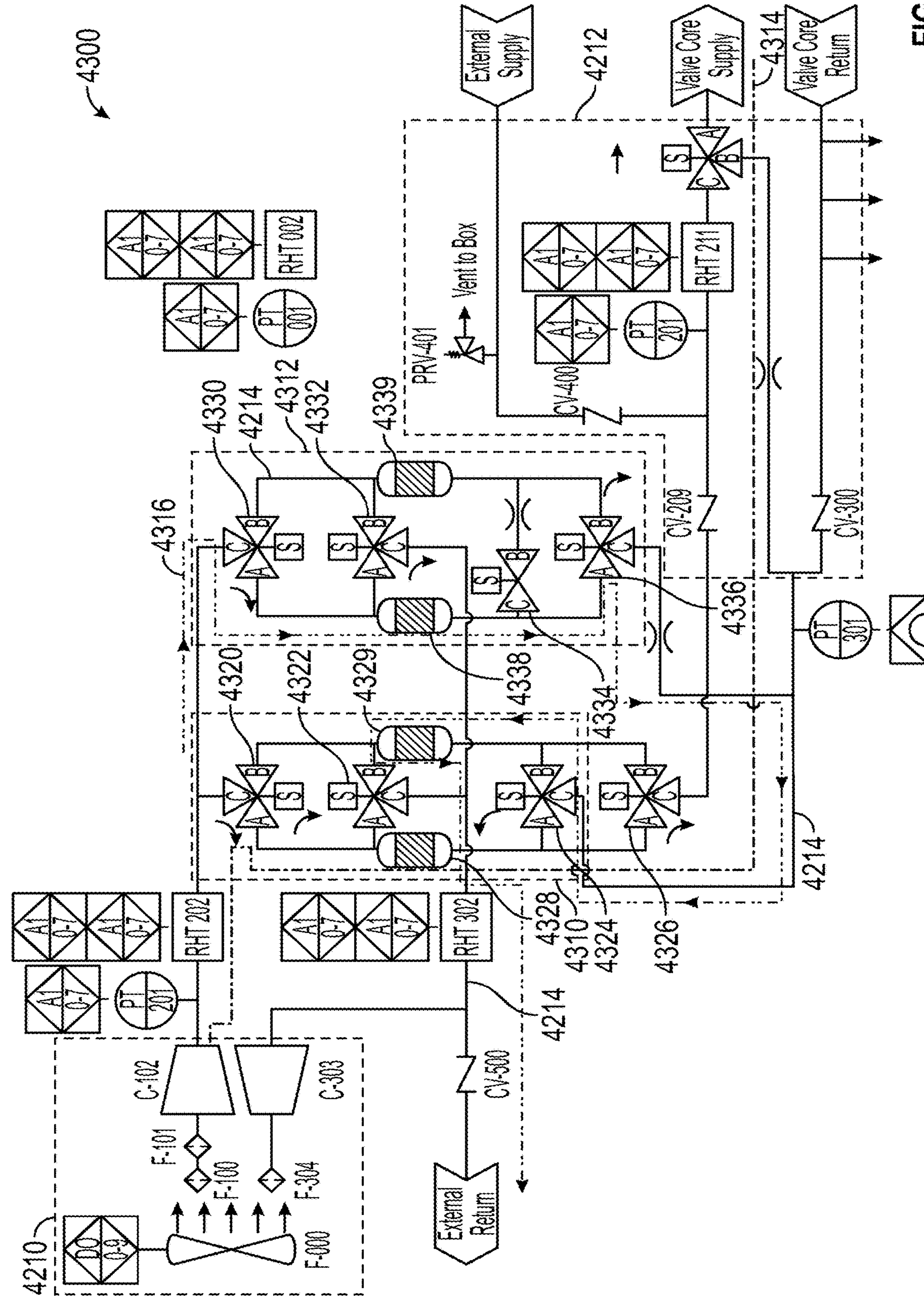

4200
4212
4214
Valve Core Supply
Valve Core Return
A1 0-7
RHT 211
PT 210
4252
4242
4220
4236
CV-209
CV-300
4230
4232
4234
S
4240
RHT 202
PT 201
RHT 301
PT 302
4250
C-101
C-303
F-100
F-304
F-000
DO 0-9
4210

FIG. 45



4300
External Supply
4212
Valve Core Supply
4314
Valve Core Return
Vent to Box
PRV-401
CV-400
CV-209
CV-300
RHT 002
RHT 211
RHT 202
RHT 302
PT 001
PT 201
PT 301
A1 0-7
DO 0-9
4330
4214
4312
4332
4339
4316
4336
4338
4334
4320
4322
4329
4328
4310
4324
4326
C-102
C-303
F-101
F-100
F-304
F-000
CV-500
External Return
4210

FIG. 46

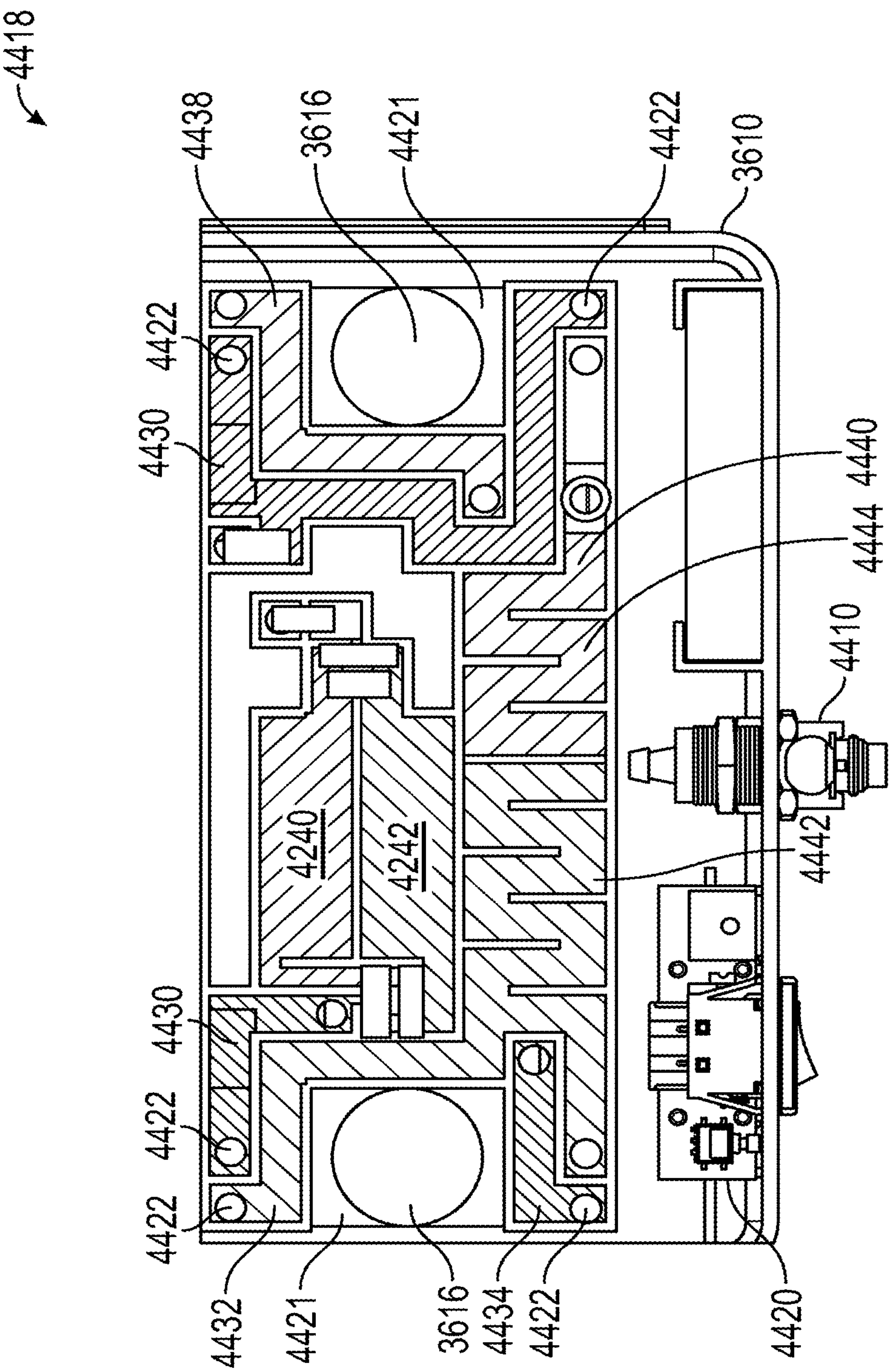

4418
4438
3616
4421
4422
3610
4422
4430
4440
4444
4410
4240
4242
4442
4430
4422
4422
4432
4421
3616
4434
4422
4420

FIG. 47

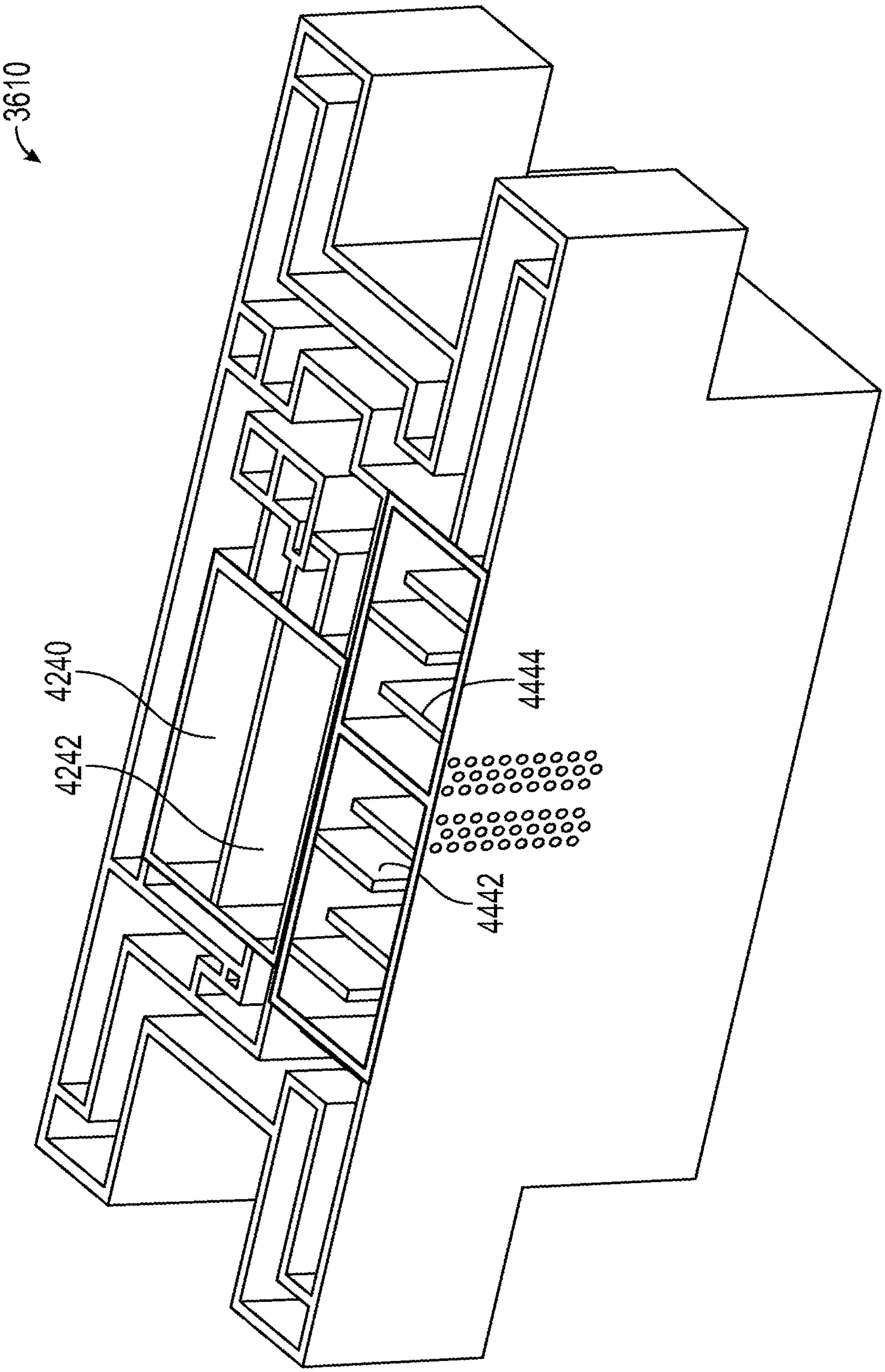


FIG. 48

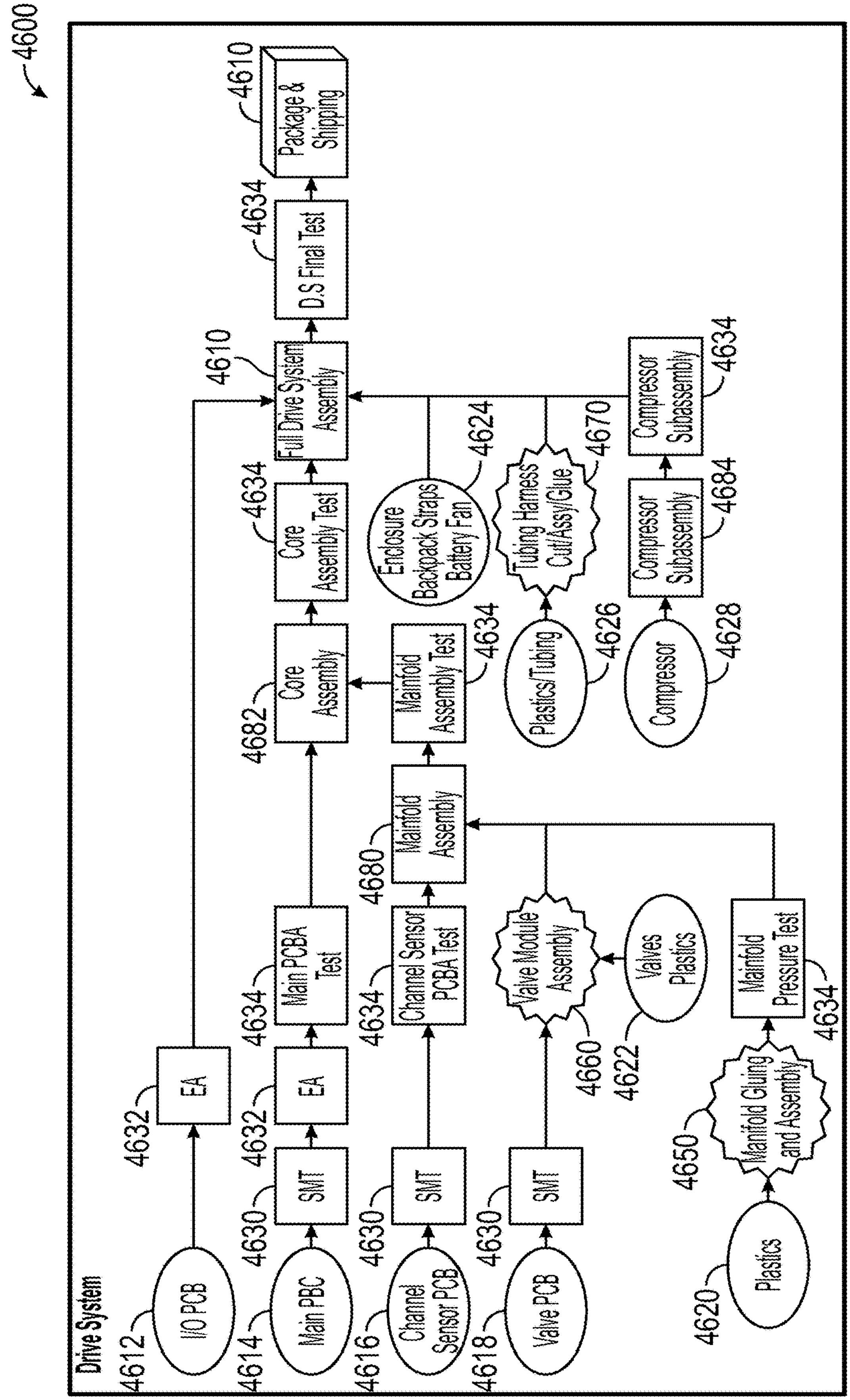

4600
Drive System
4612
I/O PCB
4614
Main PBC
4616
Channel Sensor PCB
4618
Valve PCB
4620
Plastics
4630
SMT
4632
EA
4634
Main PCBA Test
Channel Sensor PCBA Test
4650
Manifold Gluing and Assembly
Manifold Pressure Test
4660
Valve Module Assembly
4622
Valves Plastics
4680
Mainfold Assembly
Mainfold Assembly Test
4682
Core Assembly
Core Assembly Test
4626
Plastics/Tubing
4628
Compressor
4624
Enclosure Backpack Straps Battery Fan
4670
Tubing Harness Cut/Assy/Glue
4684
Compressor Subassembly
4610
Full Drive System Assembly
D.S Final Test
Package & Shipping

FIG. 49

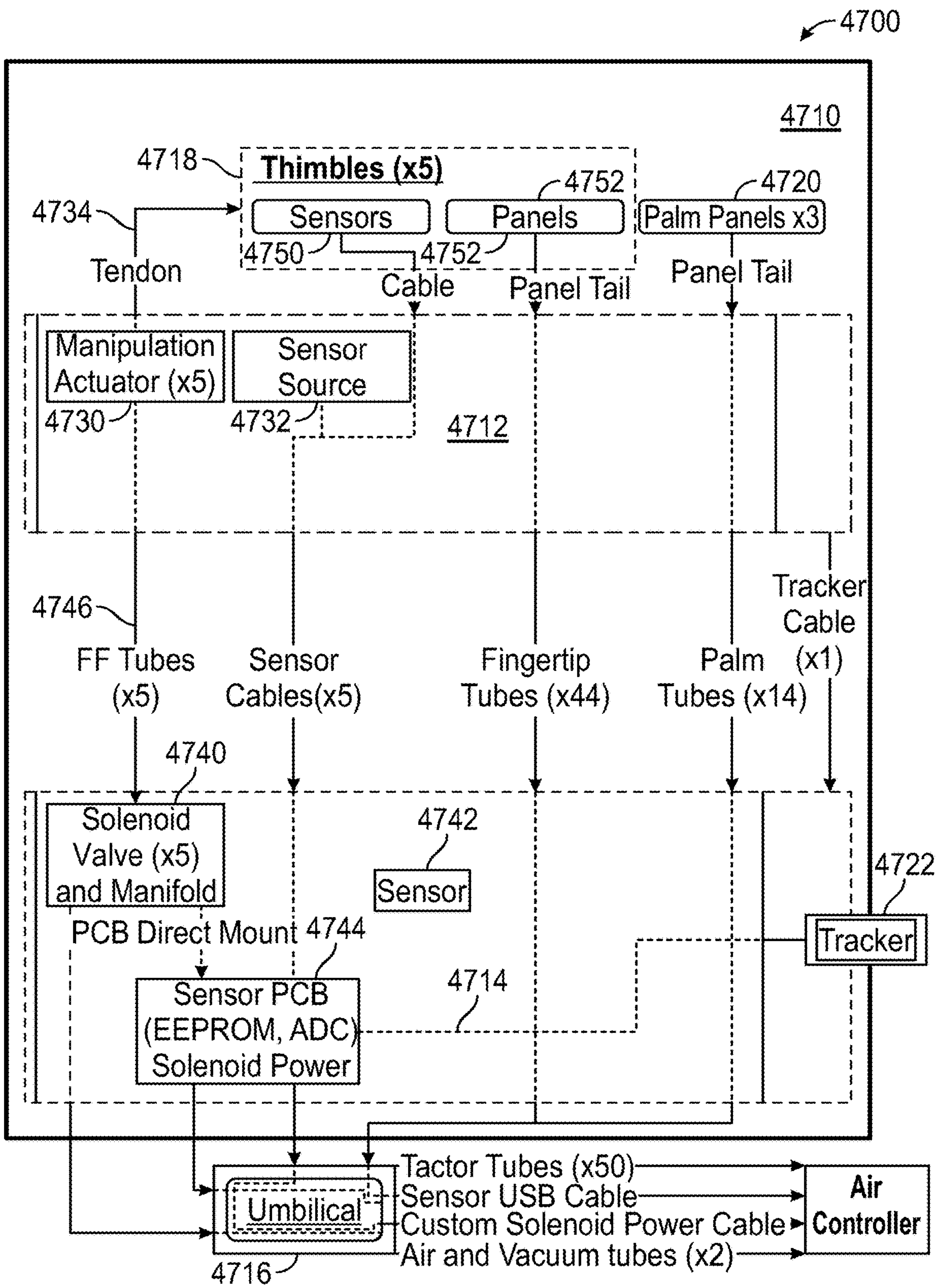

4700
4710
4718
Thimbles (x5)
4752
4720
4734
Sensors
Panels
Palm Panels x3
4750
4752
Tendon
Cable
Panel Tail
Panel Tail
Manipulation Actuator (x5)
Sensor Source
4730
4732
4712
4746
Tracker Cable (x1)
FF Tubes (x5)
Sensor Cables(x5)
Fingertip Tubes (x44)
Palm Tubes (x14)
4740
4742
Solenoid Valve (x5) and Manifold
Sensor
4722
PCB Direct Mount
4744
Tracker
4714
Sensor PCB (EEPROM, ADC) Solenoid Power
Tactor Tubes (x50)
Sensor USB Cable
Umbilical
Custom Solenoid Power Cable
Air Controller
Air and Vacuum tubes (x2)
4716

FIG. 50

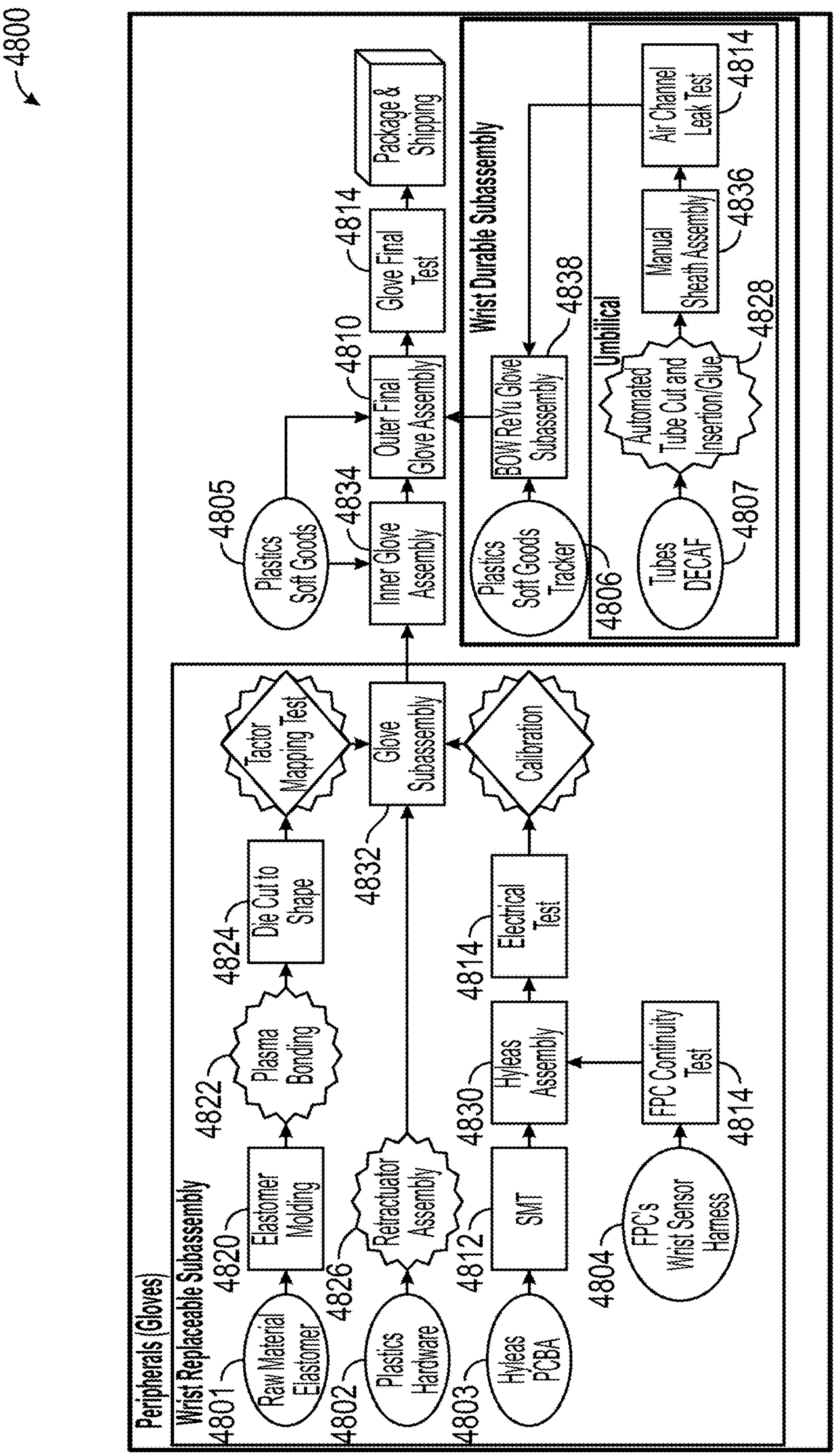

4800
Peripherals (Gloves)
Wrist Replaceable Subassembly
Raw Material Elastomer
4801
Elastomer Molding
4820
Plasma Bonding
4822
Die Cut to Shape
4824
Tactor Mapping Test
Plastics Hardware
4802
Retractuator Assembly
4826
Glove Subassembly
4832
Hyleas PCBA
4803
SMT
4812
Hyleas Assembly
4830
Electrical Test
4814
Calibration
FPC's Wrist Sensor Harness
4804
FPC Continuity Test
4814
Plastics Soft Goods
4805
Inner Glove Assembly
4834
Outer Final Glove Assembly
4810
Glove Final Test
4814
Package & Shipping
Wrist Durable Subassembly
Plastics Soft Goods Tracker
4806
BOW ReYu Glove Subassembly
4838
Umbilical
Tubes DECAF
4807
Automated Tube Cut and Insertion/Glue
4828
Manual Sheath Assembly
4836
Air Channel Leak Test
4814

FIG. 51

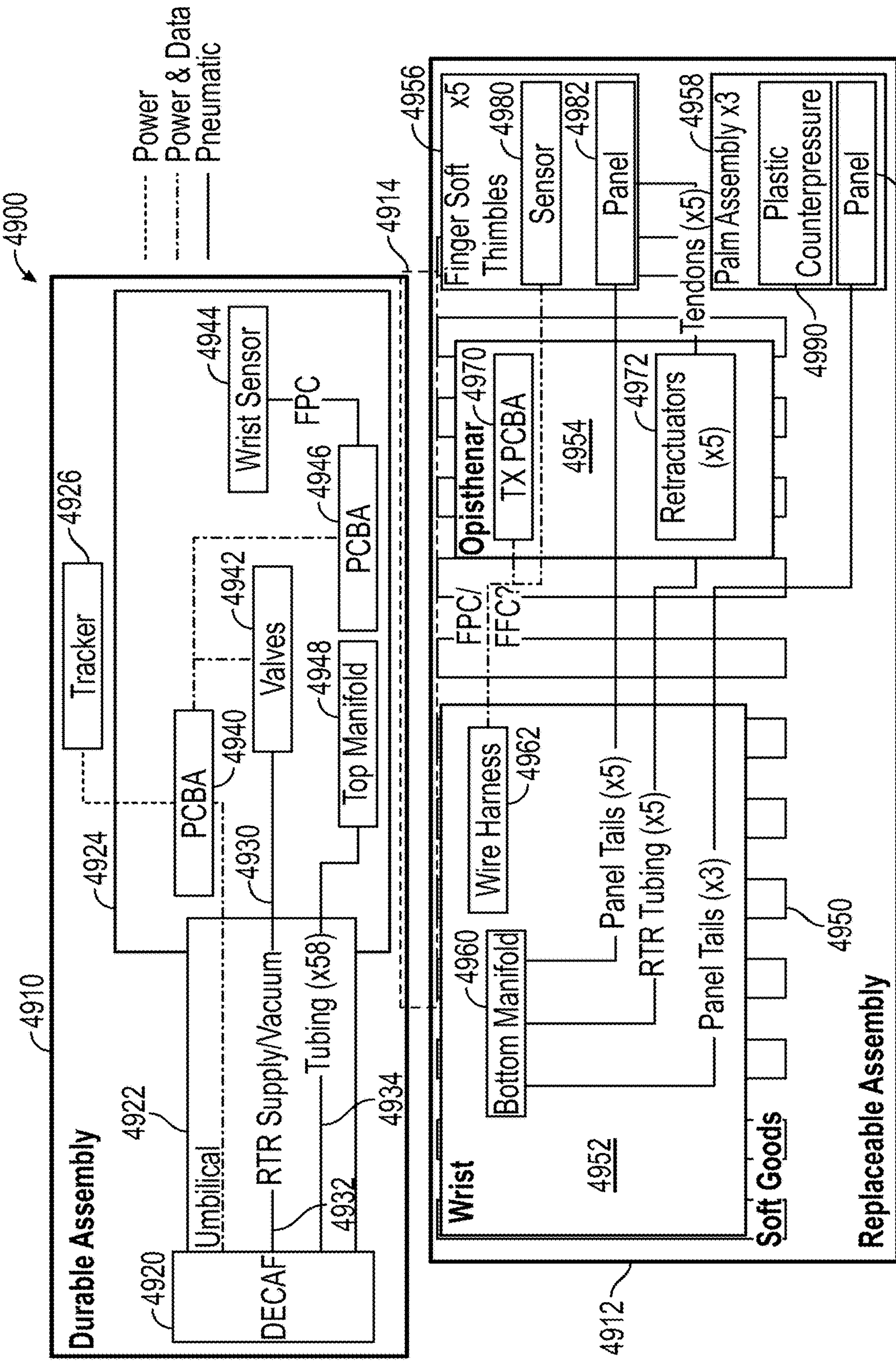

4900
Power
Power & Data
Pneumatic
Durable Assembly
4910
4920
DECAF
4922
Umbilical
RTR Supply/Vacuum
Tubing (x58)
4932
4934
4930
4924
PCBA
4940
Tracker
4926
Valves
4942
Top Manifold
4948
PCBA
4946
FPC
Wrist Sensor
4944
4914
4912
Replaceable Assembly
Soft Goods
Wrist
4952
Bottom Manifold
4960
Wire Harness
4962
Panel Tails (x5)
RTR Tubing (x5)
Panel Tails (x3)
4950
FPC/
FFC?
Opisthenar
4970
TX PCBA
4954
4972
Retractuators
(x5)
Tendons
Tendons (x5)
4990
Finger Soft
Thimbles
x5
4956
Sensor
4980
Panel
4982
Palm Assembly x3
4958
Plastic
Counterpressure
Panel
4992

FIG. 52

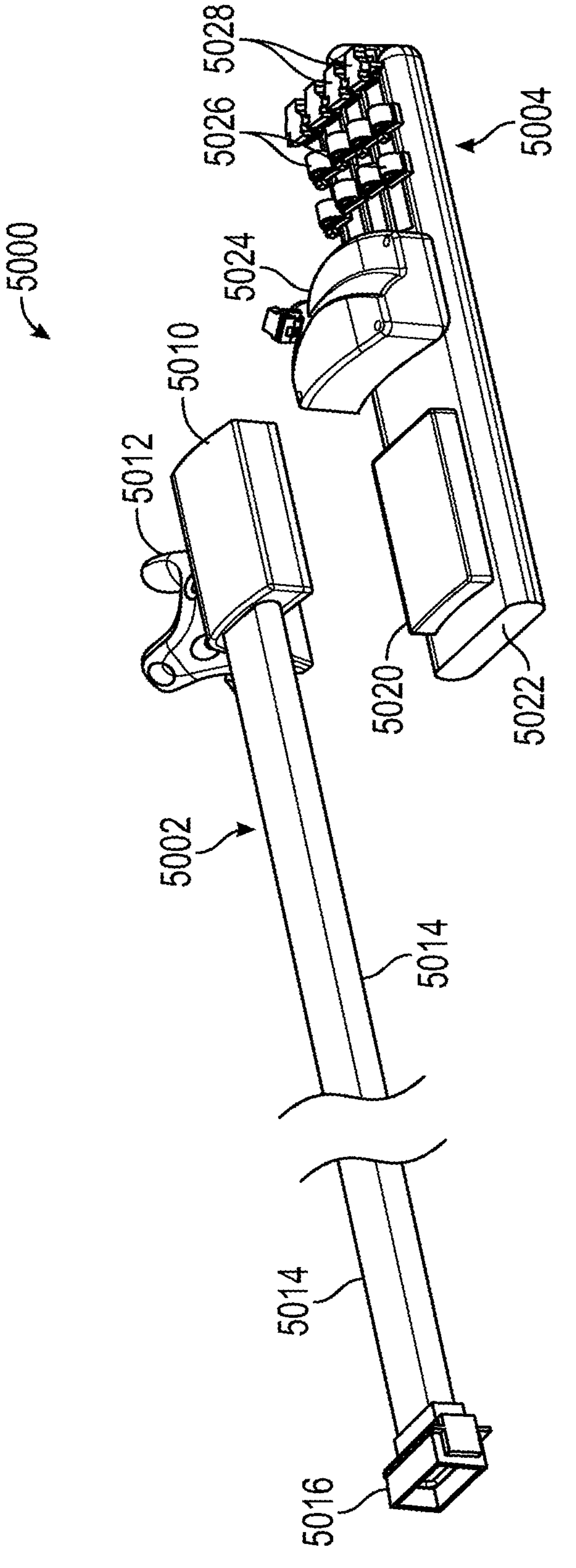

5000
5028
5026
5004
5024
5010
5012
5020
5022
5002
5014
5014
5016

FIG. 53

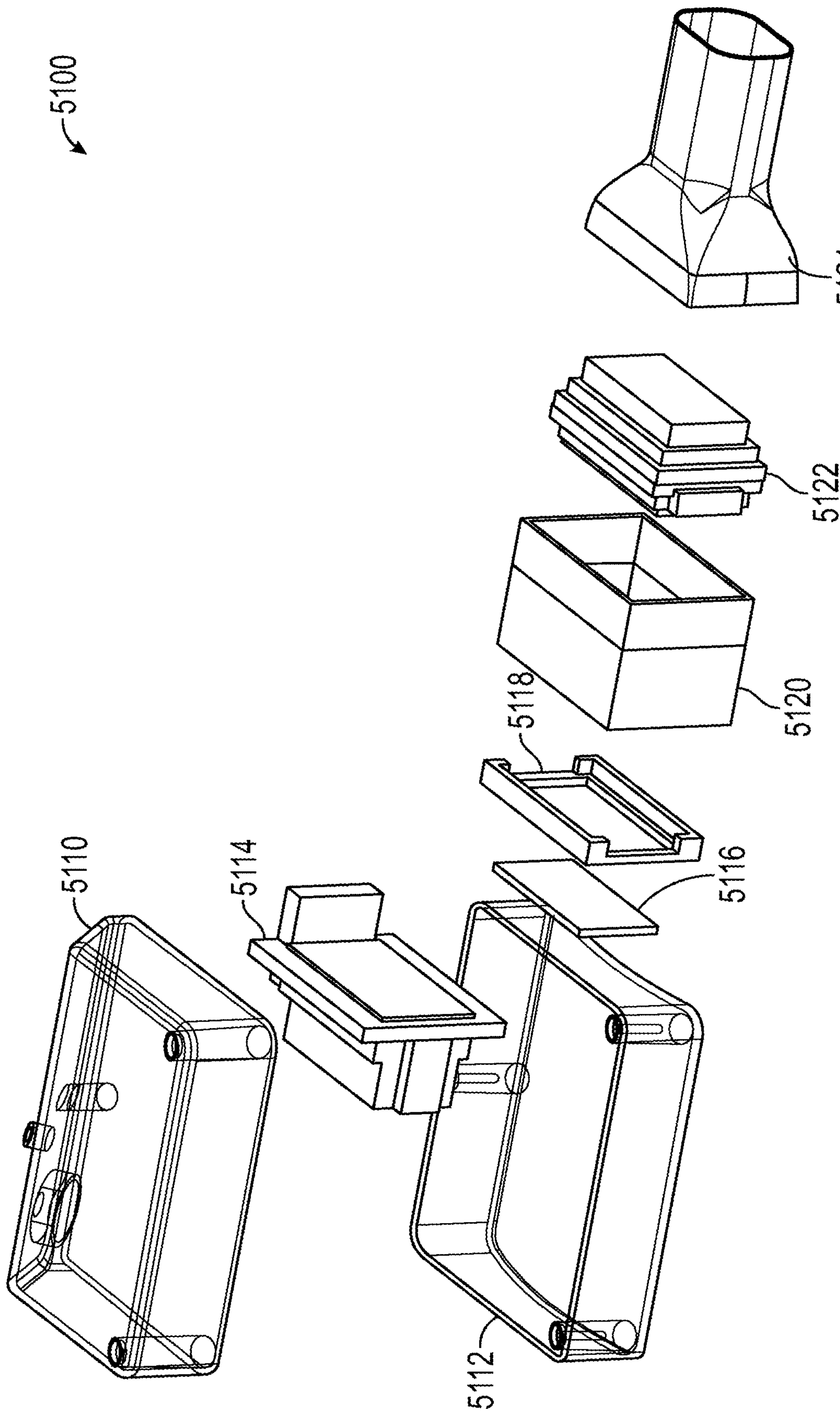


FIG. 54

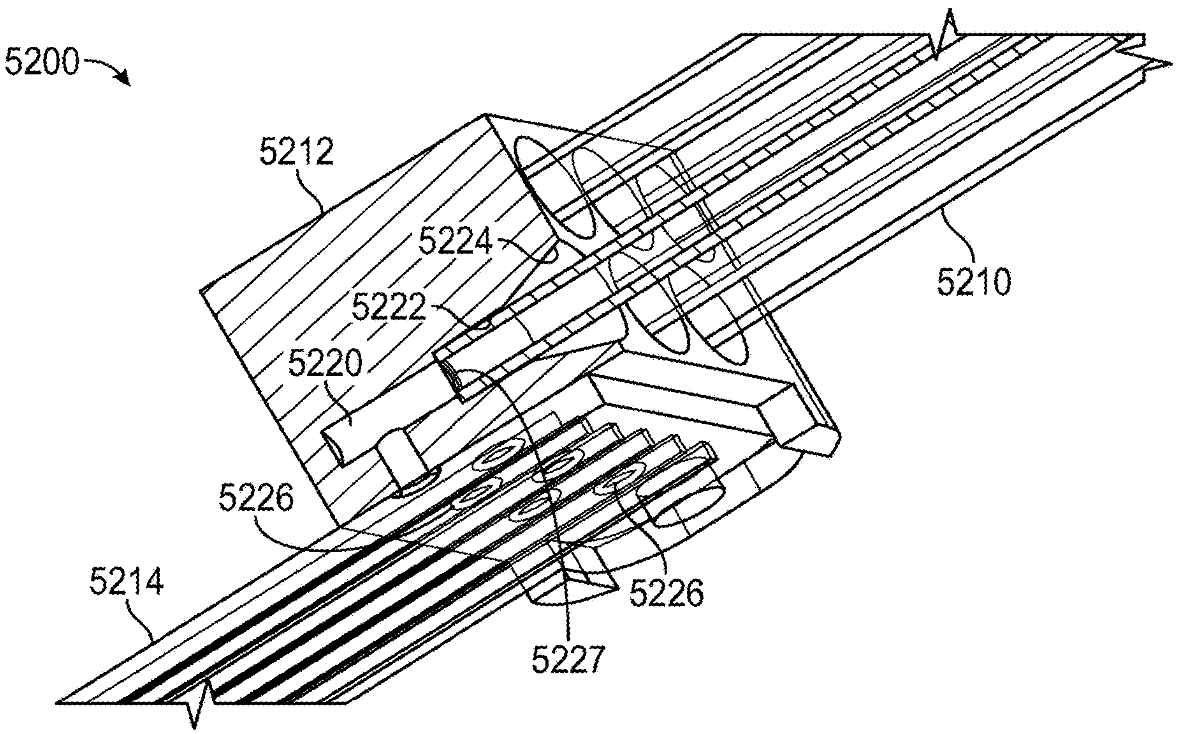

5200
5212
5224
5222
5220
5210
5226
5214
5226
5227

FIG. 55

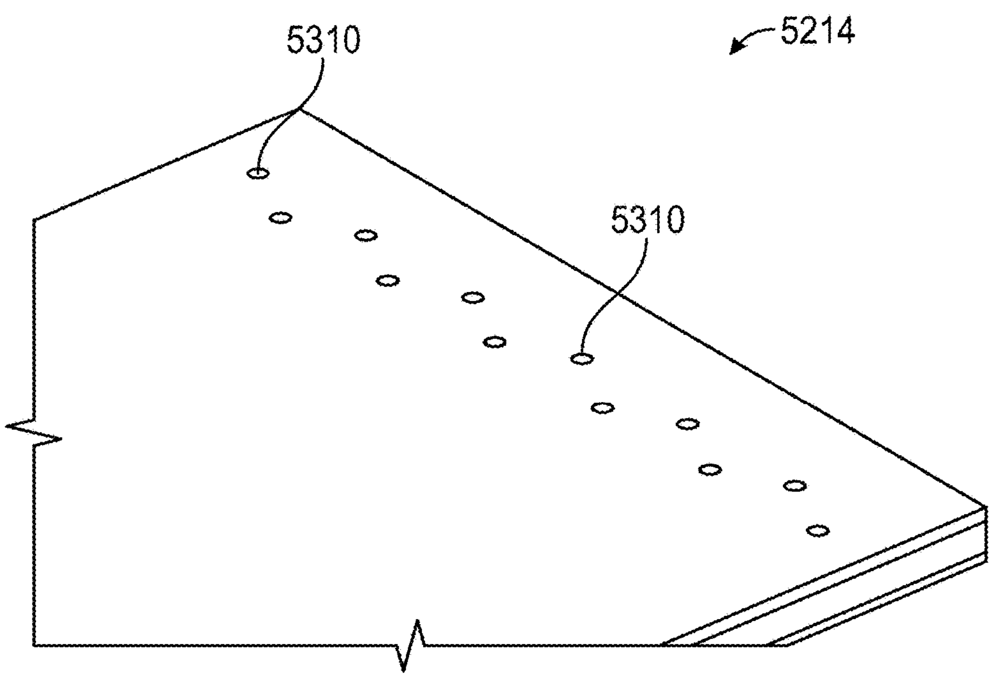

5310
5214
5310

FIG. 56

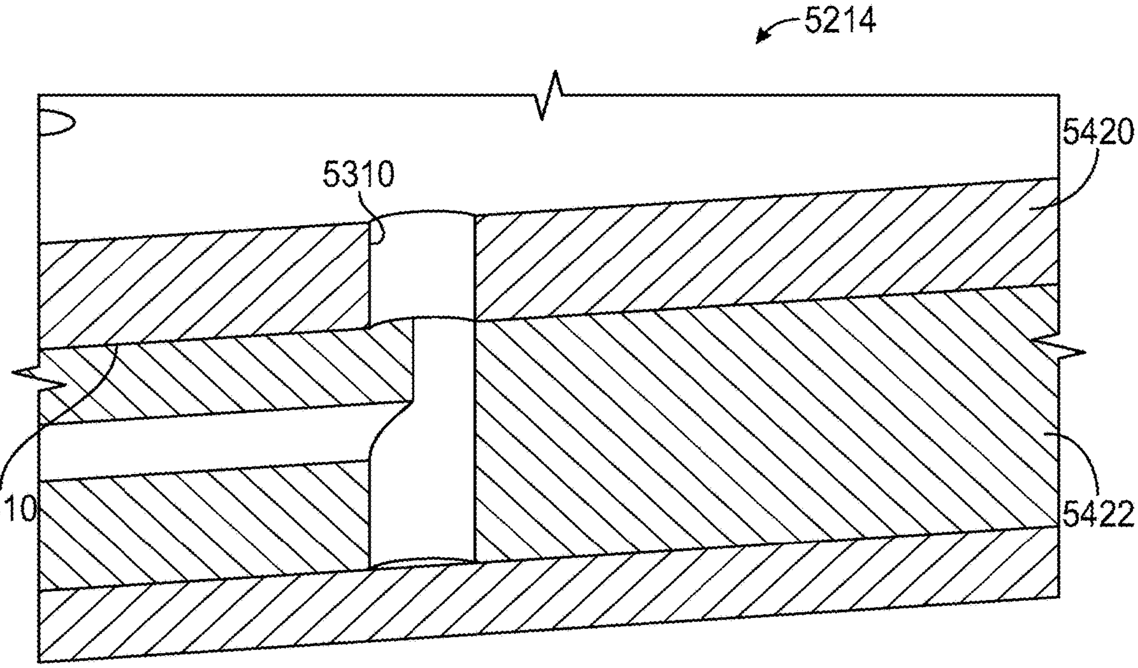


FIG. 57

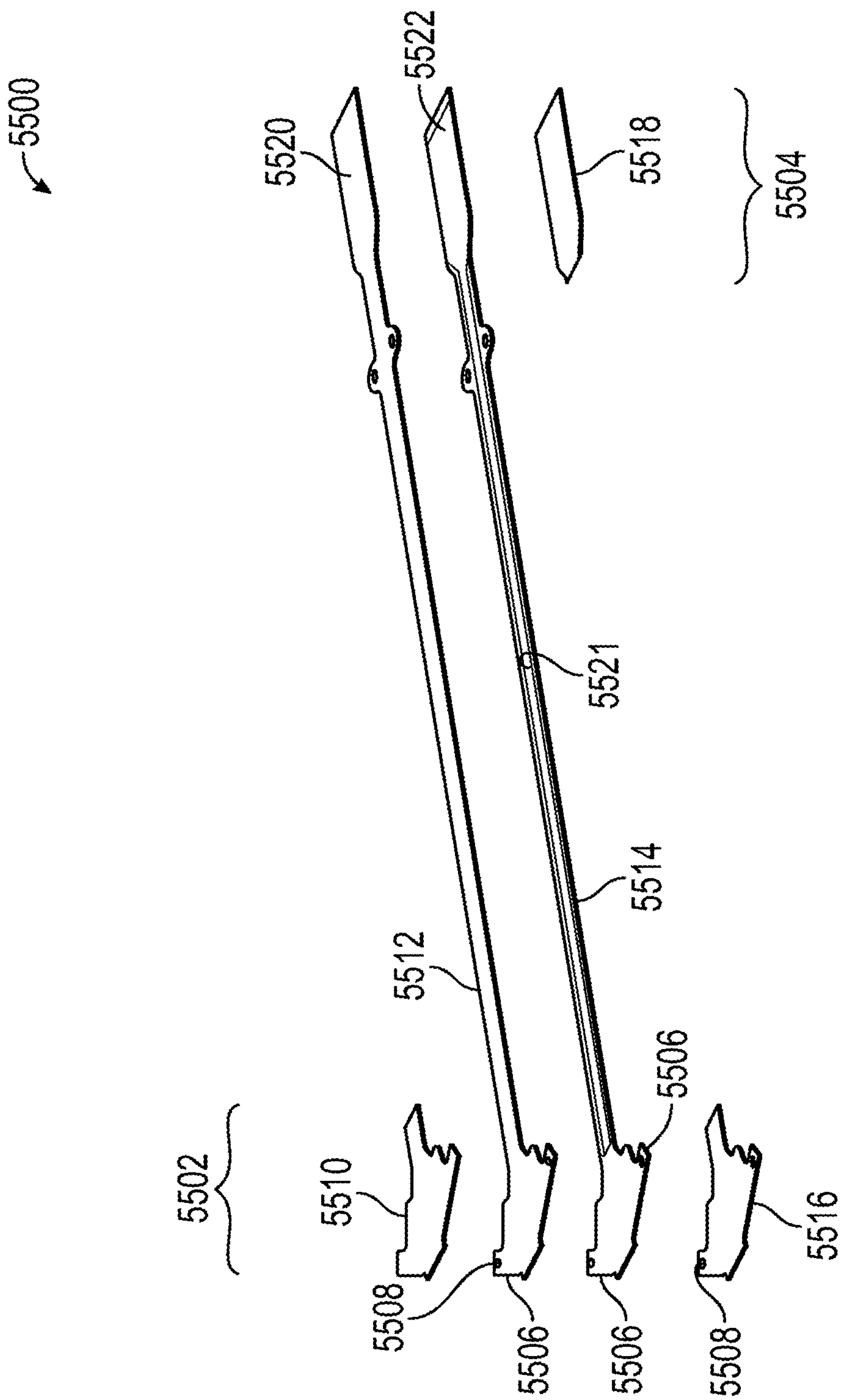
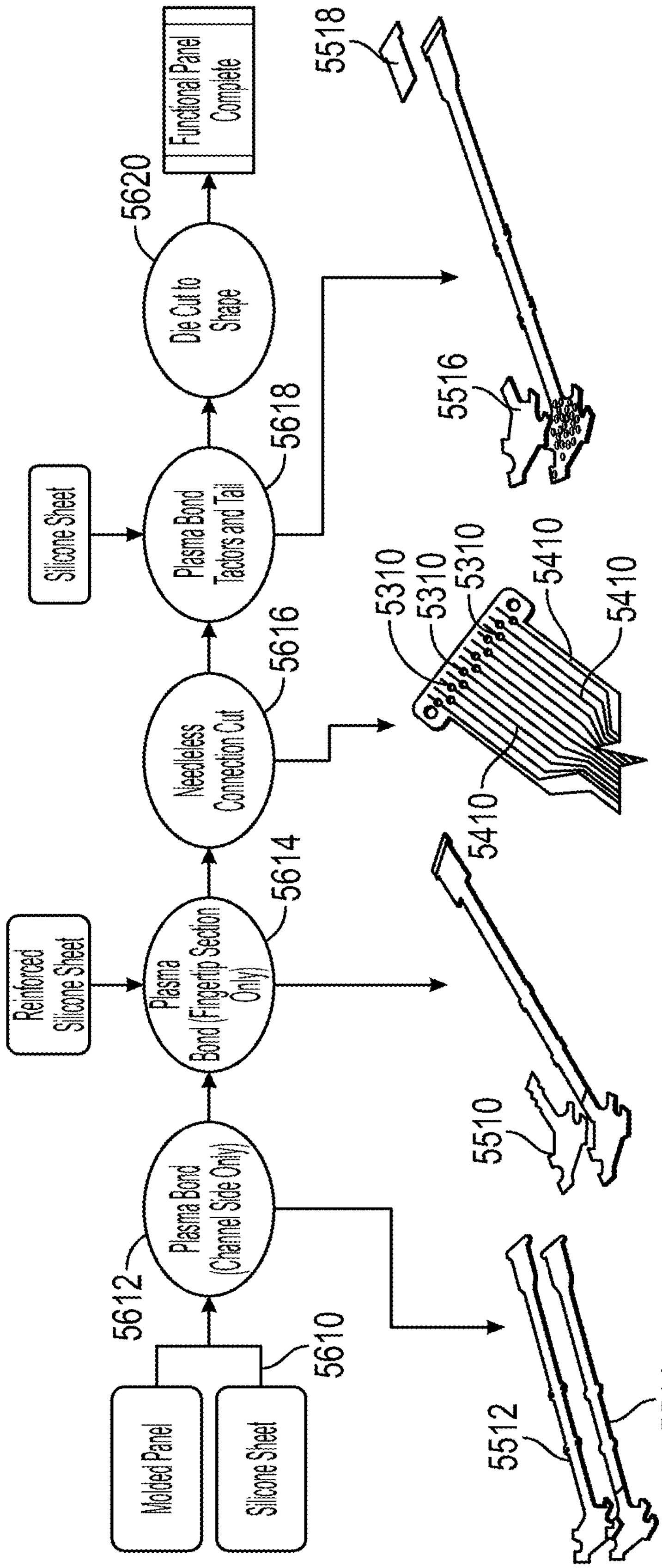

5500
5520
5522
5518
5504
5521
5514
5512
5506
5502
5510
5516
5508
5506
5506
5508

FIG. 58



5600
Molded Panel
Silicone Sheet
5610
5612
Plasma Bond (Channel Side Only)
Reinforced Silicone Sheet
Plasma Bond (Fingertip Section Only)
5614
Needleless Connection Cut
5616
Silicone Sheet
Plasma Bond Tactors and Tail
5618
Die Cut to Shape
5620
Functional Panel Complete
5512
5514
5510
5410
5410
5410
5310
5310
5310
5410
5516
5518

FIG. 59

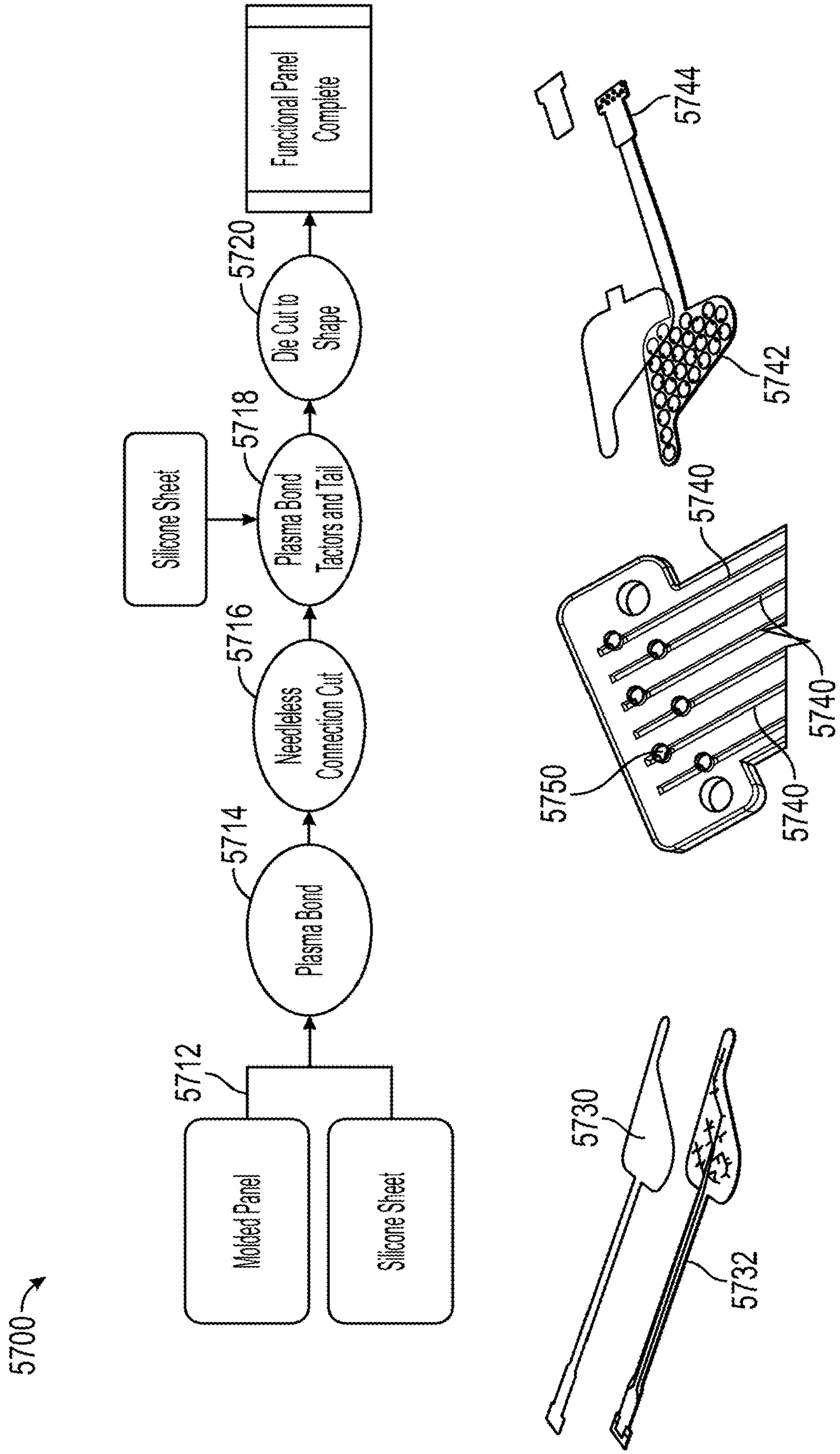

5700
Molded Panel
Silicone Sheet
5712
Plasma Bond
5714
Needleless Connection Cut
5716
Silicone Sheet
Plasma Bond Tactors and Tail
5718
Die Cut to Shape
5720
Functional Panel Complete
5730
5732
5750
5740
5740
5740
5742
5744

FIG. 60

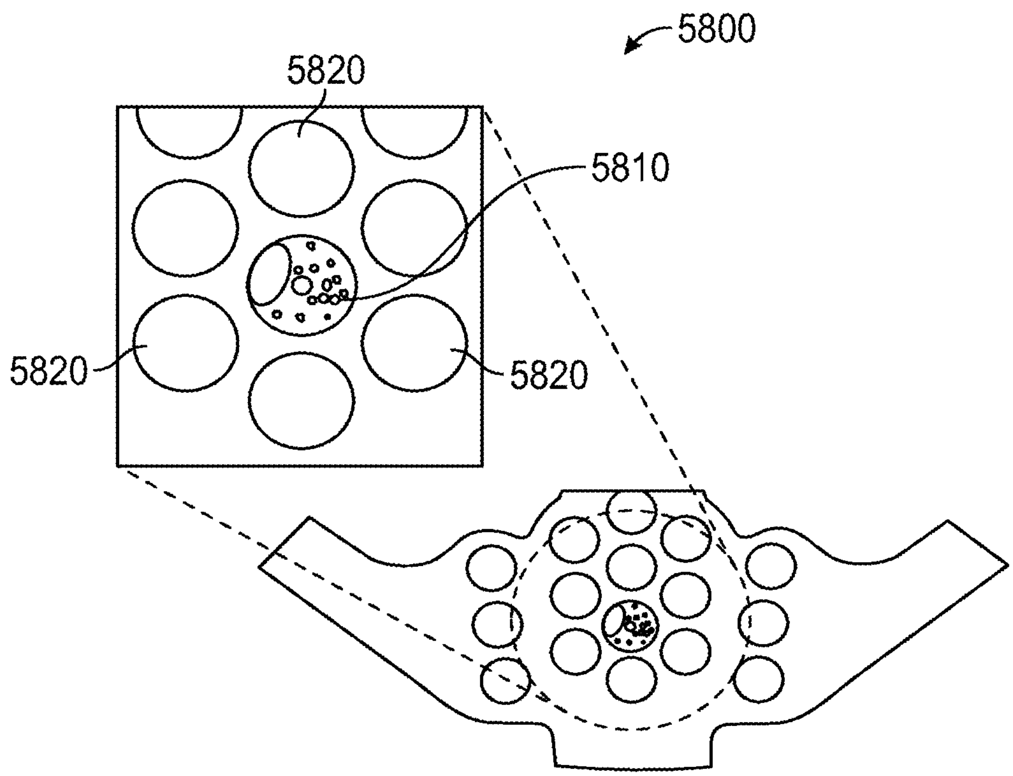


FIG. 61

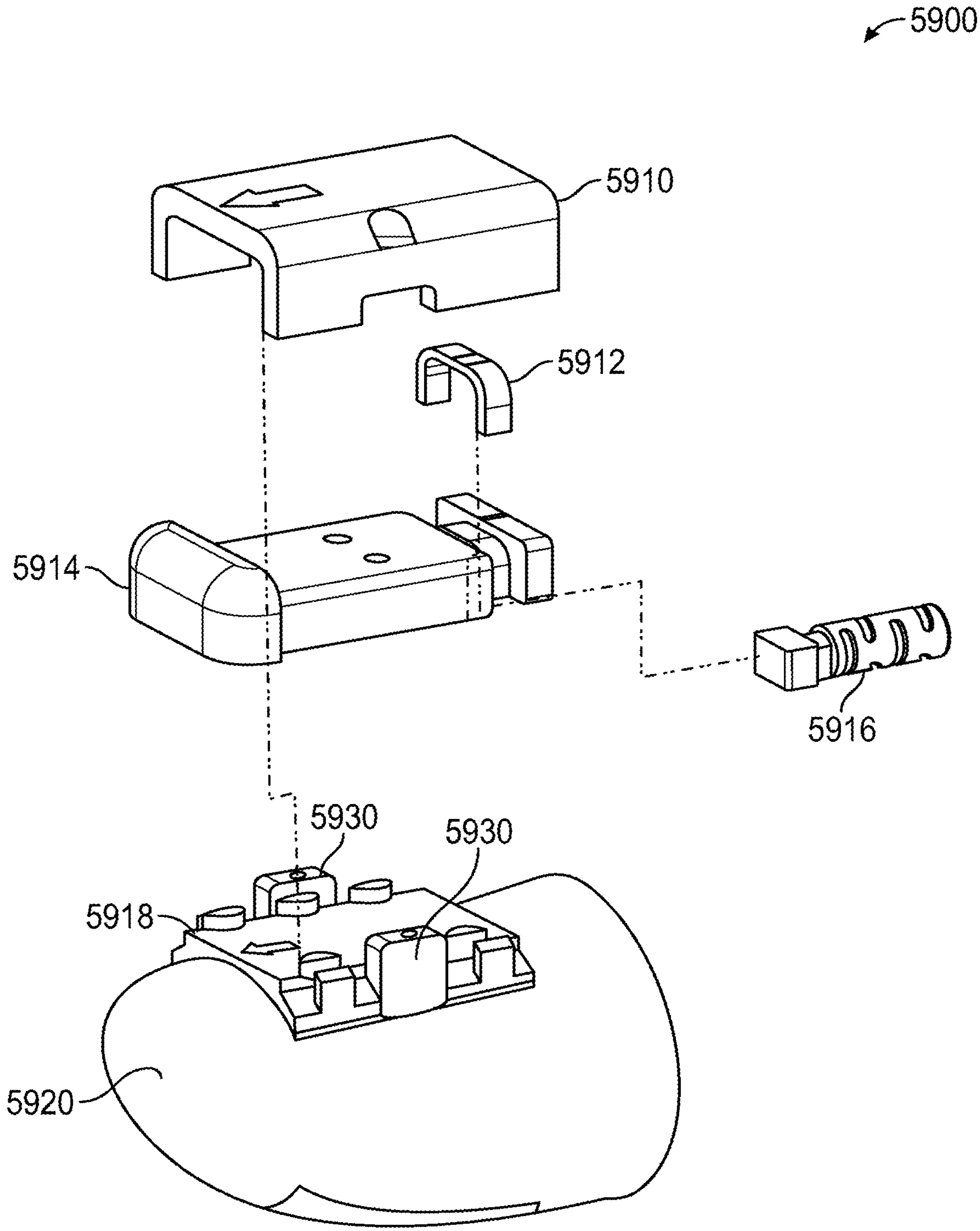


FIG. 62

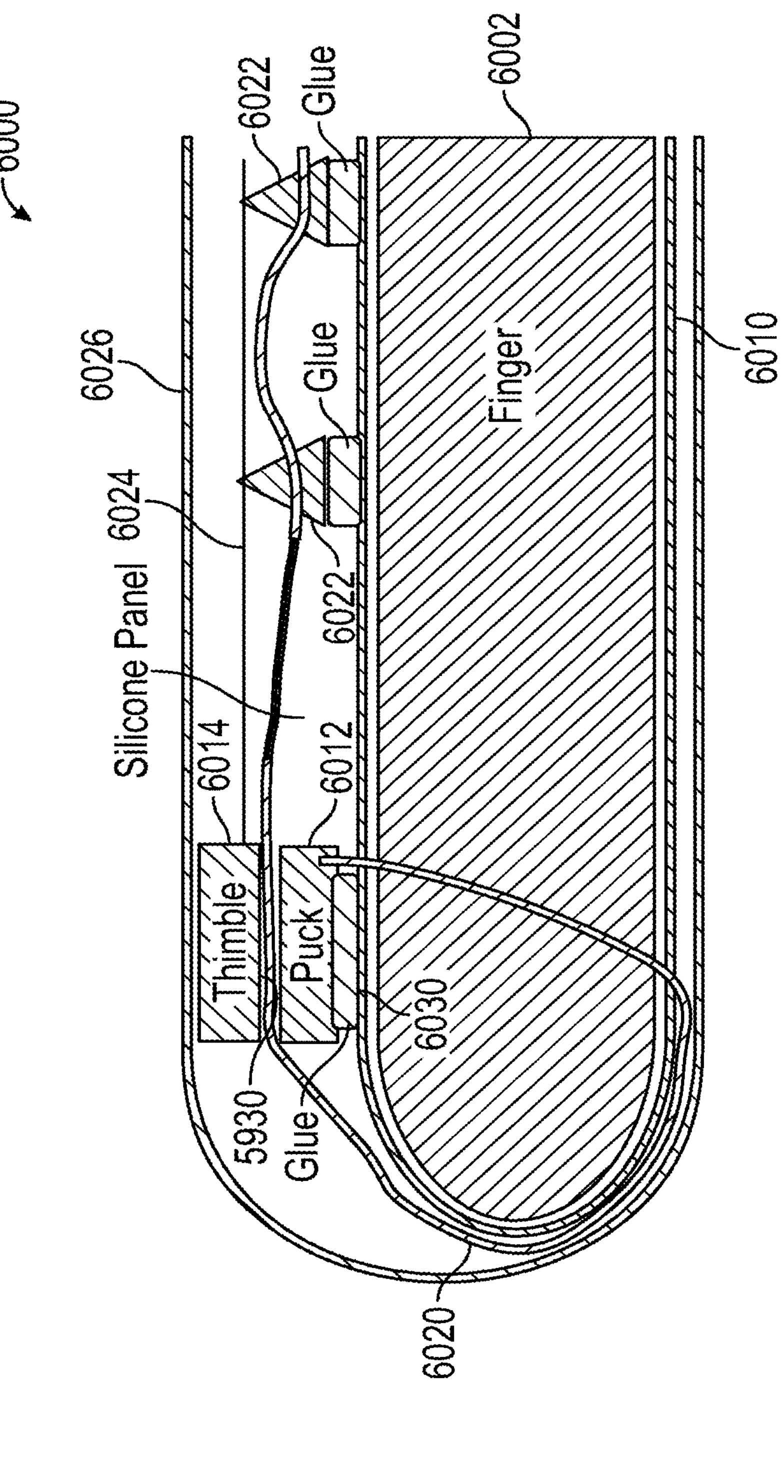

6000
6026
6022
Glue
6002
Glue
Finger
6010
Glue
Silicone Panel 6024
6022
6014
6012
Thimble
Puck
6030
5930
Glue
6020

FIG. 63

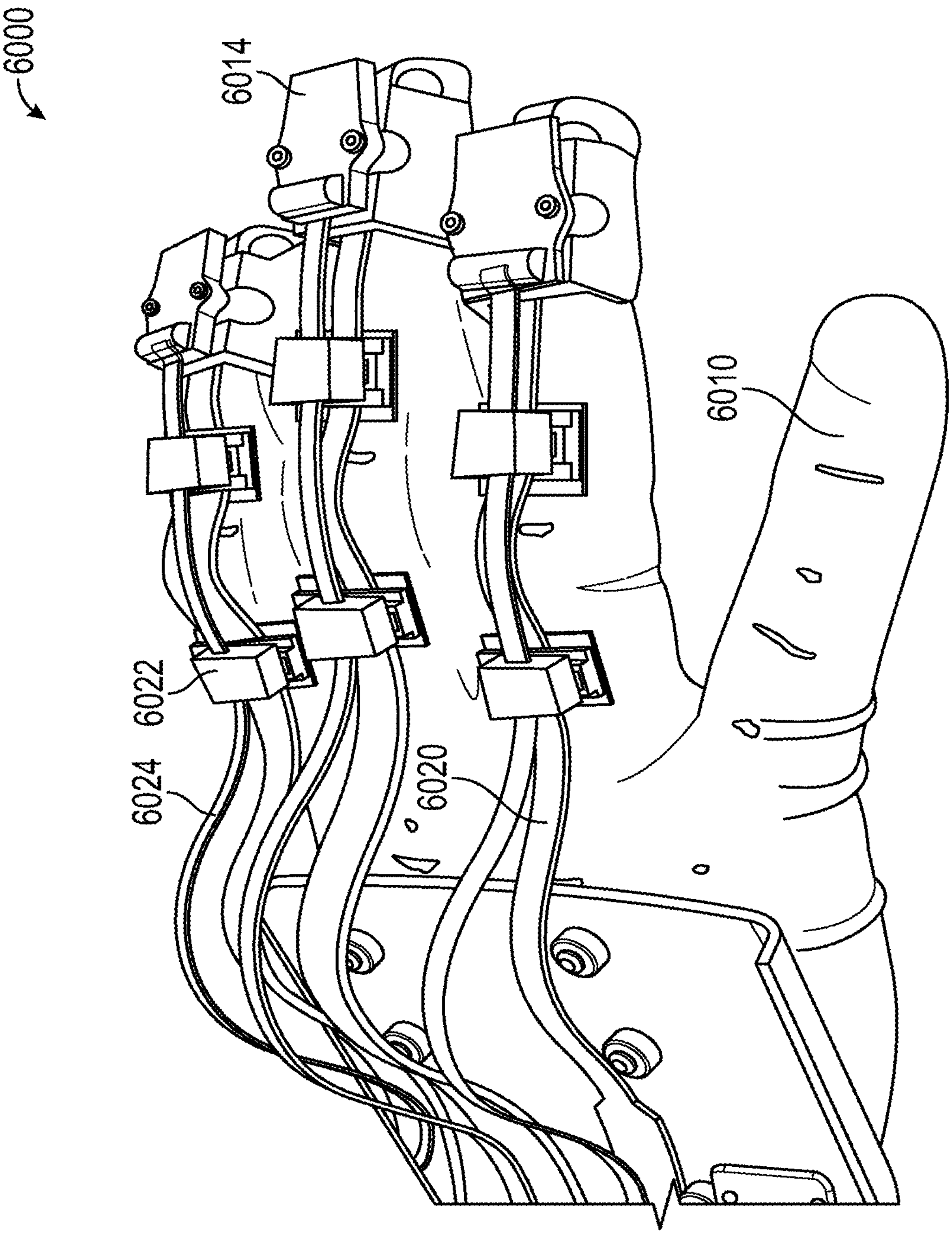


FIG. 64

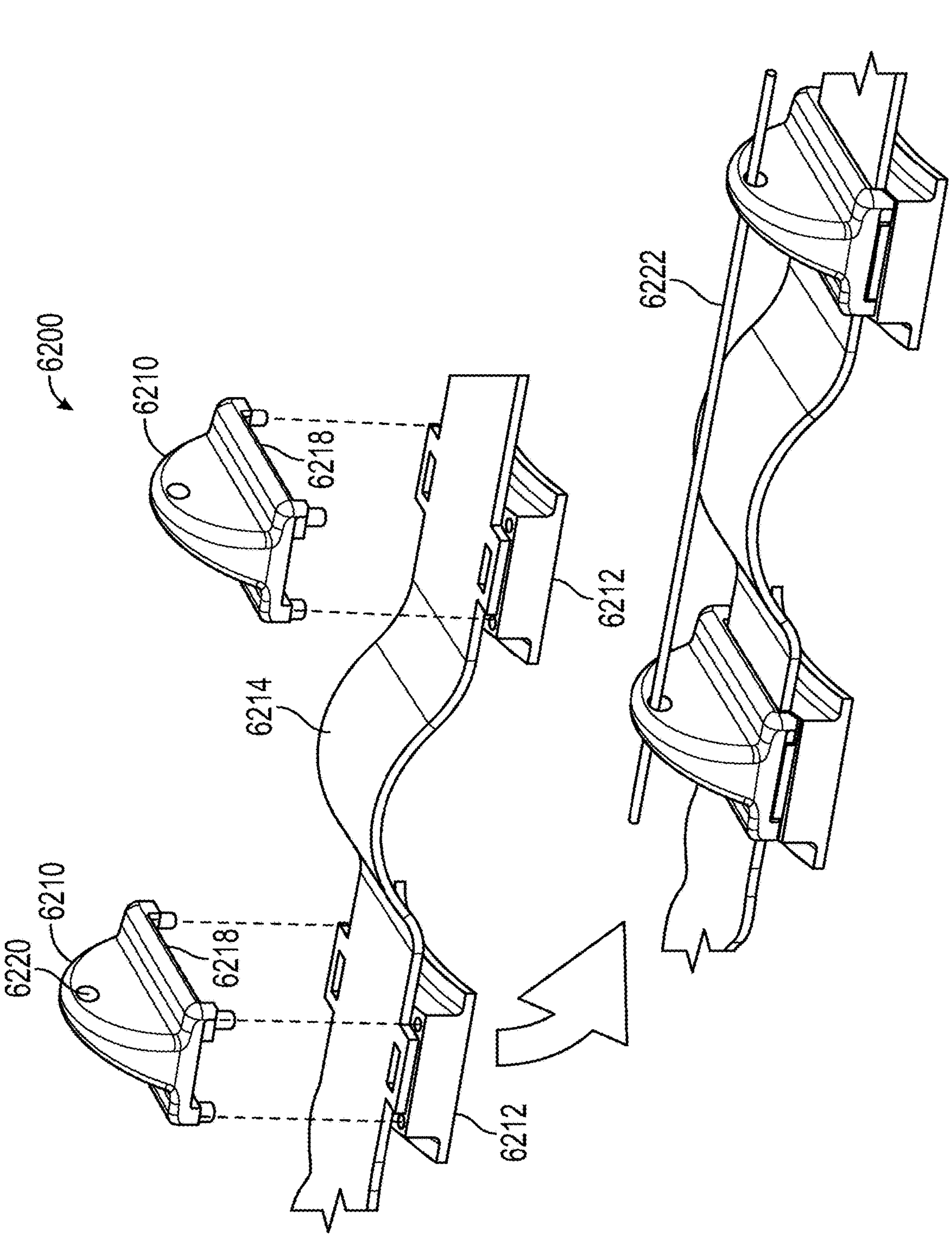


FIG. 65

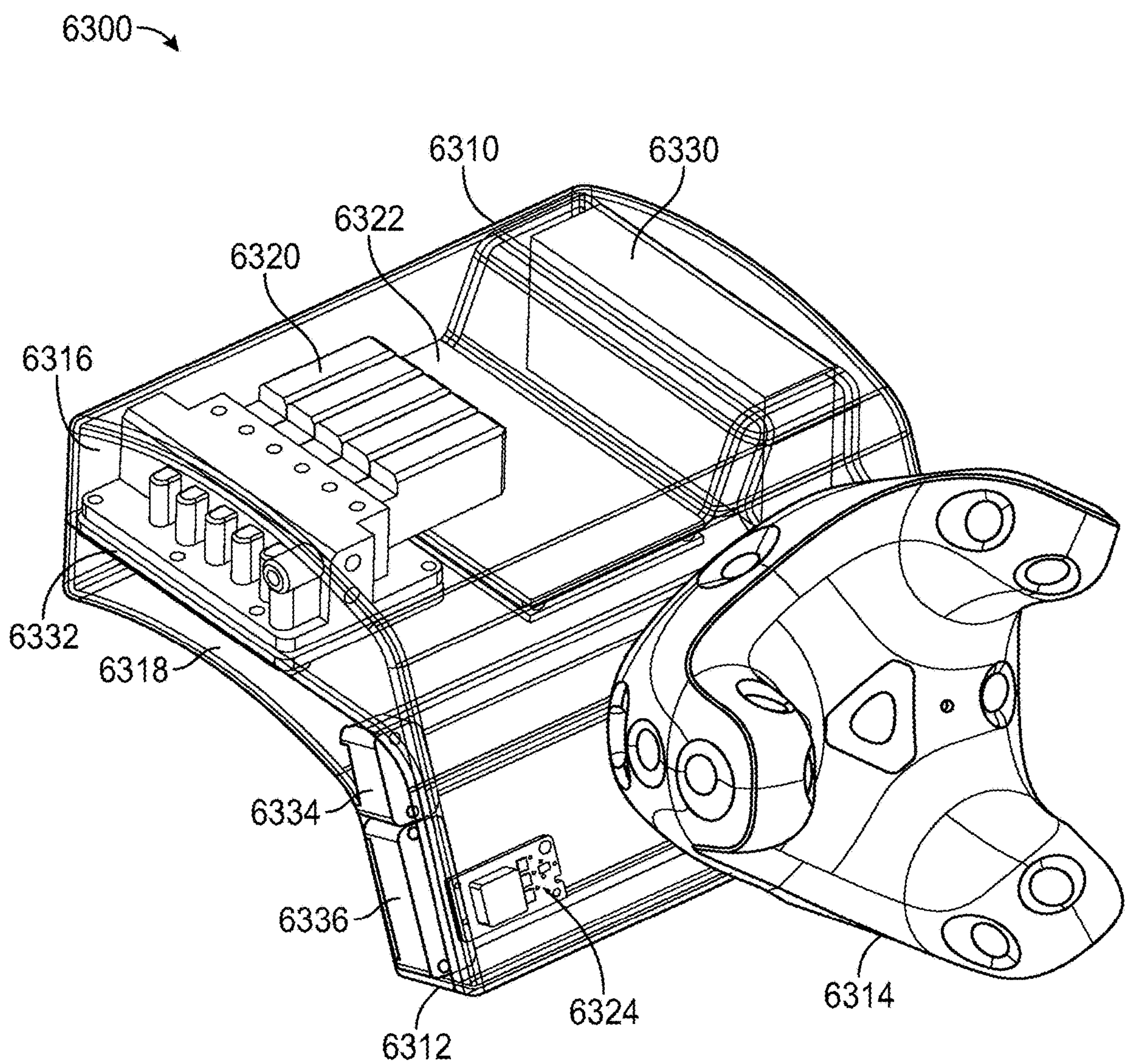

6300
6310
6330
6322
6320
6316
6332
6318
6334
6336
6312
6324
6314

FIG. 66

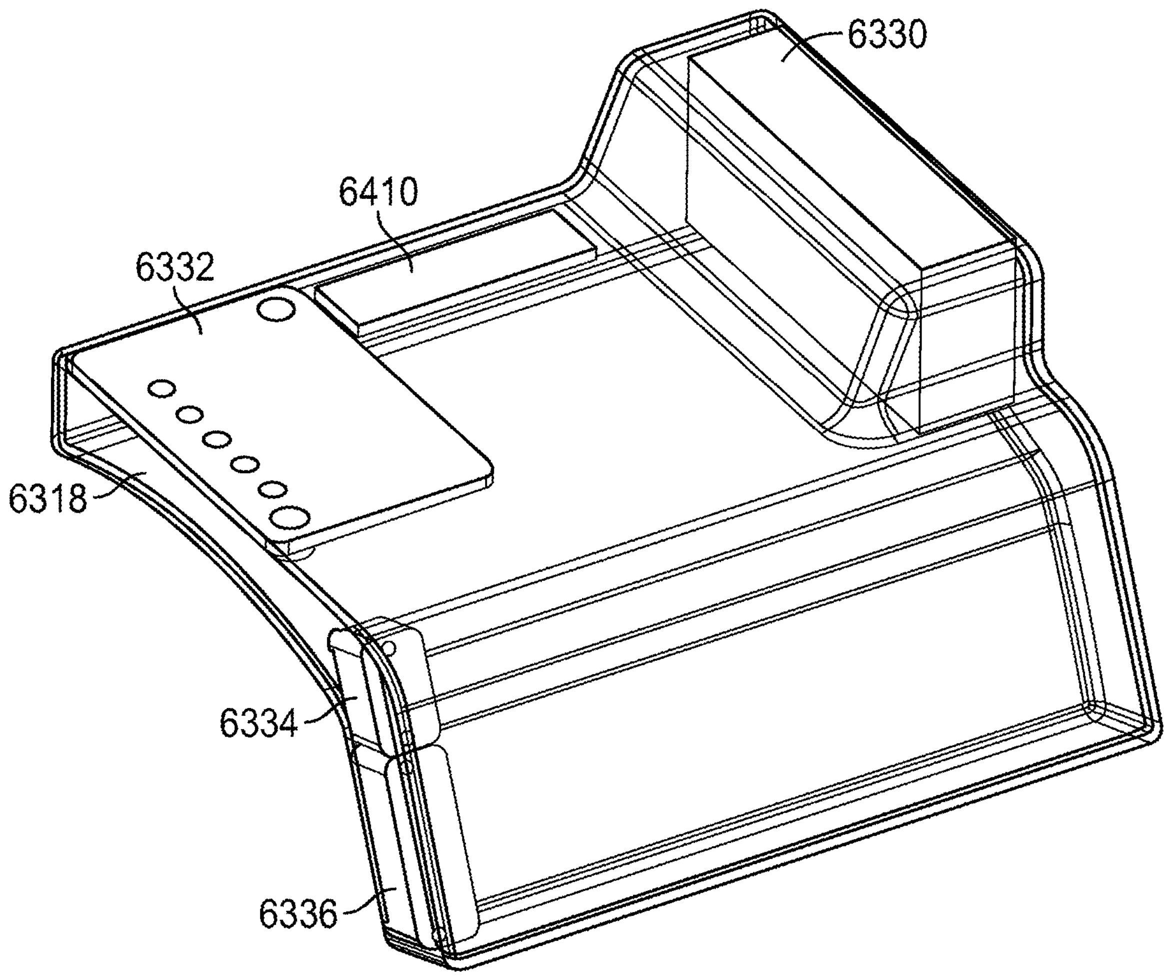


FIG. 67

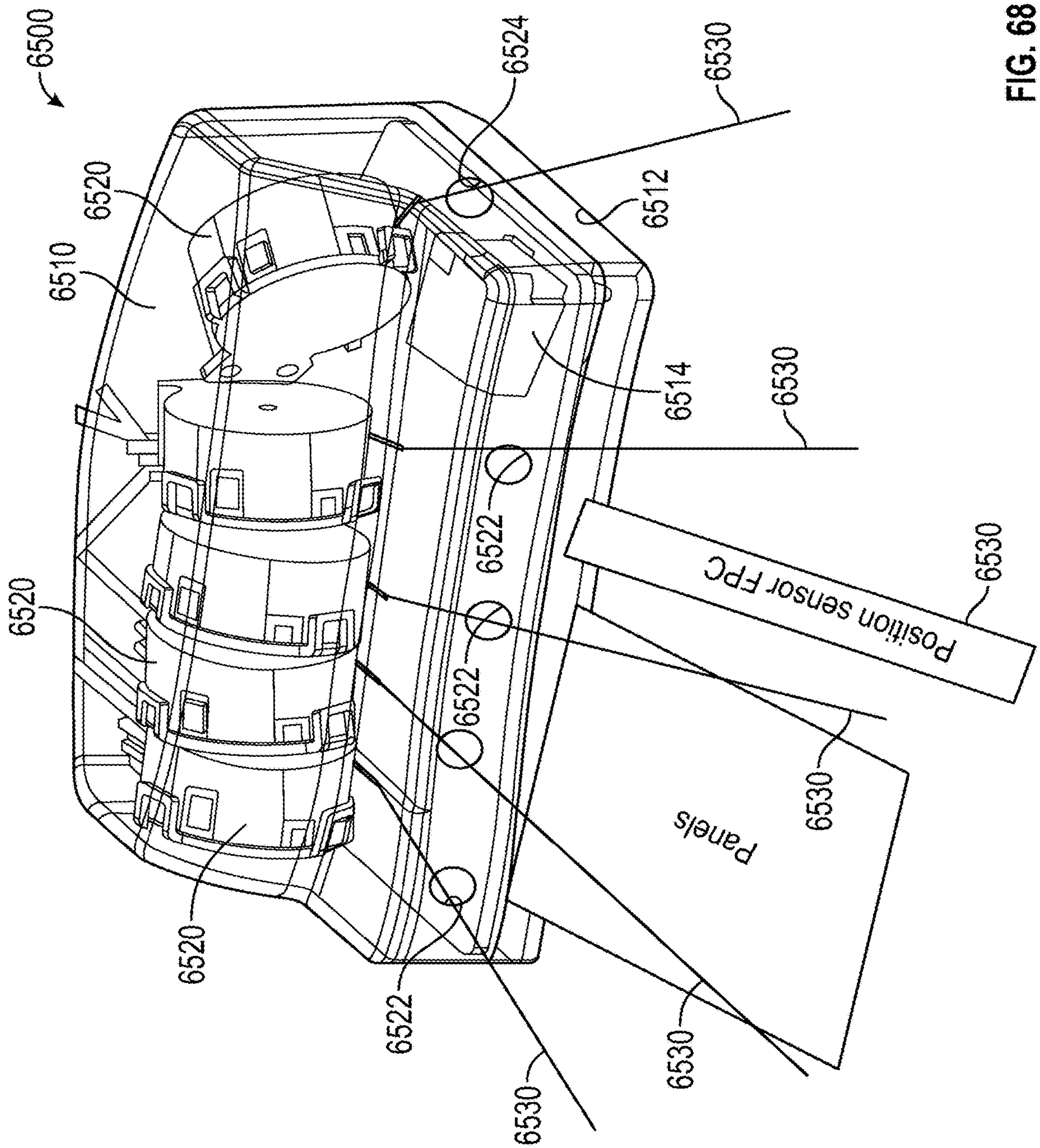

6500
6510
6512
6514
6520
6520
6520
6522
6522
6522
6522
6524
6530
6530
6530
6530
6530
6530
Position sensor FPC
Panels

FIG. 68

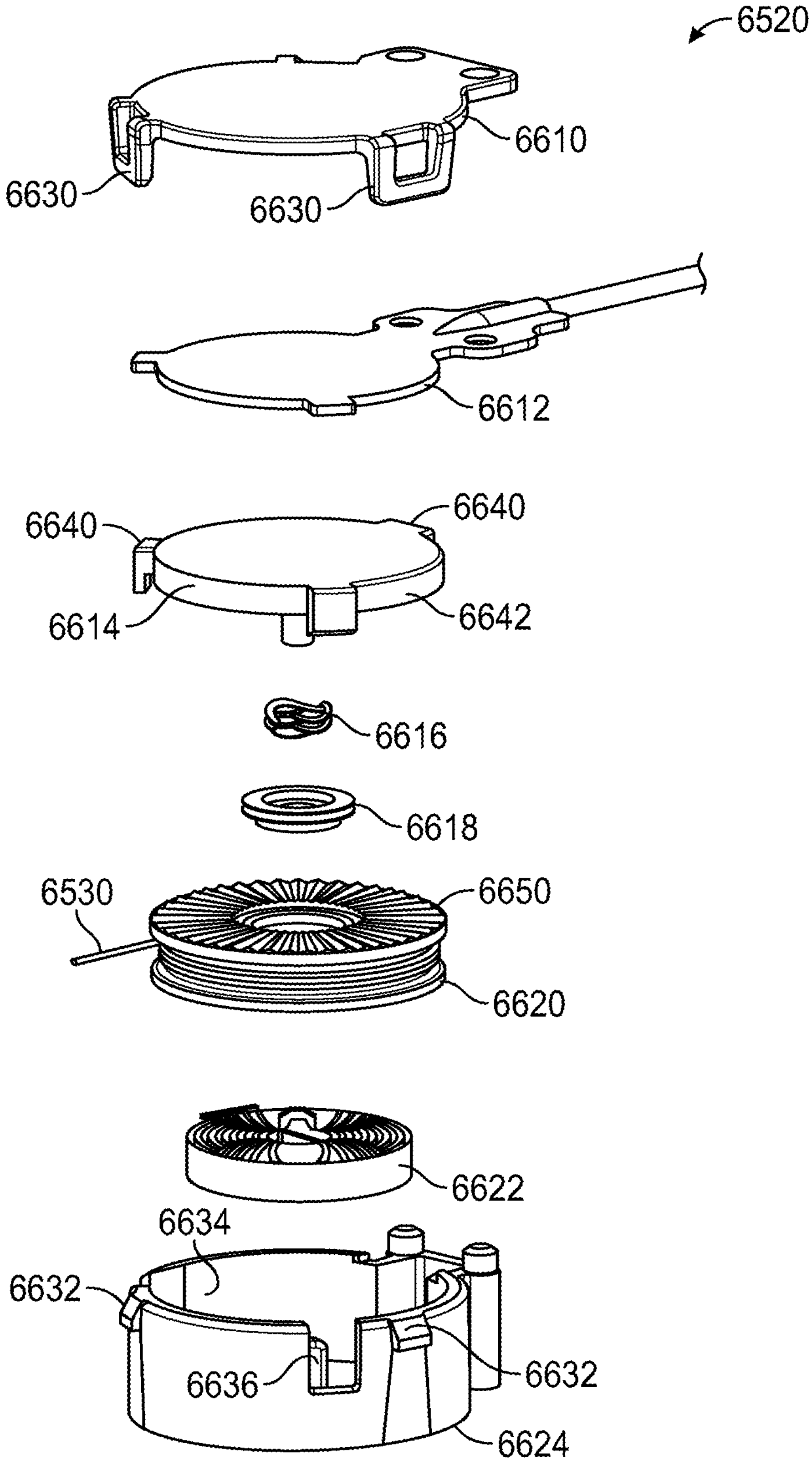

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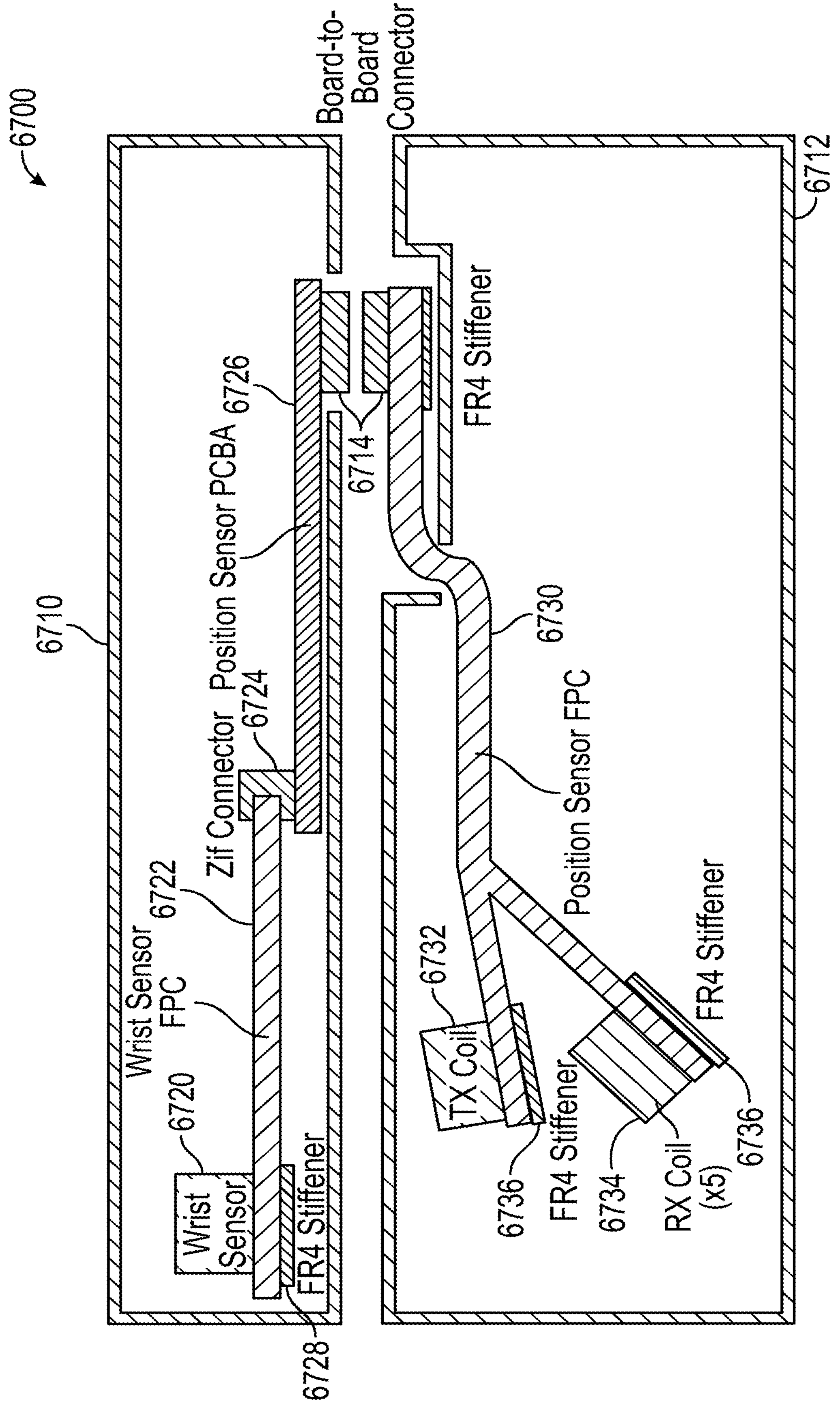
FIG. 69


6700
6710
6712
Board-to-Board Connector
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Position Sensor PCBA
6714
FR4 Stiffener
6730
6724
Zif Connector
Position Sensor FPC
6722
Wrist Sensor FPC
6720
Wrist Sensor
FR4 Stiffener
6728
6732
TX Coil
Position Sensor FPC
FR4 Stiffener
6736
FR4 Stiffener
6734
RX Coil (x5)
6736

FIG. 70

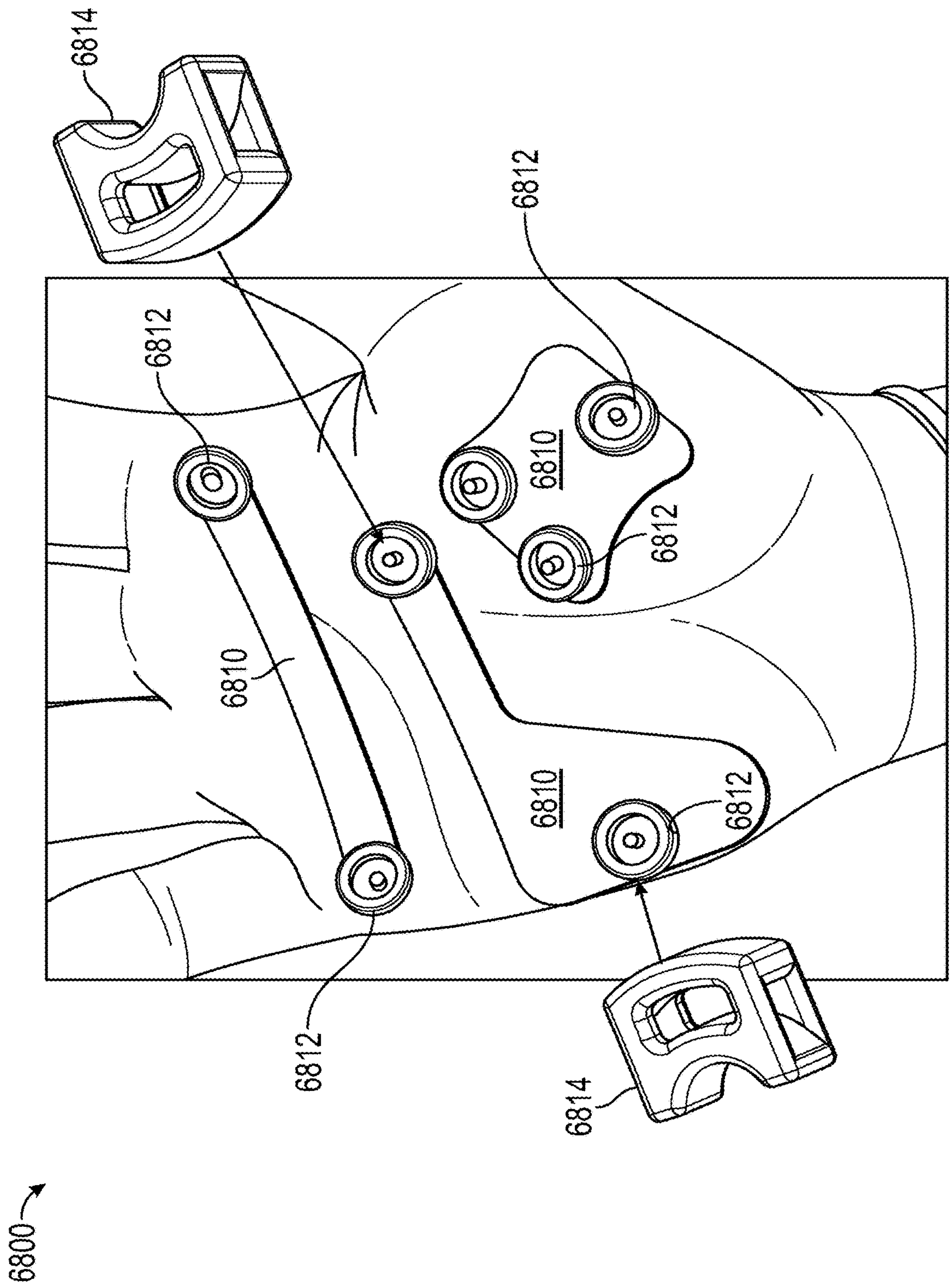

FIG. 71

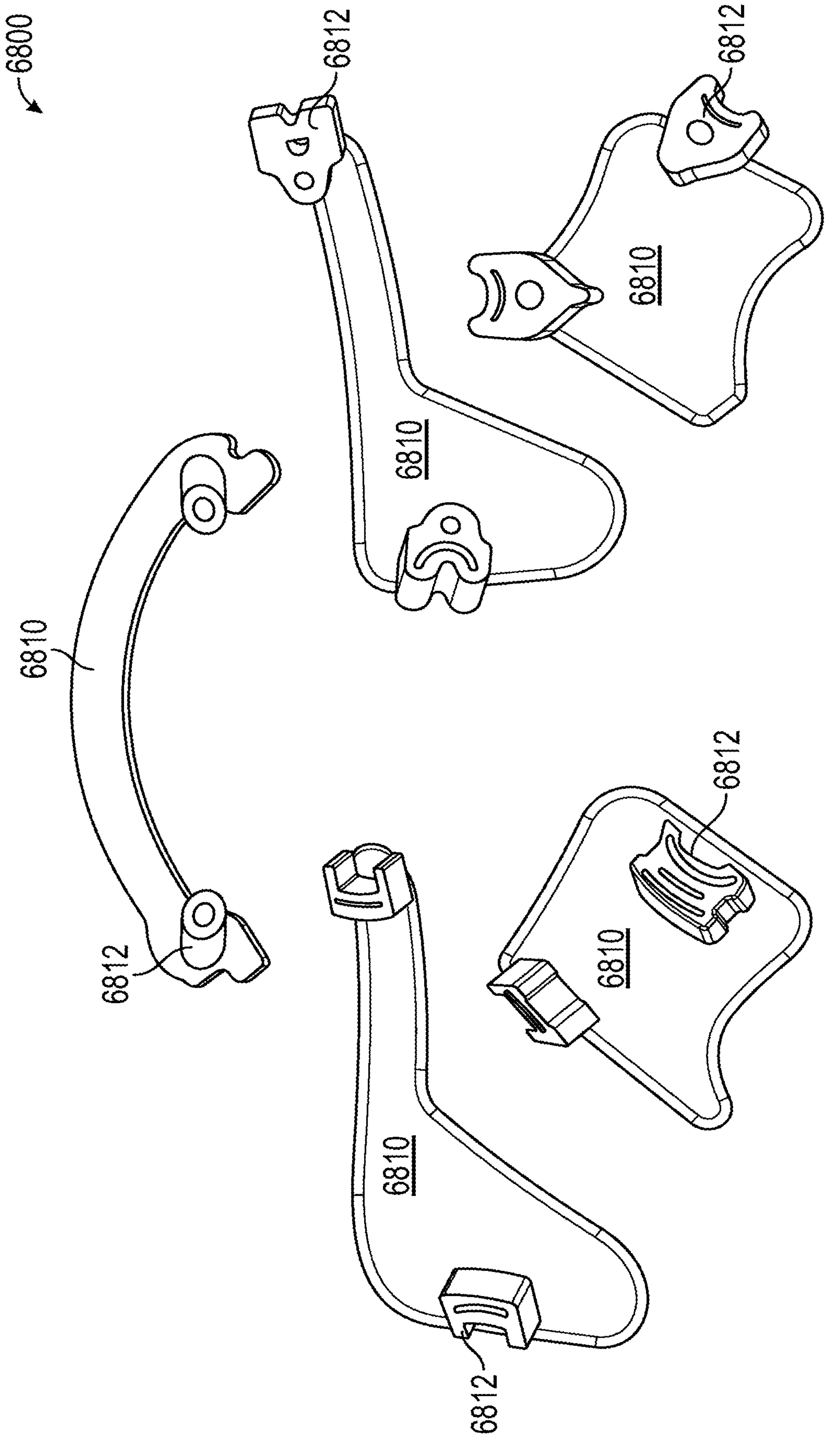

FIG. 72

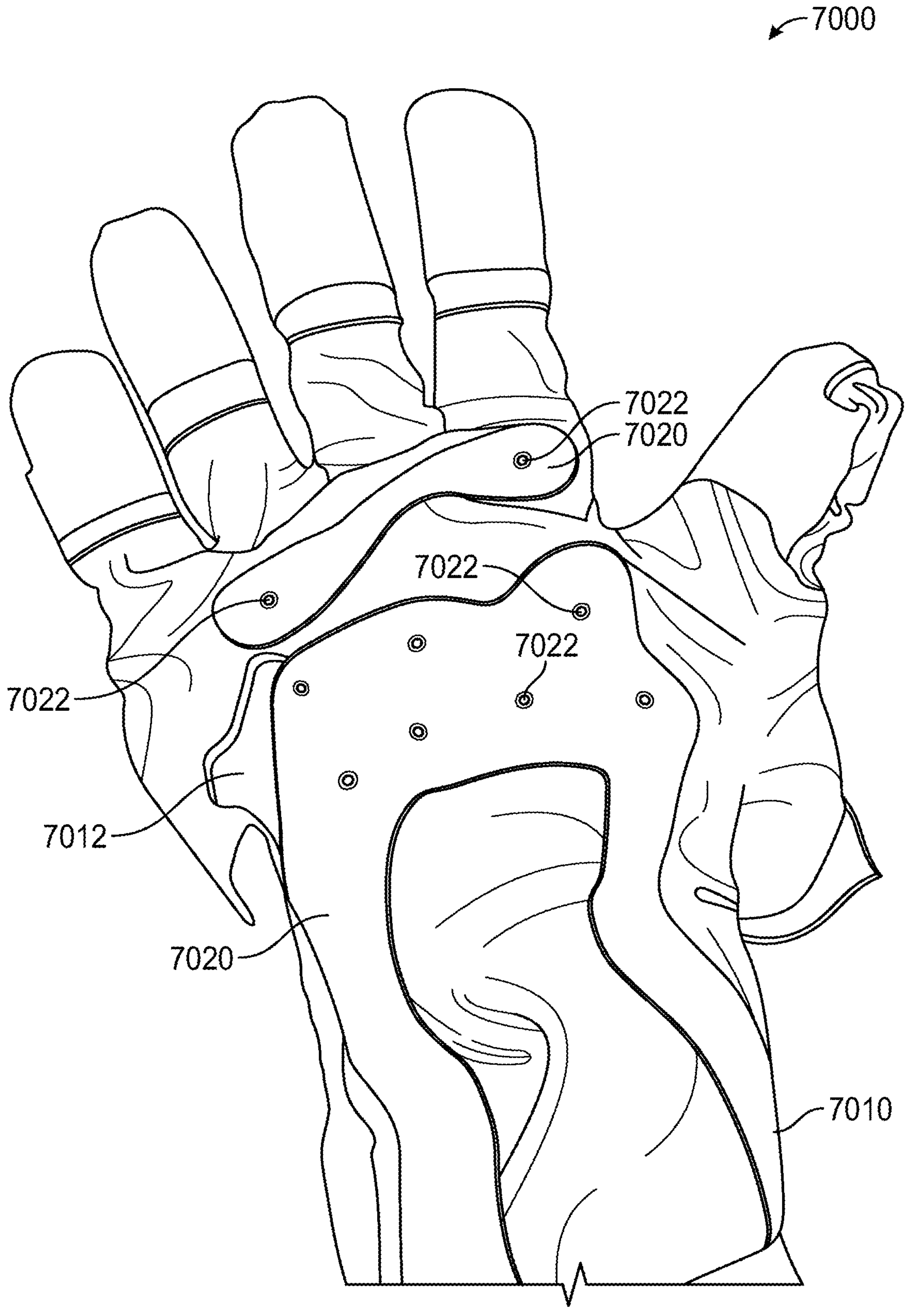

FIG. 73

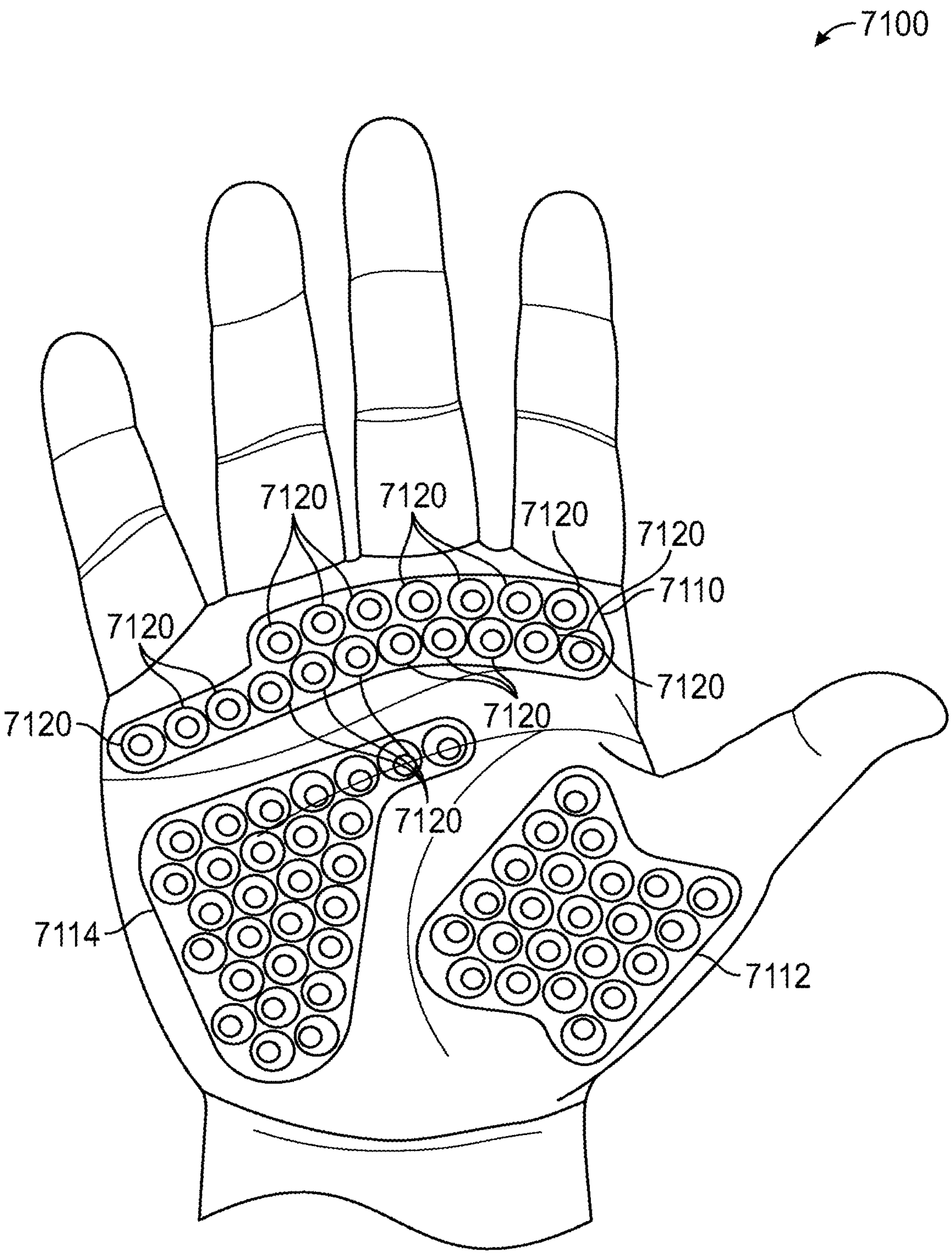
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FIG. 74

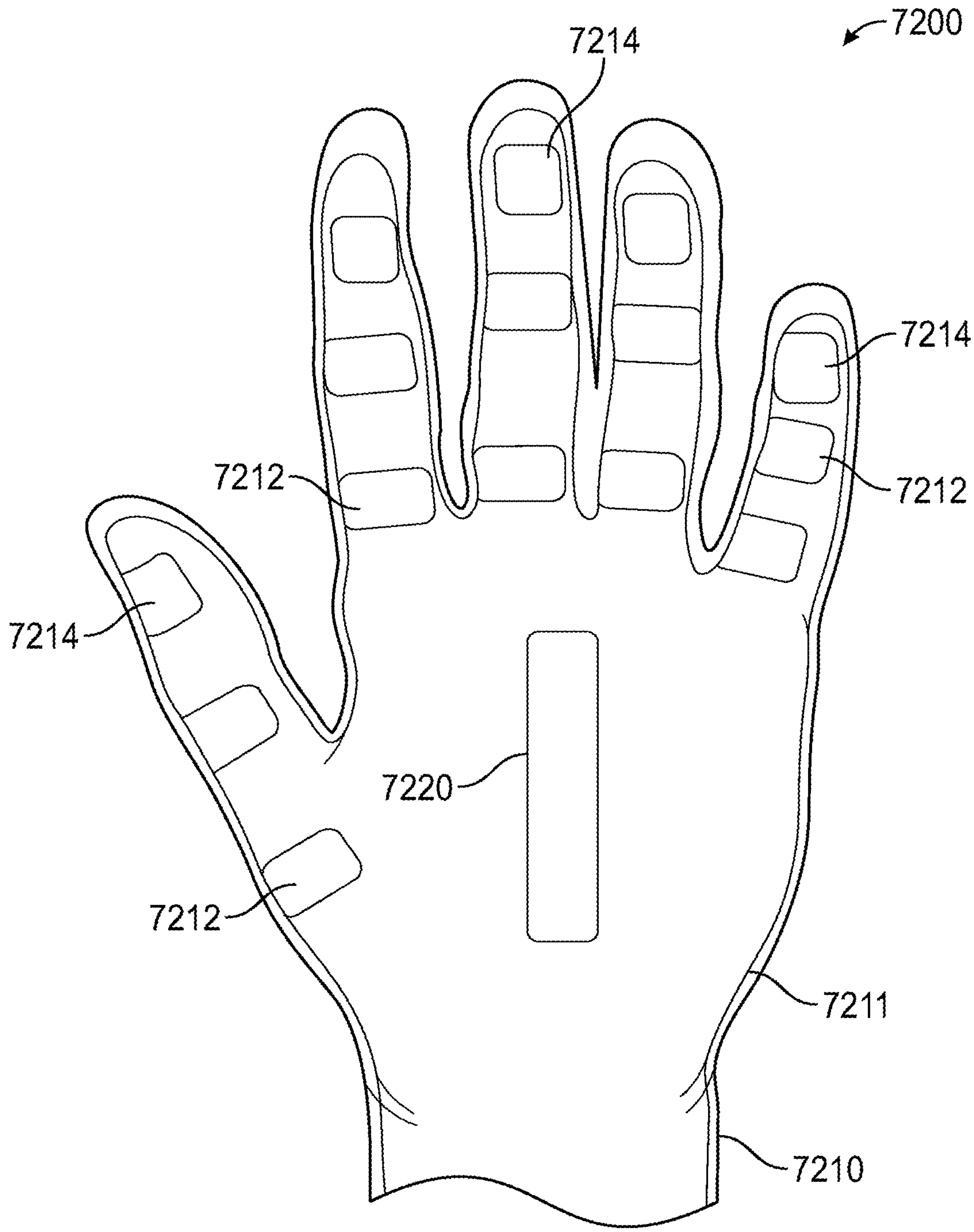

FIG. 75

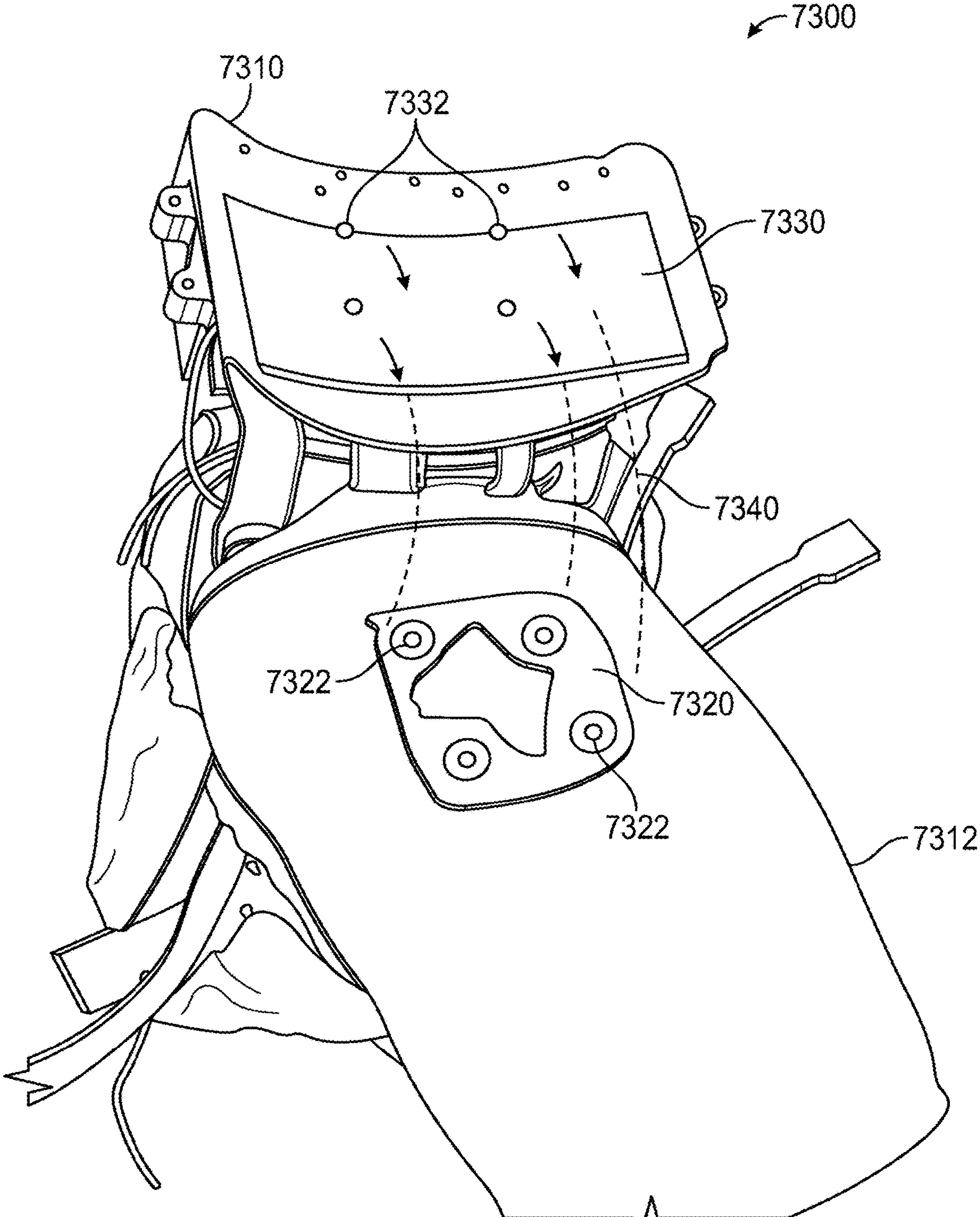

FIG. 76

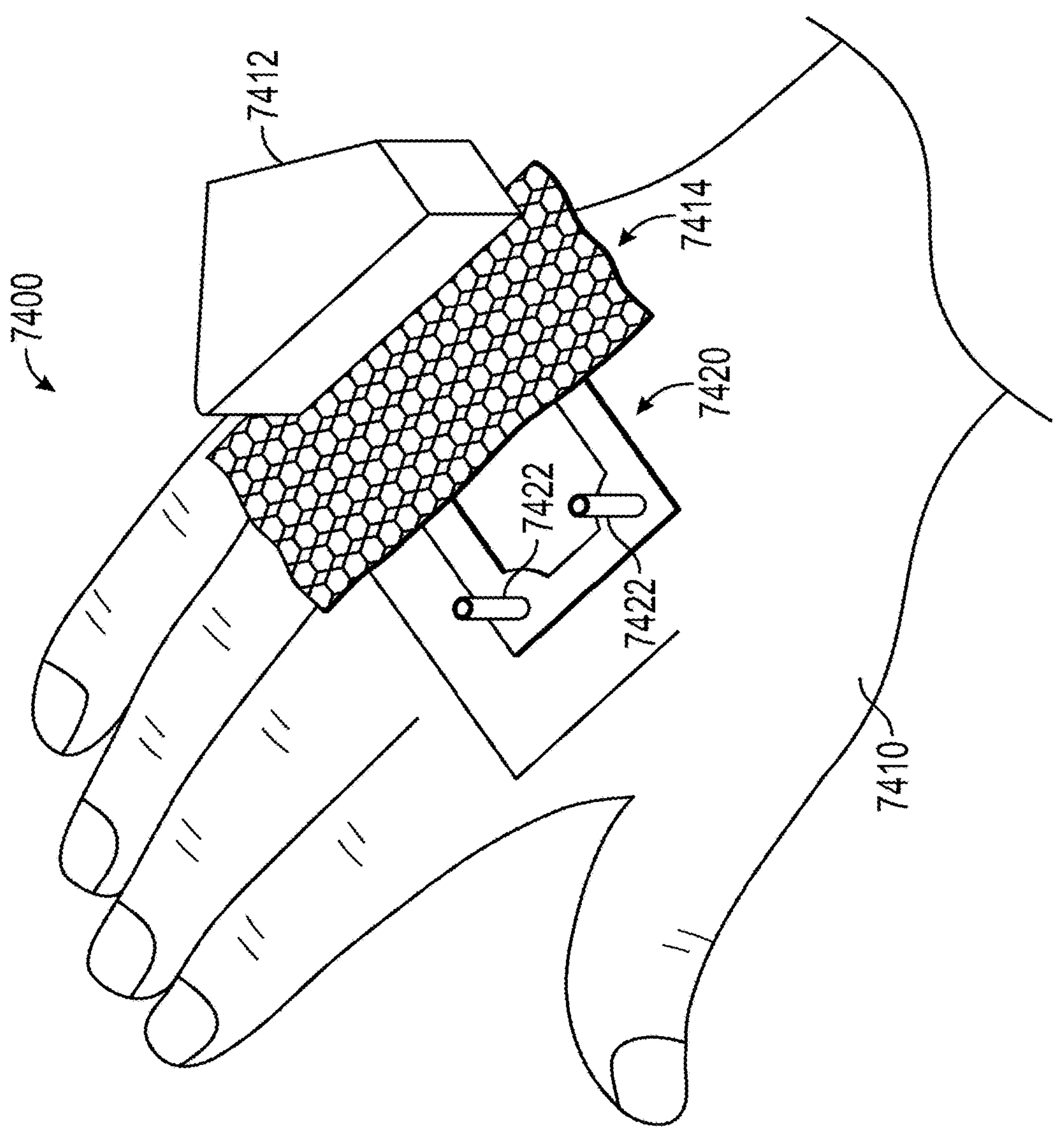

FIG. 77

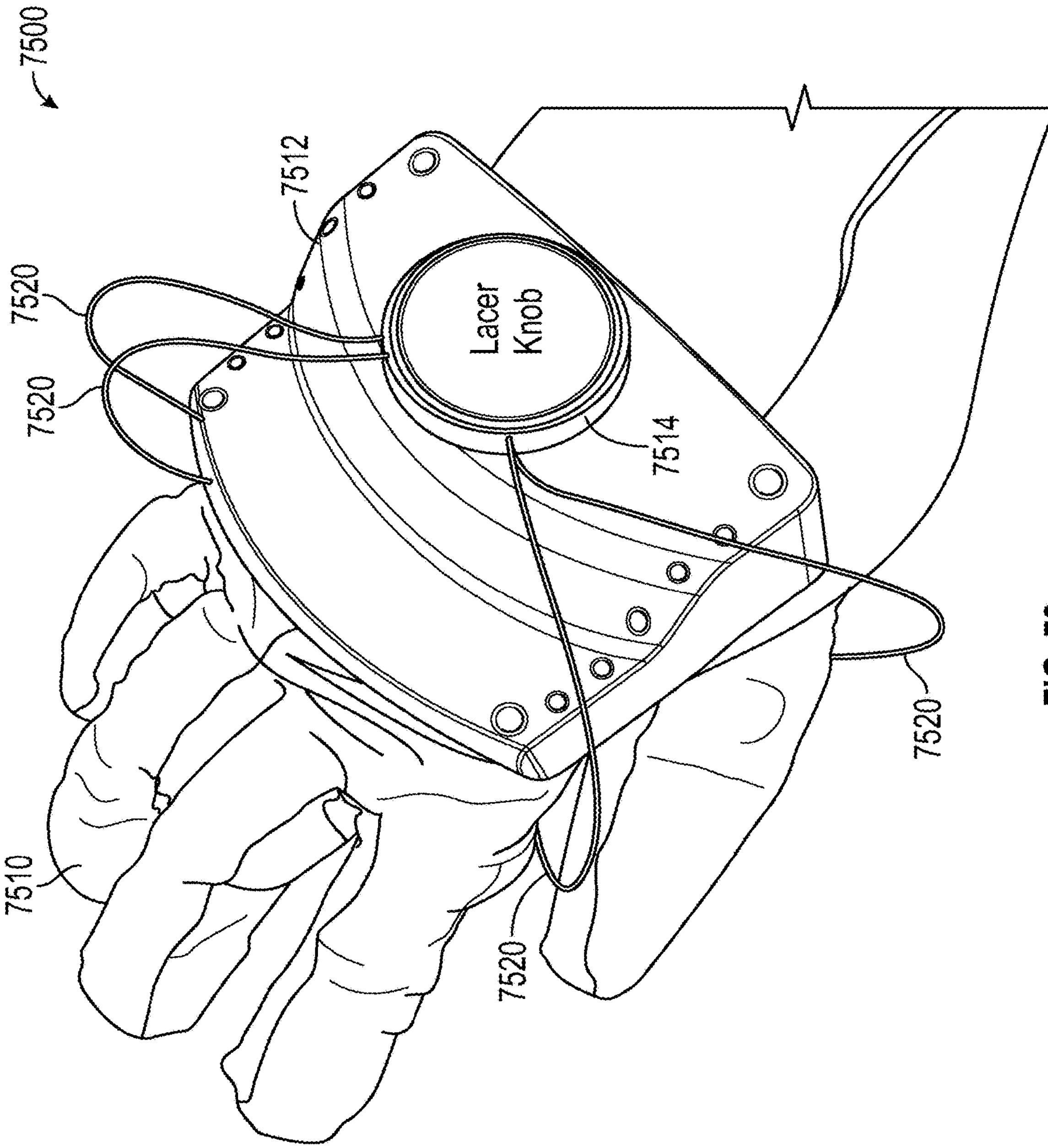

FIG. 78

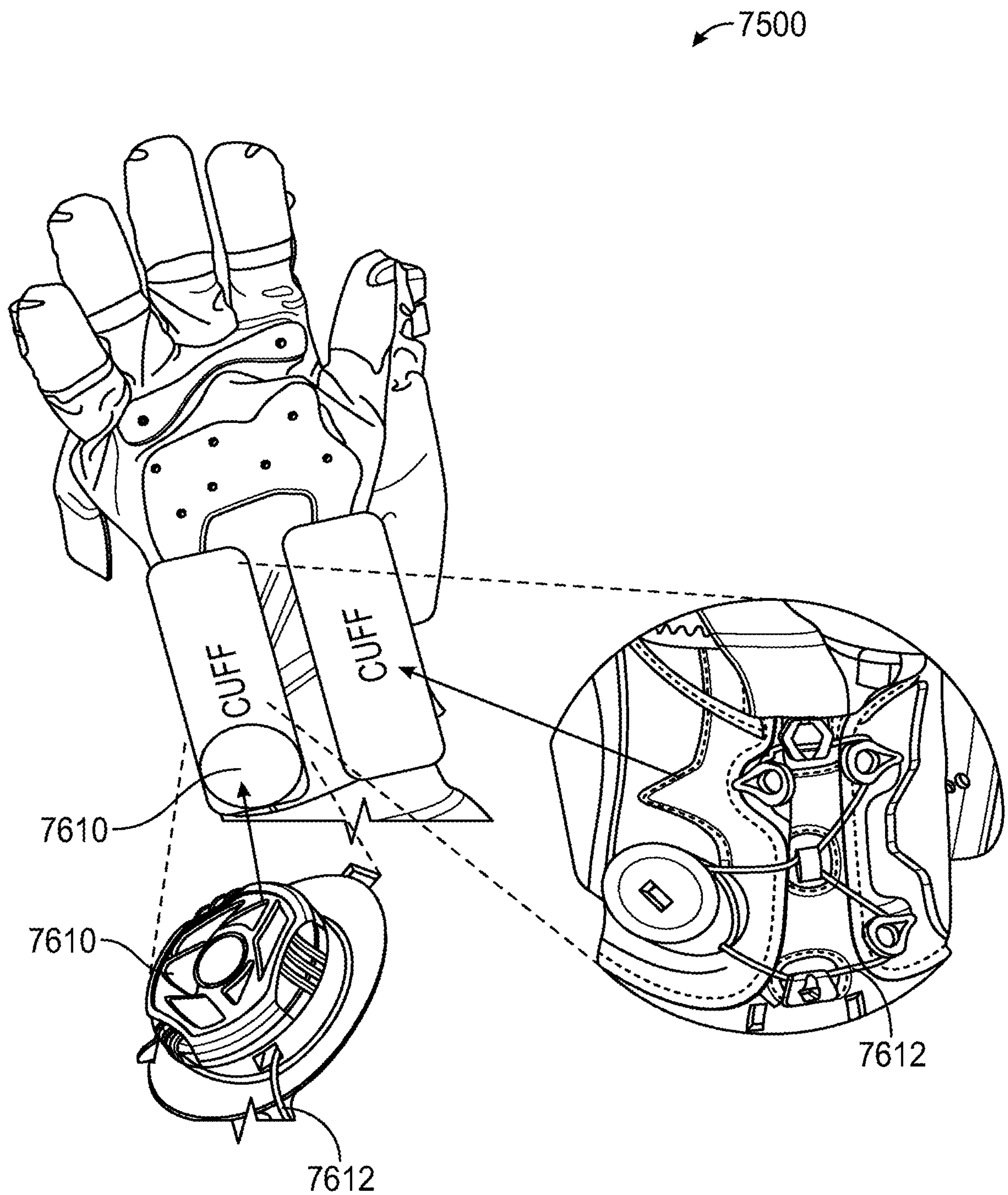

FIG. 79

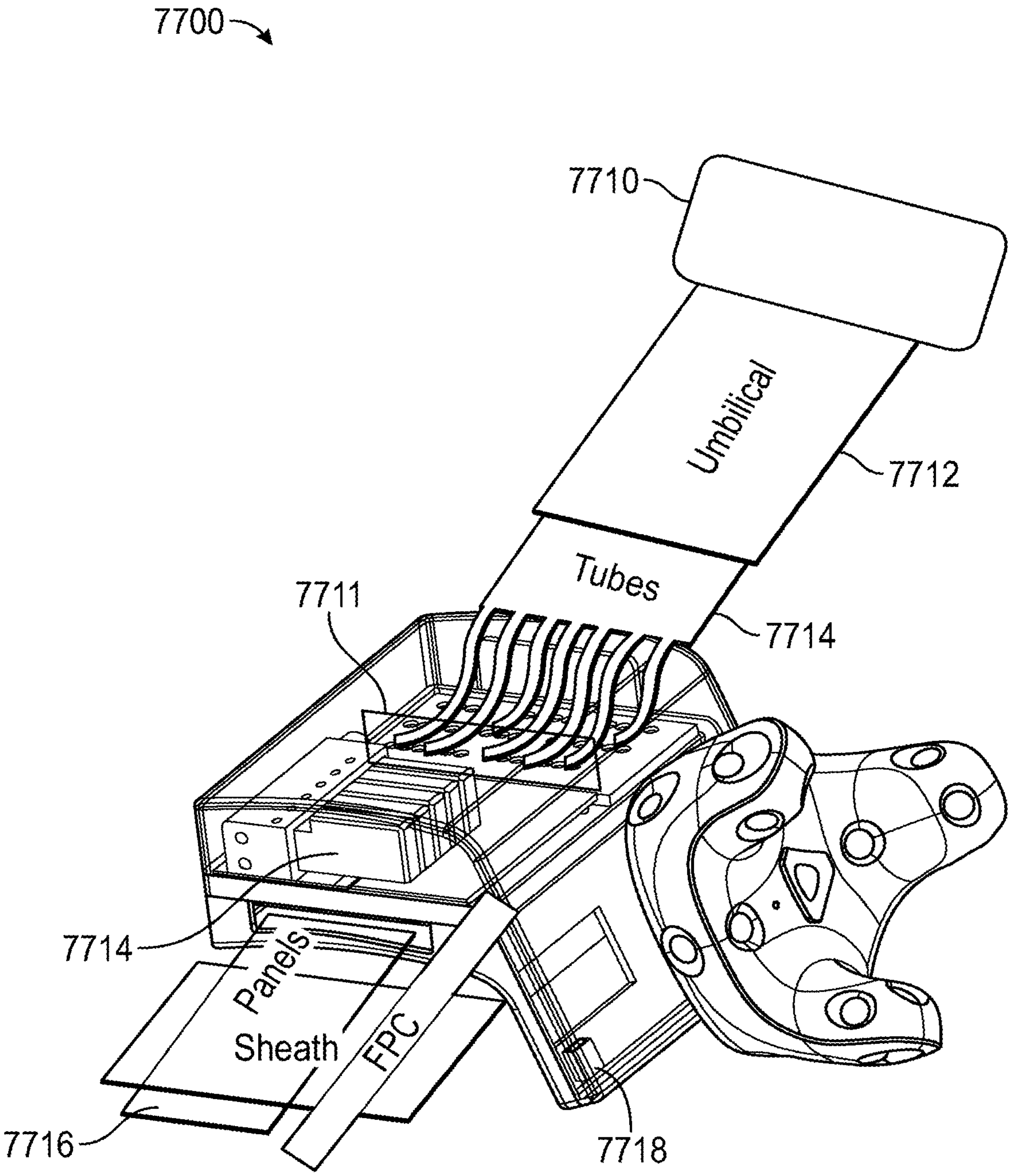

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FIG. 80

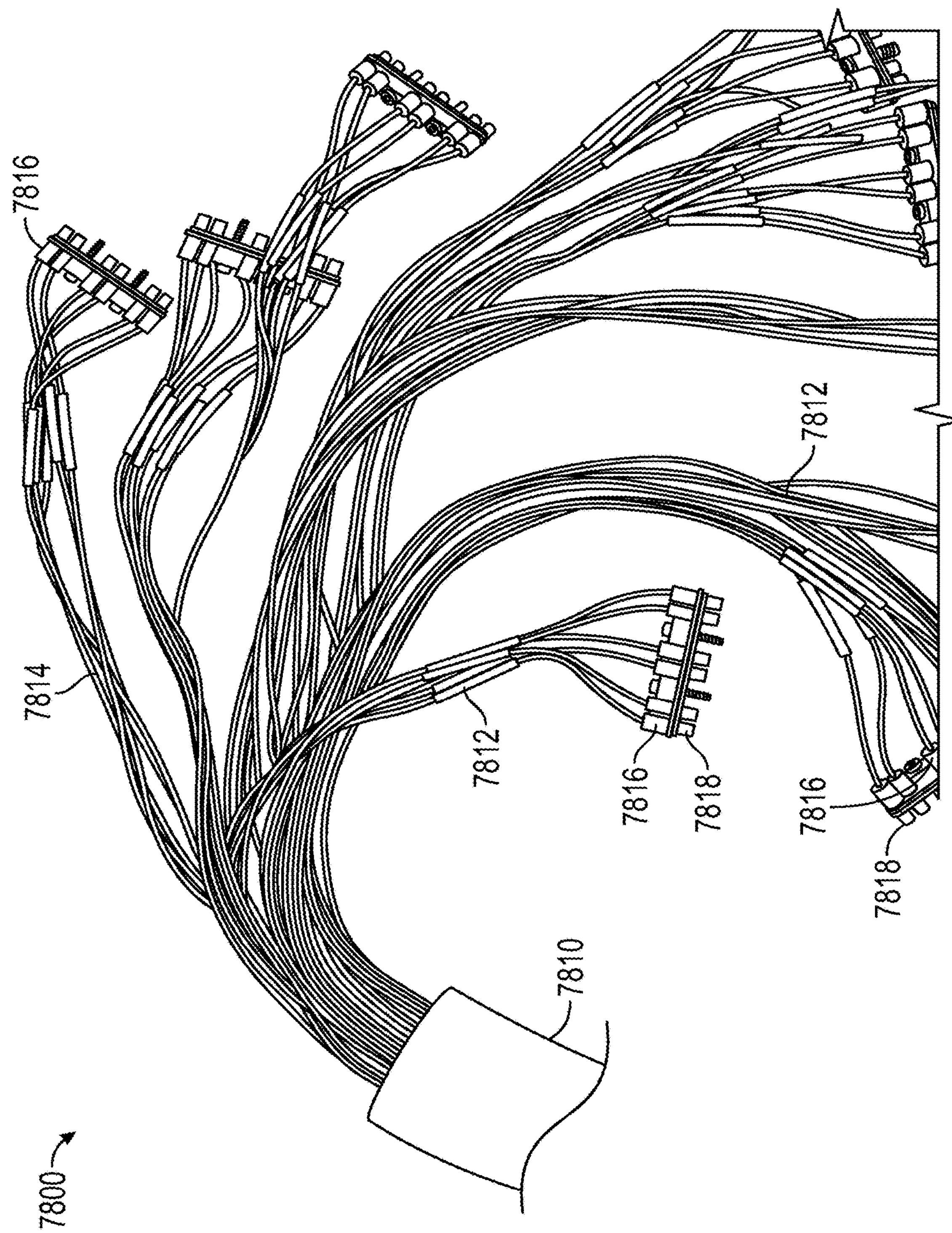

FIG. 81

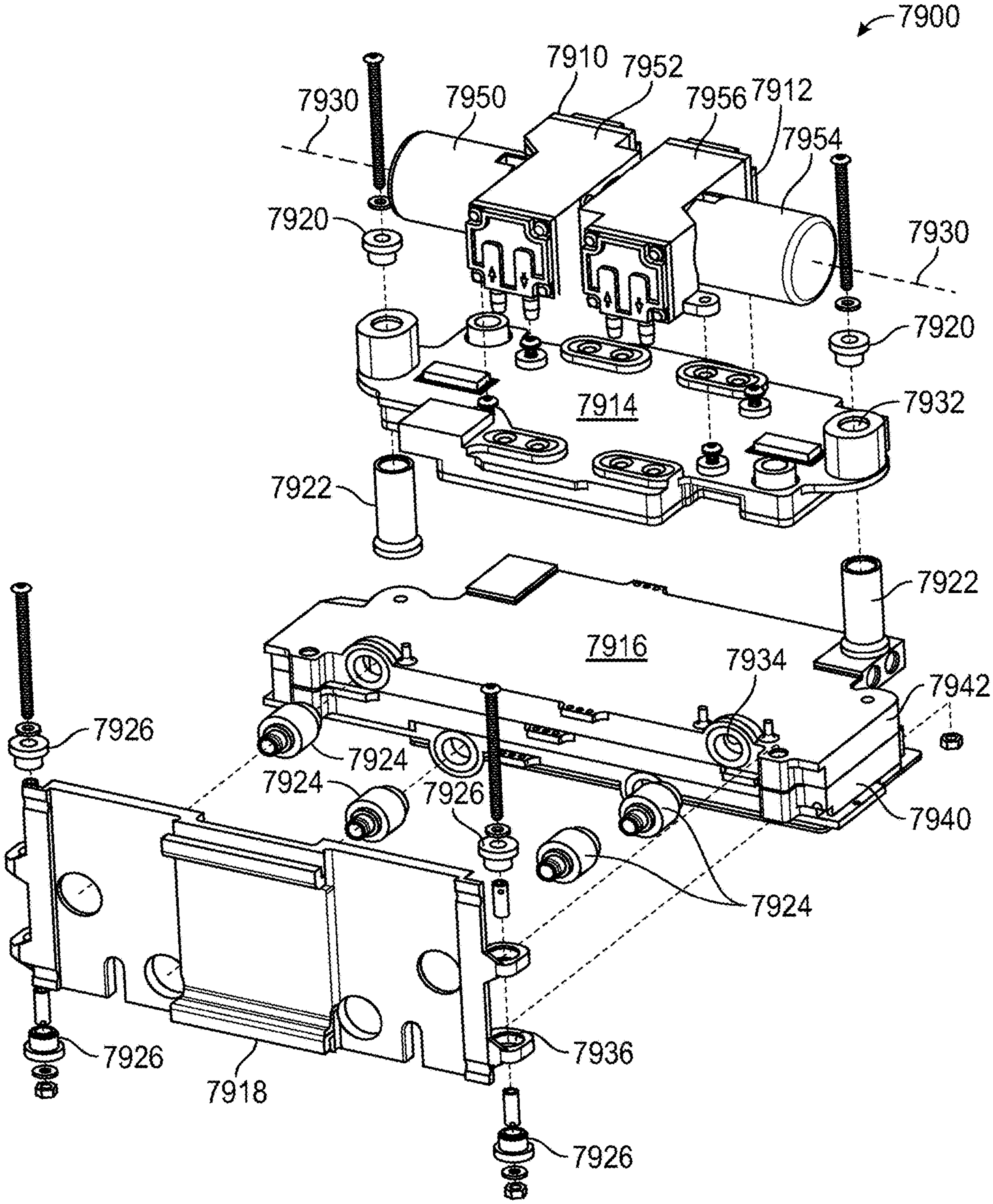
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FIG. 82

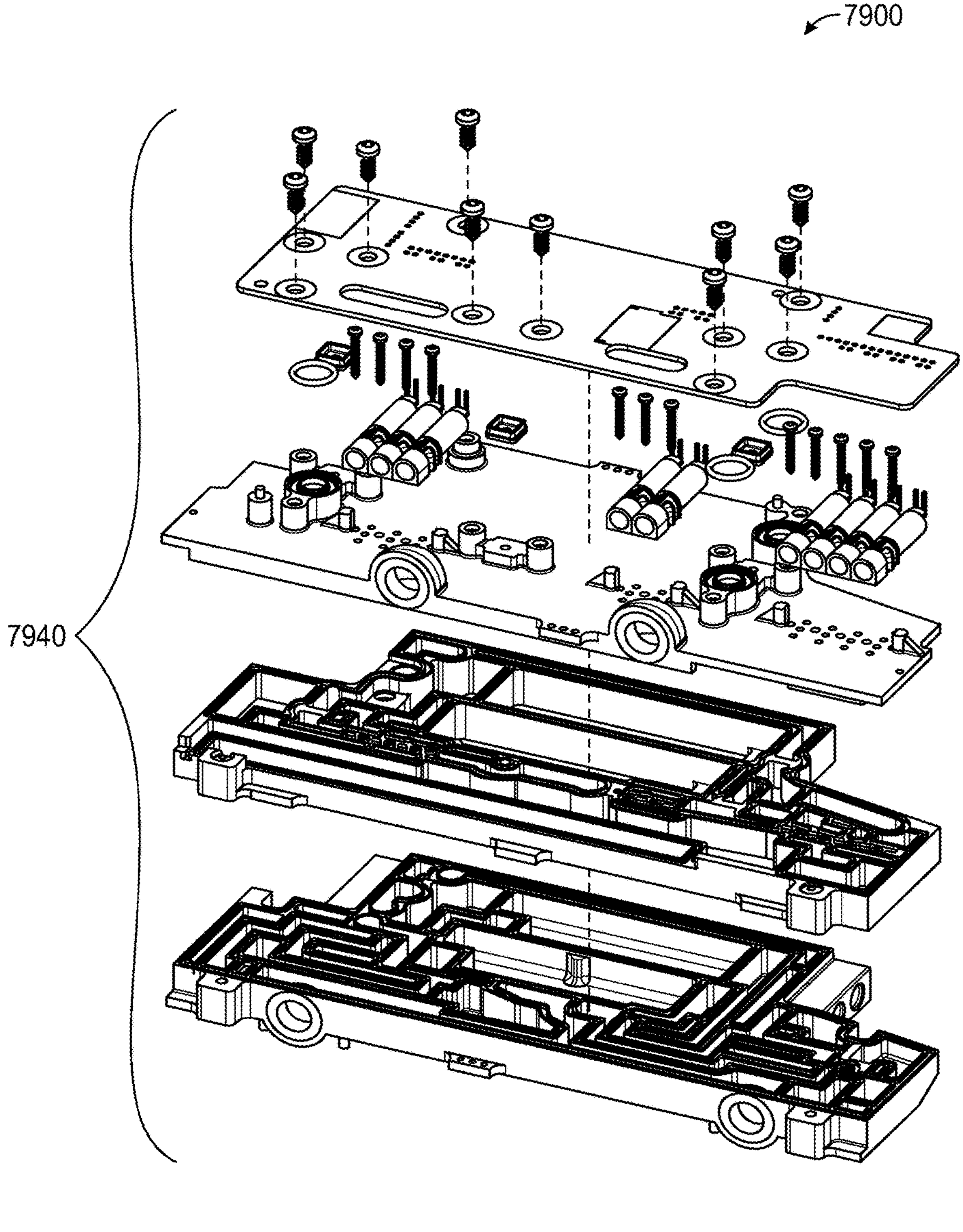

FIG. 83

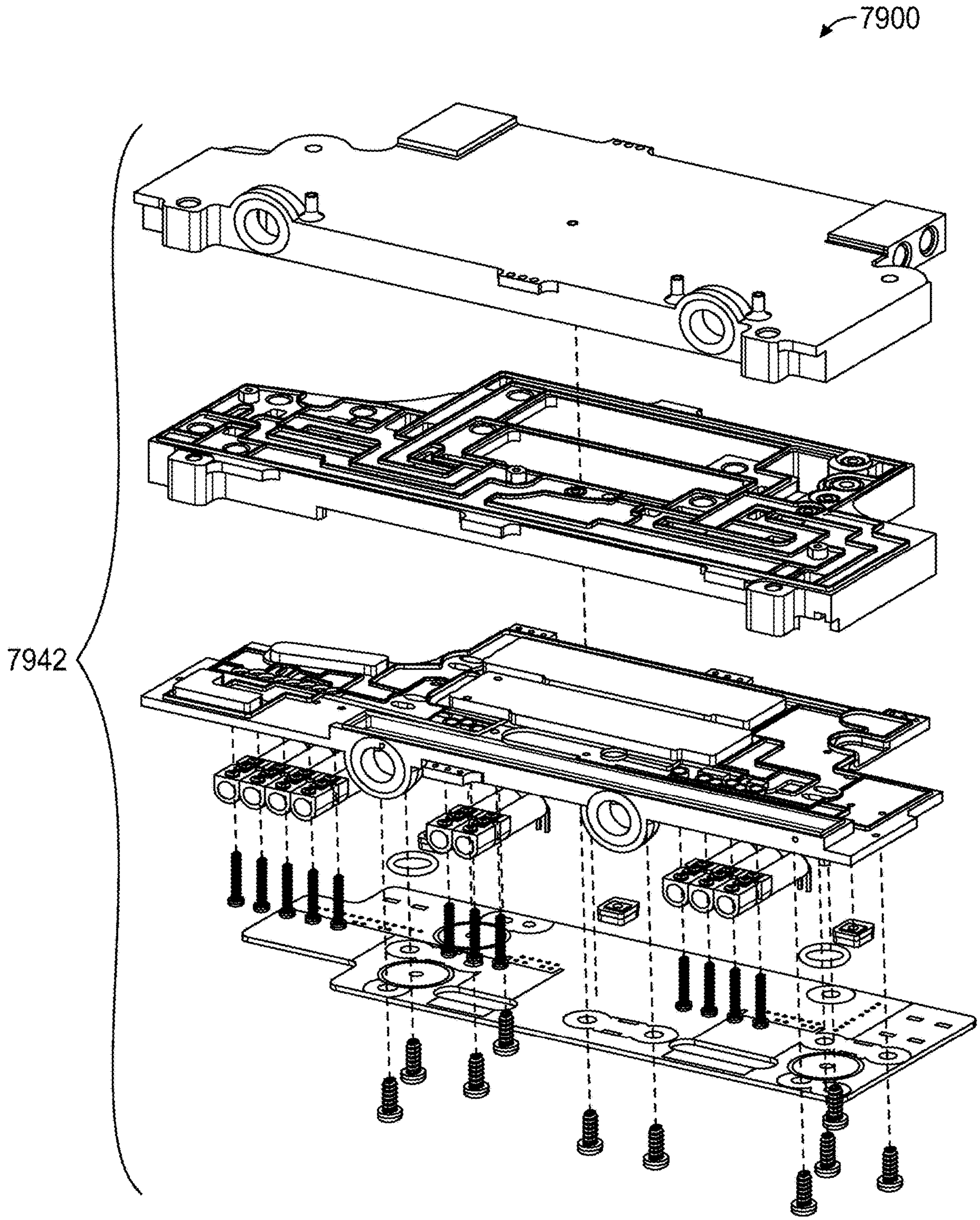

FIG. 84

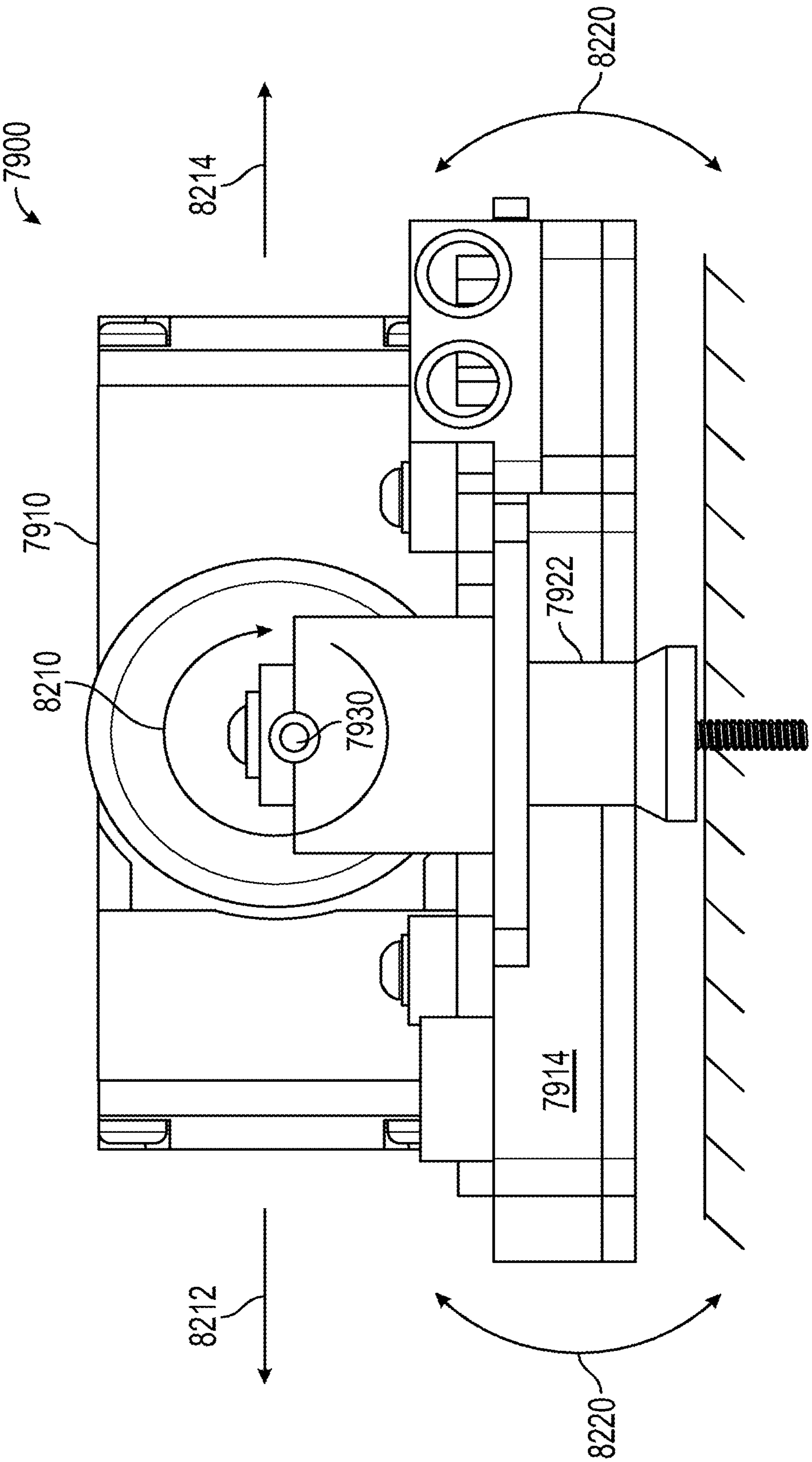

FIG. 85

HAPTIC PLATFORM AND ECOSYSTEM FOR IMMERSIVE COMPUTER MEDIATED ENVIRONMENTS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of priority to U.S. Provisional Application Nos. 63/334,010, filed Apr. 22, 2022; 63/338,010 filed May 4, 2022; 63/395,747, filed Aug. 5, 2022; 63/464,118, filed May 4, 2023; and 63/467,560, filed May 18, 2023, as well as International Patent Application Nos. PCT/US23/19494, filed Apr. 21, 2023; and PCT/US23/21015, filed May 4, 2023. The entire disclosures of the above applications are incorporated by reference.

BACKGROUND

[0002] Many conventional haptic systems are primitive both from a hardware and software/firmware perspective. Moreover, many existing haptic solutions provide limited use cases, such as inducing a symbolic vibration upon occurrence of an event, rather than realistically simulating a touch experience. Also, many existing haptic solutions do not integrate with a wide variety of applications and hardware. Although virtual reality (VR) and other types of simulations are a promising use case, existing haptic solutions often do not provide a high level of immersion into these simulations such that users can truly feel the presence of objects and other entities within the simulation via the coordination of various types of haptic feedback with other sensory interactions. Additionally, techniques for capturing motion tracking inputs from users are primitive and error prone. As a result of these and other factors, many of the most powerful use cases for haptic hardware and software remain unrealized.

[0003] Solutions that move the field of haptics forward by providing a higher level of immersion, better integration between haptic interfaces and applications, and a platform containing reusable components for rapid prototyping and development, among other uses are needed.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The accompanying drawings, which are included to provide a better understanding of the disclosure, illustrate embodiment(s) of the disclosure and together with the description serve to explain the principle of the disclosure. In the drawings:

[0005] FIG. **1** is a schematic illustrating an example haptic platform according to some embodiments of the disclosure.

[0006] FIG. **2** is a schematic illustrating a whole-body immersion system according to some embodiments of the disclosure.

[0007] FIG. **3** is a data diagram illustrating a user profile for personalizing the operation of a haptic platform according to some embodiments of the disclosure.

[0008] FIG. **4** is a schematic illustrating a wearable haptic user interface according to some embodiments of the disclosure.

[0009] FIG. **5** is a schematic illustrating a wearable haptic glove according to some embodiments of the disclosure.

[0010] FIG. **6** is a schematic illustrating a wearable haptic exoskeleton according to some embodiments of the disclosure.

[0011] FIG. **7** is a schematic illustrating an electrical system for a haptic interface according to some embodiments of the disclosure.

[0012] FIG. **8** is a schematic illustrating a network connecting several devices of a haptic platform within a haptic ecosystem according to some embodiments of the disclosure.

[0013] FIG. **9** is a schematic illustrating details of an example application programming interface according to some embodiments of the disclosure.

[0014] FIG. **10** is a schematic illustrating an example contact interpreter according to some embodiments of the disclosure.

[0015] FIG. **11** is an example method of performing motion capture according to some embodiments of the disclosure.

[0016] FIG. **12** is an example method of providing haptic feedback according to some embodiments of the disclosure.

[0017] FIG. **13** is an example method of performing grasp detection according to some embodiments of the disclosure.

[0018] FIG. **14** is a schematic illustrating an example telerobotic platform according to some embodiments of the disclosure.

[0019] FIG. **15** is a perspective view of a whole-body haptic device in accordance with an example embodiment in an exemplary pose illustrating operation of a simulated vehicle.

[0020] FIG. **16** is a perspective view of a whole-body haptic device in accordance with an example embodiment.

[0021] FIG. **17** is a perspective view of a motion platform in accordance with an example embodiment.

[0022] FIG. **18** is a perspective view of a lower-body exoskeleton in accordance with an example embodiment.

[0023] FIG. **19** is a perspective view of an actuated degree of freedom configured to permit translation along a frontal axis of a user's body and an actuated degree of freedom configured to permit translation along the sagittal axis of a user's body of the lower-body exoskeleton in accordance with the example embodiment of FIG. **18**.

[0024] FIG. **20**A is a perspective view of actuated degrees of freedom of the lower-body exoskeleton of the example embodiment of FIG. **18** configured to permit rotation about a longitudinal axis passing through the center of rotation of the user's foot, an actuated degree of freedom configured to permit rotation about a sagittal axis passing through the center of rotation of the user's foot, and an actuated degree of freedom configured to permit rotation about the frontal axis passing through the center of rotation of the user's foot in accordance with an example embodiment.

[0025] FIG. **20**B is a perspective view of the underside of a top portion of a platform of the lower-body exoskeleton in accordance with the example embodiment of FIG. **20**A.

[0026] FIG. **21**A is a perspective view of an actuated degree of freedom of a torso exoskeleton configured to permit translation along the longitudinal axis of a user's body in accordance with an example embodiment.

[0027] FIG. **21**B is a perspective cutaway view of the actuated degree of freedom of the torso exoskeleton in accordance with the example embodiment of FIG. **21**A.

[0028] FIG. **22**A is a perspective view of an upper-body exoskeleton in accordance with an example embodiment.

[0029] FIG. **22**B is a rear perspective cutaway view of the upper-body exoskeleton in accordance with the example embodiment of FIG. **22**A.

[0030] FIG. **23** is a perspective view of another whole-body haptic device in accordance with an example embodiment in an exemplary pose illustrating operation of a simulation.

[0031] FIG. **24** is a perspective view of another motion platform of the whole-body haptic device of FIG. **23** in accordance with an example embodiment.

[0032] FIG. **25** is a perspective view of a motion platform motor of the motion platform of FIG. **24** in accordance with an example embodiment.

[0033] FIG. **26** is a perspective view of a base assembly of the whole-body haptic device of FIG. **23** in accordance with an example embodiment.

[0034] FIG. **27** is a perspective view of a gantry and a footplate form in accordance with an example embodiment.

[0035] FIG. **28** is a perspective view of a foot platform actuator of FIG. **27** in accordance with an example embodiment.

[0036] FIG. **29** is a perspective view of actuated degrees of freedom of the lower-body exoskeleton in accordance with an example embodiment configured to permit rotation about a longitudinal axis passing through the center of rotation of the user's foot, actuated degrees of freedom configured to permit rotation about a sagittal axis passing through the center of rotation of the user's foot, actuated degrees of freedom configured to permit rotation about the frontal axis passing through the center of rotation of the user's foot, and actuated degrees of freedom configured to permit translation along a longitudinal axis of the user's foot in accordance with an example embodiment.

[0037] FIG. **30** is a perspective view of a core assembly of FIG. **23** in accordance with an example embodiment.

[0038] FIG. **31** is a perspective view of an interface portion of the core assembly of FIG. **30** in accordance with an example embodiment.

[0039] FIG. **32** is a perspective view of a torso portion of the core assembly of FIG. **30** in accordance with an example embodiment.

[0040] FIG. **33** is a perspective view of a pneumatic actuator of the core assembly of FIGS. **30** and **32** in accordance with an example embodiment.

[0041] FIG. **34** is a block diagram of a human-computer interface, according to some embodiments.

[0042] FIG. **35** is a schematic of an electronic system suitable for use with a haptic glove, according to some embodiments.

[0043] FIG. **36** is a simplified diagram of a production test flow, according to some embodiments.

[0044] FIG. **37** is an exploded view of components of a drive system, according to some embodiments.

[0045] FIG. **38** is a block diagram of wiring and electronics of a drive system, according to some embodiments.

[0046] FIGS. **39-40** are exploded views of a valve core-compressor assembly, according to some embodiments.

[0047] FIG. **41** is a perspective view of a strap system, according to some embodiments.

[0048] FIGS. **42-44** illustrate an exploded view of a manifold assemblies of a vale core, according to some embodiments.

[0049] FIGS. **45-46** are pneumatic diagrams for a pneumatic circuit, according to some embodiments.

[0050] FIG. **47** is a top view of a compressor, according to some embodiments.

[0051] FIG. **48** is a perspective view of a pneumatic circuit, according to some embodiments.

[0052] FIG. **49** is a process diagram of testing and assembly in a drive system, according to some embodiments.

[0053] FIG. **50** is a simplified block diagram of a glove assembly, according to some embodiments.

[0054] FIG. **51** is a testing and assembly process for a glove assembly, according to some embodiments.

[0055] FIG. **52** is a simplified block diagram of a replaceable-durable assembly, according to some embodiments.

[0056] FIG. **53** is a perspective view of a replaceable-durable assembly, according to some embodiments.

[0057] FIG. **54** is an exploded view of a multi-channel pneumatic connector, according to some embodiments.

[0058] FIGS. **55-57** illustrate a cutaway view of a multi-channel pneumatic connector, according to some embodiments.

[0059] FIG. **58** is an exploded view of a tactile panel assembly, according to some embodiments.

[0060] FIG. **59** is a flow diagram of a finger panel fabrication process, according to some embodiments.

[0061] FIG. **60** is a flow diagram of a palm panel fabrication process, according to some embodiments.

[0062] FIG. **61** is a simplified diagram of a tactile panel, according to some embodiments.

[0063] FIG. **62** is an exploded view of a fingertip assembly, according to some embodiments.

[0064] FIGS. **63-64** are a simplified cutaway side view and a perspective view of a glove assembly, according to some embodiments.

[0065] FIG. **65** is an exploded view of a tendon guide assembly, according to some embodiments.

[0066] FIGS. **66-67** are perspective views of a wrist assembly, according to some embodiments.

[0067] FIG. **68** is a perspective view of an opisthenar, assembly according to some embodiments.

[0068] FIG. **69** is an exploded view of a manipulation actuator, according to some embodiments.

[0069] FIG. **70** is a simplified block diagram of glove electronics, according to some embodiments.

[0070] FIG. **71-72** are perspective views of a palm assembly, according to some embodiments.

[0071] FIG. **73** is a perspective view of an outer glove, according to some embodiments.

[0072] FIG. **74** is a front view of a palm tactile, according to some embodiments.

[0073] FIG. **75** is a simplified diagram of a glove assembly, according to some embodiments.

[0074] FIGS. **76-77** are perspective views of an opisthenar and glove assembly, according to some embodiments.

[0075] FIG. **78-79** is a perspective view of a glove assembly, according to some embodiments.

[0076] FIG. **80** is a simplified diagram of a pneumatic routing assembly, according to some embodiments.

[0077] FIG. **81** is a perspective view of a tubing harness, according to some embodiments.

[0078] FIGS. **82-84** are exploded views of a compressor assembly, according to some embodiments.

[0079] FIG. **85** is a side view of the compressor of FIG. **82**, according to some embodiments.

DETAILED DESCRIPTION

[0080] In some aspects, the techniques described herein relate to a method of providing haptic feedback to one or more haptic interface devices, the method including: receiving, by a haptic feedback controller: first data indicating properties of one or more objects within a computer-mediated environment including an avatar corresponding to a user wearing the one or more haptic interface devices; and second data indicating a haptic effect associated with a current state of the computer-mediated environment; processing, by the haptic feedback controller, the first data and the second data to determine haptic feedback for the user based on an amount and type of contact of the avatar with the one or more objects within the computer-mediated environment and the haptic effect associated with the current state of the computer-mediated environment; generating, by the haptic feedback controller a series of haptic frames based on the determined haptic feedback, each haptic frame specifying a plurality of displacement distances for simulating the amount and type of contact and the haptic effect at a particular point in time; generating, by the haptic feedback controller, a series of actuator frames based on the series of haptic frames, each actuator frame specifying respective amounts of pressure to be provided to tactile actuators of the one or more haptic interface devices to provide a high precision simulation of the amount and type of contact and the haptic effect; and transmitting a plurality of instructions to respective actuator controls of the haptic interface devices, wherein the plurality of instructions are configured to cause actuation of the tactile actuators using the specified respective amounts of pressure to provide the high precision simulation of the amount and type of contact and the haptic effect to the user.

[0081] In some aspects, the techniques described herein relate to a system for interfacing between a haptic glove and a computer-mediated environment generated by an application, the system including: a motion capture module configured to perform steps including: receiving sensor data from a plurality of sensors arranged on the haptic glove and at least one tracked location determined by a motion tracker arranged on the haptic glove; processing the sensor data to generate relative location data specifying relative locations and orientations of a plurality of components of the haptic glove; generating a series of motion capture frames based on the at least one tracked location and the relative locations and orientations, wherein each motion capture frame indicates a spatial arrangement of the plurality of components of the haptic glove at a specific point in time; generating a series of kinematic frames based on the series of motion capture frames, wherein the series of kinematic frames map the spatial arrangement of the plurality of components of the haptic glove to a hand model corresponding to an avatar within the computer-mediated environment; and outputting the series of kinematic frames to the application, wherein the series of kinematic frames cause the application to reposition a hand of the avatar within the computer-mediated environment; and a haptic feedback module configured to perform steps including: receiving data indicating properties of one or more objects within the computer-mediated environment; processing the received data to determine an amount and type of contact of the hand of the avatar with the one or more objects within the computer-mediated environment; generating a series of haptic frames, each haptic frame specifying a plurality of displacement distances for simulating the amount and type of contact at a particular point in time; generating a series of actuator frames based on the series of haptic frames, each actuator frame specifying: respective amounts of pressure to be provided to tactile actuators of the haptic glove; and respective amounts of resistance to be applied to actuate finger brakes of the haptic glove; and transmitting a plurality of instructions to respective actuator controls of the haptic glove based on the series of actuator frames, wherein the plurality of instructions are configured to cause actuation of the tactile actuators using the specified respective amounts of pressure and the finger brakes using the specified amount of resistance to provide a high precision simulation of the amount and type of contact to the user.

[0082] In some aspects, the techniques described herein relate to a system for interfacing between a haptic glove and a computer-mediated application for controlling a telerobot, the system including: a motion capture module configured to perform steps including: receiving sensor data from a plurality of sensors arranged on one or more haptic interface devices and at least one tracked location determined by respective motion trackers arranged on the one or more haptic interface devices; processing the sensor data to generate relative location data specifying relative locations and orientations of a plurality of components of each respective haptic interface device; generating a series of motion capture frames based on the at least one tracked location and the relative locations and orientations, wherein each motion capture frame indicates a spatial arrangement of the plurality of components of each respective haptic interface device at a specific point in time; generating a series of kinematic frames based on the series of motion capture frames, wherein the series of kinematic frames map the spatial arrangement of the plurality of components of the respective haptic interface device to a model of an avatar of the user; and outputting the series of kinematic frames to the application, wherein the series of kinematic frames cause the application to reposition the avatar within a haptic digital twin of the environment of the telerobot based on the series of kinematic frames, wherein the application is further configured to transmit commands to reposition the telerobot based on the avatar; and a haptic feedback module configured to perform steps including: receiving data that is descriptive of one or more objects within an environment of the telerobot, wherein the data is generated by the application using the haptic digital twin of the environment of the telerobot; processing the received data to determine an amount and type of simulated contact of the avatar with the one or more objects; generating a series of haptic frames, each haptic frame specifying a plurality of displacement distances for simulating the contact at a particular point in time; generating a series of actuator frames based on the series of haptic frames, each actuator frame specifying respective amounts of pressure to be provided to tactile actuators of the one or more haptic interface devices to provide a high precision simulation of the contact; and transmitting a plurality of instructions to respective actuator controls of the haptic interface devices, wherein the plurality of instructions are configured to cause actuation of the tactile actuators using the specified respective amounts of pressure to provide the high precision simulation of the contact to the user.

[0083] In some aspects, the techniques described herein relate to a compressor assembly, including: a first compressor with a first motor having an axis of rotation; a second compressor with a second motor, the second motor opposing and coaxial with the first motor by having the axis of rotation; a pneumatic circuit body in pneumatic communication with the first compressor and the second compressor; an assembly mounting plate coupled to the pneumatic circuit body; at least one compressor damper coupled to damp vibration between the pneumatic circuit body and both the first compressor and the second compressor; and at least one pneumatic body damper coupled to damp vibration between the pneumatic circuit body and the assembly mounting plate.

[0084] In some aspects, the techniques described herein relate to a method of controlling a compressor assembly, the method including: commanding a first compressor to operate at a first frequency; commanding a second compressor vibrationally coupled with the first compressor to operate at the first frequency; phase shifting of second compressor components of the second compressor relative to the first compressor by commanding the second compressor to operate at a second frequency that is different from the first frequency; aligning the second compressor components to be substantially opposing momentum from operation of first compressor components of the first compressor using the phase shifting; and operating the first compressor and the second compressor at the first frequency in response to aligning the second compressor components to be substantially opposing momentum from operation of first compressor components.

[0085] In some aspects, the techniques described herein relate to a method of performing motion capture on sensor data received from a wearable haptic interface, the method including: receiving, by a simulation engine, the sensor data from the wearable haptic interface, wherein the sensor data indicates one or more locations associated with the wearable haptic interface; processing, by the simulation engine, the sensor data to generate a motion capture frame, wherein the processing includes: mapping the location data to a model corresponding to an avatar that is simulated by the simulation engine; and adjusting the mapped data to compensate for the wearable haptic interface; and modifying the simulated avatar based on the motion capture frame.

[0086] In some aspects, the techniques described herein relate to a method of providing haptic feedback to a wearable haptic interface, the method including: receiving, from a simulation application, first data indicating a state of an avatar and second data indicating a state of a simulated environment containing the avatar; processing the first data and the second data to detect a contact of the avatar with an entity that exists within the simulated environment; calculating an estimated soft-body deformation resulting from the contact; and actuating a plurality of tactile actuators of the wearable haptic interface based on the estimated soft-body deformation.

[0087] In some aspects, the techniques described herein relate to a method of providing interaction assistance within a simulated environment for a user of a wearable haptic interface, the method including: determining, based on first data indicating a state of an avatar corresponding to the user and second data indicating a state of a simulated entity within the simulated environment, an intended interaction of the avatar with the simulated entity; adjusting one or more simulated physics properties within a region of the simulated environment that corresponds to the avatar and the simulated entity; determining an end of the intended interaction of the avatar with the simulated entity; and resetting the one or more adjusted simulated physics properties.

[0088] In some aspects, the techniques described herein relate to a method of providing haptic feedback to a wearable haptic interface, the method including: receiving, from a simulation application, first data indicating a state of an avatar and second data indicating a state of a simulated environment containing the avatar; processing the first data and the second data to detect a contact of the avatar with an entity that exists within the simulated environment; determining haptic feedback instructions to provide to the wearable haptic interface based on a simulated force associated with the contact; determining, based on one or more governance standards, to modify the haptic feedback instructions in order to reduce an amount of haptic feedback applied to a user of the wearable haptic interface; and providing the modified haptic feedback instructions to the wearable haptic interface.

OVERVIEW

[0089] According to embodiments of the present disclosure, a haptic device hardware, software and intelligence platform (referred to herein for simplicity as the "haptic platform") may be provided for interfacing with, exchanging data with, and otherwise coordinating a set of haptic interface devices to enable various use cases within an ecosystem that may include various other devices and systems. Haptic interface devices may include devices that capture data based on user interactions with the haptic interface devices (e.g., a user moving a hand inserted into a haptic glove, a user moving while wearing a haptic exoskeleton, a user pressing buttons on VR controllers, etc.) and use the captured data as input to an immersive haptic system. Additionally or alternatively, haptic interface devices may provide haptic feedback data (e.g., actuation of tactile actuators, force feedback, thermal feedback, etc.) to users based on a state of an application (e.g., to simulate the experience of a virtual avatar in a VR application). Haptic interface devices may also be used for telerobotic applications in which a user may use the haptic interface devices to control a telerobot (e.g., a robot in a different location from the user) and/or receive haptic feedback based on data sensed by the telerobot. Examples of haptic interface devices may include haptic gloves (also referred to as a wearable haptic glove or haptic feedback glove), haptic exoskeletons, and whole-body haptic systems, among others. Other example configurations of haptic interface devices may be found in the application below.

[0090] As used herein, the term "exoskeleton" or "haptic exoskeleton" may refer to a wearable haptic interface that may include a plurality of components for interfacing with various user body parts (e.g., arm interfaces, hand interfaces, leg interfaces, torso interfaces, etc.) that are configured to track user movements for motion capture purposes and to provide different forms of haptic feedback, including force feedback, tactile feedback, etc. The exoskeleton may use one or more of various types of actuators, including magnetorheological (MR) actuators, hydraulic actuators, pneumatic actuators, electric actuators, and/or other types of actuators. It should be noted that an exoskeleton may include haptic gloves as part of the exoskeleton. Accordingly, when the specification refers to a haptic exoskeleton and/or haptic gloves, it should be understood that the haptic gloves may be used alone (e.g., without using an exoskeleton) and/or as part of an exoskeleton.

[0091] As used herein, the term "whole-body haptic interface" (or whole-body haptic device or system) may refer to an exoskeleton connected to various physical and/or application interfaces. These interfaces may include a motion platform that allows for increased freedom of movement and/or allows haptic feedback to be used to simulate various actions such as climbing, flying, falling, etc. In embodiments, a whole-body haptic interface may include an interface to an application that may provide a virtual environment, a haptic digital twin of a telerobot, and/or any other haptic interface application. An example whole-body haptic interface (including the associated exoskeleton, mechanical ground, etc.) is described in more detail below.

[0092] As used herein, the term "whole-body immersion system" may refer to system comprising an exoskeleton and/or whole-body haptic system, one or more VR/AR headsets, additional immersion systems (e.g., headphones), and/or an application for simulating a virtual environment and/or interacting with a telerobot. The various components of the whole-body immersion system in combination may enable realistic and immersive engagement with a virtual environment generated by an application and/or a telerobot controlled by an application, as described in further detail herein.

[0093] As used herein, the term "haptic digital twin" may refer to a computer-mediated environment that may include one or more virtual objects or surfaces, where at least some of the virtual objects or surfaces are generated based on sensor information received from a telerobot. For example, the virtual objects or surfaces may correspond to real objects or surfaces within sensor range of the telerobot as indicated by the sensor information. In some cases, the virtual objects or surfaces may be accurate representations of the real objects or surfaces. However, the haptic digital twin may also include virtual objects or surfaces that are generated by transforming sensor data indicating a real object or surface (e.g., to scale the virtual object up or down compared to the real object, change the location/state of a virtual object to a predicted new location/state, change one or more other attributes of the virtual object, and otherwise modify the object/surface in various ways that are further explored below).

[0094] Additionally or alternatively, in some cases, a haptic digital twin may include virtual objects or surfaces that do not correspond to any real object or surface. Thus, a haptic digital twin may blend virtual representations of real objects with virtual representations of simulated objects (e.g., for AR/MR use cases that mix simulations with virtual representations of real environments). Regardless of how the virtual objects/surfaces are generated by the haptic digital twin, a user of a whole-body haptic system and/or whole-body immersion system may be able to interact with the environment within the haptic digital twin and receive haptic feedback based on the interactions as described in more detail below. The haptic platform may support the development, operation, monitoring, analysis, and refinement of these and other applications described herein. According to embodiments described herein, reusable and configurable software/hardware modules may be provided to enable and simplify the creation of haptic interface systems that provide highly immersive simulations, control of telerobots, and other computer mediated environments. Accordingly, several techniques for improving the operation of haptic interface systems to provide highly immersive applications are described herein. Additionally, reusable and configurable software/hardware modules may be provided to enable immersive applications for specific use cases, such as training, gaming, simulation, telerobotics, and other use cases.

[0095] Furthermore, techniques described herein enable and simplify the provision of monitoring and analysis capabilities for observing user behaviors and operations of haptic interface systems, analyzing the behaviors and operations, and effectively using the results of the analysis. For example, user behaviors may be used to generate user profiles that may be used to personalize the operations of a haptic interface system. Additionally or alternatively, artificial intelligence systems may be used to monitor user performance in training simulations to determine, for example, if training is effective for a particular user and/or how a training environment can be improved. These and other monitoring and analysis capabilities are described in detail herein.

[0096] In embodiments described herein, a whole-body immersion system may generate a virtual environment to create an immersive "holodeck" in which a user can be fully immersed within a computer-mediated environment. In these embodiments, the whole-body immersion system may allow a user to interact with objects or other entities within a computer-mediated environment (e.g., avatars controlled by other users or entities coupled to real-world systems such as teleoperated robotic systems) via haptic interface devices such that a user can feel the weight and pressure of the objects, the texture of the objects, the temperature of the objects, and/or otherwise realistically interact with the objects/entities in the environment.

[0097] In embodiments described herein, techniques are described for improving the operation of haptic interface devices, such as a wearable haptic glove and/or wearable haptic exoskeleton. Additionally, software development kits (SDKs) that are configured to provide an interface between the haptic interface devices and a haptic system application (e.g., a game engine or simulation of a virtual environment, an application for controlling a telerobot, etc.) are described in detail. Furthermore, techniques for improving interfaces (e.g., application programming interfaces) between the haptic interface devices and the haptic system applications are described.

[0098] In embodiments described herein, techniques for improving the operation of electrical systems used to power haptic interface devices are described. Additionally, networking techniques for improving the interoperability of various devices in the haptic platform and/or within a haptic ecosystem are described.

[0099] In embodiments, various use cases are described for a haptic platform within a haptic ecosystem. A haptic platform may enable a plurality of non-telerobotic use cases including training, gaming, and other virtual reality/augmented reality/mixed reality (VR/AR/MR/XR) use cases. Additionally or alternatively, a haptic platform may enable a variety of uses cases in conjunction with the control of a telerobot.

[0100] FIG. **1** illustrates an example haptic platform **100** for developing, deploying, and operating haptic interface systems. The haptic platform **100** may include a development environment **110** that may be used to develop and deploy one or more haptic interface systems **230**, which may include one or more interface devices **232**, one or more haptic system applications **250**, and/or an SDK **242** for interfacing between the interface devices **232** and the haptic system applications **250**. In embodiments, the haptic platform **100** may include an operating platform **160**, which may include systems, libraries, and components for monitoring, analyzing, supervising, and/or otherwise managing the operation of the haptic interface systems **230**.

[0101] As shown in FIG. **1**, the operating platform **160** may be a separate system from the SDK **242** that sits between the interface devices **232** and haptic system applications **250**. Additionally or alternatively, the haptic operating platform **160** may implement elements configured or generated by the SDK **242**, such that the functions ascribed to the SDK **242** may be performed by the operating platform **160** or vice versa. In other words, in some embodiments (not shown in FIG. **1**), the haptic operating platform **160** may "sit between" the interface devices **232** and the haptic system applications **250**, applying various processing operations to data as it flows from the interface devices to the haptic system applications (e.g., input operations) and/or to data as it flows from the haptic system applications to the interface devices (e.g., haptic feedback operations). Such operations may include modifying the data to enhance an immersive simulation, monitoring/analyzing operations (e.g., monitoring and analyzing performance of a user in a training simulation), applying safety or governance-based controls (e.g., ensuring force feedback is kept below safe limits, ensuring users do not break operator rules in a multi-user simulation, etc.), mapping user movements to control of a telerobot, using telerobot sensor data to generate haptic feedback, and/or other operations for integrating a haptic interface with a computer-mediated environment that are described in more detail herein.

[0102] The development environment **110** may include one or more software/firmware components **112** that may be configured and deployed to the interface devices **232**, the haptic system applications **250**, the SDK **242**, and/or one or more systems/modules/libraries of the operating platform **160**. For example, a sensory simulation component **114** may provide software and/or firmware (e.g., software modules, libraries, deployable firmware, etc.) for simulating various sensory experiences, such as functions for translating data for haptic system applications **250** into haptic feedback data for actuating an interface device **232**, video data for display via a AR/VR/MR/XR device **240**, etc. Similarly, a sensory coordination component **116** may provide software and/or firmware for coordinating multiple sensory experiences with each other (e.g., generating and synchronizing audio feedback with haptic feedback, etc.), as described in more detail below.

[0103] The development environment **110** may further include a multitude of software/firmware components **112** of various types, such as various computing services **118**, data services **120**, and/or intelligence systems **122**. In embodiments, some of these components may be specific to certain use cases, such as telerobotic use cases, VR games, training simulations of various types, and/or the like. Various software/firmware functionalities described herein in connection with various embodiments may thus be configured and/or deployed from a development environment **110**, where they may be stored as various computing services **118**, data services **120**, and/or intelligence systems **122**. The software/firmware functionalities may be deployed to configure any of the devices described herein as appropriate for each use case described herein.

[0104] In embodiments, the software/firmware components **112** may include functionalities that may be deployed to multiple of the various haptic interface systems **230** and/or various components of the operating platform **160**. For example, sensory simulation components **114** may include a first software component for translating application data to digital haptic feedback data, and a second firmware component that may be deployable (e.g., to a haptic glove **236** and/or exoskeleton **238**) in order to translate the digital haptic feedback data to outputs for controlling microfluidics actuators, exoskeleton actuators, force feedback actuators, and/or the like. Additionally or alternatively, a first sensory simulation component **114** may be used to provide haptic feedback data for a first haptic device (e.g., a first haptic glove **236**), while a second sensory simulation component **114** may be used to provide haptic feedback data for a second haptic device (e.g., a second haptic glove **236**).

[0105] In embodiments, the development environment **110** may further include one or more hardware components **130**, which may be combined together and/or with other components to provide functionalities to one or more haptic interface systems **230**. For example, the hardware components **130** may include one or more motion capture/tracking component **132**, which may include hardware configured to track users and/or other hardware components (e.g., cameras, photovoltaic sensors, infrared sensors, etc.) and/or hardware configured to allow other components to track themselves (e.g., magnetic sensors, lasers, and/or other emitters that may be used for orientation and/or positioning). The hardware components **130** may include fluidics-based hardware (e.g., microfluidics hardware), such as fluid transport **134** components and/or fluid actuation components **136** for routing and re-routing fluids (e.g., to actuate haptic actuators). The hardware components **130** may further include various networking components **138** (e.g., networking interfaces, routers, etc.), energy management components **140** (e.g., battery charging components, power distribution components, and/or the like), computing components **142** (e.g., processors, controllers, integrated circuits, etc.), thermal management components **144** (e.g., heat sinks, fans, active and/or passive cooling systems, etc.), sensors **146** (e.g., positioning sensors, pressure sensors, temperature/environmental sensors, force sensors, vibration sensors, piezo sensors, fluidics sensors, photo sensors, and/or any other type of sensor) imaging components **148** (e.g., image and/or video cameras), scanning components **150** (e.g., sonar, radar, LIDAR, etc.), data management components **152** (e.g., various types of memory, caches, cables, etc.), and/or other such components.

[0106] In embodiments, the components may be configured, combined, and/or deployed to create and/or configure various haptic interface systems **230** to enable various haptic use cases and user experiences. For example, various software/firmware components **112** may be configured and deployed to a whole-body haptic interface **234**, such as various sensory simulation components **114** and/or sensory coordination components **116** (e.g., for generating, coordinating, and/or supplementing sensory feedback to a user of the whole-body haptic interface **234**), various computing services **118** (e.g., for calculating positioning data based on sensors of the whole-body haptic interface **234**, for converting force feedback signals into instructions to reposition components of the whole-body haptic interface **234**, etc.), various intelligence systems **122** (e.g., which may be used to modify and/or predict a state of a haptic system application **250** and/or the feedback provided to a user of the whole-body haptic interface **234**), and/or the like. Additionally or alternatively, various hardware components **130** may be used by the whole-body haptic interface **234** (e.g., the hardware components **130** may be provided to a manufacturer of the whole-body haptic interface **234**), such as motion capture/tracking components **132** for tracking user movements, fluid transport **134** components and/or fluid actuation components **136** for providing a microfluidics-based haptic interface of the whole-body haptic interface **234**, networking components **138** for allowing the whole-body haptic interface **234** to communicate with other whole-body haptic interfaces **234**, a haptic operating platform **160**, and/or various haptic system applications **250**, energy management components **140** for distributing power to the components of the whole-body haptic interface **234**, computing components **142** for processing data generated and received by the whole-body haptic interface **234**, thermal management components **144** for heating or cooling components of the whole-body haptic interface **234** (e.g., including for use in a thermal feedback interface), various sensors **146** for use in the whole-body haptic interface **234**, various imaging components **148** and/or scanning components **150** for obtaining data about a user of the whole body haptic interface (e.g., for positioning purposes), data management components **152** for transmitting, receiving, and storing data generated by and/or received by the whole-body haptic interface **234**, and/or the like. In a similar manner, some or all of the software/firmware components **112** and/or hardware components **130** may be used to configure and/or manufacture a haptic glove **236** (or other wearable item), a haptic exoskeleton **238**, various AR/VR/MR/XR devices **240**, and/or other interface devices.

[0107] Various software/firmware components **112** may also be deployed to generate a configured SDK **242**, which may be used to interface between the interface devices **232** and the haptic system applications **250**. The SDK **242** may include one or more application programming interfaces (APIs) **244**, which may include functions for converting interface device **232** data to input data for the haptic system applications **250** (e.g., a motion capture API for translating movements captured by an interface device **232** to movements of a user avatar within a haptic system application **250**), converting state data provided by the haptic system application **250** to the interface device **232** (e.g., a haptics API for translating state data from the haptic system application **250** into haptic feedback data for the interface device **232**), and/or the like. The SDK **242** may further include a haptic application plugin **246** configured to interface with various simulation engines (e.g., gaming engines). The functions of the API **244** and haptic application plugins **246** are described in more detail below.

[0108] In embodiments, haptic system applications **250** may include a variety of applications for different use cases. Telerobotic applications **252** may allow an interface device **232** to control and receive feedback from a robotic device in a remote location. Training applications **254** may provide various haptic simulations that may be used to train users to perform various skilled tasks that benefit from expert touch, such as flying aircraft or piloting other vehicles, performing dangerous activities (e.g., surgeries, working with hazardous materials, military operations), performing tasks that benefit from “muscle memory” (e.g., movements in sports), and/or the like. Gaming applications **256** may provide various virtual reality, augmented reality, or other gaming experiences for users. Product design applications **258** may allow users to design, model, interact with, test, and/or revise various real or simulated physical objects (e.g., tools, furniture, buildings), machines (e.g., vehicles), or the like within a computer-mediated environment. In embodiments, product design applications **258** may provide a virtual equivalent of a computer-aided drafting (CAD) tool. Additionally or alternatively, the product design applications **258** may integrate with external CAD tools (e.g., to generate virtual objects based on CAD data, allow testing and editing of virtual objects based on CAD data, export created or edited virtual objects to CAD formats, etc.). In embodiments, the haptic systems application **250** may leverage various software components **112**, which may provide functionality for receiving interface device **232** inputs and/or outputting state data that may be converted to haptic or other feedback data for interface devices **232**. These and other use cases are described in more detail below.

[0109] In embodiments, the haptic interface systems **230** may be managed and/or supported by a haptic operating platform **160**. The haptic operating platform **160** may include various modules and/or components for performing system management components **162**, various libraries **180** that may be accessed to provide various functionalities, various system services **192**, and/or various configured intelligence services **204**. In embodiments, the operating platform **160** may perform monitoring and reporting **222** and may further leverage the monitoring and reporting **222** to perform analysis, feedback, and optimization **224**. In embodiments, the monitoring and reporting **222** may receive inputs generated by various system management modules/components **162**, and may provide inputs to the analysis, feedback, and optimization **224**. In embodiments, the analysis, feedback, and optimization **224** may be used to further train/optimize the configured intelligence services **204** in order to improve their operation over time. The haptic operating platform **160** may further communicate with any of the haptic interface systems **230** to perform various functions described herein.

[0110] In embodiments, the system management modules/components **162** may monitor and/or manage various functionalities for the haptic interface systems **230**. In embodiments, a UI/UX module **164** may monitor and/or manage user interactions with interface devices **232** and/or haptic systems applications **250**, such as by monitoring and/or managing user interactions with virtual interactive objects provided by haptic system applications **250**. For example, a UI/UX module/component **164** may collect data on user interactions with various virtual objects, assist certain users in interacting with virtual objects (e.g., providing interactive assistance to impaired users, modifying physics simulations to make objects easier to pick up, etc.), and/or the like. In embodiments, a human factors module **166** may provide various data for adapting the operation of various haptic interface systems **230**. For example, a human factors module **166** may use various factors to configure interface devices **232**, such as estimating the size of a user’s hand to configure a haptic glove **236**, observing user movement within a haptic exoskeleton **238** to determine whether a user has any impairments or range of motion limitations, and/or the like. In embodiments, a user feedback control module **168** may personalize and/or limit various aspects of feedback that may be provided by interface devices **232**, such as a maximum amount of force feedback that a user may receive, various amounts of force feedback that may be applied to various body parts, temperature limitations for thermal interfaces, whether a VR display is allowed to flash images at high frequencies, and/or the like. In embodiments, a personalization module **170** may define various preferences for generating a user avatar, customizing interactions with the haptic system applications **250**, and/or the like. In embodiments, the human factors module **166**, user feedback control module **168**, and/or personalization module **170** may receive and/or leverage a user profile that is tailored to a particular user, type of user, and/or the like. In embodiments, user profiles may be stored (e.g., at least temporarily) in the haptic operating platform **160** (e.g., in data stores **196**), may be generated based on inputs provided by users, which may include direct input of settings and/or implicit determination of user preferences based on other user inputs, observed ranges of motions, observed reactions to stimuli, etc. In embodiments, a fleet management module **172** may monitor and/or manage groups of interface devices **232**, such as by monitoring a number of times a particular interface device **232** has been used, a number of users that have interacted with an interface device **232**, a status of an interface device **232** (available, being repaired, etc.), by assigning a particular interface device **232** to a particular user, and/or the like.

[0111] In embodiments, various libraries **180** may be used to provide functionalities to haptic interface system **230**. For example, a sensory simulation library **182** and/or sensory coordination library **184** may provide various functionalities for simulating various sensory experiences and/or for coordinating multiple sensory experiences during an immersive simulation. In embodiments, the sensory simulation library **182** and/or sensory coordination library **184** may be configured for particular applications (e.g., flight simulation, training environments, vehicle design testing environments, video game environments, etc.). In embodiments, environment libraries **186** may be used to provide functionalities for modeling specific environments (e.g., flight simulation, training environments, vehicle design testing environments, video game environments, etc.). The environment libraries **186** may include audio, visual, tactile, and/or other suitable sensory content/properties that may be used to provide a multi-sensory computer-mediated environment. For example, an environment library **186** for providing vehicle design testing environment may include shapes, surfaces, and other visual properties for 3D rendering of the interior of a vehicle, 3D renderings of environments (e.g., road/landscape) in which the vehicle can be driven, tactile properties of the various components in the interior of the vehicle (e.g., density, texture, temperature, and/or other tactile properties of a seat, steering wheel, airbags, dashboard, control buttons, and/or the like). In embodiments, governance libraries **188** may include code and/or parameters that may be leveraged to limit or otherwise control the haptic response of the haptic interface systems **230**, such as to monitor, detect, and prevent dangerous conditions, monitor, detect, and prevent illegal activity, and/or the like.

[0112] In embodiments, the haptic operating platform **160** may include various system services **192** for storing various data and providing various operating functions. The system services **192** may include storage services for storing data in storage, such as library stores **194** (e.g., for storing the various libraries **180**) and data stores **196** (e.g., for storing various data leveraged by the operating platform **160**, such as user profiles, various monitoring and reporting data, training data for intelligence services, etc.). The system services **192** may further include data processing **198** services and/or data aggregation services **200**, which may provide processing (e.g., for analysis tasks such as those performed at **224**) and/or aggregation functionalities (e.g., for monitoring and reporting **222**).

[0113] In embodiments, the haptic operating platform **160** may include various configured intelligence services **204**, which may include intelligence functions **206** and/or an intelligence controller **216**. The intelligence functions **206** may include various machine learning and/or other artificial intelligence components for providing functionalities for the haptic operating platform **160**. AI data analysis models **208** may be trained to perform data analysis tasks, such as analyzing user behavior and/or interactions with an interface device **232**, user behavior within a haptic system application **250**, and/or the like. In embodiments, the intelligence functions **206** may further include a recommendation engine **210** for generating recommendations. The intelligence functions **206** may further include an imaging engine **212** that may be configured for object recognition (e.g., for telerobotic use cases in which a telerobot includes an image sensor). The configured intelligence services **204** may further include an intelligence controller **216**, which may include analysis **218** and governance/regulatory **220** components.

Whole-Body Immersion System

[0114] FIG. **2** illustrates an example whole-body immersion system **300** that may be configured to provide immersive full-body control of an avatar within a computer-mediated environment to a user of a whole-body haptic interface **234** (e.g., a full-body interface) and/or various other haptic and/or sensory interfaces, such as haptic gloves **236**, an exoskeleton **238**, AR/VR/MR/XR interfaces **240**, other sensory interfaces **338**, etc. The whole-body immersion system **300** may comprise a wearable haptic interface **330** (which may be a whole-body haptic interface **234** or any components thereof) including one or more devices that may be configured to interface with an immersive haptic system **302**. In embodiments, the whole-body haptic immersion system may further include a "mechanical ground" (not shown in FIG. **2**, but shown and described in more detail below) that anchors at least a portion of the user to the system in order to allow simulations of weight, solidity, and inertia of objects in the computer-mediated environment. In embodiments, the whole-body haptic immersion system may further include a motion platform (not shown in FIG. **2**, but shown and described in more detail below) as part of the mechanical ground in order to allow simulations of running, climbing, flying, driving, or other such movements that would be difficult or impossible to simulate without the motion platform and mechanical ground. For example, if a user were testing the design of a car, the whole-body immersion system may be capable of positioning the user in a seated position (e.g., by arranging the components of a haptic exoskeleton, which may be connected to the mechanical ground (which may or may not include a motion platform), such that the user is a seated position), simulating acceleration forces via the motion platform (e.g., by tilting the user in various directions), and the like. Similarly, if a user were testing a simulated airplane, the whole-body immersion system (if it includes a motion platform) may be capable of rotating the user to simulate the pitch, roll, etc. of an aircraft. A detailed description of example hardware featuring a mechanical ground and motion platform is described below.

[0115] In embodiments, the immersive haptic system **302** may use an API **244** to interface between the wearable haptic interface **330** and one or more haptic system applications **250** (e.g., haptic system application **250**A). In embodiments, the wearable haptic interface **330** may be a full-body interface wearable by a user. For example, the wearable haptic interface **330** may be a component of a whole-body haptic interface **234** that further includes various other components as described elsewhere herein. In other examples, the wearable haptic interface **330** may be a partial-body interface (e.g., covering a torso and arms of the user or any other subset of haptic interfaces). In embodiments, a content management system **370** may further communicate with the immersive haptic system **302** to store, manage, and authorize access to content, among other functions.

[0116] In embodiments, the immersive haptic system **302** may execute one or more haptic systems applications **250**. As used herein, a haptic systems application **250** (or "haptic application") may be any application that is configured to output data to a wearable haptic interface (e.g., wearable haptic interface **330**) that is used by the wearable haptic interface to provide haptic feedback (e.g., tactile feedback, force-feedback, and/or thermal feedback). Examples of haptic applications may include, but are not limited to telerobotic applications **252**, training applications **254**, gaming applications **256**, product design applications **258**, metaverse applications, and/or any other type of haptic system applications **250**. For example, a first haptic system application **250**A may be a VR game, a second haptic system application may be a telerobotic application that allows a user of the whole-body immersion system to control one or more remote robots, a third haptic system application may be a training application that simulates a training environment with which one or more users interact, and the like. The hardware and software capabilities of the whole-body immersion system **300** may enable the haptic system applications **250** to provide high quality haptic feedback, which allows for the development of muscle memory and therefore allows training, prototyping, and other simulations in a safe and controlled environment. In the illustrated example, the first haptic system application **250**A includes environment data **304** (e.g., objects, surfaces, entities, etc. that a user may interact with in the computer-mediated environment via the full-body interface) and avatar data **306** representing the user within the environment. However, other types of applications may be used with the whole-body immersion system **300**, such as telerobotic applications, virtual applications without a user avatar, etc.

[0117] In embodiments, the immersive haptic system **302** may further include a governance enforcement **308** that enforces one or more governance policies relating to the sensory experiences that are defined in a respective policy library. In embodiments, a policy library may include governance policies that define safety and/or legal limits on different types of haptic feedback (e.g., force-feedback, thermal feedback, and/or tactile feedback). In some embodiments, a governance enforcement **308** may be configured to enforce safety standards that may define, for example, one or more rules specifying maximum amounts of force feedback that may be applied to various body parts, maximum ranges of motion per joint or body part, maximum amounts of acceleration that may be applied per joint or body part, and/or the like. For example, the governance enforcement **308** may limit the amount of force feedback that may be applied to a joint or other body part of a user so that the force feedback does not break the joint or other body part. Additionally or alternatively, the governance enforcement **308** may be configured to enforce safety standards that may define, for example, one or more rules specifying limits on thermal feedback and/or tactile feedback that may be applied to a user. For instance, if a haptic application is configured to train a firefighter the haptic feedback may include thermal feedback that simulates heat transfer from a fire and tactile/force-feedback that simulates forces associated with fighting a fire (e.g., explosions, items falling on the firefighter, using firefighting equipment, and the like). In this example, the governance enforcement **308** may be configured with limits on the temperatures that a user may be exposed to and the physical forces that may be applied to a user in response to the state of the application (e.g., limiting the amount of tactile force that may be applied to the user if a large piece of wood falls on a firefighter in a simulation). In embodiments, the various policy libraries containing the safety standards may be obtained from a haptic operating platform **160** (e.g., libraries **180**). It is noted that in some embodiments, the policy libraries may be defined for different types of scenarios and/or different types of users. For example, a governance policy associated with training simulations for training a professional (e.g., pilot, firefighter, military professional, or the like) in a simulated environment may define higher limits for the magnitudes of haptic feedback that a user may be subjected to than the limits for the magnitudes of haptic feedback defined in a governance policy associated with similar environments but for non-professionals. For instance, if simulating a fire-fighting scenario for trained firefighters, a first governance policy may allow a user to be exposed to temperatures that do not exceed 55° C., while if simulating a fire-fighting scenario for non-firefighters a second governance policy may allow a user to only be exposed to temperatures that do not exceed 45° C. A governance policy may thus provide haptic feedback configurations that are customized by role, by degree of experience, by type of user, or based on the personal characteristics of a user.

[0118] In embodiments, whole-body immersion system **300** may include a high-quality haptic interface featuring a plurality of microfluidics tactile actuators that may be variably actuated to provide precise contact simulations using different pressure amounts at different locations. In some cases, the fluid may be variable temperature (e.g., it may be mixed using varying proportions of hot and cold fluid) to simulate the feeling of contact with objects of varying temperature and material (which may affect thermal flux). Additionally or alternatively, the haptic interface may feature passive and/or active force feedback that may accurately simulate the feel of interacting with solid objects made of varying materials and weights. In embodiments described herein, the haptic interfaces may be configured to receive an output state from an immersive haptic system **302** (e.g., one or more of haptic data **360**, audio data **362**, video data **364**, sensor data **366**, etc.) and to actuate a set of haptic feedback components based on the output state, wherein the magnitude of the haptic response by the set of haptic feedback components is selectively adjusted to comport with a set of safety standards. In embodiments, the haptic interfaces described herein may include a flexible textile laminate that includes a microfluidic actuation layer that is configured to provide realistic haptic response (e.g., tactile and/or thermal feedback) with minimal latency.

[0119] In embodiments, an immersive haptic system **302** may include an artificial intelligence module **310**, which may be leveraged to provide functionalities used by the API **244**, as discussed in more detail below. The immersive haptic system **302** may further include a user profile **312**, which may be used to personalize an application in accordance with preferences, abilities, and/or other settings corresponding to specific users, as discussed in more detail below.

[0120] In embodiments, the immersive haptic system **302** may interface with a haptic operating platform **160**, and accordingly may leverage functionality of the governance enforcement **308** and artificial intelligence **310** (e.g., using system management components **162**, libraries **180**, and/or configured intelligence services **204**). For example in some embodiments, the governance enforcement **308** and/or artificial intelligence **310** of the immersive haptic system **302** may be configured by the haptic operating platform **160** (e.g., by downloading libraries therefrom). Additionally or alternatively, the haptic operating platform **160** may be in real-time communication with an immersive haptic system, such that the haptic operating platform **160** is responsible for enforcing governance policies and/or providing certain intelligence services to the immersive haptic system **302**.

[0121] In embodiments, the wearable haptic interface **330** may include several components, such as one or a pair of haptic gloves **236**, one or more other interfaces of a haptic exoskeleton **238** such as arm/leg/torso interfaces, AR/VR/MR/XR interfaces **240** (e.g., a VR headset), other sensory interfaces **338**, and/or one or more motion trackers **340**. In some embodiments, the haptic exoskeleton **238** may include large-muscle haptics that may provide haptic feedback to various parts of a user's limbs, torso, etc.

[0122] In embodiments, the API **244** may include and/or be implemented by one or more modules, systems, and/or controllers for interfacing between the haptic system applications **250** and the various components of the wearable haptic interface **330**. Some or all of the modules/systems/controllers of the API **244** may transmit and process data generated by the wearable haptic interface **330** so that the data may be provided as inputs to a haptic system application **250**. Additionally or alternatively, some or all of the modules/systems/controllers of the API **244** may transmit and process data generated by the haptic system applications **250** so that the data may be provided as feedback to one or more components of the wearable haptic interface **330**. In embodiments, the modules/systems/controllers of the API **244** may process any of the data based on the user profile **312**, which may be used to customize various aspects of the data flowing in either direction. Additional details regarding functionalities of the API **244** are provided with respect to FIG. **9** below.

[0123] In embodiments, the immersive haptic system **302** may provide interaction data **352** to the haptic system applications **250** via the API **244** and may provide state data **354** from the haptic system applications **250** to the API **244**. Interaction data **352** may specify movement and/or positioning data for an avatar, interactions between the avatar and the environment, a status of the avatar, and/or the like. For example, the interaction data **352** may specify position data for various parts of a user avatar, speed/direction data for moving the avatar, status data indicating current actions for an avatar (e.g., that a particular hand is performing a grasp), contact points for parts of an avatar (e.g., that one hand is touching the other hand), user status data (e.g., that a user is applying pressure in a particular direction but being prevented from moving by force feedback), user limits data (e.g., limits of a user's range of motion), and/or any other such data that may be used to render the avatar and/or interactions between the avatar and the environment. Additionally, although examples herein may refer to a user avatar, it should be understood that a haptic system application **250** may be used to control a telerobotic system or the like such that interaction data **352** may be used to control a telerobot instead of or in addition to an avatar.

[0124] In embodiments, the interaction data **352** may be generated by a motion capture system **314** and/or other components of the API **244**. The motion capture system **314** may receive six degree of freedom (6DOF) positioning data **358** from the wearable haptic interface **330** (e.g., from one or more motion trackers **340** and/or from sensors integrated into the haptic gloves **236**, haptic exoskeleton **238**, AR/VR/MR/XR interfaces **240**, other sensory interfaces **338**, etc.). The positioning data **358** may include relative positioning data, absolute positioning data, orientation data, velocity/acceleration data, pressure data, force feedback data, and/or other such data that may be measured by various sensors of the components of the wearable haptic interface **330**. For example, the haptic gloves **236** may measure the relative position of a user's fingertips, the position and orientation of the user's hand, the angle of one or more of the user's finger joints, pressure applied to one or more parts of the haptics gloves **236**, and other such data, which may be transmitted to the motion capture system **314** as positioning data **358**. The motion capture system **314**, in turn, may use the data to develop a data structure that details the current position of the user or part of the user (e.g., the user's hand and/or each joint of the hand if the position data is received from a haptic glove **236**). In embodiments, the motion capture system **314** may use prediction models (e.g., a model trained to predict hand position based on various position sensor data, as discussed elsewhere herein), user dimensions (e.g., a size of the user hand), anatomical models (e.g., a model corresponding to a human hand), and/or the like to generate the data structure. Additionally or alternatively, a haptic exoskeleton **238** may measure various positions of the user's head, various part of the user's limbs, torso, facial expressions, etc., and supply such data to the motion capture system **314** to generate data structures describing the positioning of the user's body as a whole and/or of body parts. The motion capture system **314** may then generate interaction data **352** indicating current positioning/movement data for an avatar corresponding to the user and/or other interaction data **352** for the haptic system application **250**.

[0125] In embodiments, the motion capture system **314** of the API **244** may further receive data from haptic feedback controller **316**, which may indicate, for example, when force feedback data is being used to limit or restrain user movement. The force feedback data and other such haptic feedback data may be used to modify user positioning data or other interaction data **352**. Additional details describing motion capture techniques that may be performed by the motion capture system **314** of the API **244** are provided below with respect to FIG. **9**.

[0126] In embodiments, a haptic system application **250** may output state data **354**, which may be used by the various feedback controllers to provide haptic data **360**, audio data **362**, video data **364**, and/or sensor data **366**. In some embodiments, a haptic system application **250** may further output audio/video data **356**, which may be passed through the API **244** and rendered by an AR/VR/MR/XR interface **240** and/or may be modified by audio feedback controller **318** and/or visual feedback controller **320** before being output by the API **244** and rendered by an AR/VR/MR/XR interface **240**.

[0127] In embodiments, the API **244** and/or various controllers may coordinate variable haptic feedback data that is distributed to and from various components of the wearable haptic interface **330** in order to simulate a fully immersive interaction between the user of the wearable haptic interface **330** and the environment generated by the haptic system applications **250**. As a simple example, a user may move and maneuver the user's body (e.g., thereby moving one or more components of the haptic exoskeleton **238**) to cause the user's avatar to approach an object that exists within the environment (which the user may view via the AR/VR/MR/XR interface **240**), extend the user's arm and touch the object (e.g., by moving the haptic glove **236**). As the system detects the co-location of the position of user's hand and the position of object in the computer-mediated environment, the system may cause the user to "feel" the object by inducing haptic feedback provided by the gloves **236**. The user may then grasp the object (e.g., by moving the user's fingers within the haptic glove) and lift the object, and the haptic exoskeleton **238** and/or haptic gloves **236** may provide force feedback to simulate the weight of the object. The user may then, for example, drop the object, and if the object contacts the user's avatar as it falls within the computer-mediated environment, haptic feedback may be applied so the user feels the impact of the object. In embodiments, other cues such as audio cues, visual cues, or other sensory cues may be generated by the immersive haptic system **302**, audio feedback controller **318**, visual feedback controller **320**, and/or sensory feedback controller **322** to simulate certain interactions. For example, any of these controllers may be configured using sensory simulation components **114** and/or sensory coordination components **116**, as well as other software/firmware components **112**. Additionally or alternatively (e.g., in embodiments in which the API **244** is implemented by a haptic operating platform **160**), the feedback controllers may leverage libraries **180**, such as sensory simulation libraries **182** and/or sensory coordination libraries **184**. For example, objects may be associated with sensory data within the environment, such that if a user touches, for example, a hot object, a thermal sensory interface may provide thermal haptic feedback to the user at or nearby the simulated point of contact.

[0128] In embodiments, the haptic feedback may be variable/proportional in order to provide for a more immersive whole-body environment. For example, haptic feedback provided by microfluidic actuators of a haptic glove **236** may provide variable pressure to simulate a certain amount of force applied to a user's fingers or hands. Similarly, haptic feedback provided by a thermal actuator (which may be a microfluidic tactile actuator or a separate actuator) may be variable in order to simulate a certain amount of thermal flux dependent on simulated temperature and/or thermal conductivity properties of a virtual object. Thus, for example, a user may be able to feel the difference between virtual objects/materials in a way that simulates the texture, weight, velocity, temperature, material, and/or other properties of a virtual object or virtual environment. Moreover, the various types of haptic feedback (including microfluidics, force feedback, thermal, visual, audio, and/or other sensory experiences) may be coordinated to provide a highly immersive and realistic computer-mediated environment. As discussed in more detail below, these haptic feedback actuators may also be limited in accordance with a set of governance settings defined in a governance library and/or in accordance with other settings (e.g., preferences or other settings). The types of haptic feedback may be configured based on an understanding of how the brain reacts to sensory stimulus; for example, a change in temperature may be induced that simulates, rather than exactly matches, a change in temperature of an object. This may provide immersive and realistic computer-mediated environments while maintaining safety of the user, such as by not fully matching a change in temperature that would burn the hand of a user.

[0129] Thus, the whole-body immersion system **300** may generate a fully immersive computer-mediated environment using one or more combinations of several techniques that may vary by application, by context of a particular environment, and/or by the hardware/software/firmware functionalities that are available for a given embodiment of a haptic platform. In embodiments, the whole-body immersion system **300** combines various types of inputs and feedback data in a coordinated manner in order to generate a fully immersive environment that includes touch interactions and various other interactions between the human user and physical entities within the environment. Moreover, by coordinating positioning data **358** (including relative position data, actual position data and/or other types of motion tracking data) received from haptic gloves **236**, a haptic exoskeleton **238**, and/or other interfaces, and coordinating the positioning data to generate inputs to an immersive haptic system, a user may cause interactions within a computer-mediated environment by controlling an avatar corresponding to the user, according to some embodiments of the present disclosure. Additionally, the whole-body immersion system **300** generates haptic feedback such that the state data received from the computer-mediated environment is processed and coordinated to cause variable haptic feedback at specific locations of the wearable haptic interface **330**, including the haptic gloves **236**, haptic exoskeleton **238**, and any other haptic interfaces. In particular, the coordinated variable haptic feedback may be provided by a haptic feedback controller and may simulate interactions between the user and objects or other aspects of an environment.

[0130] In embodiments, the wearable haptic interface **330** may include one or more EEG components (e.g., sensory interfaces **338** may include EEG interfaces) configured to monitor brain activity of a user. Additionally or alternatively, the wearable haptic interface **330** may include various types of biometric sensors (e.g., eye tracking, perspiration sensors, heartbeat/pulse sensors, temperature sensors, accelerometers, audio sensors, etc.) configured to measure biometric activity of a user.

[0131] In some embodiments, the motion capture system **314** and/or some other component of the API **244** may be configured to receive and monitor EEG signals measuring user brain activity and generate interaction data **352** based on the monitored EEG signals. For example, a particular haptic system application **250** may be configured to use an EEG signal as interaction data **352** to change the location or state of a user avatar, interaction with the environment, or otherwise modify the environment data **304** and/or avatar data **306**. For example, users may be able to control an avatar using an EEG interface.

[0132] Accordingly, a wearable haptic human interface device may have a set of haptic components integrated into a wearable component of the interface device (e.g., haptic gloves **236** and/or a haptic exoskeleton **238**) that realistically simulate a human sensory interaction with a physical entity based on a set of actuation commands determined from a haptic feedback response corresponding to an interaction of an avatar and an environment, and further may have a set of EEG components that monitor brain activity of a user wearing the interface device.

[0133] In some embodiments, the EEG components may monitor the brain activity of a user and may output EEG signals to the haptic operating platform **160**, which monitors user brain activity in conjunction with the state of a computer-mediated environment (e.g., a simulated scenario that the user is presented), the haptic feedback data (e.g., what is the user being subjected to), and/or biometric signals (e.g., heartbeat, breath rate, temperature, and/or the like). In some of these embodiments, the haptic operating platform **160** may analyze the user brain activity in conjunction with the haptic feedback and biometric signals to determine acceptable levels of haptic feedback (e.g., tactile, force feedback, and/or thermal feedback). In some embodiments, the operating platform **160** may train machine-learning models to determine acceptable levels of haptic feedback for a specific user (e.g., personalized limits), class of users (e.g., trained professionals, novices, children, and/or the like), or for all users.

[0134] In some embodiments, the haptic operating platform **160** may analyze the user brain activity in conjunction with a state of a computer-mediated environment, motion capture data, the haptic feedback, and/or biometric signals to improve training outcomes. In these embodiments, the haptic operating platform **160** may monitor and analyze user responses to the current environment or state of an environment, particular locations with the environment, particular positions that a user and/or user avatar is in, amounts and/or magnitudes of haptic feedback applied, and/or other conditions of the environment, the user avatar in the environment, or the user (e.g., as measured by various sensors of a wearable haptic interface **330**). The haptic operating platform **160** may thus analyze and detect which types of conditions cause less or more brain activity, cause certain types of brain activity, and/or the like. In embodiments, the haptic operating platform **160** may therefore provide analysis and metrics for detecting a level of training or learning effectiveness (e.g., which parts of a training simulation may be most memorable), whether certain conditions may be over-stimulating or under-stimulating, and/or the like. Thus, the haptic operating platform **160** may provide analysis and metrics that may be used to further develop the effectiveness of a training simulation, improve outcomes, and/or the like.

[0135] In some embodiments, the haptic operating platform **160** may analyze the user brain activity in conjunction with motion capture data, haptic feedback data, and/or an environment that controls telerobots (which may be a haptic digital twin that includes a representation of a physical environment including the telerobot) to train robotic process automation (RPA) models. In these embodiments, the haptic operating platform **160** may monitor and analyze user responses to the current environment or state of a computer-mediated environment (e.g., a game, a simulation, and/or a haptic digital twin environment for a telerobot), particular locations of the telerobot within a simulated environment or digital twin, particular positions that a user and/or telerobot is in, amounts and/or magnitudes of haptic feedback applied to the user, and/or other conditions of the simulated environment and/or telerobot, the telerobot, or the user (e.g., as measured by various sensors of a wearable haptic interface **330**). In these embodiments, the haptic operating platform **160** may provide analysis and metrics for detecting different responses to different conditions of the telerobot, including training models for controlling a telerobot for RPA. For example, the haptic operating platform **160** may detect which types of conditions are routine or exceptional based on monitoring brain activity of a user. Thus, the haptic operating platform **160** may provide analysis, metrics, and training data that may be used to train models for RPA that may be capable of detecting certain conditions and responding accordingly.

[0136] In embodiments, the immersive haptic system **302** and/or haptic operating platform **160** may be configured to analyze various user responses (e.g., including EEG signals and other user measurements) to various conditions and to model a reaction of a human (e.g., the specific user and/or a typical user) to various types of sensory stimulus. The immersive haptic system **302** and/or haptic operating platform **160** may then determine a set of parameters for configuration of a haptic experience for the human based on the reaction parameters and/or based on a set of inputs relating to an environment or entity within the environment. In embodiments, the haptic experience may be configured using data stored in a user profile **312**, as described in more detail below.

[0137] In embodiments, the whole-body immersion system **300** may further include a content management system **370**, which may include one or more of content stores **372** for storing content (e.g., haptic system applications **250** and/or content that may be accessed via the haptic system applications **250**) and/or a content permissions module/system **374**. In embodiments, the content management system **370** serves immersive content that includes haptic, audio, visual, and/or any other sensory experience content. Additionally or alternatively, the content management system **370** (e.g., using the content permissions module **374**) manages the permissions required to execute the immersive content in the environment. In embodiments, the immersive haptic system **302** interfaces with the wearable haptic interface **330** and the content management system **370** that serves immersive content that includes haptic, audio, visual, and any other sensory experience content. In these embodiments, the content management system may manage the delivery of the content including any caching, compressing, and/or streaming of the immersive content.

[0138] In embodiments, haptic experiences may be personalized to a particular user. FIG. **3** illustrates an example user profile **312** that may be used to personalize or otherwise customize the operation of the various haptic interface systems **230** described herein, including various interface devices **232**, an SDK **242**, and/or various haptic system applications **250**. In embodiments, the user profile **312** may be set up using a user profile (which may be an auxiliary application **248**). In some embodiments, the user profile **312** may be leveraged partially or entirely by components of the SDK **242** (e.g., API **244** and/or haptic application plugins **246**) such that interface devices **232** and/or haptic system applications **250** may not need to implement personalization features. For example, an API **244** may be configured to modify data received from interface devices **232** (e.g., extending a stride length or input speed for a user who wishes to move their avatar faster) before generating inputs to haptic system applications **250** and/or may be configured to modify haptic feedback or other data output by the haptic system applications **250** (e.g., in order to limit the amount or type of force feedback that may be applied to a particular user) before providing it to interface devices **232**.

[0139] As shown in FIG. **3**, the user profile **312** may specify a variety of data that may be used to personalize any of the haptic applications described herein. In embodiments, the user profile **312** may include a variety of parameters as described in more detail below.

[0140] In embodiments, the user profile **312** may include user-provided information **402**, which may be used by haptic system applications **250** to generate a user avatar (e.g., the user avatar may be generated based on user-provided measurements, such as height, hand size, etc., and/or one or more user preferences for configuring the avatar). Additionally or alternatively, the user-provided information **402** may be used by an API **244** or haptic application plugins **246** to predict movement data for a user. For example, an API **244** may be configured to estimate the location and angle of a user's fingers (e.g., from motion capture data received from a haptic glove **236**) based in part on user-provided information **402** such as a width and length of the user's hand. In some cases, user-provided measurement information **402** may or may not correspond to a user's dimensions, but may specify one or more user preferences that may be used to configure the user's avatar. For example, a user may specify a preferred size of the avatar (and/or the avatar's hand for a haptic glove), gender, skin color, and/or any other attribute of the avatar. In embodiments, the user-provided information **402** may indicate, for example, if the user has missing or extra digits or other such user-provided information **402** for configuring a hand of the avatar.

[0141] In embodiments, the user profile **312** may include derived measurement information **404** that may be implicitly generated (e.g., without explicit input by a user) based on user interactions with one or more interface devices **232**. For example, based on a user's motions as captured by a whole-body haptic interface **234**, haptic glove **236**, and/or haptic exoskeleton **238**, user measurements such as a length of a user's limbs or fingers, a preferred range of motion, etc. may be implicitly estimated. The derived measurement information **404** may then be used by haptic system applications **250** to generate a user avatar, may be used by an API **244** to modify motion data received from interface devices **232**, etc.

[0142] In embodiments, the user profile **312** may further include range of motion limitations **406** that may limit active force feedback that may be applied to a user by one or more interface devices **232**. For example, active force feedback may be used to simulate a user's avatar and a virtual object colliding in a computer-mediated environment. In these embodiments, a haptic system application **250**, API **244** or haptic application plugins **246**, and/or interface device **232** may ensure that the force feedback is limited such that it does not strain or injure a user, such as by overextending a user's limb or causing an unnatural movement. The range of motion limitations **406** may define directions/angles that are safe and/or comfortable for a particular user's joints, limbs, and/or other body parts, such that the various haptic interface system **230** do not exceed the defined range of motion limitations **406**.

[0143] In embodiments, the user profile **312** may include force feedback limitations **408** that may further limit active or passive force feedback that may be applied to a user by one or more interface devices **232**. For example, active force feedback may be limited by the user profile specifying a maximum magnitude of force feedback to be applied, a maximum acceleration to be applied via force feedback, and/or the like. In embodiments, the maximums may be separately defined for individual body parts and may be defined in terms of force/velocity/acceleration, angular force/velocity/acceleration, and/or the like. The force feedback limitations **408** may be used by a haptic system application **250**, API **244** or haptic application plugin **246**, and/or interface device **232** in order to limit haptic feedback applied to a user.

[0144] In embodiments, the user profile **312** may include thermal limitations **410** that may limit thermal feedback that may be applied to a user by a thermal feedback interface of the one or more interface devices **232**. For example, the thermal limitations **410** may specify a maximum or minimum temperature and/or thermal flux that may be provided by the thermal interface, and/or whether thermal feedback may be provided at all, in particular situations, etc. The thermal limitations **410** may be used by a haptic system application **250**, API **244** or haptic application plugin **246**, and/or interface device **232** in order to limit haptic feedback applied to a user.

[0145] In embodiments, the user profile **312** may include assistance information **412** that may be used to modify movement and/or interaction data provided to haptic system applications **250** and/or to modify haptic feedback provided to a user of interface devices **232**. For example, the assistance information **412** may indicate that certain movements of a user of a whole-body haptic interface **234**, haptic glove **236**, and/or haptic exoskeleton **238** should be amplified. For example, if a user "walks" a one-meter stride in a whole-body haptic interface **234** or haptic exoskeleton **238**, and a user's assistance information **412** indicates a 3× movement scaling factor, the user's avatar in a haptic system application **250** may move by three meters in a computer-mediated environment. In embodiments, assistance information **412** may use separate scaling factors for vertical and horizontal movement, such that certain movements (e.g., climbing) may be made easier or harder. In embodiments, the assistance information **412** may be used to simulate reduced gravity. In embodiments, the assistance information **412** may be used to limit haptic feedback provided to a user. For example, if the assistance information **412** indicates that a user has a disability related to a particular limb or body part, haptic feedback corresponding to that body part may be reduced or eliminated. Additionally or alternatively, if the assistance information **412** is used to simulate reduced gravity, then haptic feedback may comport with the simulated reduced gravity.

[0146] In embodiments, a user profile **312** may include license user permissions/license information **414** that may authorize user access to particular content and/or types of content. The user permissions/license information **414** may include, for example, credentials for accessing protected content (e.g., subscription content, purchased content, etc.), a user account that may have permissions to modify or delete user-generated content, and/or the like. Additionally or alternatively, the user permissions/license information **414** may include content restrictions, such as parental controls, which may be used to restrict access to certain types of content.

[0147] In embodiments, a user profile **312** may include psychographic information **416** that may specify, for example, that a user is sensitive to certain haptic or visual stimuli, that a user prefers less or more haptic feedback of certain types, and/or the like. The psychographic information **416** may be used to tailor haptic feedback to a particular user (e.g., increasing certain types of haptic feedback to promote muscle memory, decreasing certain types of haptic feedback to avoid overwhelming a user, etc.).

"Point of View" Immersive Environmental Interaction

[0148] In embodiments, the wearable haptic interface **330** may include motion trackers **340** that are used to track the motion of respective body parts of a user. In some embodiments, the motion trackers **340** include magnetic motion tracking sensors with sub-millimeter tracing precision. The magnetic motion tracking sensors may be arranged into magnetic sensor arrays, which may measure angular displacement and/or relative position of components of the wearable haptic interface **330**. For example, a first set of motion trackers **340** arranged in a first magnetic sensor array may track the angular displacement and/or relative position of a first limb and/or joint of a user, a second set of motion trackers **340** arranged in a second magnetic sensor array may track the angular displacement and/or relative position of a second limb and/or joint of a user, and/or the like. Tracked joints may include a user's finger joints, shoulder/arm joints, hip/leg joints, neck, spine, and/or the like.

[0149] In embodiments, the motion trackers **340** may be magnetometers that may be positioned adjacent to corresponding magnets, such as on opposing sides of an articulation of an exoskeleton. Additionally or alternatively, the magnetometers may sense a magnetic field originating externally to a respective component of the wearable haptic interface **330**, to determine an orientation of the respective component with respect to an entity other than the wearable haptic interface **330**, such as an object, an infrastructure element, a fixture or the like. Thus, motion trackers **340** may track relative movements of the corresponding magnets and/or orientation of the corresponding component of the wearable haptic interface **330** and provide information on the relative movements and/or orientations as positioning data **358** to the API **244**. In turn, the API **244** may determine (e.g., using the motion capture system **314**) a position, orientation, and/or movement of the user (or part of the user, such as a hand joint, a limb, etc.) and corresponding interaction data **352** for an avatar, telerobot, and/or the like. The use of magnetometers as described herein may enable sub-millimeter tracking precision for orientation and/or relative movement, and thus may allow accurate and immersive positioning of an avatar within an environment.

[0150] In embodiments, the whole-body immersion system **300** may be configured to simulate a non-linear relationship between an anatomy of a human using the wearable haptic interface **330** and the anatomy of an avatar that exists within a haptic system application **250** and/or the anatomy of a telerobot. For example, an avatar and/or telerobot may be taller than the human user, may have larger fingers or hands, may have more or less digits, may have tools or accessories of various types attached to limbs instead of hands, and/or the like. In these embodiments, the immersive haptic system **302** may be configured to provide an immersive environment that allows the user to control the corresponding avatar/telerobot, interact with objects within the environment, etc. despite any dimensional, morphological, or other differences between the user and the avatar and/or telerobot. In these embodiments, the API **244** (e.g., using the motion capture system **314** or some other motion tracking component described herein) may receive sensor data and/or other positioning data **358** from the components of the wearable haptic interface **330** and may determine one or more respective joint angles of one or more respective joints (e.g., of a hand) based on the sensor data and a model of the physical and mechanical characteristics and/or other parameters of a human hand. In these embodiments, the motion tracking sensors **340** and/or motion capture system **314** may provide the one or more joint angles to the immersive haptic system **302** in lieu of or in addition to positional data of the joint and corresponding body part (e.g., the interaction data **352** may include finger joint angle data instead of or in addition to a motion capture data structure representing the position of a user avatar or hand, for example) when there is a non-linear relationship between an anatomy of the human and a corresponding anatomy of the avatar. Accordingly, the immersive haptic system **302** may be configured to map the user joint angles to the anatomy of the avatar.

[0151] In embodiments, the artificial intelligence module/system **310** may provide features and functionalities to the immersive haptic system **302** to improve the immersive environment, such as by adjusting data provided as inputs to the haptic system applications **250** and/or the wearable haptic interface **330**. For example, the artificial intelligence system **310** may be configured to adjust the output state sent to the haptic human interface components of the wearable haptic interface **330** (e.g., as haptic actuation commands or other haptic data **360** or sensor data **366**) to optimize the sensory experiences of the user of the wearable haptic interface **330** with respect to one or more factors. Additionally or alternatively, the artificial intelligence system **310** may be configured to translate raw motion and/or tracking data (e.g., corresponding to motions of a user) output by the components of the wearable haptic interface **330**) into high accuracy motion capture data.

[0152] In embodiments, the immersive haptic system **302** may model a set of attributes of a set of objects in an environment (e.g., the environment data **304**) and may determine a position of a set of elements of an avatar of a user (e.g., the avatar data **306**) in the environment based on a state of a haptic interface worn by a user. For example, the immersive haptic system **302** may, in response to detecting that a position of the avatar intersects with a position of an object within the computer-mediated environment, be configured to output a set of sensory simulation data related to the object (e.g., haptic data **360**, audio data **362**, video data **364**, and/or sensor data **366**). In some embodiments, the immersive haptic system **302** may be configured to output a set of sensory simulation data related to the object that intersects with the user avatar such that the immersive haptic system outputs are configured as a set of haptic actuation instructions (e.g., haptic data **360**). Additionally or alternatively, the immersive haptic system **302** may be configured to output a set of sensory simulation data related to the object that intersects with the user avatar such that the immersive haptic system outputs a set of thermal sensory simulation data related to the object. Additionally or alternatively, the immersive haptic system **302** may be configured to output a set of sensory simulation data related to the object that intersects with the user avatar such that the sensory simulation data include a set of deformation parameters based on which a set of haptic actuators simulate touching of the object.

[0153] In embodiments, the whole-body immersion system **300** may be configured to provide a multi-sensory user experience by providing an immersive haptic system **302** that models a set of attributes of a set of objects in an environment (e.g., environment data **304**) and determines a position of a set of elements of an avatar of a user in the environment (e.g., avatar data **306**) based on a state of a haptic interface worn by a user (e.g., haptic gloves **236**, a haptic exoskeleton **238**, and/or other components of a wearable interface **33**). In these embodiments, the immersive haptic system may be configured to detect that a position of the avatar intersects with a position of an object within the environment (e.g., when the avatar represented by the avatar data **306** is in the same space, or within a threshold distance, of an object of the environment represented by the environment data **304**). When the immersive haptic system **302** detects that the position of the avatar intersects with the position of an object (e.g., the avatar and object are touching inside the environment), the immersive haptic system **302** may output a set of sensory simulation data related to the object (e.g., haptic data **360** for allowing a user to feel the object, audio data **362** representing a sound of the collision between the avatar and the environment or data for rendering such a sound, etc.). The sensory simulation data may be coordinated in order to provide an immersive feedback experience. In embodiments, immersive haptic system outputs related to the object may be configured as a set of haptic actuation instructions (e.g., haptic data **360**) and a set of audio instructions (e.g., audio data **362**) that characterize audio characteristics of the environment. Additionally or alternatively, when an immersive haptic system **302** detects that a position of the avatar intersects with a position of an object with the environment, the immersive haptic system **302** may be configured to output sensory simulation data related to the object that includes a set of haptic actuation instructions and a set of video or imaging instructions that characterize visual characteristics of the environment. Additionally or alternatively, when an immersive haptic system **302** detects that a position of the avatar intersects with a position of an object within the environment, the immersive haptic system **302** may be configured to output sensory simulation data related to the object that includes a set of haptic actuation instructions and a set of virtual reality instructions that characterize a virtual reality representation of the environment. Additionally or alternatively, when an immersive haptic system **302** detects that a position of the avatar intersects with a position of an object, the immersive haptic system **302** may be configured to output sensory simulation data related to the object that includes haptic actuation instructions and a set of augmented or mixed reality instructions that characterize augmented reality or mixed reality elements for the environment. Additionally or alternatively, when an immersive haptic system **302** detects that a position of the avatar intersects with a position of an object, the immersive haptic system **302** may be configured to output sensory simulation data related to the object that includes a set of haptic actuation instructions and a set of taste instructions that characterize taste characteristics of the environment. Additionally or alternatively, when an immersive haptic system **302** detects that a position of the avatar intersects with a position of an object, the immersive haptic system **302** may be configured to output sensory simulation data related to the object that includes a set of haptic actuation instructions and a set of scent instructions that characterize scent characteristics of the environment.

"Skin Out" Wearable Technology and Tactile Surfaces

[0154] In embodiments, a haptic platform may use one or more haptic user interface devices that provide direct haptic feedback through tactile actuators, thermal actuators and/or other actuators (e.g., electrical, biochemical, chemical, or the like) in contact with a user's skin. Software, firmware, data and intelligence techniques for leveraging these technologies to generate immersive computer-mediated environments are described in more detail herein.

[0155] FIG. **4** illustrates an example haptic interface device **450**, which may be any component of a wearable haptic interface **330** or any other device that leverages smart surface technology to receive haptic system inputs and/or provide haptic feedback. The haptic interface device **450** may, in embodiments, be configured as haptic gloves **236**, a haptic exoskeleton **238**, or any other haptic interface device. In embodiments, the haptic interface device **450** may include a plurality of haptic components **452** that may include a first glove component, a second glove component, a haptic vest, haptic arm/leg components, other smart surface components, etc., where each haptic component may feature a plurality of microfluidics tactile actuators as described herein. In other words, the haptic interface device **450** may be implemented as various haptic user interface form factors by using different hardware components in different combinations as desired. Thus, the haptic interface device **450** may have one or more haptic components **452**, which may be different combinations of components in different embodiments.

[0156] In embodiments, the haptic interface device **450** may include one or more feedback controllers **454**, which may receive haptic data **456** as inputs and generate actuation commands **458** for controlling actuators of the haptic component(s) **452** to simulate a sensation of interacting with a physical object. In embodiments, if the haptic interface device **450** includes a plurality of haptic components **452**, then each haptic component **452** may be associated with a corresponding feedback controller. Additionally or alternatively, multiple haptic components **452** may be controlled by a single feedback controller **454** (e.g., a single feedback controller **454** may generate actuation commands **458** for two different glove components).

[0157] In embodiments, haptic data **456** may be received (e.g., from an immersive haptic system **302**) and processed by the feedback controller **454** to generate actuation commands. The haptic data **456** may include one or more of haptic data **360**, audio data **362**, video data **364**, sensor data **366**, and/or other output data that represents some aspect of a current state of an immersive haptic system. Accordingly, haptic actuation commands **458** may be generated based on processing any of the data output by an immersive haptic system (e.g., immersive haptic system **302**). Additionally or alternatively, in embodiments a feedback controller **454** may implement any of the features ascribed to an API **244** herein. In other words, the functions of an API **244** may be implemented in full or in part by an immersive haptic system **302**, and/or in full or in part by a feedback controller of a haptic interface device **450**.

[0158] Power may be supplied to various components (e.g., the feedback controllers **454**, the haptic components **452**) by an electrical system **460**. Functions and capabilities of the electrical system **460** are discussed in more detail below with respect to FIG. **7**.

[0159] In embodiments, the haptic interface device **450** may include user tracking sensor(s) **462** for tracking the location or status of the user or various joints/body parts of the user (e.g., finger tracking sensors). The user tracking sensors **462** may detect absolute or relative position or orientation, velocity/acceleration data, pressure data, force feedback data, temperature data, EEG signals, and/or other such data, and may transmit corresponding sensor data **468** to one or more input state processors **464**. The input state processors, in turn, may determine and send input state data **466** indicating a state of the user or parts of the user based on the sensor data **468**. The input state data **466** may include, for example, positioning data **358**, a data structure indicating a pose of the user, location of the user, motion of the user, temperature of the user, and/or any other such data describing the user. In embodiments, the input state data **466** may be output to an immersive haptic system **302**. In embodiments, the input state processors **464** may use various techniques to determine the input state, as described in more detail elsewhere herein. In some embodiments, the input state processors **464** may perform motion detection and/or other functionalities that may be ascribed to the API **244** elsewhere herein. Additionally or alternatively, the input state processor **464** may partially process the sensor data **468** before outputting the input state data, and the motion capture system **314** or some other module of the API **244** may complete the processing of the sensor data **468**.

[0160] Accordingly, a haptic interface device may have a human computer interface terminal that includes input transducers that receive a set of sensor values from a set of sensors disposed within a wearable interface and determine a user input state based thereon (e.g., input state data **466**). Additionally or alternatively, a haptic interface device may have a human computer interface terminal that includes output transducers that receive a user output state from a computing device executing a VR simulation (e.g., haptic data **456**) and that actuates one or more actuators of a haptic component based on the user output state.

[0161] In embodiments, a haptic interface device (e.g., the haptic interface device **450**) may include a set of haptic components (e.g., haptic components **452**) that realistically simulate a human sensory interaction with a physical entity of a haptic virtual application (e.g., an entity that corresponds to data within a haptic system application **250**) based on a set of actuation commands (e.g., actuation commands **458**) determined from a haptic feedback response (e.g., haptic data **456**) corresponding to an interaction of an avatar (e.g., as represented by avatar data **306**) and an environment (e.g., as represented by environment data **304**).

[0162] In embodiments, the haptic interface device may include a haptic component that is in contact with a skin surface of a human user, and a feedback controller that controls actuation of the haptic component in response to receiving an output state from an immersive haptic system (e.g., haptic data **456**) that indicates an interaction between an avatar of the user with an object in an environment. In some of these embodiments, one or more of the haptic components **452** that are in contact with a skin surface of a human user may provide multi-resolution sensory stimulation. In these embodiments, the feedback controller may be configured to determine a respective magnitude, resolution (e.g., spatial granularity, timing granularity, or the like) with respect to each type of sensory stimulation that is used for the haptic feedback for each region of the haptic interface device based on the output state from the immersive haptic system. Additionally or alternatively, the magnitude, resolution or the like of a respective section of the haptic component may be based on portion of the human user that it contacts. In other words, the multi-resolution sensory stimulation may be position aware. For example, sensory stimulation for the fingertips may be provided with very high spatial granularity, reflecting the high density of nerve endings there, while sensory stimulation for the torso, shins, or the like may be provided with less granularity, reflecting lower density of nerve endings. Additionally or alternatively, sensory stimulation may be provided with magnitude or resolution that is related to the purpose of the simulation; for example, higher granularity may be provided to the feet in a simulation that is intended to provide training in footwork, or the like.

[0163] Additionally or alternatively, the haptic component that is on contact with a skin surface of a human user may provide multiple modes of sensory stimulation. In these embodiments, the multiple modes may include one or more of pressure stimulation, thermal stimulation, electrical stimulation, chemical stimulation, biochemical stimulation and/or frictional stimulation. Thus, the actuation commands **458** may include various haptic modes generated based on the feedback controller **454** processing the haptic data **456**. Thus, contact with various objects with differing properties may be simulated by providing corresponding actuation commands **458** to the haptic components **452**.

[0164] In embodiments, the haptic component **452** that is in contact with a skin surface of a human user may be configured to simulate temperature flux. In embodiments, simulation of temperature flux may involve providing a change in temperature of a haptic component that is monotonically related to, but different from, the actual temperature change of an object that is being simulated, such as based on an understanding of perceptual contrast (i.e., how the brain perceives changes in sensory parameters like temperature and sound more readily than it perceives absolute levels of such parameters). Such techniques allow for certain efficiency advantages in haptic hardware system design. For example, a system can be designed to render highly immersive environments without requiring the fluid conduction system to be capable of generating very large heat flux (which could otherwise require large volumes of heavy fluids and/or large, expensive heating elements). Additionally or alternatively, the haptic component that is in contact with a skin surface of a human user may be configured to simulate the absolute temperature of a physical entity. Thus, for example, contacts with objects with various thermal properties (e.g., varying temperatures, varying thermal conductivity, etc.) may be realistically and immersively simulated via a haptic component **452**.

[0165] In embodiments, the haptic component **452** that is in contact with a skin surface of a human user may be configured to simulate the weight of a physical entity. Thus, by varying the pressure applied, for example, contacts with objects of varying weight may be realistically and immersively simulated via a haptic component **452**. In embodiments, the haptic component **452** may simulate the perceived experience of bearing weight (such as to train a user in weight-bearing tasks like carrying a fire hose) while providing actual forces that are lower than would be experienced by a real physical interaction with the object. Again, awareness of how the brain experiences perceptual contrast can be used to provide a realistic experience (e.g., one that generates valuable muscle memory) without requiring the hardware system design to generate very large forces. This can provide advantages in system design (allowing for less expensive components) and safety (e.g., allowing forces to be used that are safe for the user). In other embodiments, the haptic component **452** can in fact provide highly accurate rendering of forces that precisely match the forces a user would experience in the real world when interacting with the object. In embodiments the haptic component **452** can be configured, customized and/or personalized as noted above, such as to generate forces that correspond to a particular user profile, a profile for a role or type of user, or other characteristics.

[0166] In embodiments, the haptic component that is in contact with a skin surface of a human user may be configured to simulate the surface texture of a physical entity (e.g., by applying more or less pressure via various actuators to simulate details of an object's texture). Additionally or alternatively, the haptic component that is in contact with a skin surface of a human user may be configured to simulate the flexibility or rigidity of a physical entity (e.g., by applying varying levels of feedback when a user manipulates the entity). Additionally or alternatively, the haptic component that is in contact with a skin surface of a human user may be configured to simulate the frictional characteristics of a physical entity (e.g., by applying more or less pressure when the object slides against a user's skin in the environment). Thus, for example, contacts with objects of varying properties and textures may be realistically and immersively simulated via a haptic component **452**.

[0167] In embodiments, a haptic component that is in contact with a skin surface of a human user may provide multi-resolution sensory stimulation, and the finger portions of the haptic component have the highest respective resolution sensory simulation. Because fingers have a large number of specially adapted sensory nerve endings and receptors, high resolution haptic data may provide particularly effective immersion via haptic systems that provide haptic feedback to the fingers.

[0168] In embodiments, a haptic interface device **450** may be any type of haptic interface, including a non-wearable haptic interface. For example, one or more haptic components (e.g., featuring microfluidic tactile actuators and/or thermal actuators, as described in more detail below) may be arranged on a surface of a theater chair or other chair to provide haptic feedback that may be synchronized to a movie, game, or other interactive experience. Additionally or alternatively, one or more haptic components may be used for a tactile display, which may use tactile actuators to simulate controls such as buttons and/or various features such as raised surfaces. The tactile display may also use the techniques described herein to provide haptic feedback to simulate button presses or other interactions. In embodiments, haptic interface devices **450** that are configured as tactile displays may include additional components, such as a backlight and/or a proximity sensor, that may help tailor the haptic interface device **450** for use as a tactile display.

[0169] FIG. **5** illustrates an example embodiment where the haptic interface device **450** of FIG. **4** is a wearable haptic glove **236**. In the example embodiment shown at FIG. **5**, the wearable haptic glove **236** receives haptic data **456**, generates actuation commands **458**A-B using a feedback controller **454**A, and provides the actuation commands **458**A-B to respective haptic components **452**A-B. One or more of the haptic components (e.g., haptic component **452**A) may include an interface garment **502** that includes a plurality of actuators in contact with various regions of a user's hand and/or arm and/or a microfluidic system **510** for interfacing with the interface garment **502**. For example, a first subset of the actuators may be disposed to be in contact with a user's first finger, a second subset of the actuators may be disposed to be in contact with a user's second finger, and the like, with multiple subsets of actuators arranged to provide an immersive tactile experience. Additionally or alternatively, one or more of the haptic components (e.g., haptic component **452**B) may include brakes **508** that may be used to provide feedback to simulate interactions with objects or other entities within a computer-mediated environment.

[0170] In embodiments, an interface garment **502** may comprise a plurality of microfluidic tactile actuators **504** and/or thermal actuators **506**. In some embodiments, an actuator may be both a microfluidic actuator **504** and a thermal actuator **506**. In other words, a single actuator may provide both tactile (e.g., pressure) and temperature feedback to the skin of a user (e.g., by using temperature-variable fluid). Additionally or alternatively, instead of using microfluidic actuators **504**, other types of actuators may be used.

[0171] Accordingly, in embodiments, a wearable glove may have a haptic component that includes a set of specifically located microfluidic tactile actuators. Each subset of microfluidic tactile actuators may be configured to simulate a tactile sensation at a specific contact point of the hand of the user.

[0172] In embodiments, a wearable glove may have a haptic component that includes an interface garment that includes a set of specifically located microfluidic tactile actuators. In some embodiments, each subset of tactile actuators may be configured to simulate a tactile sensation at a specific contact point of the hand of the user. Additionally or alternatively, each subset of microfluidic tactile actuators may be configured to simulate a tactile sensation at a specific contact point of the hand of the user. Additionally or alternatively, the set of specifically located microfluidic tactile actuators may include a set of thermal actuators that control a localized thermal flux at respective contact points with hands of the user.

[0173] The tactile actuators **504** may be arranged such that they are grouped using a varying tactile density with a spacing that is sufficiently small to be within a somatosensory "two-point threshold" for various parts of, for example, a user hand. The two-point threshold is a measure of the minimum distance at which the human skin can discriminate between two separate points, and varies for different regions (e.g., about 1 centimeter at the palm of the hand and several millimeters at the fingertip). If two pressure points are applied at a distance below the two-point threshold distance, the two pressure points will be perceived as a single larger pressure point rather than two distinct points. Thus, by arranging actuators according to the two-point threshold (e.g., using varying density corresponding to different parts of a user hand, for example), the tactical actuators may be configured to simulate contact with a single solid surface, multiple objects, or permutations in between).

[0174] Accordingly, in embodiments a haptic glove interface may have a plurality of tactile actuators arranged to oppose a hand of a user of the haptic glove and having a varying tactile density with spacing between adjacent tactile actuators based on a somatosensory two-point discrimination of a human hand at a corresponding location in the haptic glove. In embodiments the somatosensory two-point discrimination may be based on a model of the human hand or based on the measured characteristics of the hand of a specific user.

[0175] In embodiments, the wearable haptic glove **236** may include hand and/or finger tracking sensor(s) **532** for tracking the location or status of the user's hands and/or fingers. The hand/finger tracking sensors **532** may detect absolute or relative position or orientation of the hands and/or fingers, velocity/acceleration data for the hands and/or fingers, pressure data for the hands and/or fingers, force feedback data for the hands and/or fingers, temperature data for the hands and/or fingers, and/or other such data, and may transmit corresponding sensor data **538** to one or more hand state processors **534**. The hand state processors, in turn, may determine and output hand state data **536** indicating a state of the user or parts of the user based on the sensor data **538**.

[0176] In embodiments, fluidics-based actuators of the interface garment **502** may be supplied with fluid by a microfluidics system **510** that is connected via one or more fluid conduits **520** with the one or more microfluidic tactile actuators **504**. The tactile actuators may use fluids and/or air provided by the microfluidic system **510**. In embodiments, the microfluidics system **510** may include a fluid supply **512** (e.g., an air supply if the microfluidics system uses air) containing a pressurized working fluid, a compressor **514** for pressurizing the working fluid, a plurality of valves **516** (e.g., piezo valves) operatively coupled to the supply **512** and the fluid conduits **520** for selectively actuating the microfluidics actuators, and a manifold **518** coupled to each of the valves **516**. In embodiments, the fluid supply **512** may include separated hot and cold fluid supplies that may be used to provide variable temperature fluids for thermal feedback. Although the microfluidics system **510** may be a component of a wearable haptic glove **236**, in some embodiments some or all of the components of the microfluidics system **510** may instead be part of a separate component that is connected to the wearable haptic glove via one or more conduits **520**. For example, a compressor **514** may be separate from the wearable haptic glove **236** (e.g., it may be contained within a wearable backpack) and connected to the fluid supply **512** via a conduit **520**.

[0177] Accordingly, in embodiments a haptic glove interface may have a set of tactile actuators with a microscale configuration, a pressurized fluid supply, a fluid conduit coupled for fluid communication between the set of tactile actuators and the pressurized fluid supply, and a set of valves operatively coupled with the pressurized fluid supply and the fluid conduit to selectively actuate the set of tactile actuators with a working fluid from the pressurized fluid supply.

[0178] In embodiments, a haptic interface device may have a plurality of tactile actuators, a plurality of fluid conduits each coupled with at least one of the plurality of tactile actuators, a plurality of valves each coupled to at least one of the plurality of fluid conduits, a manifold coupled to each of the plurality of valves, and a wearable compressor configured to supply a pressurized working fluid to the manifold.

[0179] In embodiments, the wearable haptic glove **236** may further comprise one or more additional haptic components (e.g., haptic component **452**B) that comprise one or more brakes **508**A-N. In embodiments, the brakes **508** may be tendon type resistance brakes, magnetorheological brakes, or other types of brakes that may be used to limit user movement. The brakes may be coupled with various parts of a user's hand to provide passive and/or variable feedback to restrain a user's fingers or other parts of a user's hand (e.g., a wrist). Thus, separate brakes may be provided at least for each finger of a user's hand. For example, a brake may be activated to prevent the movement of a finger when the finger of the corresponding avatar touches or applies pressure to an object or entity within a computer-mediated environment (e.g., if the object or entity is rigid). Thus, the sensation of gripping, holding, squeezing, or otherwise applying pressure to physical objects with the fingers may be realistically simulated. The brakes may be capable of binary (e.g., on/off) and/or variable (e.g., proportional) braking. In embodiments, instead of or in addition to using brakes **508**, active force feedback actuators may be used, which may be capable of providing force feedback to a user's fingers (e.g., to simulate the active feedback pressure applied to a user's fingers when squeezing a spring or rubber ball, for example). However, in other embodiments the haptic interface device (e.g., haptic glove) may have a tendon type resistance brake coupled with a finger of a user and configured to selectively resist movement of the finger.

[0180] In embodiments, the wearable haptic glove **236** may include hand and/or finger tracking sensor(s) **532** for tracking the location or status of the user's hands and/or fingers. The hand/finger tracking sensors **532** may detect absolute or relative position or orientation of the hands and/or fingers, velocity/acceleration data for the hands and/or fingers, pressure data for the hands and/or fingers, force feedback data for the hands and/or fingers, temperature data for the hands and/or fingers, and/or other such data, and may transmit corresponding sensor data **538** to one or more hand state processors **534**. The hand state processors, in turn, may determine and output hand state data **536** indicating a state of the user or parts of the user based on the sensor data **538**.

[0181] Although FIG. **5** shows an example embodiment in which the haptic interface device **450** is a wearable haptic glove **236**, the components shown in FIG. **5** may be used in other embodiments of a haptic interface device **450**. For example, a microfluidics system **510** connected via fluid conduits **520** to an interface garment **502** having one or more microfluidic tactile actuators **504** and/or one or more thermal actuators **506** may also be part of any type of haptic interface device **450**, such as an exoskeleton. Thus, an interface garment **502** may be arranged such that the actuators of the textile layer provide tactile and/or thermal feedback at any body part of a user (e.g., head, torso, limbs, etc.) or may be arranged on other surfaces that a user may contact. Moreover, as shown at FIG. **4**, a haptic interface device **450** may include multiple haptic components **452** (e.g., one per body part).

[0182] Accordingly, in embodiments, a wearable haptic system may have a haptic component that includes an interface garment that includes a set of specifically located microfluidic tactile actuators. Each subset of microfluidic tactile actuators may be configured to simulate a tactile sensation and/or control a localized thermal flux at a specific contact point of a body part of a user.

[0183] FIG. **6** illustrates an example embodiment where the haptic interface device **450** of FIG. **4** is a haptic exoskeleton **238**. In embodiments, an exoskeleton **238** may include one or more haptic gloves **236**. In other words, the glove(s) **236** (not shown in FIG. **6**) may be subcomponents of the exoskeleton **238**. In the example embodiment shown at FIG. **6**, the haptic exoskeleton **238** receives haptic data **456**, generates actuation commands **458**A-D using a feedback controller **454**A, and provides the actuation commands **458**A-D to respective haptic components **452**A-D. Additionally or alternatively, multiple feedback controllers **454** may be used to provide respective actuation commands to each haptic component. The haptic exoskeleton **238** may include one or more haptic components that include an interface garment **502** (e.g., haptic component **452**A) in communication with a microfluidics system **610** via one or more fluid conduits **620**, which may be configured as discussed above for FIG. **5**. Additionally or alternatively, the haptic exoskeleton may include one or more haptic components that include force feedback actuators **604** (e.g., magnetorheological (MR) actuators, pneumatic actuators, electric actuators, and/or other types of force feedback actuators). In embodiments, the various force feedback actuators **604** (e.g., MR actuators, pneumatic actuators, electric actuators, etc.) may be configured to provide force feedback to various body parts (e.g., arms, legs, etc.) in order to realistically simulate a human touch sensory system interaction with a physical object or entity. In embodiments, this sensory system interaction may be further enhanced by the use of tactile actuators as part of the interface garment **602**. For example, when a user's avatar (e.g., as represented by avatar data **306**) braces their arm and leans against a wall in an environment (e.g., as represented by environment data **304**), a force feedback actuator may provide force feedback that correspondingly allows the user to brace the user's arm against an arm component of the exoskeleton. At the same time, the tactile actuators of the interface garment **602** may realistically contact with various surface features of wall (e.g., based on the texture or geometry of the wall). Thus, the haptic actuators (e.g., including the tactile and force feedback actuators) may provide a synchronized simulation of an interaction with a wall. Similar interactions between a user's avatar and an environment may be realistically simulated by force feedback actuators in contact with other body parts of a user (e.g., force feedback to a user's foot when the user's avatar kicks a ball within the environment or the like), either alone or in combination with tactile actuators (e.g., depending on body part and coverage of an interface garment). In embodiments, data representing the interactions between the avatar and the environment are provided by the immersive haptic system as haptic data **456**, and the one or more feedback controllers **454** of the wearable haptic exoskeleton may determine and synchronize the haptic actuation commands to be distributed to the various components of the haptic exoskeleton in order to realistically simulate the interactions.

[0184] Accordingly, in embodiments, a wearable haptic exoskeleton device may have a set of haptic actuators that realistically simulate a human touch sensory system interaction with a physical entity (e.g., an object/entity within a computer-mediated environment). In some of these embodiments, the haptic actuators may be force feedback actuators that may be MR haptic actuators, pneumatic actuators, electrical actuators, and/or other types of actuators. Additionally or alternatively, the haptic actuators may be tactile actuators. In some embodiments, at least a subset of the haptic actuators of the exoskeleton system may be in contact with the arms of the wearer. Additionally or alternatively, at least a subset of the haptic actuators of the exoskeleton system may be in contact with the legs of the wearer. Additionally or alternatively, at least a subset of the haptic actuators of the exoskeleton system may be in contact with the neck of the wearer. Additionally or alternatively, at least a subset of the haptic actuators of the exoskeleton system may be in contact with the head of the wearer. Additionally or alternatively, at least a subset of the haptic actuators of the exoskeleton system may be in contact with the torso of the wearer. Additionally or alternatively, at least a subset of the haptic actuators of the exoskeleton system may be in contact with at least one foot of the wearer. In any of these embodiments, the haptic actuators may realistically simulate a human touch sensory system force feedback and/or tactile interaction with a physical entity.

[0185] In embodiments, the haptic exoskeleton and motion platform may support and move a user's center of mass (e.g., the exoskeleton may be able to lift the user, move the user, and/or orient the user in various ways) via one or more contacts with the user's limbs and/or torso. In these embodiments, the haptic exoskeleton and motion platform may be capable of realistically simulating a range of movements and/or accelerations through an environment such as flying, falling, climbing, and other such movements/accelerations. Accordingly, the wearable haptic exoskeleton device may have a set of force feedback actuators that, in conjunction with a motion platform, generate a set of movements that help simulate a movement or acceleration of a center of mass of the body of the wearer.

[0186] In embodiments, a wearable haptic interface may have an exoskeleton connected to a motion platform, one or more force feedback actuators **604** (e.g., magnetorheological actuators or other types of actuators), one or more microfluidic layers (e.g., interface garments **602**), and/or one or more microfluidic tactile actuators (e.g., as shown in FIG. **5**, but not shown in FIG. **6**). The microfluidic layer(s) may be coupled to the exoskeleton large-scale actuator(s) and/or the microfluidic actuator(s) (e.g., such that they form a single haptic exoskeleton **238**).

[0187] In embodiments, the haptic exoskeleton **238** may include body part tracking sensor(s) **632** for tracking the location or status of various parts and/or joints of the user's body. The body part tracking sensors **632** may detect absolute or relative position or orientation, velocity/acceleration data, pressure data, force feedback data, temperature data, EEG signals, and/or other such data, and may transmit corresponding sensor data **538** to one or more body state processors **634**. The hand state processors, in turn, may determine and output body state data **636** indicating a state of the user or parts of the user based on the sensor data **638**.

[0188] FIG. **7** illustrates an example electrical system **460** of a haptic device (which may be a wearable haptic interface **330**, a haptic interface device **450**, a wearable haptic glove **236**, a haptic exoskeleton **238**, and/or some other haptic device described herein) together with example devices that may receive power from the electrical system **460**. The electrical system **460** may be battery-powered, may be chargeable using an energy storage system **710**, may manage energy and/or power delivery via energy management system **708** and/or power management system **704**, and may include an electrical harness **702** for carrying power to electrically powered devices, such as one or more exoskeleton actuators **720**, fluidic valves **516**, a wearable compressor **514**, and/or any other electrically powered device that is described herein.

[0189] In embodiments, the electrical system **460** may include a wearable battery **712**, which may be a conformal wearable battery (e.g., a flexible and wearable battery). In embodiments, the wearable battery **712** may receive charge via an energy storage system **710**, which may receive power (e.g., from an external charging connection that is connected to the electrical system **460** when not in use by a user) and manage the delivery of power to the wearable battery **712** to recharge the battery. In embodiments, the energy storage system **710** may be configured to determine whether to charge the wearable battery **712**, how much current and/or power to deliver, and/or the like. The energy storage system **710** may manage charging of the wearable battery **712** in order to prioritize different objectives, such as quick charging, battery longevity, and/or the like. In embodiments, the wearable battery **712** may be connected to an electrical harness **702**, which may include electrical conductors shaped and routed to avoid interference with the movements of a user of the wearable haptic interface that includes the electrical system **460**. The electrical harness **702** may include a plurality of distinct electrically conductive pathways for routing power to various components, including the components described herein that promote immersive haptics. The electrical harness **702** may further be configured to limit immersion interference through physical forces between the electrical harness and body borne portions of the wearable user interface.

[0190] In embodiments, the output of the wearable battery **712** may be managed by the energy management system **708** and/or the power management system **704**. The energy management system **708**, for example, may prioritize delivery of power to specific high-priority components and/or for specific high-priority operations if/when the wearable battery **712** is not capable of supplying enough energy to all of the components (e.g., because of a low charge falling below a threshold, because of a current/power draw exceeding the output capability of the wearable battery **712**, and/or the like). Additionally or alternatively, the energy management system **708** may cause the electrical system to operate in various modes based on a haptic operation to be performed by the wearable interface, such as a high-power operation mode and a high voltage transient operation mode.

[0191] Additionally or alternatively, a power management system **704** may limit the delivery of power to certain components using a power output governor to ensure safe operation of the wearable haptic interface, such as by limiting the delivery of power to exoskeleton actuators **720** to avoid applying too much force feedback to a user of the exoskeleton. In embodiments, the power management system **704** may receive immersive haptic system data **722**, which it may process to determine whether any feedback to be delivered to a haptic device may exceed a safety threshold such that the power output governor **706** may limit the delivery of power to the corresponding haptic device. Additionally or alternatively, the power management system **704** may monitor the capability and/or status of a power source (e.g., the wearable battery **712**). For example, the power management system **704** may determine whether the power source is capable of exceeding a safety margin or threshold. Thus, the power management system **704** may manage power distribution based on a status of a power supply of the wearable user interface and output states received from an immersive haptic system that generates an environment. In embodiments, the power output governor **706** may use a set of rules (which may be default rules and/or user-specific rules that may be obtained configured based on data within a user profile **312**) to determine a maximum amount of force that may be applied to various body parts of a user via an exoskeleton or any other haptic component with a force feedback component.

[0192] In embodiments, power may be supplied (e.g., via the electrical harness **702**) to one or more exoskeleton actuators **720**, which may move components connected to various body parts of a user to simulate interactions with an environment or entities within the environment, provide force feedback, simulate different gravity configurations, etc. Additionally or alternatively, power may be supplied from the wearable battery **712** (e.g., via the electrical harness **702**) to one or more valves **516**A-N (e.g., piezo valves of a microfluidics system **510**) that selectively control the delivery of working fluid to a microfluidic actuator. Additionally or alternatively, power may be supplied from the wearable battery **712** (e.g., via the electrical harness **702**) to a wearable compressor **514**, which may compress a working fluid for a microfluidics system. In embodiments, any other electrically powered components may also receive power from the electrical system **460**.

[0193] Although the example electrical system **460** shows a wearable battery **712**, in embodiments the electrical system **460** may operate without a battery (e.g., using mains power) and/or may operate in either battery-power mode and/or via a connection to another power source. For example, a wearable haptic interface **330** for a whole-body haptic system may operate in a fixed position and therefore lack the need for a wearable battery **712**.

[0194] Accordingly, a haptic interface device may have a plurality of electrically actuated valves, a body borne electric compressor, and an electrical harness including a plurality of distinct electrically conductive pathways routed to promote immersive haptics and limit immersion interference through physical forces between the electrical harness and body borne portions of the wearable user interface.

[0195] In embodiments, a haptic interface device may have an energy storage system and an energy management system. The energy management system may be programmed to dynamically configure the energy storage system for one of a high-power operation and a high voltage transient operation based on a haptic operation to be performed by the wearable interface, and the energy management system may be configured to prioritize predefined high priority haptic operations when energy stored in the energy storage system falls below a threshold amount.

[0196] In embodiments, the haptic interface device may have a wearable component and a conformable battery incorporated into the wearable component.

[0197] In embodiments, the haptic interface device may have a power management system that manages power distribution to components of the wearable user interface based on a status of a power supply of the wearable user interface and output states received from an immersive haptic system that generates an environment.

[0198] In embodiments, the haptic interface device may have an exoskeleton and a power management system. The exoskeleton may be configured for applying forces to a body of the user and the power management system including a power output governor configured to limit a power output to the exoskeleton based on a set of rules and a portion of the body of the user to which the forces will be applied.

[0199] FIG. **8** shows an example embodiment of an immersive haptic system **800** in which various components, including one or more immersive haptic systems **302**, one or more wearable haptic interfaces **330**, and/or one or more external resources **808** are connected via one or more networks **810**. Each immersive haptic system **302** may have a network interface **802** that facilitates communications with other devices via the network(s) **810**. The wearable haptic interface **330** may include one or more haptic interface devices **450** (e.g., haptic gloves **236**, a haptic exoskeleton **238**, AR/VR/MR/XR interfaces **336**, other sensory interfaces **338**, motion tracking sensors **340**, etc.), each of which may also have a network interface **802** for communicating with other devices via the network(s) **810**. In embodiments, although network(s) **810** may be used to connect some or all of the separate devices (including the devices shown in FIG. **8** and/or any other devices described herein), the immersive haptic system **302** may also be capable of communicating with a wearable haptic interface **330** using a low-error ultra-low-latency network **812**, which may be a short range link, rather than routing communications through a network **810**. The use of the ultra-low-latency network **812** may provide a more immersive experience by providing a better connection between user movements and feedback provided to the wearable haptic interface **330** (e.g., audio/video and/or force feedback). For example, networks that provide less than one millisecond (1/1000 second) latency may be used as the ultra-low-latency network **812**.

[0200] In embodiments, the haptic interface devices **450** may comprise a low-error ultra-low-latency component network **814** for interconnecting various components of the haptic interface devices **450**, such as interfaces, actuators, processors, and/or related components. For example, the ultra-low-latency component network **814** may be used to communicate actuation commands **458** from a feedback controller **454** to a haptic component **452**, as discussed above for FIGS. **4-6**.

[0201] Various immersive haptic systems **302** may connect to and communicate with each other via the networks **810**, for example, to facilitate a multi-user haptic system application **250**, such as a multiplayer game, a collaborative simulation, and the like. The immersive haptic systems **302** may communicate with each other and/or a centralized server to synchronize a state of the haptic system application **250** among the immersive haptic systems. For example, one of the immersive haptic systems **302** may act as the host and other immersive haptic systems **302** may act as clients for the host. Additionally or alternatively, the immersive haptic systems **302** may communicate with an external host (e.g., an external resource **808** that includes a host system).

[0202] Similarly, multiple wearable haptic interfaces **330** may communicate with each other and/or with external resources to provide multi-user immersive experiences. In example embodiments, user data (e.g., user input data captured by one or more haptic interface devices **450** based on user movements) may be transmitted directly to other wearable haptic interfaces **330**, to other immersive haptic systems **302**, to external resources **808**, and/or the like.

[0203] In embodiments, multi-user haptic applications may allow one or more user(s) to interact with the application using wearable haptic interfaces **330** while one or more other user(s) may interact with the application without using a haptic interface. For example, a training application may allow one or more trainers to engage with a computer-mediated environment and/or a user avatar within the computer-mediated environment without using a haptic interface. Additionally or alternatively, some of the users of the multi-user haptic application may be non-human agents, such as artificial intelligence agents.

[0204] In embodiments, external resources **808** may include devices that may provide data to the immersive haptic system **302**. For example, the external resources **808** may include an edge networking system that receives data from IoT edge devices. The IoT edge devices may provide data about an environment that may be incorporated into the environment by the immersive haptic system **302**. Such data may represent, for example, locations and/or states of real objects or entities that may be rendered as objects or entities in the environment. As a specific example, IoT edge devices may be attached to doors of a physical environment, such that the state of the real doors (e.g., open, closed, locked) may be correctly rendered in a VR simulation or haptic digital twin, thus allowing users to interact with the real doors while participating in the VR simulation and/or controlling a telerobot.

[0205] In embodiments, the network **810** may be an RF network, a hard-wire network, an edge network connecting IoT devices (e.g., the external resources **808** that may be IoT edge devices), a 5G/cellular network, a mesh network, a peer-to-peer network, and/or the like. In embodiments, the network **810** may include a networking system that may optimize and/or re-route communications between devices. For example, such a networking system may select an optimum network and/or routing scenarios for a given set of available networks in real-time. The network(s) **810** may connect to external networks. The networks(s) **810** may include at least one ultra-low-latency network. The network interfaces **802** may include systems that are configured to support one or more of the various types of networks as needed for any given embodiment.

[0206] Accordingly, an immersive haptic system may have a low-error-ultra-low-latency network that connects a set of interfaces of an immersive haptic system to a corresponding set of interfaces of a wearable haptic interface, such that packet latency is less than one millisecond (1/1000 of a second). Additionally or alternatively, a haptic interface device may have a low-error-ultra-low-latency network that interconnects a set of interfaces, actuators, processors, or and related components of a wearable haptic device, such that packet latency is less than one millisecond (1/1000 of a second).

[0207] In embodiments, an immersive haptic system may have a radio frequency (RF) network that interconnects one or more immersive haptic systems, one or more haptic interface devices, and/or one or more external resources for RF communications. Additionally or alternatively, an immersive haptic system may have a fixed networking system that interconnects one or more immersive haptic systems, one or more haptic interface devices, and/or one or more external resources for hard-wire communications. Additionally or alternatively, an immersive haptic system may have a 5G/cellular networking system that interconnects one or more haptic interface devices with one or more of an immersive haptic system and one or more external networks. Additionally or alternatively, an immersive haptic system may have a mesh networking system that interconnects one or more immersive haptic systems, one or more haptic interface devices, and/or one or more external networks. Additionally or alternatively, an immersive haptic system may have a peer-to-peer networking system that interconnects one or more immersive haptic systems, one or more haptic interface devices, and/or one or more external networks.

[0208] In embodiments, an immersive haptic system may have an immersive haptic system that interfaces with a wearable haptic interface and with an edge networking system that provides data relating to an environment obtained from IoT devices within the environment to the immersive haptic system, wherein the immersive haptic system generates the environment corresponding to the edge networking system and updates the environment based on the data received from the edge networking system.

[0209] In embodiments, an immersive haptic system may have a networking system that adaptively selects an optimum network and routing scenarios for a given set of available networks in real-time, wherein the networks interconnect one or more immersive haptic systems, one or more haptic interface devices, and/or one or more external networks.

SDK and Intelligence Platform

[0210] In embodiments, an immersive haptic system may leverage an SDK **242** that includes and/or integrates with the API **244** and that is configured to provide an interface for integrating a wearable haptic interface with one or more other systems, such as other haptic system applications **250** (e.g., game engines, robotics systems, simulation systems, AR/VR systems, training systems, enterprise database systems, or many others), as well as for designing applications, user experiences and use cases involving such other haptic system applications **250**. The foregoing may be deployed in an ecosystem that benefits from various haptic interactions, use cases and applications, where the SDK **242** and/or API **244** provide a set of modules, components and capabilities for linking and integration with other systems to enable them. FIG. **9** illustrates an example diagram showing additional functions of the API **244** that sits between one or more haptic system applications **250** and a wearable haptic interface **330**. The API **244** includes a plurality of various functional units, which may be embodied as hardware and/or software modules. The API **244** may perform motion capture ("Mocap") functions for converting motion data into inputs to the haptic system application **250**, may perform haptic functions for converting haptic system application **250** state data into haptic feedback, and may perform other functions. Thus, the API **244** may be bidirectional, providing the translation (e.g., including extraction, transformation, loading, normalization and the like) between sensory stimulation information of a wearable or other haptic system and information about objects and entities from a computer-mediated environment. In embodiments, the motion capture functions may be performed by a motion capture system **314** (e.g., components **902**, **904**A-N, **906**, **908**, and/or **910** may be part of or executed by a motion capture system **314**) and the functional units that perform haptic feedback functions may be performed by a haptic feedback controller **316** (e.g., components **930**, **932**, **934**, and/or **936** may be part of or executed by a haptic feedback controller **316**). The API **244** may be responsible for a variety of functions beyond those illustrated in FIG. **9**. The wearable haptic interface **330** may comprise one or more motion tracking sensors **340**, any type of haptic interface device **450**, a wearable haptic glove **236**, a haptic exoskeleton **238**, and/or the like.

[0211] In embodiments, the API **244** may be configured to receive data from various sensors (e.g., motion tracking sensors **340**, which may be optical sensors or other types of motion tracking sensors, and/or sensors that are part of a haptic interface device **450**, a wearable haptic glove(s) **236**, and/or a haptic exoskeleton **238**) and process the data to generate animated frame data **958** that may be used to position one or more entities that exist inside the haptic system application **250** (e.g., a user avatar). Furthermore, the API **244** may be configured to receive world information **962** (e.g., data describing the state of avatar and/or objects within an environment) and/or haptic effects **964** from the haptic system application and process the data to generate microfluidic data **970** for providing tactile feedback and/or force feedback to various actuators of the haptic interface device **450**, the wearable haptic glove(s) **236**, the haptic exoskeleton **238**, and/or the like. In embodiments, the API **244** may be configured to interface with haptic system applications **250**, for example by providing animated frame data **958** in a format understandable by the application and/or a plugin for the application and by receiving and processing world information **962** and/or haptic effects **964** from the application and/or a plugin to the application. For example, the API **244** may leverage a haptic application plugin **246** to input data to and/or receive data from a telerobotics application **252**, training application **254**, gaming application **256**, product design application **258**, and/or any specific other type of application **250**.

[0212] Although the API **244** is shown as existing between the wearable haptic interface **330** and the haptic system application **250**, in other example embodiments various functions attributed to the API **244** may be performed by components of the wearable haptic interface **330** and/or by the haptic system application **250**. For example, some components and functionalities of the API may be executed by a plugin to a haptic system application (e.g., a game engine plugin). Additionally or alternatively, some components and functionalities of the API may be executed by hardware onboard a wearable haptic device. Accordingly, it should be understood that the various functionalities described herein may be located on different devices in different implementations.

[0213] In embodiments, the API **244** may include one or more sensor data processors **902** for processing raw motion data **954** receiving from one or more hardware sensors and executing sensor processing workflows to process the raw motion data **954** into relative location data **953**. The raw motion data **954** may comprise a plurality of data streams received from different sensors, including low-level motion capture data. For example, a first sensor data processor **902** may be configured with a sensor processing workflow for processing raw motion data captured by finger-tracking sensors of a haptic glove, a second sensor data process may be configured with a sensor processing workflow for processing raw motion data captured by arm tracking sensors of a wearable exoskeleton, etc. Various types of motion sensors that output raw motion data **954** in various formats may be used by the wearable haptic interface, and the sensor data processors **902** may accordingly use various workflows to perform various processing functions. The sensor data processors may perform error checking, formatting, normalization, prediction, integration, and/or other types of functions for converting raw motion data **954** into relative location data **953**. The relative location data **953** may include relative positioning, orientation, velocity, and/or acceleration data for one or more body parts (e.g., fingers, arms, legs, head, torso, etc.) and/or other tracked locations (e.g., one or more locations of a prop held by a user).

[0214] In some embodiments, sensors of the wearable haptic interface **330** may perform onboard sensor data processing. In the illustrated example of FIG. **9**, the motion tracking sensors **340** may perform onboard sensor data processing such that absolute location data **952** may be output to the API. In embodiments, the absolute location data **952** may include a tracking data stream that is indicative of an absolute location of the user with respect to the 3D environment of the user, as detected by the motion tracking sensors **340**. Thus, in some embodiments, data received from haptic hardware may not need to be processed using some or all of the sensor data processing workflows.

[0215] The API **244** may further include one or more motion processors **904** for generating mocap data **956** from the absolute location data **952** and/or relative location data **953**. The mocap data **956**, for example, may include a data structure describing the positioning and/or motion (e.g., velocity, acceleration, etc.) of the user and/or various part of the user. Thus, for example, the motion processors **904** may generate mocap data **956** that relates the positioning of various body parts (or other tracked locations, such as for props) to each other (e.g., based on the absolute location data **952** and/or relative location data **953**). In embodiments, the mocap data **956** may include a skeleton data structure indicating how the user is positioned.

[0216] In some embodiments, the one or more motion processors **904** may optionally use artificial intelligence and/or machine learning (AI/ML) techniques to generate models that predict the mocap data **956** based on the absolute location data **952** and/or relative location data **953**. For example, the location data may be taken from a set of sensors that does not exactly measure every body part and/or joint of the user. Accordingly, an ML model (e.g., a neural network) may be trained to output a data structure indicating a user's current and/or future position (e.g., a skeletal data structure) based on the location and/or raw motion data. For example, the model may predict the current angle of each joint of a user's hand based on location data indicating the location of a user's fingertips and the location of the palm of the hand. Such a model may be trained based on training data and/or may use rules that define, for example, how human hand joints typically or most likely bend to obtain various positions. Similar AI/ML models and techniques may be developed and deployed for other body parts and/or joints, such as leg joints (e.g., hips, knees, ankles, etc.), arm joints, and the like.

[0217] In some embodiments, the mocap data **956**A-N may optionally be adjusted using one or more compensators, such as a glove slip compensator **906** and/or a touch compensator **908**. For example, the glove slip compensator **906** may adjust the motion capture data based on glove slip caused by the glove shifting position on the user's hand. Furthermore, the touch compensator **908** may adjust the motion capture data to detect whether a user intends to touch one hand or finger to another, for example. In this example, the user may be prevented from bringing a sensor on one hand or finger to within a minimum distance of a sensor on another hand or finger (e.g., because of the thickness of the wearable haptic gloves **236** and/or the positioning of the sensors thereon). The touch compensator **908** may thus determine that a user intends to touch one body part (or a tracked prop) to another body part even when the user is prevented from doing so by the haptic interface device **450**. In embodiments, the slip compensator **906** and/or touch compensator **908** may use machine learning algorithms to adjust the motion capture data. Additional details regarding the operation of the compensators are provided below.

[0218] Finally, the animation frame generator **910** may generate and output animated frame data **958**, which may include structured location and/or motion data that may be used to modify a user avatar that exists within the haptic system application **250**. In some embodiments, the animated frame data **958** may be generated by reformatting the compensated motion capture data (e.g., after processing by the slip compensator **906** and/or touch compensator **908**) into a format that is compatible with the haptic system application. The animated frame data **958** may include, for example, positioning and/or motion data for each part of a user's body. As one example, the animated frame data **958** may include skeletal animation data that may be used for positioning the user avatar. In embodiments, the animation frame generator **910** may compensate for dimensional And/or skeletal differences between the user and the user avatar using animation retargeting. For example, if a user avatar is taller and has longer limbs than the user, the motion capture data may be adapted to fit the dimensions of the avatar. As another example, if the user avatar has a different skeletal structure than the user (e.g., the user avatar may be an animal, robot, etc. with different numbers of arms, legs, digits, etc. than the user), the animation frame generator **910** may use various techniques for mapping between human skeletal data and avatar skeletal data.

[0219] The above-described process for converting sensor data received from a wearable haptic interface **330**, such as raw motion data **954** and/or location data **952**, into more structured data for input into a haptic system application **250**, such as a game engine, may be referred to herein as a "Mocap stack."

[0220] The API **244** may receive data from the haptic system application **250** to generate haptic feedback. For example, a contact interpreter **930** may receive state information from the haptic system application **250**, such as world information **962** and/or haptic effects **964**. The world information **962** may include environment data **304** and/or avatar data **306**. In these embodiments, the contact interpreter **930** may calculate whether a part of the user avatar is in contact with (and/or nearby) any objects or entities in the environment and/or with another part of the avatar (e.g., the avatar's hands may be in contact with each other). The contact interpreter **930** may further calculate a relative force applied to the avatar by the contact. Additionally or alternatively, the haptic system application **250** may determine and output such contact information. Furthermore, in some embodiments, the haptic system application may perform some or all of the haptic feedback processing described herein natively and may output the haptic data as haptic effects **964**.

[0221] In embodiments, the contact interpreter **930** may output a haptic frame **966** indicating where haptic feedback should be applied to a user (e.g., a force vector specifying a magnitude and direction of force feedback, actuation patterns specifying multiple points of contact with a user hand or other location and an amount of pressure applied at each point, etc.) and/or which actuators should be activated. In some embodiments, the haptic feedback locations of the haptic frame may not necessarily correspond to the locations of actuators on the wearable haptic interface. For example, a haptic frame may indicate a particular haptic feedback location (e.g., for simulating a virtual "needle prick") that may not correspond to any tactile actuator. In this example, the haptic feedback may be provided to a user using multiple actuators near the corresponding location. Additionally or alternatively, even if a haptic feedback location corresponds to an actuator, multiple other nearby actuators may be actuated to provide a stronger haptic feedback response. In other examples, a haptic frame may specify haptic feedback over an area, which may be simulated by actuating multiple actuators that may be within the corresponding area and/or outside the corresponding area. Accordingly, it should be understood that the haptic feedback specified by the haptic frame may not map exactly to the locations of the tactile actuators. In embodiments, the contact interpreter **930** may be configured to extract contour information from the world information **962** at a sufficient rate to perform real-time haptic control. The contact interpreter may use various strategies to transform the contour data extracted from the world information into actuation patterns. These strategies may be based on various rules and/or algorithms for simulating human tactile perception in a way that may be used to maximize the realism of a sensation, as discussed in further detail below. Additionally or alternatively, the haptic feedback may be provided in a way that departs from a realistic simulation of a sensation. For example, an application may use one or more haptic effects to rapidly turn actuators on and off (e.g., "flutter" the actuators) to provide a signal to the user for various reasons.

[0222] An actuation calculator **932** may then determine an actuator frame **968** including data for causing various components and/or actuators of the wearable haptic interface **330** to actuate based on the haptic frame **966**. In embodiments, the actuation calculator **932** may use simulated peripheral data from a simulated peripheral database **934** to perform the generation of the actuator frame **968**. The actuation controller **936** may then output actuator command data **970** (e.g., various actuation commands addressing individual actuators) to various components of the wearable haptic interface). In some embodiments, the actuation commands may be used to adjust microfluidic actuators. Additionally or alternatively, the actuation commands may be used to adjust MR actuators, pneumatic actuators, finger brakes, or other types of force feedback actuators. In embodiments, the actuator command data **970** may cause actuation of one or more tactile actuators **504** and/or thermal actuators **506** of an interface garment **502**. Additionally or alternatively, the actuator command data **970** may cause actuation of one or more brakes **508**. Additionally or alternatively, the actuator command data **970** may cause actuation of one or more force feedback actuators **604**.

[0223] In embodiments, instead of outputting actuator command data **970** to the wearable interface devices, the actuation controller may cause a separate microfluidic system to actuate external microfluidic valves that are connected (e.g., via fluid/air conduits) directly to the wearable haptic interface components.

[0224] In embodiments, the actuation calculator **932** and/or the actuation controller **936** may output data to a hardware state visualizer (HSV) controller **938**, which may output to a hardware state visualizer **940**. The hardware state visualizer may be an example of an auxiliary application **248**. The hardware state visualizer may allow viewing the state of any of the components of the wearable haptic interface **330** (e.g., the current or past positioning of the component, current or past actuation of the valve and/or actuators, etc.), which may be used for monitoring, debugging, and/or other such purposes. In embodiments, the actuation controller **936** may also output haptic feedback data back to the haptic system application **250** (e.g., so that the haptic system application **250** may use the haptic feedback data for in-engine purposes).

[0225] The above-described process for converting application data received from a haptic system application **250**, such as world information **962** and/or haptic effects **964**, into data for providing haptic feedback to a wearable haptic interface **330** may be referred to herein as a "Haptic stack."

[0226] Accordingly, in embodiments, an immersive haptic system may have an SDK (e.g., including an API and/or haptic application plugin) that is configured to process raw motion data received from a wearable human interface into structured motion data using a set of sensor data processing workflows, the structured motion data being input into a game engine/simulation engine/haptic digital twin, and to process an output state of an environment generated by the game engine/simulation engine/haptic digital twin into a set of haptic control commands that are input to the wearable human interface.

[0227] In embodiments, an immersive haptic system may leverage an SDK including an API that performs a set of sensor data processing workflows on sensor data received from a set of sensors of a wearable haptic interface to obtain structured motion data and inputs the structured motion data that indicates a motion of a user wearing the wearable haptic interface to a haptic application.

[0228] In embodiments, an immersive haptic system may have an SDK including an API that performs a set of sensor data processing workflows on a set of data streams to obtain structured motion data and inputs structured motion data that indicates a motion of a user wearing the wearable haptic interface to a haptic application. The set of data streams may include a low-level motion capture data stream that is indicative of a relative position of specific body parts of a user wearing the wearable haptic interface (and/or a tracked prop) and a tracking data stream that is indicative of an absolute location of the user with respect to the 3D environment of the user. In some of these embodiments, the SDK/API may be configured with a set of modules for determining an intent of the user's motion based on the absolute location and the relative location of the user. Additionally or alternatively, the SDK/API may include a machine-learning component that determines an intent of the user's motion based on the absolute location and the relative location of the user.

[0229] The touch compensator **908** may be configured to detect a self-touch state (e.g., an intent to touch one finger to another finger) and adjust the motion capture data accordingly as mentioned earlier. In some cases, a user wearing the wearable haptic interface **330** may be physically unable to touch two body parts together due to the thickness of the wearable haptic interface **330** and/or movement restrictions imposed by the wearable haptic interface **330**. For example, when wearing a haptic glove **236**, a user's fingertips may be placed inside thimbles that are connected to fingertip tracking sensors. Accordingly, if the user tries to perform a pinch motion by contacting two fingers together, the thimbles on the two fingers may prevent the fingertip sensors from coming closer than a minimum distance depending on the thickness of the thimbles. In this example case, the touch compensator **908** may be configured to detect that the user wishes to pinch two fingertips together even when the fingertip sensors indicate that the fingertips are some distance apart. For example, the touch compensator **908** may detect that two fingertips are approaching each other and subsequently that they stop moving towards each other when they reach a minimum possible distance. The touch compensator **908**, in such a situation, may adjust the motion capture data so that the motion capture data indicates that the two fingers are touching.

[0230] Similarly, the touch compensator **908** may be configured to detect that two hands are touching each other even though hand sensors indicate that the hands are some distance apart (e.g., due to the thickness of the wearable haptic gloves **236**). As another example, the touch compensator **908** may be configured to detect that a hand is touching the user's body even though the motion capture data indicates that the hand is some distance from the body (e.g., due to the thickness of the haptic glove **236** and/or the thickness of other components of a haptic exoskeleton **238**). In embodiments, the touch compensator **908** may leverage a neural network or some other machine learning algorithm to detect a self-touch. The neural network may be trained using data that correlates motion capture data to target data indicating a self-touch state and/or a type of the self-touch state (e.g., a pinch or other thimble collision state, a clap or other hand-to-hand contact state, hand-to-body contact state, etc.). Thus, the trained neural network may be able to detect various types of self-touch states and then apply compensation when they are detected. Additionally or alternatively, non-ML based approaches may be used to detect self-touch (e.g., a non-ML algorithm that monitors distances between certain sets of sensors).

[0231] Accordingly, an immersive haptic system that interfaces with a wearable user interface may have an SDK including an API that performs a set of sensor data processing workflows on a set of data streams to obtain structured motion data and inputs structured motion data that indicates a motion of a user wearing the wearable haptic interface to a haptic application. The set of sensor data processing workflows may include detecting a self-touch state when one or more of the data streams indicates that a first portion of the user is physically unable to contact another portion of the user, and in response to detecting the self-touch state, compensating the structured motion data to indicate that the first portion and the other portion of the user are in contact.

[0232] In embodiments, the immersive haptic system may interface with a wearable haptic interface that includes at least a glove that includes a set of finger thimbles that respectively track motions of a respective finger. In these embodiments, the immersive haptic system may have an SDK including an API that performs a set of sensor data processing workflows on a set of data streams to obtain structured motion data and inputs structured motion data that indicates a motion of a user wearing the wearable haptic interface to a haptic application. The set of sensor data processing workflows may include a thimble compensation that detects a thimble collision state when one or more of the data streams indicates that a first finger of the user is physically unable to contact a second finger of the user do to the physical thickness of the finger thimbles of the wearable haptic interface and in response to detecting the thimble collision state, compensates the structured motion data to indicate that the first portion and the other portion of the user are in contact.

[0233] In embodiments, wearable user interface that interfaces with an immersive haptic system may include at least a first glove and a second glove. In these embodiments, the immersive haptic system may have an SDK including an API that performs a set of sensor data processing workflows on a set of data streams to obtain structured motion data and inputs structured motion data that indicates a motion of a user wearing the wearable haptic interface to a haptic application. The set of sensor data processing workflows may include detecting a hand-to-hand contact state when one or more of the data streams indicates that a first hand of the user is physically unable to contact the other hand of the user due to a thickness of the first and second glove, and in response to detecting the hand-to-hand contact state, compensating the structured motion data to indicate that the first hand and the other hand are in contact at a location corresponding to the hand-to-hand contact state

[0234] In embodiments, a wearable haptic interface may include an exoskeleton with at least one glove. In these embodiments, the immersive haptic system may have an SDK including an API that performs a set of sensor data processing workflows on a set of data streams to obtain structured motion data and inputs structured motion data that indicates a motion of a user wearing the wearable haptic interface to an immersive haptic system. The set of sensor data processing workflows may include detecting a hand-to-body contact state when one or more of the data streams indicates that a hand of the user is physically unable to contact the other hand of the user due to a thickness of the glove and/or a thickness of other components of the exoskeleton at the location where the hand-to-body contact is detected, and in response to detecting the hand-to-hand contact state, compensating the structured motion data to indicate that the first hand and the body of the user are in contact at a location corresponding to the hand-to-body contact state.

[0235] The API **244** may include an interaction assistant **920** that assists in simulating the interactions between an avatar and an environment. Some haptic system applications **250** may have difficulty in providing immersive and realistic interactions between an avatar and one or more objects or entities in an environment for a variety of reasons. For example, a physics engine of the haptic system application **250** may run at a lower than optimal rate due to processing constraints and/or the physics engine may use simplified physics (e.g., simulating an avatar's hand using a simplified rigid body), which may create unrealistic interactions between the user and the environment. The interaction assistant **920** may be configured to compensate for rigid-body assumptions and other limitations of such physics engines. For example, it may simulate physical interactions of the body with elements of the world based on the nature of the interaction, rather than just the geometry. The interaction assistant **920** may use a variety of techniques to improve the simulation of these interactions.

[0236] The interaction assistant **920** may receive world information from the haptic system application **250** to determine when a user avatar is interacting with an object or entity in the environment. In embodiments, the interaction assistant **920** may perform grasp detection by detecting when a user is attempting to grab an object (e.g., when the user's fingers or hands is near the object and the user is applying pressure in the direction of the object). The interaction assistant **920** may thus include a grasp detector algorithm configured to find contact forces that indicate that the user wants to pick up something (e.g., pinch, pick up or otherwise move something). Specifically, the interaction assistant **920** may detect pairs of forces (e.g., from two fingers, two hands, two parts of a hand) or more (e.g., three forces, four forces, etc.) associated with a user avatar that indicates the user wishes to pick up something.

[0237] In embodiments, the interaction assistant may use one or more AI/ML techniques to detect a grasp intent or other interaction. For example, the interaction assistant may leverage a model trained using data that correlates avatar data to data indicating an intended interaction (e.g., a grasp, etc.). Thus, the trained model may be able to detect various types of interactions states and notify the haptic system application **250** when they are detected.

[0238] After detecting a grasp intent, the interaction assistant **920** may then output interaction data **960** indicating to the haptic system application **250** that the object/entity is being grasped by the user. Additionally, the interaction assistant **920** may modify physics parameters in the vicinity of the contact points between the hand and the object/entity in order to secure the object/entity in the avatar's grasp (e.g., by increasing friction or some other motion resistance such as air resistance and/or affixing the object to a contact point of the user's hand until the grasp ends), may deform the hand model for the user avatar in order to simulate how a real hand would slightly deform when pressure is applied to an object, and/or the like. In embodiments, the modifications may include damping the contact by increasing the resistance to the motion of the object/entity being grasped (e.g., so that it is not as inclined to move or slip from the grasp). Thus, by adding artificial contact damping to the object/entity, the effects of the assumptions of the physics engine about rigidity may be mitigated and entities may become easier to pick up and hold within the environment. Accordingly, one or more forces may be applied to influence the object/entity to move in a way that matches the motion of the avatar grasping the object/entity, thus allowing the user's avatar to maintain the object/entity in the grasp.

[0239] In embodiments, contact dampening may be applied to the haptic feedback applied to a user's hand (e.g., via a wearable haptic glove **236**, also referred to herein as a haptic glove **236**) such that the dampened feedback is also felt realistically by the user.

[0240] In embodiments, the modifications performed by the interaction assistant **920** may be based on a set of characteristics of the avatar's hand (which may be customized based on a user's hand) and a set of characteristics of the object/entity (which may have varying hardness/softness, slipperiness, etc.). The interaction assistant may also perform other types of interaction assistance in addition to grasp detection, such as improving the interaction between a user's avatar and a simulated clothing item or accessory worn by the avatar or otherwise attached to the avatar. In these embodiments, the interaction assistant may use a set of characteristics of various parts of the human body (e.g., either default characteristics that correspond to an avatar or custom characteristics specified by a user profile **312**) depending on the interaction.

[0241] In embodiments, the interaction assistant may use an on-board physics engine (discussed elsewhere herein) to improve an interaction simulation. For example, a haptic system application **250** may use rigid physics models, which may not always accurately simulate interactions involving a human hand or other human body parts. In one example, a user may move a haptic glove **236** in a way such that the corresponding avatar's hand would go through an object in the environment. In this case, an on-board physics engine may control the virtual avatar hand using a spring damper constraint (e.g., a virtual spring that controls the position and a virtual damper that controls velocity and instability). Thus, if the user causes the virtual hand to move to a position that is incompatible with the virtual environment, the user may see (e.g., on a VR headset) the virtual hand rest on the surface as determined by the soft-body physics engine. In this case, the physics engine may use a virtual spring force to effectively "pull" the hand back from the incompatible location. In some cases, the on-board physics engine may be a soft-body physics engine that may be capable of deforming meshes (which may include avatar meshes and/or object meshes) to simulate interactions between an avatar and one or more deformable objects/entities within the environment (e.g., depending on a set of properties associated with the objects/entities such as material, stiffness, etc.).

[0242] Accordingly, an immersive haptic system that interfaces with a wearable user interface that includes at least one glove may have an SDK including an API and/or haptic application plugin that receives an output state of an environment being generated by an immersive haptic system and executes a set of workflows to translate the output state into a set of actuation commands for a haptic controller of a haptic interface. The set of workflows may include an interaction detection stage that identifies an occurrence of an interaction between an avatar of a user wearing the haptic interface and an object in the environment and, in response to identifying the occurrence of the interaction, compensates for any rigid-body assumptions made by the immersive haptic system when simulating the interaction between the avatar and the object such that the compensation is based on a set of characteristics of the human body (e.g., based on the avatar, which may be customized based on user dimensions, preferences, etc. as described elsewhere herein) and a set of characteristics of the object. In some of these embodiments, the set of workflows may, in response to identifying that the type of interaction is a grasping interaction, compensate for any rigid-body assumptions made by the immersive haptic system when simulating the grasping of object by the avatar such that the compensation is based on a set of characteristics of the human body and a set of characteristics of the object. In these embodiments, the compensation may include providing a physical constraint to the immersive haptic system instructing the immersive haptic system to influence a location of the object within the environment relative to a location of a hand of the avatar (e.g., fixing the object to the hand or otherwise influencing its location and behavior) for a duration of the grasping interaction. Additionally or alternatively, the set of workflows may, in response to identifying that the type of interaction is a hand contact interaction with an object, compensate for any rigid-body assumptions made by the immersive haptic system when simulating the contact with the object by the avatar such that the compensation is based on a set of characteristics of the human body and a set of characteristics of the object. In these embodiments, the compensation may correspond to a dampening of the feedback force applied by the haptic interface to a hand of the user to account for the non-rigidity of the human hand.

[0243] FIG. **10** shows a more detailed view of example components of a contact interpreter **930**. A body contact detector **1002** may detect contact points between a user avatar and one or more objects/entities within a computer-mediated environment. For example, the body contact detector **1002** may use a ray cast mechanism to determine contact occurrences and locations (e.g., where contact may be based on an actual overlap and/or based on the avatar being within a certain proximity of the objects/entities) and/or may use haptic effects **964** information generated by the haptic system application **250**.

[0244] In example embodiments, the body contact detector **1002** may generate a field of ray traces, where each ray trace originates from a plane attached to the avatar. The plane may be referred to herein as a "sampling plane." In embodiments, the spatial resolution of the ray traces may be adjusted to achieve different levels of fidelity. Each ray trace may extend forward until either contacting another body or reaching a maximum ray trace length. For those ray traces that contact another body (e.g., an object within the environment), the contact interpreter **930** may record the length along that trace between the contour of the body and the contour of the avatar. This distance information may be collected in a structure which may be referred to as a "separation field." The separation field therefore may include a plurality of measurements of the distance between the avatar and a second body (e.g., object/entity). A measurement of zero may indicate that the avatar and the second body are directly contacting one another along that sample. There may be only one shared collision point (e.g., because game engines may natively model virtual objects as rigid).

[0245] In embodiments, a soft-body physics engine **1010** and/or haptic frame generator **1012** may process the body contacts using simulations that provide more accurate and immersive contact simulations than a rigid body physics engine (which may be used by the haptic system application **250**). The soft-body physics engine **1010** may simulate, for example, an interaction between a virtual hand of the user's avatar and an object in the environment using soft body physics, as indicated by world information **962**. The engine **1010** may thus determine how a body part (e.g., a hand) would realistically be deformed by contact with an object and/or how the object would be deformed by contact with the body part and may thus generate more realistic haptic feedback data. In embodiments, the soft-body physics engine **1010** may perform a determination of how an interaction between a body part (e.g., a hand) and an object (e.g., a complex shape like a rock with many nooks and crannies) should be represented (e.g., a particular pressure on the hand). The pressure may increase as a user applies increasing force to the object (e.g., by grasping the rock harder with the user's hand), and the contact area may broaden as the soft-body physics engine **1010** calculates deformations for the avatar's hand. The engine **1010** may output the soft body contact information to the haptic frame generator **1012**.

[0246] In an example embodiment, in order to simulate flexibility of the user avatar, the soft-body physics engine **1010** and/or haptic frame generator **1012** may treat the separation between the objects as a negative measure of how much the two objects would interpenetrate were they both modeled as flexible. For example, the smallest separations would lead to the most interpenetration and the largest separations would lead to the least interpenetration. In embodiments, the avatar may be treated as flexible and the contacting objects as rigid, meaning the interpenetration value may be a measure of avatar surface deflection. Additionally or alternatively, the contact interpreter may perform real-time flexible body simulations.

[0247] In embodiments, the amount of modeled avatar contour deflection normal to a ray trace can be calculated as:

$$d(z) = \begin{cases} \frac{S_{max} - z}{k}, & z < S_{max} \\ 0, & z \geq S_{max} \end{cases}$$

[0248] In this example equation, d is the avatar's deflection at the given sample, z is the ray trace length at the given sample, S_{max} is the largest separation at which the avatar's contour should be deflected, and k is some constant that represents the modeled stiffness of the avatar. In embodiments, if a single actuator corresponds to the calculated sample, the actuator may be instructed to press down a distance d on the user's skin based on the calculation above.

[0249] In many cases, a single actuator may not correspond to a single sample. In these cases, a function may be used that provides virtual separation data and converts it into displacement commands for actuators. Many strategies may be used to accomplish this function. In an example approach, an "influence list" may be built for each actuator that contains all ray trace samples that are geometrically closer to that actuator than to any other actuator. In embodiments, if any actuator is left without an associated sample, the contact interpreter **930** may associate the actuator with the sample closest to it. When calculating the displacement d for an actuator, the haptic frame generator **1012** may supply as the argument the smallest separation of all samples in that actuator's influence list.

[0250] In some embodiments, the contact interpreter **930** may calculate an actuator displacement as a weighted sum of separations, for example. Additionally or alternatively, the contact interpreter may scale the displacement by the magnitude of the virtual reaction force at the interface of the avatar and the second body.

[0251] In embodiments, approaches and techniques for translating contact into haptic actuation patterns may be modified by the use of various techniques, including digital filtering (e.g., using a low pass filter on body contour data of an entity to emphasize large detail, using a high pass filter on body contour data of an entity to emphasize fine detail) and other such techniques. In embodiments, these techniques may be used in different scenarios in order make elements of an object's virtual geometry more noticeable to a user. For example, these filters might be configured on a virtual-object-by-virtual-object basis. Additionally or alternatively, different filters may be used for the sake of enhancing perception for the user. For example, a user with some nerve damage may have greater difficulty detecting fine-scale features such as surface texture. Regardless of any filter associated with a virtual object being contacted, an additional high-pass filter could be used to further accentuate all details from all interactions. This could aid the hard-of-touch user to feel objects in greater fidelity than the user would be able to otherwise.

[0252] In embodiments, contacts detected by the body contact detector **1002** may also be processed by a thermal interpreter **1006**, which may use one or more thermal flux models **1008** to determine the thermal flux resulting from the contact. Additionally or alternatively, the thermal interpreter **1006** may determine an amount of thermal flux based on soft-body contact information generated by the soft-body physics engine **1010**. The thermal interpreter **1006** may determine, for each point or area of contact, how much thermal flux should be applied as haptic feedback and output this information to the haptic frame generator **1012**. The thermal interpreter **1006** may also determine that thermal feedback should be applied based on other interactions or world information **962**, such as the user avatar standing in a hot location and/or near a hot item in the environment or the sun shining on the user avatar in the environment, etc. The thermal interpreters **1006** may determine an amount of thermal flux using the thermal flux model **1008** based on thermal data associated with the user avatar, thermal data associated with an object in contact with the user avatar, and/or thermal data of other objects/entities in the environment (where thermal data may include temperature, material, or other properties that may affect thermal flux). The thermal haptic information generated by the thermal interpreter **1006** may later be used to actuate thermal actuators and/or microfluidics actuators with temperature-variable fluid, as discussed elsewhere herein.

[0253] In embodiments, a haptic interpreter **1004** may determine other types of contact that may not be accurately simulated by a physics simulation. For example, the haptic interpreter **1004** may determine that haptic feedback should be applied based on the presence of wind in the environment. As another example, the haptic interpreter **1004** may determine that haptic feedback simulating a vibration should applied based on a user avatar being nearby or in contact with a machine (e.g., the avatar may be holding a powered drill or sitting in a vehicle with a running motor). The haptic interpreter **1004** may detect these situations and determine areas of the body that would be affected (e.g., a hand and arm when a user holds a powered drill). The haptic interpreter **1004** may output the haptic feedback data to a haptic frame generator **1012**. A haptic frame generator **1012** may process, combine, and output the various data regarding haptic feedback to generate a haptic frame **966**, as described earlier.

[0254] Accordingly, a wearable human interface may have a set of sensors and an on-board processing system that is configured to execute an API and a physics engine. The API may be configured to process sensor data to obtain motion capture data indicating motion of a user wearing the wearable human interface. Furthermore, the physics engine may be configured to generate an environment that includes an avatar of the user that is controlled within the environment by the physics engine based on the motion capture data.

[0255] In embodiments, an immersive haptic system may be configured to simulate an environment including an avatar of a user of a wearable haptic interface. The immersive haptic system may execute a soft-body simulation of the environment and may determine an output state of the environment with respect to the avatar based on interactions between the avatar and the environment. Furthermore, a contact interpreter may monitor the output state of the environment to identify contact events between the avatar and an object in the environment and, in response to identifying the contact event, may determine a corresponding haptic feedback response based on the contact event, wherein the haptic feedback response is transformed into a set of actuation commands that are provided to the wearable haptic interface.

[0256] In embodiments, an immersive haptic system may have an immersive haptic system that is configured to generate an environment including an avatar of a user of a wearable haptic interface. The immersive haptic system may handle the physics-based calculations for the environment and may determine an output state of the environment with respect to the avatar based on interactions between the avatar and the environment. Furthermore, a haptic interpreter may maintain a set of physical characteristics of the environment and objects contained therein and may apply one or more haptic augmentations to the output state based on the set of physical characteristics and the interactions between the avatar and the environment.

[0257] In embodiments, an immersive haptic system may have an immersive haptic system that is configured to simulate an environment including thermal characteristics of the environment/objects in the environment and simulate an avatar of a user of a wearable haptic interface. The immersive haptic system may handle the physics-based calculations for the environment and may determine an output state of the environment with respect to the avatar based on interactions between the avatar and the environment. The output state may include one or more thermal states of the environment, of objects in the environment, and/or of the avatar. Furthermore, a haptic interpreter may maintain a set of thermal characteristics of the environment and of objects contained therein and may apply one or more thermal augmentations to the output state based on the set of physical characteristics and the interactions between the avatar and the environment. The thermal augmentations may be used to determine a thermal feedback response that is used to actuate one or more thermal actuators.

[0258] FIG. **11** illustrates an example method for implementing motion capture based on sensor data received from haptic interface devices. For example, the method of FIG. **11** may allow one or more haptic interface devices to be used as input to control an avatar in a virtual environment and/or a telerobot. The functions shown in FIG. **11** may be implemented by an application programming interface (API) that receives information from a wearable haptic interface **330**, processes the information, and provides the processed information to a haptic system application **250**. In other words, the API may provide a computer-mediated interface between a haptic interface device and a virtual environment and/or telerobot. In embodiments, the various functions of FIG. **11** may be implemented by an SDK that may integrate with a haptic system application **250** (e.g., a game engine SDK or plugin for a game application). Additionally or alternatively, the method may be implemented by the application **250** itself (e.g., as a module or plugin of the application **250**). Additionally or alternatively, the method may be implemented by a device onboard the wearable haptic interface **330** and/or components thereof. In the descriptions below, the method steps are performed by a motion capture system **314** of the API **244**, which may execute on a device that runs the application **250** and/or a separate device. The motion capture system may comprise one or more sensor data processors **902**, motion processors **904**, a slip compensator **906**, a touch compensator **908**, and/or a kinematic frame generator **910**, described elsewhere herein. However, it should be understood that the method and the functions thereof may be implemented at various hardware devices and/or sub-systems thereof.

[0259] At **1102**, the motion capture system receives sensor data and/or location data from the one or more haptic interface devices of the wearable haptic interface **330**. For example, the haptic interface devices may comprise an exoskeleton, one or more haptic gloves, and/or any other haptic interface devices. The sensor data may comprise raw sensor data from various types of sensors that may be arranged on the haptic interface, such as accelerometers, gyroscopes, magnetometers, and/or positional trackers. For example, a haptic glove may have a sensor arranged on each fingertip of the glove, on a palm of the glove, on various finger joints, etc. Similarly, sensors may be arranged at various locations on an exoskeleton or other haptic interface device. Accordingly, the sensor data may include various raw sensor data streams corresponding to a plurality of different sensors. Additionally or alternatively, the wearable haptic interface **330** may comprise motion trackers **340** that may perform sensor processing and output absolute location data (e.g., location and orientation within a 3d space around the user) using various techniques (e.g., visual tracking based on a signal emitted by a lighthouse or base station, inside-out visual tracking, etc.). Motion trackers may be arranged at various anatomical locations of the haptic interface devices (e.g., one tracker on each hand of respective gloves, one tracker on a torso of an exoskeleton, etc.). Thus, the motion capture system may receive both raw sensor data as well as absolute location data output by various trackers and/or associated lighthouses/base stations. In embodiments, the location data may be received by the motion capture system as ongoing data streams including a plurality of time-specific measurements. Thus, the data streams may, over time, indicate whether the user is moving or not. Additionally or alternatively, some of the sensors (e.g., inertial measurement units with gyroscopes) may estimate velocity directly, and thus the data streams may further include motion/velocity data.

[0260] At **1104**, the motion capture system may process any raw sensor data using various data processing techniques to generate detailed relative location data for one or more body parts, such as fingertips, finger joints, palms, one or more locations on the arms and legs, torso, head, etc. The sensor data processing may involve filtering, normalization, and/or the like to remove the influences of noise or interference, generate location data in a usable format, etc. The sensor data processing may further involve various calculations to determine relative locations depending on the type of raw sensor data (e.g., dead reckoning, trilateration, triangulation, inertial navigation, optical tracking, magnetic tracking, time of arrival or time of distance arrival measurement, and/or the like). In embodiments, the sensor processing may involve sensor fusion to determine location data based on sensor data from multiple sensors. Additionally or alternatively, the sensor processing may involve compensation for sensor slip.

[0261] In embodiments, the motion capture system may generate relative location data **953** that is indicative of a relative position and/or orientation with respect to some common reference point. For example, the motion capture system may generate location data including positions and orientations of each sensor of a haptic glove relative to a single common reference point, such as a metacarpophalangeal joint of a middle finger (which may be referred to as "MCP 3") for the hand. The motion capture system may apply a similar technique to calculate relative positions for other body parts with respect to other common points and/or to a single common point for all body parts. In other words, each tracked location may be relative to a single reference point or to different reference points that may be used for different purposes (e.g., hand tracking, body tracking, etc.).

[0262] At **1106**, the motion capture system may perform motion processing on the absolute and relative location data to integrate the location data into a single coordinate system and store the data as a motion capture data frame **956**. The motion capture frames may be time-based, such that the motion capture system continually generates a series of motion capture frames, where each motion capture frame corresponds to a particular timestamp or period of time. Each motion capture frame may include absolute and/or relative location data for various body parts corresponding to the haptic interface devices, including an absolute position and orientation of a reference point and/or a plurality of relative position and orientation data for particular key points with respect to the reference point (e.g., position offsets) and a particular time. As an example using a haptic glove, the motion capture system may generate a motion capture frame that indicates all of the tracked positions of one hand at a particular time calculated as relative positions with respect to a selected common point such as MCP 3 for that hand. A single motion capture frame may correspond to a single body part at a given time (e.g., a first motion capture frame for a left glove and a second motion capture frame for a right glove, with additional frames for legs, torso etc.). Additionally or alternatively, a motion capture frame may include motion capture data for multiple tracked body parts. The motion capture system may continuously generate a series of motion capture frames over time, where each motion capture frame corresponds to a particular timestamp and/or time period, in order to provide a series of frames that can be used to control an avatar/telerobot over time.

[0263] At **1108**, the motion capture system may compensate for interface slip (e.g., glove slip) by adjusting a generated motion capture frame based on various factors that may indicate slip. For example, a glove slip may occur when the user's hands move within a glove but the glove does not move or vice versa (e.g., the glove slips with respect to the hand). Similar interface slips may occur with respect to an exoskeleton (e.g., torso slip). The motion capture system may detect glove slip using techniques such as observing inconsistencies in motion capture frames over time (e.g., if a sensor on the back of the hand moves but the fingers do not move, a glove slip may be detected). Additionally or alternatively, the motion capture system may detect glove slip using redundant sensors or other techniques. Based on the detected glove slip, the motion capture system may adjust relative position and/or orientation data of a motion capture frame to correct for the detected slip (e.g., depending on where the slip is detected). In embodiments, the motion capture system may take user profile data into account when detecting and compensating for glove slip. For example, the motion capture system may perform glove slip compensation based on data indicating the size of the user's hand, the length of the user's fingers, and/or other factors that may affect glove slip.

[0264] At **1110**, the motion capture system may compensate for touch intents based on the haptic interfaces. For example, as described above, fingertip thimbles of a haptic glove may make it impossible for a user's fingers to physically touch when making a "pinch" motion. In this example, when the measured position of two of a user's fingertips come within a threshold distance of each other (e.g., a minimum distance that may be defined based on the size of fingertip thimbles), the relative position offsets for the two fingertips may be adjusted to indicate that they are touching (e.g., such that an avatar's fingers will touch as intended by the user). Similar touch compensations may be performed for hand-to-hand touch intents, hand to body intents, leg to leg touch intents, and/or the like. In each example, position data may be adjusted to cause a touch if two contact points are within a certain distance of each other (e.g., which may indicate a touch intent), where the distance may vary depending on various features and/or dimensions of the haptic interface (e.g., because a glove interface may be more or less thick than a leg interface for an exoskeleton).

[0265] The motion capture system may also detect other intents and cause performance of other actions (e.g., either instead of or in addition to touch compensation) in response to the detected intents. For example, certain predefined gestures (e.g., a pinch or other movement or one or more fingers) may be interpreted as a command that may cause an arbitrary action within the application. In these embodiments, the motion capture system may detect the predefined gesture and store a separate indication of the detected gesture and/or a command associated with the gesture (e.g., within the motion capture frame and/or kinematic frames provided to the application). In embodiments, the command associated with the gesture may be user-configurable (e.g., the user profile may indicate which gesture is associated with which command). In embodiments, the motion capture system may use AI and/or machine learning techniques (e.g., trained models) to detect gestures based on any of the data received by and/or generated by the motion capture system (e.g., the location data and/or raw sensor data, relative position and orientation data, motion capture frames, etc.).

[0266] At **1112**, the motion capture system may generate a series of time-based kinematic frames based on the series of time-based motion capture frames (with any relevant compensations applied), where the kinematic frames may be fed into an application to control the movement of an avatar and/or telerobot. The motion capture system may use the motion capture frame data to generate a model including an intended position and orientation of each body part that corresponds to a haptic interface device based on the user movements and positioning. For example, if the wearable haptic interface **330** comprises haptic gloves **500**, the kinematic frames may include corresponding hand models. If the wearable haptic interface **330** includes an exoskeleton **600**, the kinematic frames may include a full body model. In embodiments, the kinematic frame may further indicate any detected gestures and/or commands associated with the detected gesture.

[0267] In embodiments, the motion capture system may optimize a kinematic frame to account for any differences in a model based on the user's inputs and a model of the avatar and/or telerobot. For example, the avatar and/or telerobot may have a different hand size, different finger lengths, limb size, height, and/or the like. In these embodiments, the motion capture system may perform optimization to reconcile the differences between the models/dimensions, such as by mapping (e.g., interpolating) from the user model to the avatar and/or telerobot model. In embodiments, the mapping may be limited by one or more animation capabilities of the avatar model (e.g., extension and/or flexion limits). In embodiments, an avatar and/or telerobot may have a different morphology (e.g., different number of fingers, different number and/or type of limbs, etc.). As an example, a user may be controlling a telerobot that is a surgical instrument, robotic arm, etc. In these embodiments, the user's movements may be mapped to the telerobot using various strategies. For example, the avatar may mirror the telerobot such that the kinematic frame generator maps between the morphology of the user and the morphology of the telerobot (e.g., such that certain user motions cause certain motions of the avatar and telerobot).

[0268] Alternatively, the avatar may share a morphology of the user even when the telerobot does not. For example, an application may generate a human avatar that allows a user to simulate moving around within a haptic digital twin and perform various actions for controlling a surgical instrument telerobot. As a more illustrative example, a haptic digital twin may allow a surgeon to control a human avatar to move around within a haptic digital twin of a real body undergoing surgery, cause the avatar to touch certain tissue within the haptic digital twin to cause the surgical telerobot to touch the tissue, perform certain actions within the haptic digital twin to cause the surgical telerobot to perform corresponding actions, and/or the like. In these embodiments, because the computer-mediated haptic digital twin may change a scale of the user with respect to the avatar, for example, the surgeon may be able to simulate moving around within an organ, blood vessel, or other small space via the haptic digital twin, may very finely control motion and/or operation of the telerobot at a very small scale, etc. Although this example is illustrative, many other examples of using haptic digital twins to control telerobots with different scales and/or morphologies are enabled by the motion capture system. As another brief example, a user avatar may be "scaled up" in a haptic digital twin to allow the user to interact with a large construction site, control heavy equipment with various morphologies by moving the avatar, and/or the like. Accordingly, it should be understood that a haptic digital twin may generate a computer-mediated environment that allows a user to interact with a digital twin of a real environment using avatars that do not exist in the real world.

[0269] In embodiments, an avatar model may be generated and/or adjusted based on user profile data. For example, a user profile may indicate an actual or preferred height, size, etc. of an avatar for the user, including various user-specified dimensions for constructing the avatar. In these embodiments, the motion capture system may generate a kinematic frame using the avatar model by mapping a user model generated based on motion capture data to the avatar model. In some embodiments, as described above, this process may involve the use of ML models that may assist in mapping the user model to the avatar model. In embodiments, the mapping may involve animation retargeting or other approaches.

[0270] At **1114**, the motion capture system may transmit the kinematic frame(s) to the application for use in adjusting the avatar/telerobot. The motion capture system may send a series of kinematic frames as they are generated so that the application may continuously update the avatar/telerobot. In embodiments, the motion capture system may communicate with a plugin/module that is integrated with the application, for example by calling an update function provided by the plugin/module. In embodiments, an SDK may provide the integration for the application.

[0271] FIG. **12** illustrates an example method for implementing haptic feedback based on world information and/or haptic effects data that is received from an application. The functions shown in FIG. **12** may be implemented by the same API that implements the method of FIG. **11**. As described above, the API may provide a computer-mediated interface between a haptic interface device and a virtual environment and/or telerobot. In embodiments, the various functions of FIG. **12** may be implemented by an SDK that may integrate with a haptic system application **250** (e.g., a game engine SDK or plugin for a game application). Additionally or alternatively, the method may be implemented by the application **250** itself (e.g., as a module or plugin of the application **250**). Additionally or alternatively, the method may be implemented by a device onboard the wearable haptic interface **330** and/or components thereof. In the descriptions below, the method steps are performed by a haptic feedback controller **316** of the API **244**, which may execute on a device that runs the application **250** and/or a separate device. The haptic feedback controller may comprise a contact interpreter **930** and/or components thereof, an actuation calculator **932**, an actuation controller **936**, and/or other components as described elsewhere herein. It should be understood that the method and the functions thereof may be implemented at various hardware devices and/or subsystems thereof.

[0272] The haptic feedback method of FIG. **12** may be used to provide different types of haptic feedback depending on the haptic actuators present within the one or more haptic interface devices. For example, as described herein, a haptic glove may include a plurality of microfluidic tactile actuators and resistive brakes that provide force feedback to a user's fingers. Accordingly, the haptic feedback method may generate commands for controlling the microfluidic tactile actuators and/or activating one or more of the resistive brakes. Additionally or alternatively, an exoskeleton may include active force feedback actuators that may resist user movement and/or actively reposition a user. Accordingly, the haptic feedback method may generate tactile actuation commands, braking commands, and/or active force feedback commands for controlling one or more actuators of the exoskeleton.

[0273] At **1202**, the haptic feedback controller may receive data that is relevant to haptic feedback from an application (e.g., a game, a simulation, and/or a haptic digital twin for a telerobot). The haptic feedback controller may continuously receive haptic feedback data in order to provide continuous haptic feedback. In other words, the method of FIG. **12** may act as a pipeline, with new data continuously being received at step **1202** and processed according to the described method. The haptic data may include world info data (e.g., data about an avatar and/or objects near the avatar within a virtual environment and/or haptic digital twin) and/or haptic effects data, which may be used together to provide detailed information about the haptic state of a virtual environment generated by the application. For example, world info data may include attributes of virtual objects within the environment (e.g., within a threshold distance of a user's avatar). Each virtual object may be associated with certain physical characteristics, which may or may be not represented by the application (e.g., a plugin to the application may store and maintain the attributes, which may be used specifically for haptic feedback). These attributes may describe various characteristics of the object that may be used for generating haptic feedback, such as a compliance or softness of the object, a texture of the object, a weight of the object, etc., along with a unique ID number for lookup purposes. The world info data may further include avatar data describing, for example, objects associated with the avatar, such as fingers of the avatar, a hand of the avatar, etc.

[0274] The haptic effects data may include information related to specific haptic interactions within a computer-mediated environment. The haptic feedback data may specify object effects, such as haptic actuation patterns for controlling microfluidic tactile actuators when a user avatar contacts an object (e.g., a first waveform that mimics the feeling of holding a drill that is turned on, a second waveform that mimics other haptic patterns, etc., where the object effect may specify a preset and/or arbitrary waveform). Additionally or alternatively, the haptic feedback data may specify spatial effects that may be activated when the avatar enters a certain control volume (e.g., a waveform that simulates wind from a fan when a user is near a fan, etc.). Additionally or alternatively, the haptic feedback data may specify direct effects that indicate direct pass-through command(s) to a haptic interface (e.g., which may indicate various patterns of activations of haptic feedback actuators, such as tactile actuators).

[0275] At **1204**, the haptic feedback controller may determine contact information based on the world info data and/or the haptic effects data. For example, the haptic feedback controller may interpret the state of an environment (and/or a haptic digital twin that represents the current state of a telerobot and/or the surrounding environment) and determine any contact information for use in generating haptic feedback. The haptic feedback controller may generate the contact information using several techniques (including any of the techniques described elsewhere herein with respect to the contact interpreter and/or interaction assistant).

[0276] In embodiments, the haptic feedback controller may determine contact information based on world info **962** that indicates contact between a user avatar and one or more virtual objects within a virtual environment including the user avatar. Additionally or alternatively, when controlling a haptic digital twin for a telerobot, the haptic feedback controller may determine contact information based on world info **962** for various objects within the haptic digital twin, where the objects may be real objects detected by sensors of the telerobot. The haptic feedback controller may determine the contact information (e.g., points of contact and attributes thereof) using various techniques. For example, as described elsewhere herein, the haptic feedback controller (e.g., using a contact interpreter **930**) may use the world info to construct a local environment for the avatar (e.g., including the avatar and objects near the avatar within the virtual environment and/or haptic digital twin). Within the local environment, the haptic feedback controller may generate a field of ray traces that originate from a sampling plane associated with the avatar and thereby determine a separation field that measures the distance of variants points on the avatar to one or more objects. Additionally or alternatively, instead of or in addition to using a sampling plane, the haptic feedback controller may generate ray traces from each point of an avatar model that corresponds to a tactile actuator on a corresponding haptic interface device. For example, if a haptic glove has 20 tactile actuators in a particular arrangement on a palm of the glove, then the contact interpreter **930** may generate 20 ray traces on corresponding points of the avatar's palm to determine a distance from each point to a virtual object in order to generate contact information.

[0277] In embodiments, the contact interpreter may generate contact information that takes into account one or more haptic attributes of the virtual objects and/or haptic digital twin objects as specified by the world info **962** (e.g., attributes that indicate a compliance or softness of the object, a texture of the object, a weight of the object, etc.). In embodiments, the haptic feedback controller may also generate contact information that indicates an object's temperature, a thermal flux between the object and the avatar (which may depend on distance, a virtual material of the object, a temperature difference between the object and a temperature associated with the avatar, etc.). In these embodiments, the object properties may be generated by a virtual environment (e.g., game or simulation), sensed by a telerobot, predicted using machine learning techniques, and/or the like.

[0278] In embodiments, in some cases the haptic feedback controller may generate updated object properties based on the contact and provide the updated object properties back to the application **250**. For example, if a user avatar is contacting a "cold" object associated with a low temperature, the temperature of the item may be warmed through the contact with the avatar. In some cases, the application **250** may handle updates to the item properties on its own. Additionally or alternatively, the haptic feedback controller may update the properties and transmit the updated properties to the application.

[0279] At **1206**, based on the various contact information generated based on world info **962** and/or based on various haptic effects **964**, the haptic feedback controller may calculate an amount of deformation to be applied as haptic feedback at particular locations for a particular time and compile such information into a haptic frame, as well as any force feedback that may be applied to force feedback actuators. The amount of deformation may be measured using a distance metric (e.g., in millimeters) that is hardware-agnostic (e.g., it may not comprise specific units of pressure or instructions to provide the specified amount of pressure to a specific actuator). Thus, for example, the haptic frame may specify an ideal amount and/or type of haptic feedback that may be later translated into a specific set of actuation commands that may be used to simulate the ideal haptic feedback.

[0280] In embodiments, the haptic frame may specify deformations for the one or more points associated with the various ray traces used to determine contact information. The haptic feedback controller may determine the amount of deformation based on a distance to a virtual object (e.g., as indicated by the length of the ray trace), where the distance may be a negative distance (e.g., indicating that the object is in contact with the avatar and thus should be deforming the avatar). The haptic feedback controller may further determine the amount of deformation based on the attributes indicating, for example, a stiffness or softness of the virtual object, texture of the virtual object, etc. (e.g., where contact a more compliant virtual object may produce less deformation, etc.). In embodiments, the deformations may be calculated using any of the techniques described with respect to the contact interpreter **930** and/or interaction assistant **920** (e.g., soft-body deformation, use of an on-board physics engine, etc.).

[0281] In embodiments, the haptic feedback controller may modify the deformation amounts based on one or more haptic effects. As discussed above, some of the haptic effects may be associated with time-varying waveforms, such as sine waves, square waves, etc. In these examples, the haptic feedback controller may add or subtract different amounts of deformation to different time-based haptic frames in a series of haptic frames as indicated by the waveforms. For example, to simulate a square wave (e.g., which may simulate the feeling of holding a drill that is turned on), the haptic feedback controller may increase the amount of deformation applied to certain haptic feedback points for a first period of time, then reduce the amount of deformation applies to the haptic feedback points for a second period, increase the deformation for a third period, and so on.

[0282] In some examples, certain haptic feedback points are not in contact with any virtual objects, but are associated with a spatial effect (e.g., a "wind" spatial effect). In these examples, the haptic feedback controller may use the waveform associated with the spatial effect to determine a displacement distance for each of the points. In some examples, certain haptic feedback points are associated with deformations (e.g., due to simulated contact with a virtual objects) as well as an object effect (e.g., as in the example where the avatar's hand is holding a virtual drill object that is turned on). In these examples, the haptic feedback controller may use the waveform associated with the object effect to temporarily increase the displacement distance that simulates the contact to further simulate the object effect. In other words, the displacement that simulates an object effect may be added to the displacement that simulates contact (at least for some of the time-based haptic frames).

[0283] In some cases, a displacement that simulates a contact for a particular point may already be "maxed out" (e.g., the displacement is beyond a threshold distance that would cause a maximum haptic feedback to be applied). In these cases, to further simulate a haptic effect, the displacement that simulates the contact may be reduced prior to adding the displacement that simulates the object effect. In other words, the contact displacement may be reduced enough so that it can be temporarily increased and then decreased following an object effect waveform. Other similar strategies may be used; thus, the contact displacement and the haptic effects displacement may be combined using calculations other than simple addition (e.g., the combination may be akin to using a compression filter).

[0284] In embodiments, the haptic frame may further include an indication of thermal flux for any or all of the contact points. Thus, the haptic feedback controller may generate a haptic frame that includes, for each contact point, a displacement and/or a thermal flux.

[0285] In embodiments, the haptic feedback controller may calculate brake and/or other force feedback behavior based on contact information and/or object attributes, and add the brake and/or other force feedback behavior to the haptic frame. For example, if an avatar's hand is holding a virtual object associated with a high stiffness attribute (e.g., a billiards ball), the haptic feedback controller may generate a value indicating a maximum braking resistance for each finger in contact with the object (e.g., to prevent the user from further flexing their fingers and thereby simulate the feel of holding a billiards ball). As another example, if an avatar's hand is holding a virtual object associated with a high compliance attribute (e.g., a plush toy), the haptic feedback controller may generate a value indicating a lesser amount of braking resistance to accurately simulate the feel of holding the object. In some cases, the haptic feedback controller may generate active force feedback based on object properties (e.g., to simulate the active opposing force vector created by applying force to a spring or rubber ball, for example) and/or the movement of virtual objects into the user (e.g., if a virtual object collides with the avatar or otherwise applies force to the avatar). As discussed elsewhere herein, the haptic feedback controller may limit the force feedback based on various user preferences, safety and governance standards, and/or using other techniques to prevent discomfort or injury to the user of the wearable haptic interface. Furthermore, although some of the examples described above refer to haptic feedback to actuators (e.g., tactile actuators and/or brakes) of a haptic glove, the haptic frame may specify ideal haptic feedback for any part of a user's body associated with a haptic interface (e.g., hands, limbs, torso, etc.).

[0286] At **1208**, the haptic feedback controller may translate the haptic frame into a hardware-specific actuator frame that indicates the type of feedback that may be applied based on the actuator configuration of the haptic interface units. For example, if the haptic interface unit comprises a number of microfluidic tactile actuators that can be pressurized within a minimum and maximum range, the haptic feedback controller may generate an actuator frame specifying the amount of pressure applied to each actuator based on a corresponding displacement distance of the haptic frame. Additionally or alternatively, if the microfluidics tactile actuators use varying-temperature fluids to provide thermal feedback, the haptic feedback controller may generate an indicator of a mixture amount of various temperature fluids (e.g., hot, cold, and/or room-temperature fluids) to simulate a particular thermal flux. Additionally or alternatively, if the haptic interface device includes brakes and/or other force feedback actuators, the haptic feedback controller may generate amounts of braking (e.g., a percent resistance) and/or other force feedback (e.g., a force feedback vector and/or offset specifying a direction and/or location that may be used to apply force feedback). The haptic feedback controller may use a "peripheral" database that models the behavior of each haptic interface device (e.g., a left glove, a right glove, other components of an exoskeleton, etc.) for this purpose. For example, the database may specify, for each haptic interface device, the number and type of actuators, the minimum and maximum amount of pressure that may be applied to each actuator, an amount of thermal flux that may be simulated using the actuator, amounts of force feedback available, and/or the like.

[0287] At **1210**, the haptic feedback controller may generate individual actuator commands for specific actuators based on the actuator frame and, at **1212**, transmit the actuator commands to actuator controls (e.g., a microfluidics control unit that provide microfluid to a plurality of tactile actuators, a controller for each actuator, etc.). The haptic feedback controller may continuously transmit commands based on time-based haptic frames and corresponding actuator frames as they are generated, thereby providing continuous haptic feedback that simulates interactions with virtual objects.

[0288] FIG. **13** illustrates an example method for implementing interaction detection and assistance feedback based on avatar and/or world information that is received from an application. For example, the method may detect a grasp intent and implement grasp assistance (e.g., by adjusting one or more physics properties of a computer-mediated environment to allow the user's avatar to more easily maintain a grasp of a virtual object with an environment). Other such interaction assistance may include assistance in engaging with a virtual user interface that exists with a simulated environment and/or the like. The functions shown in FIG. **13** may be implemented by the same API that implements the method of FIGS. **11-12**. As described above, the API may provide a computer-mediated interface between a haptic interface device and a virtual environment and/or telerobot. In embodiments, the various functions of FIG. **13** may be implemented by an SDK that may integrate with a haptic system application **250** (e.g., a game engine SDK or plugin for a game application). Additionally or alternatively, the method may be implemented by the application **250** itself (e.g., as a module or plugin of the application **250**). Additionally or alternatively, the method may be implemented by a device onboard the wearable haptic interface **330** and/or components thereof. In the descriptions below, the method steps are performed by an interaction assistant of the API **244**, which may execute on a device that runs the application **250** and/or a separate device. It should be understood that the method and the functions thereof may be implemented at various hardware devices and/or sub-systems thereof.

[0289] At **1302**, the interaction assistant may receive world information **962**, which may comprise data that is indicative of the structure, location, and/or other properties of objects within an environment generated by an application **250**. The world information **962** may further include information about an avatar within the environment (e.g., a position and/or orientation of the avatar). Additionally or alternatively, the interaction assistant may receive information about the avatar in the form of one or more kinematic frames generated by a kinematic frame generator **910**, as described elsewhere herein. For example, the kinematic frames may indicate one or more positions and/or orientations associated with an avatar.

[0290] At **1304**, the interaction assistant may determine an avatar interaction intention with one or more of the virtual objects as indicated by the world information **962**. For example, if the avatar information indicates that at least two fingers of the avatar are moving inwards towards a virtual object in between the fingers (e.g., as in a grasp) and/or contacting the virtual object, the interaction assistant may detect a grasp intent. Other similar interactions, such as contacting objects at one or more locations, pushing virtual buttons, turning virtual knobs or controls, throwing virtual objects, or performing other movements in relation to virtual objects and/or contact with the virtual objects may be detected by the interaction assistant in similar ways (e.g., based on the movement of the avatar with respect to the virtual objects and/or contacts with the virtual objects). In embodiments, as described above, one or more AI-assisted and/or machine learning models may be used to detect one or more interaction intents, such as a grasp intent.

[0291] At **1306**, the interaction assistant may determine interaction data **960** comprising one or more instructions for modifying the environment generated by the application in order to assist the detected interaction. For example, for a grasp interaction, as described elsewhere herein, the interaction assistant may include contact dampening, increasing friction or similar physics parameters to make the grasp easier to maintain, and/or affixing the grasped object to the avatar. For other similar interactions, other physics parameters and/or environment parameters may be modified in appropriate ways.

[0292] In embodiments, the instructions may include instructions for modifying the positioning of an avatar, such as using a spring force to pull a part of an avatar (e.g., a hand) into a position that is compatible with the virtual environment (e.g., moving the hand so that it is resting on the surface of a wall or other surface that the hand would otherwise intersect with). In these embodiments, the interaction assistant may output avatar configuration instructions to a kinematic frame generator **910** and/or a directly to the application **250** to cause modification of the positioning of the avatar.

[0293] At **1308**, the interaction assistant may output the interaction assistance information to the application. For example, the interaction assistant may transmit instruction(s) as interaction data **960** to the application **250**, which may cause the application **250** to modify the virtual environment to assist the interaction (e.g., via contact dampening, adjustment of other physics properties, attachment of virtual objects to the avatar, movement of virtual objects, movement of parts of the avatar, and/or the like).

[0294] The immersive haptic system **302** and/or API **244** may have additional features and functionalities. For example, in embodiments the immersive haptic system **302** and/or API **244** may be configured to identify and/or predict hazard characteristics implied by interactions within an environment that contains an avatar of the user of a wearable haptic interface. This functionality may be used for governance, training, and/or injury avoidance associated with wearable device limitations. In these embodiments, the immersive haptic system may handle the physics-based calculations for the environment and may determine one or more predefined hazard output states, wherein the output state includes one or more predetermined haptic hazard warnings that are translated to the user.

[0295] Additionally or alternatively, the immersive haptic system and/or API may be specifically configured to simulate one or more medical environments that contain an avatar of the user of a wearable haptic interface and soft objects representing the subject of a medical procedure. For example, a training environment may be configured to simulate a surgery for training a surgeon. In this example use case, the training environment may be configured to include haptics that are designed to vary by tissue type, position, and/or the like, and that may be based on digital data such as CT, MRI, or other scans. Moreover, the environment may provide real-time feedback for actual procedures. Additionally or alternatively, the environment may use machine learning and/or expert training to improve various models. In these embodiments, the immersive haptic system/API may handle the physics-based calculations for the environment and may determine one or more predefined haptic output states that are translated to the user.

[0296] Additionally or alternatively, the immersive haptic system/API may be specifically configured to identify and/or predict wearable user health characteristics implied by avatar interactions within an environment. Identified and/or analyzed aspects of the user's health may include a range of motion detected based on user movements, an amount of force that the user is capable of generating with various limbs/muscle groups (e.g., in short bursts or over time), an overall fitness of the user based on an amount of activity over a period of time, and/or the like. In these embodiments, the immersive haptic system/API may handle the physics-based calculations for the environment and may determine one or more predefined health notification output states. The output states may include one or more predetermined outputs such as haptic health warnings, automated haptic parameter adjustments, and the like.

[0297] Additionally or alternatively, the immersive haptic system and/or API may be specifically configured to identify soft objects within a computer-mediated environment. In these embodiments, the immersive haptic system/API may handle the physics-based calculations for the environment and object and may determine one or more predefined haptic outputs associated with that object that are conveyed to the wearable user.

[0298] Returning to FIG. **1**, in embodiments, the operation of the haptic system application, SDK **242** and/or API **244**, wearable haptic interface **330**, and other components of haptic interface system **230** may be monitored by the haptic operating platform **160**. The haptic operating platform **160** may monitor **222** and analyze **224** the data received from the haptic interface systems **230** using one or more configured intelligence services **204**, as discussed elsewhere herein. In embodiments, the analytics may involve any of the data described herein, including motion capture data, interaction data, and/or haptic feedback data generated by the API **244**, in various contexts including training environments and scenarios, gaming, telerobotics, and other contexts described herein. The analytics may involve governance, standards, and/or other approved evaluation methods.

[0299] As one example of monitoring and analytics, the haptic operating platform **160** may be configured to score a user's performance in a training task, such as a user's performance in a medical simulation, a piloting/driving simulation, and/or the like. Additionally or alternatively, the scoring tasks may be carried out by the haptic system application **250** and/or some other component of an immersive haptic system **302**.

[0300] As another example of monitoring and analytics, the haptic operating platform **160** may be configured to gather user data for a user of the wearable haptic interface **330** or other user devices described herein to analyze and optimize haptic parameter settings for the user. Thus, for example, any data stored in a user profile **312** may be determined either explicitly or implicitly by the haptic operating platform **160** based on monitoring and analyzing a user's interactions with the haptic systems described herein. In embodiments, a predefined set of environments, scenarios, and associated haptic cues that are designed to gather wearable user data may be used to obtain the necessary data for explicitly or implicitly defining a user profile **312**.

[0301] Accordingly, an immersive haptic system may have an analytics system designed to analyze, report, and visualize the results of haptics-based training environments and scenarios.

[0302] In embodiments, an immersive haptic system may have a series of predefined environments, scenarios, and associated haptic cues that are designed to gather wearable user data, wherein the data is analyzed to identify and implement user-specific optimized haptic parameter settings.

[0303] In embodiments, an immersive haptic system may have an analytics system that automatically monitors, detects, predicts, and reports on a set of conditions of a haptic user interface device system. For example, the analytics system may generate reports for safety reasons and/or for fleet management.

[0304] Returning to FIG. **8**, in embodiments the networked immersive haptic systems **302**, wearable haptic interface **330**, and/or external resources **808** may be part of a multi-user haptic system. The multi-user haptic system may include multiple immersive haptic systems, such as immersive haptic system **302**A, immersive haptic system **302**B, and/or other immersive haptic systems. The multiple immersive haptic systems may be executed by a single device or multiple devices. In embodiments, the multiple immersive haptic systems may share resources. Additionally or alternatively, a single immersive haptic system **302** may be configured to interface with multiple wearable haptic interfaces **330** by simulating multiple user avatars in a computer-mediated environment and providing haptic feedback to multiple users corresponding to one of the multiple user avatars. In some embodiments, haptic feedback may be distributed among multiple devices based on individual user locations within an environment. For example, an event (e.g., an explosion in a gaming or training simulation) near a group of users would provide a range of individual haptic experiences to various users. In multi-user embodiments, it should be noted that different users may optionally have different interfaces with different levels of haptic capability (e.g., VR headset+haptic glove, VR headset+haptic exoskeleton+one or two haptic gloves, etc.). Additionally, some users (e.g., instructors for a training simulation, AI agents, etc.) may interact with the computer-mediate environment without using a haptic interface. Accordingly, the immersive haptic system(s) **302** may tailor the haptic feedback for events or interactions within the computer-mediated environment to the various users based on any differences in haptic capability and/or whether the users have any haptic capability at all.

[0305] In embodiments, multiple wearable haptic interfaces **330** may be connected via peer-to-peer networks such that at least one of the haptic interface devices provides sensory feedback data intended for another haptic interface device to the other haptic interface device.

[0306] In embodiments, the external resources **808** may include a configuration system that may configure one or more immersive haptic systems and/or a plurality of haptic interface devices using a set of parameters, governance libraries, policies, and other/or configuration data. For example, the configuration system may configure the wearable haptic interface using data from user profile(s) **312** for respective users of the wearable haptic interface, may configure each wearable haptic interface **330** with libraries that are specific to a haptic system application, and/or the like. The configuration system may also perform fleet configurations that include sets of parameters, governance, policies, scenarios, etc. for a specific task, scenario, and/or the like. Additionally or alternatively, the configuration system may configure a network **810** that connects the plurality of wearable haptic interfaces **330** (e.g., by configuring various network protocols, topologies, routing rules, routing priorities, and/or the like).

[0307] Accordingly, a multi-user haptic system may have an immersive haptic system that executes a multi-user simulation of an environment and interfaces with multiple wearable haptic interfaces. The immersive haptic system may determine a respective output state corresponding to each respective user. Each respective output state may be translated to provide a respective sensory feedback response that is specific to a respective user based on a state of an avatar of the user within the environment and/or based on the haptic capabilities of the wearable haptic interface **330** (if any) used by the user.

[0308] In embodiments, a multi-user haptic system may have an immersive haptic system that executes a multi-user simulation of an environment and determines a respective output state corresponding to each respective user that is translated to provide a respective sensory feedback response that is specific to the respective user. In these embodiments, a plurality of haptic interface devices may be interconnected with a peer-to-peer network such that at least one of the haptic user interface devices provides sensory feedback data intended for another haptic interface device to the other haptic interface device. Additionally or alternatively, a configuration system may configure the immersive haptic system and a plurality of haptic interface devices with a set of parameters, governance libraries, policies, and other/or configuration data. Additionally or alternatively, the configuration system may configure a network that interconnects the immersive haptic system with a plurality of haptic interface devices.

[0309] Returning to FIG. **1**, the haptic operating platform **160** may be configured to perform various regulatory, governance, and/or compliance automation tasks in conjunction with the operation of various haptic interface systems **230**. For example, the haptic operating platform **160** (e.g., using various configured intelligence services **204**, which may leverage other haptic operating platform **160** capabilities such as system management components **162**, libraries **180**, system services **192**, etc.) may perform training compliance validation by monitoring the performance of a user inside one or more haptic system applications **250**, such as training applications **254**. In embodiments, the haptic operating platform **160** may use one or more approved training libraries for a range of applications such as surgical procedures, hazardous material handling, and the like. The libraries may also include libraries for a range of medical, environmental, and/or safety standards and the like. The haptic operating platform **160** may thus be or include a compliance system that may receive tactile and/or motion capture data for a user and/or may analyze the data to validate compliance with a set of tests for a training simulation.

[0310] In embodiments, the haptic operating platform **160** may be further configured to perform health compliance monitoring (e.g., using various configured intelligence services **204**, which may leverage other haptic operating platform **160** capabilities such as system management components **162**, libraries **180**, system services **192**, etc.). For example, the haptic operating platform **160** may leverage libraries for ergonomic modeling, physical safety limit testing, and the like. Moreover, by monitoring and analyzing sensor data capture by sensors of the various interface devices **232**, the haptic operating platform **160** may be configured to monitor, analyze, and report on one or more metrics of user health. Thus, the haptic operating platform **160** may be or include a compliance system that may receive user tactile and/or motion capture data and may analyze the data to validate compliance with safety or other health standards.

[0311] In embodiments, the haptic operating platform **160** may be further configured to monitor user data for other reasons, such as for governance reasons (e.g., leveraging governance libraries **188**). Additionally or alternatively, the haptic operating platform **160** may be further configured to validate compliance with governance standards, such as safety standards, haptic feedback limits, and/or the like. For example, the haptic operating platform **160** may be configured to receive data generated by an immersive haptic system **302** and/or by a wearable haptic interface **330** (e.g., sensor data, motion data, location data, biometric data, etc.), analyze the data to monitor and/or validate compliance with safety standards, such as by monitoring and/or reducing amounts of haptic feedback, monitoring and/or limiting force feedback to certain ranges of motion, monitoring and/or prohibiting display of certain content (e.g., rapidly-flashing images), and/or otherwise monitoring, validating, and/or enforcing compliance with governance standards.

[0312] Accordingly, an immersive haptic system/platform may be configured to execute a series of approved predefined environments, scenarios, and associated haptic cues that are executed in connection with a training simulation. The system/platform may further include a compliance system that may receive data captured from a wearable user interface worn by a user and/or data captured from an immersive haptic system and that may analyze the captured data to validate compliance with a set of tests relating to the training simulation. For example, a haptic operating platform **160** may be configured to receive sensor data, motion data, location data, haptic feedback data, biometric data, and/or the like and analyze the data to monitor and/or validate compliance with various tests. In embodiments, the tests may require specific reactions, performance of specific movements, and/or the like. Additionally or alternatively, the tests may require certain states to be achieved in a haptic system application **250** (e.g., the user causes the user avatar to move to a certain position, achieve a certain goal within the application, etc.). Accordingly, the haptic operating platform **160** may monitor and validate compliance based on both immersive haptic system data as well as data received from a wearable haptic interface.

[0313] In embodiments, an immersive haptic system/platform may be configured to execute a series of approved predefined environments, scenarios, and associated haptic cues that are executed in connection with a safety training simulation. In these embodiments, the system/platform may further include a compliance system that may data captured from a wearable user interface worn by a user and/or data captured from an immersive haptic system and that may analyze the captured data to validate compliance with a set of safety standards. For example, the haptic operating platform **160** may be configured to receive sensor data, motion data, location data, haptic feedback data, biometric data, and/or the like and analyze the data to monitor and/or validate compliance with safety standards. In embodiments, the standards may specify haptic feedback limits, range of motion limits, other force feedback limits, thermal limits, audio volume limits, vision limits, and/or the like.

[0314] In embodiments, an immersive haptic system/platform may be configured to execute a series of approved predefined environments, scenarios, and associated haptic cues that are executed in connection with a training simulation. In these embodiments, the platform may further include a compliance system that may receive data captured from a wearable user interface worn by a user and/or data captured from an immersive haptic system and that may analyze the captured data to validate compliance with a set of regulatory and/or governance standards. For example, the haptic operating platform **160** may be configured to receive sensor data, motion data, location data, haptic feedback data, biometric data, and/or the like and analyze the data to monitor and/or validate compliance with various regulatory and/or governance standards. In embodiments, the standards may prohibit and/or require certain actions, prohibit and/or require certain interactions with other users, prohibit certain unsafe conditions, and/or the like. Additionally or alternatively, the tests may require certain states to be achieved in a haptic system application **250**.

[0315] In embodiments, an immersive haptic system/platform may be configured to execute a series of predefined environments, scenarios, and associated haptic cues that are designed to gather user health data from a wearable haptic human interface. In these embodiments, the platform may further include a monitoring system that may monitor a set of conditions of the user to ensure compliance with regulatory and governance standards associated with wearable haptic human interface. Additionally or alternatively, the monitoring system may monitor a set of conditions of the user to ensure the haptic interface is being used safely.

[0316] In some embodiments, a haptic operating platform **160** may be configured to provide and recommend different experiences to a user and to detect various user preferences to customize and optimize an immersive environment for a user. For example, the haptic operating platform **160** may maintain one or more libraries **180** that may be used to provide various experiences via the one or more haptic system applications **250**. In one example, various libraries provided by the haptic operating platform **160** may configure a training application **254** to provide different simulation experiences. In example embodiments, the simulation experiences include a variety of predefined simulation experiences that each include a respective environment, one or more scenarios, and one or more associated haptic cues relating to the experience.

[0317] In embodiments, the haptic operating platform **160** may be or include a recommendation system (e.g., using recommendation engine **210**) that may recommend one or more of the predefined simulation experiences to a user based on a set of learned preferences of the user. The learned preferences may include, for example, objects or entities that a user is observed to interact with in one or more simulation experiences (including, in some cases, other users of a multi-user simulation), types of interactions that the user frequently engages in, an observed health status of the user, and/or other observed user preferences.

[0318] Moreover, in embodiments the haptic operating platform **160** may include a preferences system (e.g., using a personalization module/component **170**) that is configured to learn a set of haptic feedback preferences of the user based on user status data provided by a wearable haptic interface. For example, the status data may be provided in response to the user experiencing scenarios and associated haptic cues within the environment and may indicate whether the user enjoys certain scenarios or haptic cues, frequently interacts with certain scenarios or haptic cues, and/or the like. The preferences system may be used to generate data for a user profile **312** and/or to recommend other experiences/scenarios to a user (e.g., using a recommendation system).

[0319] Accordingly, an immersive haptic platform may have an immersive haptic system that may be configured to execute simulations that are designed to provide a user wearing a wearable haptic human interface with a multi-sensory experience with respect to an environment. The platform may further include an experience library that maintains a set of different predefined simulation experiences that each include a respective environment, one or more scenarios, and one or more associated haptic cues relating to the experience. Additionally or alternatively, the platform may include a recommendation system that recommends one or more of the predefined simulation experiences to a user based on a set of learned preferences of the user. Additionally or alternatively, the platform may include a preferences system that may learn a set of haptic feedback preferences of the user based on user status data provided by the wearable haptic human interface in response to the user experiencing scenarios and associated haptic cues within the environment. For example, the platform may detect a negative response to certain haptic feedback (e.g., amounts of force/thermal feedback above a certain limit, force feedback applied when a user is extending a limb in a certain direction, etc.), as detected by biometric sensors (e.g., perspiration sensors, eye sensors, audio sensors for picking up voice cues, etc.), by a user avoiding certain situations and/or locations within an environment, and/or the like. In embodiments, the platform may then analyze the negative responses over time (e.g., using one or more AI techniques and/or machine learning models) to determine the user haptic preferences and later use the user haptic preferences to adjust haptic feedback.

[0320] In embodiments, the SDK **242** (e.g., including the API **244** and/or a haptic application plugin **246**) may be configured to allow integration of third-party content into one or more haptic system applications **250**. For example, an SDK **242** may be configured to receive third-party content from a haptic operating platform **160** and/or from some other computing device. The SDK **242** may facilitate the integration of the third-party content into one or more environments (e.g., by providing a haptic application plugin **246** for an immersive haptic system **302**). In embodiments, the third-party content may include physical attributes of respective entities in the content such that the immersive haptic system can determine a haptic response corresponding to one or more entities based on the physical attributes thereof in response to the user interacting with the one or more entities in an environment.

[0321] In some embodiments, the third-party content may include computer-aided design (CAD) objects. Thus, for example, a haptic system application **250** may be configured to render and allow a user avatar to interact with virtual buildings, items, and other such CAD objects. Moreover, in some embodiments, the user may be able to virtually edit the CAD objects. An SDK **242** (e.g., using a haptic application plugin **246**) may allow third-party CAD objects to be integrated into the haptic system application **250** and rendered therein such that the user can interact with objects and receive haptic feedback relating to the interactions. Moreover, the user may be able to change the CAD objects, and the SDK **242** may cause the changes to be transmitted to a third-party content repository. In embodiments, the third-party CAD objects may include physical attributes that may be used in generating a haptic feedback response. In embodiments, the physical attributes may include the density of objects, the texture of objects, the malleability of objects, and/or the like.

[0322] Accordingly, an immersive haptic platform may have an immersive haptic system that is configured to execute a series of predefined environments, scenarios, and associated haptic cues that are designed to provide a user wearing a wearable haptic human interface with a multi-sensory experience with respect to the environment. The platform may further include an SDK that facilitates integration of third-party content into one or more environments. The third-party content may include physical attributes of respective entities represented in the content such that the immersive haptic system can determine a haptic response corresponding to one or more entities based on the physical attributes thereof in response to the user interacting with the one or more entities in an environment. In some embodiments, the third-party content may include CAD objects that include a set of physical attributes of a respective entity represented by the CAD object such that the immersive haptic system can determine a haptic response corresponding to the CAD object based on the physical attributes thereof in response to the user interacting with the respective CAD object in a computer-mediated environment.

[0323] In embodiments, user profiles **312** may be used to customize motion capture operations. For example, a slip compensator **906** and/or touch compensator **908** may use user profile **312** data indicating dimensions of a user's body, hands, etc. in order to better determine when slip or touch compensation is needed. Similarly, motion processors **904** may use measured, estimated, or provided data about a user's hand size, arm length, leg length, height, etc. to estimate a position of one or more user body parts, an angle of one or more joints, and/or the like using the absolute location data **952** and/or relative location data **953** received from sensors.

[0324] In embodiments, the API **244** may be configured to estimate various body measurements based on absolute location data **952** and/or relative location data **953** after asking the user to assume different poses or make different motions. For example, by asking or requiring the user to assume a starting pose and then an ending pose, the API **244** may collect location data describing the movement from one to the other and use a model of the relationship connecting certain poses to the positioning of various sensors, brakes (e.g., which may unspool by certain amounts when a user flexes a joint), and the like. In a specific example, the API **244** may use a model of the relationship connecting hand pose to how a brake wheel rotates to measure finger length by recording the change in spool rotation between known postures. A process for measuring the hand in this way may include a user donning a haptic glove **236** and holding their fingers out straight (e.g., a first pose) such that the positions of the brake drums may be recorded. Next, the user may curl their index through pinky fingers as much as possible simultaneously (e.g., a second pose) such that positions of the brake drums at the point of greatest fingertip angle may be recorded. Next, the user may curl their thumb such that a drum position may be recorded at the most extreme thumb tip angle. Then, using a model of how the amount of string drawn via the brake correlates to finger length via tendon guide heights, the API **244** may calculate the lengths of all user fingers, including the thumb. Such a process may then be repeated for the opposite hand. Similar processes may be used to measure the length of user limbs or other body parts.

[0325] Returning to FIG. **9**, in embodiments, an API **244** may use user profiles **312** in a user profile storage **912** to customize one or more haptic feedback operations of the API **244**. Although FIG. **9** illustrates the user profile storage **912** within the API **244**, the user profiles may be stored in storage that is separate from the API **244** (e.g., in a different module or different device). For example, when the API **244** detects a contact between the user avatar and an object/entity within the environment, the API **244** may generate haptic feedback data taking into account a set of physical traits of the user. This customized feedback response may be provided for safety reasons (e.g., to avoid providing too much haptic feedback to a user who is injured or impaired in some way), to account for sizing issues of the wearable haptic interface **330**, and/or the like. Additionally or alternatively, the API may individualize the haptic feedback based on one or more stimulus response traits of the user. For example, a user profile **312** may indicate that a user is sensitive to certain amounts or forms of haptic feedback such that the haptic feedback should be reduced, or vice versa that a haptic feedback response should be increased for certain types of haptic feedback. Additionally or alternatively, the API may individualize the haptic feedback based on one or more psychographic traits of the user (e.g., whether the user detects, responds, and/or learns from certain types of feedback better than others), preference traits of the user (e.g., explicit settings provided by the user for haptic feedback response), and/or stimulus behavioral traits of the user (e.g., whether the user reacts more to certain types of haptic feedback than others).

[0326] Accordingly, a system coordinating haptics in a simulated sensory experience may have an immersive haptic system that may model a set of attributes of a set of objects in an environment and may determine a position of a set of elements of an avatar of a user in the environment based on a state of a haptic interface worn by a user. In these embodiments, in response to detecting that a position of the avatar intersects with a position of an object, the immersive haptic system may output a set of sensory simulation data related to the object. In some of these embodiments, the immersive haptic system may determine an individualized haptic response for the user based on the attributes of the intersected object and a set of physical traits of the user. Additionally or alternatively, the immersive haptic system may determine an individualized haptic response for the user based on the attributes of the intersected object and one or more of a set of stimulus response traits of the user, a set of psychographic traits of the user, a set of preference traits of the user, a set of stimulus behavioral traits of the user, and/or the like.

Non-Telerobotic Use Cases

[0327] The haptic operating platform **160** and/or haptic interface system **230** may be applied to a variety of use cases. These use cases may generally be grouped into telerobotic uses cases (e.g., use cases that involve controlling a remote robot from a distance and/or interacting with a haptic digital twin generated based on data received from the robot) and non-telerobotic uses cases, such as virtual reality applications, non-telerobotic training simulations, non-telerobotic gaming, and/or the like. Within non-telerobotic use cases, the various applications may operate using purely virtual environments (e.g., where the computer-mediated environment is not a digital twin of any real-world environment) and/or AR/MR use cases, where a user may interact with both real objects in the user's environment and simulated objects (e.g., such as where a virtual user interface with virtual controls overlays a real physical object). For example, a user may have a wearable haptic interface **330** that allows the user to move within a real environment (e.g., the user may be wearing a AR/MR headset and haptic gloves **236** that allow the user to move freely without any tethers) that may be enhanced with simulated objects, such that the user can physically interact with the real environment and interact via a simulated environment with the simulated objects using motion capture and haptic feedback.

[0328] As described above, the platform and/or interface systems may provide integration of CAD objects into computer-mediated (e.g., VR/AR/MR/XR) environments. For example, an SDK **242** may be configured to interpret CAD objects data and generate object/entity data therefrom in a format that may be used by a haptic system application **250** (e.g., a game engine) to render the CAD object in an environment, allowing a user to virtually observe and interact with the CAD object using a haptic interface, receive haptic feedback based on interactions with the CAD object, and/or the like. For example, such an application may be used to test ergonomics of tools and/or physical interfaces by allowing a user to interact with a CAD object simulating the ergonomics of the tool or physical interface. As a specific example, a user may be able to hold and operate a handheld tool within the environment (e.g., an electronic device such as a laptop or game controller, a construction tool, a firearm, etc.), with the environment providing a realistic sense of weight, tactile feedback when operating physical buttons or other controls, tactile feedback when gripping the tool in various positions, and otherwise allowing ergonomic testing. As a second example, a user may be able to operate a simulated driver's or pilot's interface of a vehicle, with the platform and/or interface allowing a user to sit in a simulated seat (e.g., with an exoskeleton providing force feedback to lock certain actuators, such as magnetorheological actuators or pneumatic actuators, so that the user can sit), providing realistic interaction with a simulated steering wheel or flight yoke, providing tactile feedback for various physical controls, and/or the like. In some of these embodiments, the haptic system application **250** may provide controls for creating and/or editing the tool and/or interface such that the user can easily generate, prototype, and test alternate designs (e.g., such as by moving buttons or controls within the environment, increasing or decreasing the size, resistance, responsiveness, etc. of the physical controls, and/or the like). Additional details concerning control of a vehicular telerobot are provided below.

[0329] In embodiments, CAD objects that are interpretable by the SDK **242** may include various properties that may be used to render the CAD object in the environment and that may affect haptic feedback responses generated based on a user avatar interacting with the rendered CAD object. For example, the CAD object may include physical attributes indicating, for example, a type of material used by the object or a component of the object, a surface texture of the object or component of the object, a temperature of the object or component of the object, different states of the object (e.g., on/off for a CAD object representing a machine), various properties associated with different states (e.g., a CAD object representing an item with a motor may generate noise and vibration when in an on status), different interactive components that may be used to adjust the state (e.g., interactive switches or other buttons on the CAD object), and/or the like.

[0330] In embodiments, the SDK **242** may further allow integration with mechanical and industrial design tools. For example, the API **244** may receive user inputs (e.g., via a motion capture stack of the API **244**) and may use the inputs to create and/or edit the CAD object via a mechanical or industrial design tool. Thus, for example, the SDK **242** may act as a bridge between an environment generated by a haptic system application **250** and a CAD tool, allowing outputs to flow from one to the other to create an immersive CAD interface.

[0331] Accordingly, an immersive haptic platform may have an immersive haptic system that is configured to generate a set of environments, scenarios, and associated haptic cues that are designed to provide a user wearing a wearable haptic exoskeleton interface with a multi-sensory experience with respect to the environment. The platform may further include an SDK that may facilitate integration of computer-aided design (CAD) objects into one or more environments. In embodiments, each respective CAD object may model an ergonomic design of an item and may include a set of physical attributes of the item such that the immersive haptic system allows a user wearing the wearable haptic exoskeleton interface to test the ergonomics of the item in an environment, modify any of the attributes or other properties of the CAD object, etc.

[0332] In embodiments, the platform and/or interface systems may provide a training environment, such as training for surgical tools, medical or hazardous materials handling, and/or the like. In embodiments, a training application **254** may provide an environment for training in various scenarios and using various haptic cues. Moreover, the training application **254** may be configured using libraries of files that specify data for a specific training application, scenario, and/or environment. In embodiments, the SDK **242** may be configured to receive the libraries (e.g., from a haptic operating platform **160**) and provide the libraries to a haptic system application **250** in order to configure the haptic system application **250** to provide the training.

[0333] In embodiments, the SDK **242** may provide data to the haptic operating platform **160** on one or more user interactions that take place within the training scenario (e.g., tactile and motion data generated by the API **244** and/or captured from a wearable interface) so that the haptic operating platform **160** may act as a testing system to monitor and report on the training (e.g., using monitoring and reporting **222**), score the user's performance in the training (e.g., using analysis **218** to validate a user's compliance with training standards), generate feedback to refine the training scenarios (e.g., using various configured intelligence services **204**), and the like.

[0334] Accordingly, an immersive haptic platform may have an immersive haptic system that is configured to execute a set of predefined environments, scenarios, and associated haptic cues that are executed in connection with one or more surgical training simulations. In these embodiments, the platform may further include a testing system that analyzes data captured from a wearable user interface worn by a user and/or data captured from an immersive haptic system to validate compliance with a set of standards associated with the surgical training simulations. For example, the testing system may analyze whether a user performed steps in a certain order, whether some or all of a set of assigned tasks were performed, whether any negative outcome states were detected by the simulation (e.g., whether a simulated nerve or artery was damaged), whether the simulated patient achieved a positive outcome, etc.

[0335] In embodiments, the immersive haptic platform may have an immersive haptic system that is configured to execute a set of predefined environments, scenarios, and associated haptic cues that are executed in connection with one or more training simulations relating to handling of hazardous materials. In these embodiments, the platform may further include a testing system that analyzes data captured from a wearable user interface worn by a user and/or data captured from an immersive haptic system to validate compliance with a set of standards associated with the handling of the hazardous materials. For example, the testing system may analyze whether a user properly handled the simulated hazardous materials, whether any simulated hazardous materials failed to be contained according to a specified procedure, whether a certain set of assigned tasks were performed in a certain sequence, and/or the like.

[0336] In testing embodiments as well as other embodiments, a user's performance may be observed by another user acting as a trainer, helper, or instructor. The trainer/helper/instructor may engage with the immersive simulation using a separate wearable haptic interface, which may be used to control another avatar, thus creating a multi-avatar simulation. Additionally or alternatively, the trainer/helper/instructor may engage with the immersive simulation using a non-haptic interface (e.g., a VR/AR/MR/XR headset, a computing device that renders the environment via an ordinary display, etc.). In some cases, one or more user avatars (e.g., a training avatar) may be controlled by an AI agent. Additionally or alternatively, an avatar may be controlled directly by a human user (e.g., using a wearable haptic interface or otherwise) at some times and by an AI agent at other times.

[0337] In embodiments, the platform and/or interface systems may provide immersive VR/AR/MR/XR simulations for various settings that may include enterprise settings (e.g., virtual meetings), retail settings (e.g., virtual shopping), and/or entertainment (e.g., a virtual theme park, theater, event, etc.). Again in these situations, different users may engage using different types of interfaces; for example, some users (e.g., customers) may use a wearable haptic interface **330**, whereas other users (e.g., employees, salespeople, helpers, trainers, instructors, etc.) may use other interfaces, leverage AI agents (e.g., at least some of the time), etc.

[0338] In embodiments, by providing multi-user simulation capabilities that allow multiple users to interact with each other and with an environment, the platform beneficially allows for highly immersive remote meetings. The haptic platform may use various technologies discussed herein, including avatar rendering based on motion capture, interaction assistance, and/or haptic feedback, among other technologies discuss herein, to provide remote meetings that allow for simulating personal interaction, collaboration on various tasks (e.g., maintenance, document creation and editing, ideation, product assembly and repair, training, provision of customer assistance, and/or the like).

[0339] In embodiments, the platform may use the technologies described herein to provide immersive VR/AR/MR/XR simulations that allow businesses (e.g., retail businesses) to provide a range of services to help consumers with repair and maintenance tasks, product evaluation and purchasing decisions, enhanced virtual vacation experiences, etc., and/or other metaverse experiences. Additionally or alternatively, the platform may use the technologies described herein to provide immersive VR/AR/MR/XR simulations that allow for simulated theme park experiences (e.g., simulated roller coaster rides with multiple users, simulated theme park locations, etc.), event experiences, and/or the like. These various simulations and experiences may be provided by configured haptic system applications **250**, which may be executed by the immersive haptic system **302**.

[0340] Accordingly, an immersive haptic platform may have an immersive haptic system that is configured to execute a set of predefined environments, scenarios, and associated haptic cues that are executed in connection with an enterprise setting. In these embodiments, the immersive haptic system may interface with a plurality of haptic interface devices such that users of the haptic interface devices interact virtually in the enterprise setting.

[0341] In embodiments, an immersive haptic platform may have an immersive haptic system that is configured to execute a set of predefined environments, scenarios, and associated haptic cues that are executed in connection with a retail setting. In these embodiments, the immersive haptic system may interface with a plurality of wearable haptic interfaces such that users of the wearable haptic interfaces interact virtually with commercial goods in the enterprise setting, such that the immersive haptic system simulates interaction with virtual representations of physical goods. Additionally or alternatively, a haptic operating platform may monitor and/or analyze the interactions between users and the virtual representations of physical goods, for example to detect what goods or types of goods users are drawn to, interact with, frequently purchase, and/or the like. Accordingly, by receiving and analyzing data generated by an immersive simulation of a shopping experience, detailed and high-quality interaction data may be generated and used to improve product experiences, such as product packaging, organization of products, arrangement of shopping spaces, and/or the like.

[0342] In embodiments, an immersive simulation platform may have an immersive haptic system that is configured to execute a set of predefined environments, scenarios, and associated haptic cues that are executed in connection with a set of theme park entities. In these embodiments, the immersive haptic system may interface with one or more wearable haptic interfaces such that users of a respective wearable haptic interfaces can select a simulation from a set of predefined simulation themes for the theme park entities.

[0343] In embodiments, various haptic kits may be provided for specific industries and use cases. The haptic kits may include hardware items (e.g., one or more interface devices **232**) and configured software, such as a configured SDK **242** for use in an immersive haptic system **302**, one or more haptic system applications **250**, and/or the like. The kits may be targeted to different industries and use cases by providing various haptic system applications **250**, corresponding haptic application plugin **246**, and/or the like. In embodiments, a purchaser or licensor of the kit may be licensed to interactively select and receive various libraries **180** from a haptic operating platform **160** that may configure one or more haptic system applications **250** to provide various use cases and experiences. Additionally or alternatively, the purchaser or licensor of the kit may be licensed to use monitoring and reporting **222** and/or analysis, feedback, and optimization **224** services provided by the haptic operating platform **160**.

[0344] Accordingly, an immersive haptic platform may have a software development kit that includes a set of user experience design interfaces by which a user may select, configure and integrate a set of haptic sensory stimulus interactions and a set of environment entity interactions into an immersive multi-sensory simulation experience.

[0345] In embodiments, an immersive haptic platform may have a software development kit that includes a set of user experience design interfaces by which a user may select, configure and integrate a set of haptic sensory stimulus interactions and a set of telerobotics interactions into an immersive multi-sensory simulation experience.

Telerobotic Platform

[0346] FIG. **14** illustrates an example telerobot platform **1400** in which the wearable haptic interface **330** is used to control a telerobot **1406** and receives haptic feedback based on the sensor data **1410** received from the telerobot **1406**. It should be noted that a telerobot platform **1400** may be an example of a whole-body immersion system **300**. In other words, the telerobot platform may include the API **244**, implement the methods of FIGS. **11-13**, etc. In embodiments, the wearable haptic interface **330** may interface with an immersive haptic system **302** running a haptic system application **250** as discussed above. The haptic system application **250** may simulate a state of the telerobot and an environment containing the telerobot using one or more haptic digital twins, such as an environment digital twin **1402** (e.g., a digital twin of the environment that the telerobot is in, which may be pre-stored and/or generated based on sensor/state data **1410** received from the telerobot **1406**) and/or a robot digital twin **1404** (e.g., a digital twin of the telerobot **1406** that is synchronized to a current state of the telerobot **1406**). In embodiments, a telerobot controls system **1412** may maintain synchronization between the haptic system application **250** and the telerobot **1406** by generating control data **1408** to control the telerobot and by receiving sensor/state data **1410** from the telerobot. Thus, the immersive haptic system **302** may provide a computer-mediated environment for both the telerobot **1406** and a haptic system (such as a wearable system), using various methods, systems and capabilities similar or identical to the ones noted throughout this disclosure. For example, the motion capture capabilities, the SDK **242** and the API **244** may provide for multi-directional translation of motion data among the haptic interface, the telerobot **1406**, and the haptic system.

[0347] In embodiments, the term “robot” as used herein may refer to robots that are able to move, sense, and/or otherwise interact with a surrounding environment. For example, robots may include a simple single degree of freedom actuator, a stationary robotic arm, a device with legs, wheels, rotors, wings, propulsion systems for moving on the ground, in the air, through water, etc. Additionally or alternatively, the robots may have any number of sensors for sensing a surrounding environment and/or any number of tools or appendages for interacting with a surrounding environment. In specific embodiments, the telerobot may be a vehicle (e.g., a boat, plane, car, etc.) piloted by a user of a wearable haptic interface, may include one or more robotic arms adapted to perform certain tasks (e.g., surgery, industrial tasks, etc.), and/or the like.

[0348] In embodiments, a telerobot may be a humanoid robot that may be directly controlled by a user of a wearable haptic interface in at least some situations. For example, a user of the wearable haptic interface may directly control the humanoid robot, such that motion capture inputs are translated (e.g., by a haptic digital twin) into commands that cause the humanoid robot to perform the same motions as the user. At the same time, interactions between the humanoid robot and its surrounding environment may be used (e.g., by the haptic digital twin) to generate haptic feedback to provide a sensory simulation of the interactions for the user. In some embodiments, the humanoid robot may operate autonomously at least part of the time. For example, a user of a wearable haptic interface may directly control/pilot a humanoid robot during a training phase to repeatedly perform a task. The haptic operating platform **160** may gather and store sensor data from the humanoid robot and/or instructions for controlling the humanoid robot during the training phase to create a training data set. The haptic operating platform **160** (e.g., using one or more intelligence services **204**) may then train one or more intelligence models to control the humanoid robot to autonomously perform the task using the training data set. Additionally or alternatively, humanoid robots may operate autonomously in some situations but may switch to human-piloted operations in other situations. For example, in some embodiments a fleet of humanoid robots may operate largely autonomously, but human users with wearable haptic interfaces **330** may be able to begin piloting the humanoid robots when needed or desired (e.g., if a robot gets stuck, is unable to perform some task, if an emergency occurs, etc.).

[0349] In embodiments, the haptic digital twin may be a computer-mediated environment where at least a subset of the virtual objects or surfaces are generated based at least in part on sensor data received from a telerobot. However, the computer-mediated environment may also include virtual objects that do not correspond to sensor data received from the telerobot (e.g., for AR and/or MR applications). The computer-mediated environment may further include a virtual representation of a telerobot, which may be generated based on a current state of a telerobot (e.g., a detected position, orientation, etc.). Moreover, in some cases, any of the objects or surfaces (e.g., including the telerobot) may be updated based on predicted future behavior. For example, when a haptic application is controlling a telerobot in a remote location, latency issues may cause delays when commanding a telerobot to perform some action. For example, a user of a wearable haptic interface may command a telerobot to move, interact with an object, etc. If the latency is high (e.g., because the telerobot is on another continent, in space, on another planet, etc.), then there may be a delay in receiving sensor data indicating that the telerobot moved, sensor data indicating the results of the interaction, etc. Therefore, the haptic digital twin may compensate for latency by predicting the outcome of an action (e.g., by modeling the robot moving within the environment before receiving sensor data indicating the robot has actually moved, by modeling the robot’s interaction with a virtual representation of an object before receiving sensor data indicating the outcome of the real interaction, etc.). Thus, it should be understood that the haptic digital twin may be “scaled” in time with respect to a real environment by predicting a future state of the environment. In these cases, when sensor data indicating the outcome of some movement or interaction is later received, the haptic digital twin may then be reconciled to the actual outcome.

[0350] In embodiments, a haptic digital twin may include an avatar controlled by a user of the wearable haptic interface. In some embodiments, the avatar within the computer-mediated environment may correspond to the telerobot, for example by sharing a perspective of the telerobot (e.g., a video feed captured by one or more cameras of the telerobot **1406** may be displayed via an AR/VR/MR/XR interface **336** of the interface **330**), by controlling movement of the telerobot based on motion capture inputs received from the wearable haptic interface **330**, by providing haptic feedback based on sensor data received from the telerobot, etc. However, in some cases the avatar may be temporarily separated from the telerobot. For example, a user that is “seeing” from the perspective of the telerobot may, in some cases, be able to reposition the avatar’s perspective within the environment without causing any corresponding movement of the telerobot. For example, when a telerobot is looking at an object, a user of a wearable haptic interface may use motion capture to control the avatar to “look around” the object in order to see behind it, for example. In these examples, the user may be able to see from differing perspectives because the haptic digital twin may include virtual representations of objects/surfaces that the telerobot has seen before (e.g., even if the telerobot cannot currently see the objects/surfaces) and/or because the immersive haptic system **302** may use machine learning model and/or environment models to predict additional objects/surfaces that are not in view.

[0351] Alternatively, an avatar may be fully separated from a telerobot such that a user may use a wearable haptic interface to move the avatar without necessarily moving the telerobot, may interact with objects/surfaces in the environment that the telerobot is not interacting with, etc. For example, a user may control the avatar to move about freely within a computer-mediated environment generated based on data sensed by the telerobot (e.g., a telerobot may be used for generation of a haptic digital twin even if a user is not directly controlling the telerobot). In some of these embodiments, a user may still perform actions that cause the telerobot to perform actions. For example, the user may issue commands that cause the telerobot to move to a particular location with an environment of the telerobot by pointing to a virtual representation of the location within the haptic digital twin and issuing a command (e.g., a gesture command, a voice command, a command issued via a virtual menu interface, etc.). In other cases, the telerobot may move or interact with objects based on the behavior of the user. For example, a telerobot may "follow" a user may moving within an environment such that it can use its sensors to update the haptic digital twin based on where the user moves within the environment.

[0352] In some cases, the avatar and/or environment may be scaled with respect to data sensed by the telerobot. For example, virtual objects and/or surfaces that are generated based on real objects/surfaces sensed by the telerobot may be scaled up or down with respect to a user avatar. Thus, for example, a user may be able to experience and interact with a computer-mediated environment at a much smaller or larger scale (e.g., such that the user may view and interact with the environment from the perspective of a giant or at a microscopic level). In one example embodiment, a surgeon may control a scaled-down avatar to move about within a computer-mediated environment containing a virtual representation of real tissue captured by a telerobotic surgical instrument. In this example, the user avatar may be much smaller than the user such that the user can more easily view and interact within a small and/or space-constrained environment. Moreover, the haptic digital twin may include transformations of objects/surfaces sensed by the telerobot. For example, the haptic digital twin may generate a virtual representation of plaque within a blood vessel that is sensed by a telerobot such that the virtual representation has a different color or shape (e.g., the plaque may appear as stalactites that a surgeon may "mine" with a virtual pickaxe, which may instruct the telerobot to perform an operation to remove the plaque at a spot the surgeon mines).

[0353] As another example, a haptic digital twin may transform a virtual representation of a real object to allow a user better control over interactions between a telerobot and the real object. For example, if a user is using motion capture to control a telerobot that is much stronger than a human, then a weight or inertia of a virtual representation of an object may be adjusted so that it feels much lighter or easier to move when haptic feedback is provided to the user (thereby allowing the user to experience the increased strength of the telerobot). Continuing the example, the haptic digital twin may also transform a virtual representation of a real object to make it less rigid, thus allowing the user to feel that they can easily damage or warp the object with the strength of the telerobot.

[0354] In embodiments, a haptic system application **250** may provide a virtual reality, augmented reality, and/or mixed reality (VR/AR/MR/XR) environment that allows a user to control the telerobot **1406** in real life and/or control the haptic robot digital twin within the environment provided by the haptic environment digital twin **1402**. In embodiments, the telerobot **1406** and the robot digital twin **1404** are kept synchronized by a telerobotic control system **1412**. Furthermore, the haptic system application **250** may update the digital twin(s) based on user inputs and/or motion capture data received from the wearable haptic interface **330** and may provide haptic feedback based on a state of the digital twin(s) and/or data captured by a respective robot (e.g., motion and/or sensor data captured by the respective robot). The haptic feedback may be generated by the API **244** as discussed elsewhere herein.

[0355] In embodiments, a telerobotic platform **1400** may use configured wearable haptic interfaces to **330** to provide more immersive control over the telerobot **1406** and feedback from the telerobot **1406**. For example, if the telerobot is a vehicle, the wearable haptic interface **330** may be configured to position the user as if the user were sitting in the driver's seat and/or cockpit of the vehicle. In embodiments, an environment digital twin **1402** and/or a robot digital twin **1404** may simulate a set of real-life controls (e.g., a steering wheel and pedals for a car, a yoke, throttle, pedals etc. for an airplane, and the like) such that the user may rely on muscle memory already developed for controlling a vehicle in real life, develop new muscle memory, and the like. Additionally or alternatively, the haptic system application **250** may simulate alternate controls (e.g., controls that a user may be more accustomed to, such as a different size/shape/type of steering wheel or flight yoke than one provided in a corresponding vehicle or plane).

[0356] In embodiments, the immersive haptic system **302** (e.g., via the haptic system application **250** and/or API **244**) may be configured to generate haptic or other feedback based on the control data **1408** sent to the telerobot **1406** and/or the sensor/state data **1410** received from the telerobot **1406**. For example, if the control data instructs the telerobot **1406** to accelerate rapidly and/or the sensor/state data **1410** indicates the telerobot **1406** is accelerating rapidly, the immersive haptic system **302** may provide haptic feedback simulating the acceleration forces to the wearable haptic interface **330**. In another example, if the control data instructs the telerobot **1406** to drill into an object and the sensor data/state data received from the telerobot **1406** indicates that the density of the object changes, the immersive haptic system **302** may provide haptic feedback simulating the change in density. In embodiments, the haptic system application **250** may synchronize one or more digital twins with the state of the telerobot and/or its surroundings in real time to provide immersive control of and feedback from the telerobot **1406**.

[0357] Accordingly, a telerobotic platform may have a telerobotic control system that controls a device, a telerobotic immersive haptic system that executes a digital twin of an environment of the device, and a haptic interface device that presents the digital twin to a user in a multi-sensory manner and provides a multi-medium user interface for controlling the device via the digital twin. In these embodiments, the telerobotic immersive haptic system may update a state of the digital twin of the environment in real-time based on user actions detected by the wearable haptic interface and sensor data received from the telerobotic control system.

[0358] In embodiments, a telerobot **1406** may have a plurality of sensors including cameras, force sensors, mapping/scanning sensors, location sensors, acceleration sensors, orientation sensors, and other types of sensors that may recognize a surrounding environment of the telerobot **1406** and/or a state of the telerobot **1406**. These sensors may be built into the telerobot **1406** and/or may be provided by sensors that are in the environment of the robot (not shown in FIG. **11**), such as base stations, networked cameras, IoT devices, and/or the like. In embodiments, the telerobot control system **1412** may process the sensor/state data received from the telerobot (and/or other devices in the environment of the telerobot) and provide the processed data to the haptic system application **250**, which may correspondingly update the environment digital twin **1402** and/or robot digital twin **1404**. Haptic feedback may then be provided (e.g., by the API **244**) based on the environment, robot, or user avatar interfacing with the robot.

[0359] In some embodiments, the telerobot **1406** and/or immersive haptic system **302** may use real-time scanning (e.g., laser, white light, etc.) techniques and/or object recognition (artificial intelligence, machine learning, etc.) techniques to accurately map the environment of the telerobot and provide appropriate haptic feedback accordingly. Additionally or alternatively, artificial intelligence and/or machine learning may be applied by the immersive haptic system to generate control data **1408** for semi-autonomously controlling the telerobot **1406**.

[0360] Accordingly, a telerobotic platform may have a set of digital scanning and image recognition tools that are used to construct digital surfaces and produce and recognize digital facsimiles of real environments and objects within the environment. The digital facsimiles may include associated haptic cues that are conveyed to the wearable user interface.

[0361] In embodiments, the immersive haptic system (e.g., the telerobotic control system **1412**, haptic system application **250**, and/or API **244**) may leverage various libraries to improve feedback, reduce processing, provide consistency, facilitate governance, and/or perform other functions. As one example, a library for generating an environment based on scanning data may be leveraged by the haptic system application **250** to construct the environment digital twin. In embodiments, the libraries may be stored and retrieved from a haptic operating platform **160**.

[0362] Accordingly, a telerobotic platform may have a set of object and environment libraries that directly translate real-world characteristics to haptic cues for a wearable user interface.

[0363] In embodiments, the telerobotic platform **1400** may leverage various toolkits to provide direct translation of specific sensor assemblies into appropriate haptic feedback cues. For example, such toolkits may reduce latency and/or certain processing requirements. The toolkit may be leveraged for use with structured and/or known real environments.

[0364] Accordingly, a telerobotic platform may have one or more purpose-built sets of remote sensors installed in the robot that may be built to understand and interpret certain environments, objects, and interactions. In these embodiments, haptic cues associated with the environments and objects are provided to the wearable user interface directly without additional processing.

[0365] In embodiments, the haptic system application **250** may leverage various libraries or other data collections for various environments, objects within an environment, sensor groupings, etc. to generate and update an environment digital twin **1402** and/or a robot digital twin **1404**.

[0366] Accordingly, a telerobotic platform may have a telerobotic control system that controls a device, a telerobotic immersive haptic system that executes a digital twin of an environment of the device, and a haptic interface device that presents the digital twin to a user in a multi-sensory manner and provides a multi-medium user interface for controlling the device via the digital twin. In these embodiments, the telerobotic immersive haptic system may be configured with collections of different digital twin libraries that are used to model the physical properties of the environment and objects depicted within the library. The telerobotic immersive haptic system may further be configured to update a state of the digital twin of the environment in real-time based on user actions detected by the wearable haptic interface, sensor data received from the telerobotic control system, and the digital twin libraries.

[0367] In embodiments, a wearable haptic interface **330** may include one or more sensors for receiving voice commands from a user (e.g., microphones) and the immersive haptic system **302** may be configured to interpret the voice commands and use them to update the environment digital twin **1402** and/or robot digital twin **1404** (and therefore the control the synchronized telerobot **1406**), adjust operating parameters, adjust a haptic feedback response, or otherwise modify the environment.

[0368] Accordingly, a telerobotic platform may have a telerobotic control system that controls a device, a telerobotic immersive haptic system that executes a digital twin of an environment of the device, and a haptic interface device that presents the digital twin to a user in a multi-sensory manner and provides a multi-medium user interface for controlling the device via the digital twin. In these embodiments, the telerobotic immersive haptic system may update a state of the digital twin of the environment in real-time based on voice-based commands detected by the wearable haptic interface and sensor data received from the telerobotic control system.

[0369] In embodiments, the immersive haptic system **302** may be configured (e.g., via the API **244**) to interpret certain movements or gestures of the user wearing the wearable haptic interface **330** as commands to take some action, adjust an operating parameter, otherwise update an environment digital twin and/or robot digital twin inside a haptic system application **250**.

[0370] Accordingly, a telerobotic platform may have a telerobotic control system that controls a device, a telerobotic immersive haptic system that executes a digital twin of an environment of the device, and a haptic interface device that presents the digital twin to a user in a multi-sensory manner and provides a multi-medium user interface for controlling the device via the digital twin. In these embodiments, the telerobotic system may update a state of the digital twin of the environment in real-time based on gesture-based commands detected by the wearable haptic interface and sensor data received from the telerobotic control system.

[0371] In embodiments, various types and combinations of networking technologies may be used to network the telerobot **1406**, immersive haptic system **302**, and wearable haptic interface **330**. In embodiments, a single network may be used to network all of the devices. In other embodiments, different networking technologies may be used to communicate between different combinations of devices/systems (e.g., when a telerobot **1406** is remote from an immersive haptic system **302** and/or wearable haptic interface **330**).

[0372] In embodiments, any combination of radio frequency (RF) communication hardware and protocols may be used to interconnect the robotic and wearable haptic devices to systems, sensors, processors, other haptic devices and robots, or external networks. Additionally or alternatively, any combination of hard-wired communication hardware and protocols may be used to interconnect robotic and wearable haptic devices to systems, sensors, processors, other haptic devices and robots, or external networks. Additionally or alternatively, any combination of 5G communication hardware and protocols may be used to interconnect robotic and wearable haptic devices to systems, sensors, processors, other haptic devices and robots, or external networks. Additionally or alternatively, any combination of peer and mesh network communication hardware and protocols may be used to interconnect robotic and wearable haptic devices to systems, sensors, processors, other haptic devices and robots, or external networks. Additionally or alternatively, any combination of low-latency network communication hardware and protocols may be used to interconnect robotic and wearable haptic devices to systems, sensors, processors, other haptic devices and robots, or external networks. In embodiments, a low latency networking system may route data among the telerobotic control system, the haptic interface, and the immersive haptic system using a combination of different networks.

[0373] Accordingly, a telerobotic platform may have a radio frequency (RF) network that interconnects a telerobotic control system that controls a device, a telerobotic immersive haptic system that executes a digital twin of an environment of the device, and a haptic interface device that presents the digital twin to a user in a multi-sensory manner and provides a multi-medium user interface for controlling the device via the digital twin. Additionally or alternatively, the platform may include a fixed networking system that interconnects a telerobotic control system that controls a device, a telerobotic immersive haptic system that executes a digital twin of an environment of the device, and a haptic interface device that presents the digital twin to a user in a multi-sensory manner and provides a multi-medium user interface for controlling the device via the digital twin. Additionally or alternatively, the platform may include a 5G/cellular networking system that interconnects a telerobotic control system that controls a device, a telerobotic immersive haptic system that executes a digital twin of an environment of the device, and a haptic interface device that presents the digital twin to a user in a multi-sensory manner and provides a multi-medium user interface for controlling the device via the digital twin. Additionally or alternatively, the platform may include a mesh networking system that interconnects a telerobotic control system that controls a device, a telerobotic immersive haptic system that executes a digital twin of an environment of the device, and a haptic interface device that presents the digital twin to a user in a multi-sensory manner and provides a multi-medium user interface for controlling the device via the digital twin. Additionally or alternatively, the telerobotic platform may include a low latency networking system that interconnects a telerobotic control system that controls a device, a telerobotic immersive haptic system that executes a digital twin of an environment of the device, and a haptic interface device that presents the digital twin to a user in a multi-sensory manner and provides a multi-medium user interface for controlling the device via the digital twin. In any of these embodiments, the telerobotic immersive haptic system may update the digital twin of the environment in real-time based on sensor data received from the telerobotic control system and actions taken by the user via the wearable haptic interface.

[0374] In embodiments, the telerobotic platform may leverage (alone or in combination) edge processing, data services, specialized processors, and other such networking techniques to optimize both communications and processing tasks to minimize I/O and processing latency for haptic response and associated robotic actions. For example, an intelligence service, purpose-built devices and chips, or other combinations may be used.

[0375] Accordingly, a telerobotic platform may have an edge networking system that interconnects a set of IoT devices with a telerobotic control system that controls a device, a telerobotic immersive haptic system that executes a digital twin of an environment of the device, and a haptic interface device that presents the digital twin to a user in a multi-sensory manner and provides a multi-medium user interface for controlling the device via the digital twin. In these embodiments, the telerobotic control system and the IoT devices may provide sensor data indicating a state of the device and/or the environment of the device to the telerobotic immersive haptic system and the telerobotic immersive haptic system may update the digital twin of the environment in real-time based on the sensor data.

[0376] In embodiments, the wearable haptic interface **330** may include one or more haptic components that engage a user's hands, torso, full body, etc. (e.g., a haptic exoskeleton **238** and/or haptic gloves **236**). Inputs may be received from the haptic exoskeleton **238**, processed by a motion capture stack of an API **244**, and used to modify an environment digital twin **1402** and/or robot digital twin **1404**. A telerobotic control system **1412** may then send control data **1408** to control a telerobot **1406** based on the state of haptic system application **250** and/or inputs from the haptic exoskeleton **238**. In embodiments, the immersive haptic system **302** may be integrated into the haptic exoskeleton **238** (e.g., the immersive haptic system **302** may be executed by a computing device attached to the haptic exoskeleton **238**).

[0377] Accordingly, a wearable haptic exoskeleton may have an interface to a telerobotic control system that controls a robotic device and to a telerobotic immersive haptic system that executes a digital twin of an environment of the robotic device. The exoskeleton may further include a haptic interface system that presents the digital twin to a user in a multi-sensory manner and provides a multi-medium user interface for controlling the robotic device via the digital twin. The exoskeleton may further include a set of tactile haptic actuators that realistically simulate a human touch sensory system as the robotic device interacts with a physical entity. Additionally or alternatively, at least a subset of the actuators of the exoskeleton are large-scale haptic actuators (e.g., magnetorheological actuators) configured to provide force feedback in response to a various forces detected by the robotic device.

[0378] In embodiments, the wearable haptic interface **330** may include an AR/VR/MR/XR interface **336** for viewing the environment digital twin and/or video received by sensors of the telerobot and one or more haptic gloves **236** for controlling the telerobot **1406** as described herein throughout.

[0379] In embodiments, a wearable haptic glove may have an interface to a telerobotic control system that controls a robotic device, a motion-based user interface for controlling the robotic device, and a set of haptic actuators that realistically simulate a human touch sensory system as the device interacts with a physical entity. In these embodiments, at least a subset of the actuators of the haptic glove are configured to provide force feedback resistance in response to a resistive force being detected by the robotic device.

[0380] In embodiments, a telerobotic platform **1400** may be used for robotic process automation. Robotic process automation may refer to a process whereby interactions with objects, machines, systems, or other entities in an environment may be automated by an artificially intelligence (AI) system that may observe human interactions and use the human interactions to train the AI system to perform similar interactions. For example, a human may use the telerobotic platform **1400** to control a robotic arm in an industrial setting (e.g., an assembly line) while an AI system observes the interaction to learn how to operate the robotic arm without human control. The telerobotic platform **1400** provides a tight coupling between the telerobot **1406** and the wearable haptic interface **330**, enabling an AI system (e.g., part of an immersive haptic system **302** and/or a haptic operating platform **160**) to learn how to autonomously or semi-autonomously operate the telerobot **1406**. For example, the robotic system sensors and vision systems may operate with the intelligence system to identify objects, environments, etc. and to define appropriate haptic cues for virtual interaction with identified elements. Moreover, the human-provided operating inputs may be correlated with success metrics as part of a machine learning and/or training system. The trained model/system may enable an autonomous operation that may require limited or no user direction. In embodiments, the system may operate autonomously or semi-autonomously while still providing haptic feedback to a user wearing a wearable haptic interface **330** (e.g., such that the user may take over when necessary). In embodiments, the artificial intelligence functions may be performed by one or more configured intelligence functions **206** of the haptic operating platform **160**.

[0381] Accordingly, a telerobotic platform may have a machine-learning component that monitors user actions via a wearable haptic human interface device that presents a digital twin of an environment of a telerobotic-controlled device via a multi-medium user interface for controlling the device via the digital twin of the environment and that outputs multi-sensory feedback responses based on output received from the digital twin. In these embodiments, the machine-learning component may train a robotic agent based on outcome data sets that respectively indicate a state of the device and/or a proximate environment of the device, a user actions performed by the user in response to the state, and a corresponding outcome of the user action.

[0382] In embodiments, the addition of a real physical system (e.g., the telerobot **1406** and its environment) into the haptic experience loop may be improved by using a high-rate physics system incorporated into the telerobot **1406** (or the immersive haptic system **302**) that can be directly coupled or otherwise integrated with the haptic physics engine to produce the corresponding haptic cues with minimum latency.

[0383] Accordingly, a telerobotic platform may have a telerobotic control system that controls a device based on multi-medium input provided by a user via a wearable haptic interface. In embodiments, the telerobotic platform may include a high update-rate physics system that interfaces with a high update-rate haptic physics engine to produce the corresponding haptic cues that correspond to sensor data collected by the telerobotic control system with minimum latency.

[0384] In embodiments, the telerobotic platform **1400** may be configured for a variety of different use cases. In embodiments, libraries may be provided to the telerobotic platform (e.g., from the haptic operating platform **160**) to configure the immersive haptic system **302** for different use cases. For example, the libraries may configure the telerobotic control system **1412** to interface with a variety of different types of robots, to receive and interpret data from a variety of different types of sensors, to use various protocols, etc. Additionally or alternatively, different haptic system applications **250** may be used to configure the immersive haptic system **302** to build/update/maintain different types of digital twins for different use cases. Additionally or alternatively, different pre-generated digital twins may be used to configure the immersive haptic system **302**. Additionally or alternatively, an SDK **242** and/or an API **244** with different functionalities may be used to configure the immersive haptic system **302**. In embodiments, data may be supplied by the haptic operating platform **160** to a telerobotic platform **1400** upon request depending on a use case associated with a telerobotic platform **1400**.

[0385] In embodiments, a telerobotic platform **1400** may be used for remote maintenance. For example, an immersive haptic system **302** may be configured to interface with telerobots **1406** used for remote maintenance, to read and interpret diagnostics data taken by the telerobot **1406** (e.g., using sensors data captured by the telerobot, analyzing images taken by the telerobot **1406** to identify various conditions, etc.), to control the telerobot **1406** to carry out standardized maintenance actions of one or more maintenance workflows (e.g., automatically or upon request), and/or the like.

[0386] In embodiments, a telerobot platform **1400** may be used for combat or perimeter security workflows. For example, an immersive haptic system **302** may be configured to interface with telerobots **1406** to automatically analyze and identify potential threats (e.g., by analyzing images taken by the telerobot **1406**), to perform patrols of scheduled areas or surveillance of particular areas, to interface with security systems, and/or the like.

[0387] In embodiments, a telerobot platform **1400** may be used for manufacturing workflows. For example, an immersive haptic system **302** may be configured to operate within an assembly line, maintain a telerobot **1406** in compliance with safety and other governance requirements, perform various standardized manufacturing actions, and/or the like.

[0388] In embodiments, a telerobot platform **1400** may be used for hazardous material handling workflows. For example, an immersive haptic system **302** may be configured to interpret data from hazardous materials sensors, maintain a telerobot **1406** in compliance with safety and other governance requirements, perform various hazardous materials handling actions, and/or the like. Hazardous materials handling may particularly benefit from remote supervision, piloting, and/or control of telerobots, including human robots, described in more detail above.

[0389] In embodiments, a telerobot platform **1400** may be used for first responder workflows. For example, an immersive haptic system **302** may be configured to receive and interpret health data from medical sensors and equipment, identify and prioritize potential response actions, perform various medical treatments, and/or the like.

[0390] Accordingly, a telerobotic platform may have a set of remote, immersive AR/VR interfaces and a set of wearable haptic interfaces to a telerobotic system. The haptic interfaces may include a set of actuators that simulate a set of fine motor or large-motor sensory experiences. Additionally or alternatively, the haptic interfaces may include a set of actuators that simulate a set of fine motor or large-motor sensory experiences for a set of remote maintenance workflows. Additionally or alternatively, the haptic interfaces may include a set of actuators that simulate a set of fine motor or large-motor sensory experiences for a set of combat or perimeter security workflows. Additionally or alternatively, the haptic interfaces may include a set of actuators that simulate a set of fine motor or large-motor sensory experiences for a set of manufacturing workflows. Additionally or alternatively, the haptic interfaces may include a set of actuators that simulate a set of fine motor or large-motor sensory experiences for a set of hazardous material handling workflows. Additionally or alternatively, the haptic interfaces may include a set of actuators that simulate a set of fine motor or large-motor sensory experiences for a set of first responder workflows.

Whole-Body Haptic Interface

[0391] A whole-body haptic device **2000**, which may be an example embodiment of a whole-body haptic interface **234**, is described below. The whole-body haptic device **2000** may be used as (or instead of) a wearable haptic interface **330**, wearable haptic glove **236**, wearable haptic exoskeleton **238**, and/or any other haptic interface described above.

[0392] FIG. **15** shows a whole-body haptic device **2000** (may also be referred to as a whole-body haptic system **2000** or a full-body haptic system or device) in accordance with an example embodiment. To better illustrate the motion of the various degrees of freedom of the example embodiment of FIG. **15**, the whole-body haptic device **2000** may be configured in an exemplary pose demonstrating operation of a simulated vehicle by a user **2002**.

[0393] In example embodiments, the whole-body haptic device or system **2000** may be a holodeck type of system that may include a motion platform and a lower body exoskeleton that may provide various degrees of freedom. It will be appreciated in light of the disclosure that the whole-body haptic system **2000** and various portions contained therein and combinations thereof contain ornamental features individually and in combination that are separate and apart from the many functional aspects disclosed herein.

[0394] The whole-body haptic device **2000** may include a motion platform **2030**, interface garments **2006**, **2008** (e.g., haptic glove **2008**), a lower-body exoskeleton **2046**, a torso exoskeleton **2160**, and an upper-body exoskeleton **2200**. User **2002** wears a head-mounted display device **2004** to provide audiovisual feedback.

[0395] Motion platform **2030** is coupled to the lower-body exoskeleton **2046** by a structural frame **2010**. In an example embodiment, the structural frame **2010** may include extruded aluminum structural members **2012** coupled by gussets **2016** and reinforced by diagonal braces **2014**. In other example embodiments, the structural frame **2010** may include any suitably rigid structural member, including metals such as aluminum, steel, and titanium, and polymers or polymer composites, such as glass or carbon fiber reinforced polymers.

[0396] In example embodiments, the lower-body exoskeleton **2046** may include a footplate **2074** (as shown in FIGS. **18** and **20**A). The footplate **2074** may include the platform **2128** as well as the actuated degrees of freedom between the gantry **2048** and the platform **2128**. For example, the lower body exoskeleton **2046** may include the gantry **2048** and the footplate **2074**, where the footplate **2074** may further include several actuated degrees of freedom that may collectively control the position of the platform **2128**. In some example embodiments, the platform **2128** may have both a bottom portion **2130** and an upper portion **2134** that may be magnetically coupled through a membrane (as shown in FIGS. **18** and **20**A and described in the disclosure). In other example embodiments (e.g., as shown in FIGS. **23-33**), the platform **2128** may be monolithic.

[0397] The structural frame **2010** may be coupled to the torso exoskeleton **2160**, which is in turn coupled to the upper-body exoskeleton **2200**. The upper-body exoskeleton **2200** may include a first manipulator **2240***a* and second manipulator **2240***b*, each of which may be coupled to the haptic glove **2008** (e.g., as also referred to as one of the interface garments **2008**).

Enclosure

[0398] FIG. **16** shows a whole-body haptic device (e.g., the whole-body haptic device **2000**) with an enclosure **2018** in accordance with an example embodiment. The enclosure **2018** may enhance user safety by physically isolating a user from potentially dangerous mechanisms of the whole-body haptic device without impeding the operation of the haptic device. The enclosure **2018** may also enhance aesthetic appeal of the whole-body haptic device.

[0399] In the example embodiment of FIG. **16**, the whole-body haptic device may be configured to be partially recessed beneath a load-bearing surface **2026** capable of supporting the weight of a human. In an example embodiment, the load-bearing surface **2026** may include a modular access floor of the type commonly used in data centers, technical rooms, and offices of commercial buildings. Referring now to FIG. **15**, the motion platform **2030**, the structural frame **2010**, and the lower-body exoskeleton **2046** except for an upper portion **2134** of a platform **2128** (e.g., foot platform **2128**) may be recessed beneath the load-bearing surface **2026** (e.g., as shown in FIG. **16**).

[0400] Referring again to FIG. **16**, the load-bearing surface **2026** may be coupled to an outer membrane **2024**, which is in turn coupled to the structural frame **2010**. The outer edge of an inner membrane **2022** is coupled to the structural frame **2010** to form a continuous surface in combination with the outer membrane **2024** and the load-bearing surface **2026**.

[0401] The membranes **2022**, **2024** may include an elastomer, such as silicone rubber. The membranes **2022**, **2024** may optionally include geometric features, such as corrugations, that may increase their displacement in the manner of a bellows. In another example embodiment, the membranes **2022**, **2024** may include a substantially inelastic material, such as steel, aluminum, or fabric that may include a geometric feature of the type described in the disclosure permitting increased displacement.

[0402] The membranes **2022**, **2024** may have a sufficiently high enough modulus of elasticity to allow a user to comfortably walk across their surface (e.g., when entering or exiting the whole-body haptic device), but a sufficiently low enough modulus of elasticity to permit displacement by the motion platform **2030** and the lower-body exoskeleton **2046** during operation.

[0403] In an example embodiment, the outer membrane **2024** may have a relatively higher modulus of elasticity as compared to a modulus of elasticity of the inner membrane **2022**. The outer membrane **2024** may be primarily displaced by operation of the motion platform **2030**, whereas the inner membrane **2022** may be primarily displaced by operation of the lower-body exoskeleton **2046**. Given the substantially higher force output of a typical motion platform, an increased elastic modulus is appropriate for the outer membrane **2024**. This increased modulus may be achieved by use of a sufficiently more rigid material or alternate geometry, such as by sufficiently increasing thickness.

[0404] The inner membrane **2022** may include a first portion having a relatively increased elastic modulus and a second portion having a relatively reduced elastic modulus. In example embodiments, a membrane (e.g., the inner membrane **2022**) may include a first and second portion having substantially different elastic moduli. In the example embodiment of FIG. **16**, inserts **2020** may include a substantially inelastic but flexible material with a geometry approximating the workspace of each foot of a user. The inserts **2020** may include a material with a sufficiently high yield strength and a sufficiently low magnetic susceptibility to avoid interfering with a magnetic coupling **2146** (e.g., as shown in FIG. **20**B) of the foot platform **2128** (e.g., as shown in FIG. **20**A) of the lower-body exoskeleton **2046**. Suitable materials for inserts **2020** may include, but may not be limited to, thin sheets of substantially non-magnetic grades of stainless steel, such as 316 stainless steel, carbon, or glass fiber reinforced polymer sheets, and/or durable synthetic fabrics, such as nylon or polypropylene.

[0405] The upper portions **2134** (e.g., upper portions **2134***a*, **2134***b* as shown in FIG. **16**) of the foot platforms **2128** (e.g., foot platforms **2128***a*, **2128***b* as shown in FIG. **18**) of the lower-body exoskeleton **2046** may each include a cover **2136** (e.g., lower-body cover). The cover **2136** may include a material, such as silicone rubber, that may be substantially elastic. Furthermore, the edges of the upper portions **2134***a*, **2134***b* may be rounded or tapered, as shown in FIG. **16**, such that an object impacting the edges may tend to be deflected up and away from the surface of the inserts **2020** and/or the membranes **2022**, **2024**. The combination of such a geometry with the substantially elastic material of the cover **2136** may help to enhance user safety by reducing the maximum amount of energy that upper portions **2134***a*, **2134***b* may transfer to a body part of a user in case of accidental contact with the sides of the upper portions **2134***a*, **2134***b* during operation. In example embodiments, as shown in FIG. **16**, there are various covers **2028** (e.g., upper-body covers) for structural components of the upper-body exoskeleton **2200**. In example embodiments, the platforms (e.g., foot platforms **2128**) may include a relatively soft cover (e.g., providing sufficient flexibility when the platforms are in use by the user) as described in the disclosure. In example embodiments, the platforms **2128** may include a tapered edge or a rounded edge (e.g., edges of the upper portions **2134***a*, **2134***b*) as described in the disclosure.

Motion Platform

[0406] FIG. **17** shows the motion platform **2030** of the whole-body haptic device **2000** in accordance with an example embodiment. In the example embodiment of FIG. **17**, the motion platform **2030** may be of a "Stewart-type" geometry that is known to those skilled in the art. Stewart-type motion platforms may provide high acceleration, force output, and/or positioning accuracy due to their parallel kinematic architecture. Stewart-type motion platforms may also be readily available commercially, making them particularly suitable for the whole-body haptic device **2000**.

[0407] The motion platform **2030** may include a base **2032**, coupled to brackets **2034**, each of which may be coupled to universal joints **2036** (e.g., first universal joints **2036**). The first universal joints **2036** may be coupled to ball-screw actuators **2038**, which may drive structural members **2040**. The structural members **2040** may be coupled to universal joints **2042** (e.g., second universal joints), which may in turn be coupled to a frame **2044**. The combined motion of ball-screw actuators **2038** may control all six degrees of freedom of the frame **2044** within the mechanism's workspace.

[0408] Various alternate actuation mechanisms may be contemplated for the motion platform **2030**, including but not limited to, lead screw and crank-based mechanisms. The motion platform **2030** may be driven by any appropriate actuator familiar to those skilled in the art, such as electromechanical, pneumatic, and/or hydraulic actuators.

[0409] In example embodiments, the motion platform **2030** may include at least an actuated degree of freedom configured to permit rotation about the longitudinal axis (or vertical axis) of a user's body. This degree of freedom may be provided to enable a user to change direction naturally during ambulation or other locomotion. As a user's center of mass rotates, this degree of freedom may be actuated such that components of a whole-body haptic device distal to the motion platform **2030** may remain substantially aligned with the longitudinal orientation of the user's center of mass.

[0410] In example embodiments, an actuated degree of freedom of the motion platform **2030** may be configured to permit rotation about the longitudinal axis of a user's body by including a rotary actuator at the distal end of the kinematic chain of the motion platform **2030** configured to permit a rotation of at least about 720 degrees. In other example embodiments, the degree of freedom may be configured to permit continuous rotation. In some example embodiments, the motion platform **2030** may include a rotary electrical coupling or connector, such as a slip ring, to enable power and data to pass through to other elements of a whole-body human-computer interface. For example, in some example embodiments, the motion platform **2030** may include the rotary electrical coupling or connector that may be configured to permit the actuated degree of freedom to continuously rotate at least about 720 degrees. In example embodiments, the actuated degrees of freedom (e.g., fourth and fifth actuated degrees of freedom as described in the disclosure) of the lower-body exoskeleton **2046** may include a parallel mechanism. The parallel mechanism may include a rotary actuator coupled to a crank. In example embodiments, as shown in FIG. **15**, the rotary actuator may be a rotary actuator motor **2270** that provides the rotation-related one or more degree(s) of freedom corresponding to the rotational movement of the motion platform **2030**.

[0411] In example embodiments, a whole-body haptic device having a motion platform with a single longitudinal degree of freedom, as described in the disclosure, may be sufficient to permit basic ambulation on a predominantly flat surface. However, additional degrees of freedom may be required to accurately simulate ambulation on slopes or uneven surfaces, as well as vehicle operation and other commercially important forms of non-ambulatory locomotion. Accordingly, in example embodiments, the motion platform **2030** of a whole-body haptic device (e.g., whole-body haptic device **2000**) may include at least two additional actuated degrees of freedom, including: a first actuated degree of freedom configured to permit rotation about a sagittal axis of a user's body, and a second actuated degree of freedom configured to permit rotation about a frontal axis of a user's body.

[0412] In example embodiments, the sagittal and frontal degrees of freedom of the motion platform **2030** may be actuated such that the orientation of a top surface of the motion platform **2030** (e.g., the top surface of the frame **2044** of the motion platform **2030**) may substantially match the orientation of a corresponding section of terrain of a computer-mediated environment during user ambulation. For example, the frame **2044** may be tilted so as to present a simulated uphill slope that matches a virtual hill encountered in a computer-mediated environment. In example embodiments, the sagittal and frontal degrees of freedom of the motion platform **2030** may be further actuated to emulate acceleration of a simulated vehicle via a three degree of freedom acceleration cueing algorithm of the type commonly known to those skilled in the art.

[0413] In an example embodiment, as shown in FIG. **17**, a motion platform further includes: a first actuated degree of freedom configured to permit translation along the sagittal axis of a user's body, a second actuated degree of freedom configured to permit translation along the frontal axis of a user's body, and a third actuated degree of freedom configured to permit translation along the longitudinal axis of a user's body. The additional degrees of freedom, while not strictly necessary for simulating natural user locomotion in a computer-mediated environment, substantially improve the fidelity of the resulting simulation, particularly for motions that involve rapid accelerations. Six degrees of freedom motion cueing algorithms, as known to those skilled in the art, may be utilized to permit optimal use of both linear degrees of freedom (e.g., to simulate high frequency, short duration accelerations) and rotary degrees of freedom (e.g., to simulate low frequency, sustained accelerations).

Exoskeleton

[0414] Referring now to FIG. **15**, the whole-body haptic device **2000** may include the lower-body exoskeleton **2046**, the torso exoskeleton **2160**, and the upper-body exoskeleton **2200**. The lower-body, torso, and upper-body exoskeletons form a continuous kinematic chain from the motion platform **2030** to the interface garments **2006**, **2008** and ultimately to the body of the user **2002**. Thus, the exoskeletons may enable the simulation of arbitrary grounded forces acting on the portions of the body of the user **2002** to which they are coupled. In conjunction with cutaneous feedback that may be provided by interface garments **2006**, **2008** (including tactile and/or thermal feedback), at least one exoskeleton of the whole-body haptic device (e.g., whole-body haptic device **2000**) may present a complete haptic representation of any simulated objects with which the user **2002** may interact in a computer-mediated environment, closely approximating the full set of haptic sensations which may be present in an interaction with a comparable real-world object.

Lower-Body Exoskeleton

[0415] FIG. **18** shows the lower-body exoskeleton **2046** in accordance with an example embodiment. The lower-body exoskeleton **2046** may include a first gantry **2050*a*** and a second gantry **2050*b***; one for each of a user's feet. In example embodiments, as described in the disclosure, the lower-body exoskeleton **2046** may include the footplate **2074** (e.g., **2074*a***, **2074*b***). Specifically, as shown in FIG. **18**, there may be two separate footplates **2074*a***, **2074*b*** for each foot (e.g., left foot and right foot). For example, the footplates **2074*a***, **2074*b*** may be mirrored components that fit for the left and right feet, respectively.

[0416] Referring now to FIG. **19**, each gantry **2048** (e.g., first gantry **2050*a*** or second gantry **2050*b***) may include a first actuated degree of freedom **2052** (e.g., degree of freedom **2052*a***, **2052*b***) configured to permit translation along the frontal axis of a user's body, and a second actuated degree of freedom **2066** configured to permit translation along the sagittal axis of a user's body.

[0417] Actuated degrees of freedom **2052** (e.g., degrees of freedom **2052*a***, **2052*b***), **2066** of each gantry **2048** (e.g., first gantry **2050*a*** or second gantry **2050*b***) may include belt-driven actuators **2054**, **2068**. A servomotor **2058** (e.g., servomotor **2058*a***, **2058*b***) may be coupled to a gear reducer **2060**, which may be in turn coupled to an input flange **2062** of the belt-driven actuator **2054**. As the servomotor **2058** (e.g., servomotor **2058*a***, **2058*b***) rotates, it may rotate the belt of the belt-driven actuator **2054**, which may in turn be coupled to a carriage **2064** to produce controlled linear motion. This controlled linear motion may produce the degrees of freedom **2052*a***, **2052*b***. In example embodiments, the degrees of freedom **2052*a***, **2052*b*** may include an actuated degree of freedom **2052*a*** that may be coupled to a second passive degree of freedom **2052*b***. This may occur through the use of an identical belt-driven actuator **2054** and by use of a coupling shaft **2056** in example embodiments as described in the disclosure and shown in FIGS. **18-19** for each gantry **2048** (e.g., gantry **2050*a***, **2050*b***). Pairing degrees of freedom **2052*a*** and **2052*b*** may provide a stable base for the lower-body exoskeleton **2046** (e.g., as shown in FIG. **18**), permitting the exoskeleton to accommodate larger loads.

[0418] Each gantry **2048** (e.g., first gantry **2050***a* or second gantry **2050***b*) may further include an actuated degree of freedom **2066** (e.g., the second actuated degree of freedom **2066**), including at least one belt-driven actuator **2068**, the servomotor **2058** (e.g., the servomotor **2058***b*), a gear reducer **2070**, and an output carriage **2072**. Each belt-driven actuator **2068** may be coupled to the carriage **2064** of the actuated degrees of freedom **2052***a*, **2052***b* to produce a two degree of freedom planar gantry. In example embodiments, each belt-driven actuator **2068** may be coupled to the carriage **2064** and/or the output carriage **2072** to provide the actuated degree of freedom **2066** (e.g., the second actuated degree of freedom **2066**) as shown in FIGS. **18-19** and described in the disclosure for each gantry **2048** (e.g., gantry **2050***a*, **2050***b*). Referring again to FIG. **18**, in example embodiments, a third actuated degree of freedom **2158** may be configured to permit rotation about the longitudinal axis of a user's body that may be coupled to the output carriage **2072** of each gantry **2048**. In example embodiments, as shown in FIGS. **18** and **20**A, a servomotor **2076** (e.g., similar to the servomotors **2058***a*, **2058***b*) may be coupled to a gear reducer **2078**, which may be in turn coupled to a crossed roller bearing **2090** to form the third actuated degree of freedom **2158**.

[0419] In a first alternate example embodiment, the gantry **2048** (e.g., first gantry **2050***a*, second gantry **2050***b*) may include a holonomic drive system of the type commonly found in omnidirectional wheeled robots, including at least three and more preferably four independently actuated omnidirectional wheels (such as omni-wheels or mecanum wheels). In one variation of this example embodiment, the wheels of the holonomic drive system may be directly driven by actuators on board the gantry. In a second variation example embodiment, the wheels of the holonomic drive system may be driven by a mechanical transmission, such as a flex shaft or hydrostatic transmission coupled to an externally located actuator to reduce moving mass.

[0420] In another example embodiment, the gantry **2048** (e.g., first gantry **2050***a*, second gantry **2050***b*) may include a cable-driven mechanism, such as a planar parallel cable-driven mechanism including four actuated winches located approximately at the corners of a rectangle slightly larger than the desired workspace. In example embodiments, the cable-driven mechanism may be a cable-driven actuator.

[0421] Three actuated degrees of freedom per foot, as described in the disclosure, may be sufficient to allow arbitrary in-plane motion of a user. In one example, a user may be walking in a straight line along a flat simulated surface in a computer-mediated environment. Each of the user's feet may be position tracked by a six degree of freedom motion capture system (or, more generally, a multiple-degree-of-freedom motion capture system having, e.g., two, three, four, five, or six degrees of freedom). In example embodiments, the motion capture system may include a passive optical tracking system, an active optical tracking system, or an electromagnetic motion capture system. During the swing phase of a user's ambulation, the platforms **2128** (e.g., foot platforms **2128***a*, **2128***b*) may be actuated so as to substantially match the position of the user's foot projected onto the plane defined by the range of motion of the actuated degrees of freedom of the gantry **2048** (e.g., first gantry **2050***a*, second gantry **2050***b*). Simultaneously, both the swing foot and stance foot platforms **2128** may be actuated so as to substantially cancel out the user's net motion vector, as sensed by a position sensor located near a user's center of mass and/or a force sensor (e.g., a force sensor **2120** such as a single or multi-axis force sensor **2120**) in the platforms **2128**, in a manner similar to the motion of a treadmill. In example embodiments, the force sensor (e.g., force sensor **2120**) may include at least three sensed degrees of freedom. In example embodiments, the platforms (e.g., foot platforms **2128**) may be actuated to produce motion proportional to a sensed force vector of the force sensor (e.g., force sensor **2120**). The force sensor (e.g., force sensor **2120**) may be configured to indicate (e.g., output a signal that varies with) a weight of a user. In example embodiments, the force sensor may be configured to sense a contact state (e.g., in contact with or not in contact with) of a user's foot with the platforms (e.g., foot platforms **2128**). Finally, in example embodiments, the motion platform **2030** (e.g., as shown in FIG. **15**) may be actuated so as to match the acceleration the user would have experienced had their net motion not been offset. The result may be a natural sense of ambulation, closely matching the forces and accelerations the user would experience during comparable real-world ambulation, without the need for a comparable space. Arbitrary slope of the simulated locomotion plane may be accommodated by actuation of the motion platform **2030** (e.g., as shown in FIG. **15**) up to the mechanical limits of the device, as described in the "Motion Platform" section of the disclosure.

[0422] In another example embodiment, a user may make a turn while walking along the same simulated surface. In this example, the platform **2128** corresponding to the user's plant foot may need to stay fixed as the whole-body haptic device **2000** (e.g., as shown in FIG. **15**) and the rest of the user's body may rotate around it. Accordingly, the plant foot platform **2128** may be rotated and translated so as to stay materially fixed in space relative to the user's reference frame. This may typically require simultaneous actuation of all three planar degrees of freedom of the lower-body exoskeleton **2046** for the plant foot.

[0423] In another example embodiment, the displacement of platforms **2128** may be modified by at least one property based on an interface or interaction between a representation of a user's foot in a computer-mediated environment and a virtual surface or simulated surface. The representation of the user's foot along with the virtual surface or simulated surface may be represented together in the same computer-mediated environment. For example, a user (e.g., user's feet) may interact with any simulated object or terrain (e.g., flat simulated surface or an uneven surface) in a computer-mediated environment, closely approximating the full set of haptic sensations (e.g., displacement of platforms **2128**) which may be present in this interaction with a comparable real-world object or terrain. In another example, there may be a specified number of actuated degrees of freedom (e.g., three degrees of freedom) applied to each foot to allow for ambulation on a predominantly flat surface. Additional degrees of freedom may be applied for accurate simulation of uneven surface having substantial changes of slope over short distances, such as stairs or rough terrain. In example embodiments, actuated degrees of freedom may be further actuated (e.g., causing modification of the displacement of platforms **2128**) to match the orientation of the surface (e.g., terrain) in the computer-mediated environment supporting the user's feet when walking over various types of terrain. For example, displacement may be amplified to represent slipping on a low friction surface, like ice, or conversely, displacement may be reduced to represent ambulation across a sticky surface, like tar.

[0424] In example embodiments, the lower-body exoskeleton **2046** (e.g., as shown in FIG. **15**) may have three actuated degrees of freedom per foot, as described in the disclosure, which may be sufficient to permit basic ambulation on a predominantly flat surface. However, additional degrees of freedom may be required for accurate simulation of uneven surfaces having substantial changes of slope over short distance, such as stairs or rough terrain. Accordingly, in example embodiments, the lower-body exoskeleton **2046** (e.g., as shown in FIG. **15**) may include at least two additional actuated degrees of freedom, including: a first actuated degree of freedom configured to permit rotation about the sagittal axis of a user's body, and a second actuated degree of freedom configured to permit rotation about the frontal axis of a user's body.

[0425] FIG. **20**A shows actuated degrees of freedom configured to permit rotation about a vertical axis such as a longitudinal axis passing through the center of rotation of the user's foot, an actuated degree of freedom configured to permit rotation about a sagittal axis passing through the center of rotation of the user's foot, and an actuated degree of freedom configured to permit rotation about the frontal axis passing through the center of rotation of the user's foot of the lower-body exoskeleton of the example embodiment of FIG. **18**. In example embodiments, a servomotor **2076** (e.g., similar to servomotors **2058***a*, **2058***b*) may be coupled to a gear reducer **2078**, which may be in turn coupled to a crossed roller bearing **2090** to form an actuated degree of freedom **2158** (e.g., similar to the third actuated degree of freedom **2158**). The actuated degree of freedom **2158** may be coupled to mounting plates **2080**, **2084** which may be joined by fasteners **2082** and spacers **2088**. Other fasteners **2086** may join the mounting plate **2080** to the output carriage **2072** (e.g., as shown in FIG. **19**). As shown, FIG. **20**A also shows the footplate **2074** as described in the disclosure.

[0426] In example embodiments, a structural plate **2094** (e.g., base plate) may be coupled to the output flange of the crossed roller bearing **2090** and to crank actuator assemblies **2092***a* and **2092***b* by use of fasteners **2096**. Each actuator assembly **2092** (e.g., actuator assembly **2092***a*, **2092***b*) may include a mounting bracket **2098** joined to a gear reducer **2104** by fasteners **2100**. The gear reducer **2104** may be coupled to another servomotor **2108**. An output shaft **2102** of the gear reducer **2104**, driven by the servomotor **2108**, may be coupled to crank arms **2106**, **2110**, which may be in turn coupled to a shaft **2114**. The shaft **2114** may be coupled to a first ball joint rod end **2118**, permitting two passive degrees of freedom. The first ball joint rod end **2118** may be coupled to a rod **2122**, which may in turn be coupled to a second ball joint rod end **2124**. The second ball joint rod end **2124** may terminate at a clevis **2128** by way of a shaft **2126**.

[0427] Structural members **2130** and **2122**, in combination with fasteners **2124** and spacers **2126** may form a substantially rigid structural frame, which may be coupled to clevises **2128** on either end. In example embodiments, the clevises **2128** may be mounted such that they are offset in both planar axes, as shown in FIG. **20**A. In example embodiments, a universal joint **2112** may be mounted approximately in the center of the assembly of FIG. **20**A, coupling the structural plate **2094** and the structural member **2130**. In example embodiments, a single or multi-axis force sensor **2120** may optionally be mounted to the universal joint **2112** by use of a mounting boss **2116**. The actuator assemblies **2092***a* and **2092***b*, in combination with the universal joint **2112**, may form a parallel two degree of freedom mechanism providing controlled motion about the sagittal and frontal axes of a user's body.

[0428] In an example embodiment, the degrees of freedom of the motion platform **2030** (e.g., as shown in FIG. **15**) permitting rotation about the sagittal and frontal axes of a user's body may be actuated to produce a planar approximation of terrain of a computer-mediated environment and the corresponding actuated degrees of freedom of the lower-body exoskeleton, as described in the disclosure, that may be actuated to substantially match a deviation of non-planar terrain from the planar approximation. In example embodiments, the non-planar terrain that may be generated within the computer-mediated environment may be a computer-generated three-dimensional representation of a real terrain or an artistically created terrain. For example, in simulating ascending a set of stairs in a computer-mediated environment, the motion platform **2030** (e.g., as shown in FIG. **15**) may be angled to match the slope of the staircase, and as the user's feet move over the staircase, the platforms **2128** (e.g., foot platforms) of the lower-body exoskeleton **2046** (e.g., as shown in FIG. **18**) may be counter-rotated to closely approximate the stair treads. This additive approach to synthesis of uneven terrain may minimize the required range of motion of out-of-plane degrees of freedom of the lower-body exoskeleton **2046** (e.g., as shown in FIG. **18**), enabling a substantially less complex, less expensive, and safer device.

[0429] In another example embodiment, as shown in FIG. **15**, degrees of freedom of the lower-body exoskeleton **2046** permitting rotation about the sagittal and frontal axes of the foot of the user **2002** may be employed to simulate foot controls of a simulated vehicle such as a ground, sea, or air vehicle. For example, the platforms **2128** (e.g., foot platforms **2128**) of the lower-body exoskeleton **2046** may be positioned to simulate the gas (e.g., gas pedal), brake (e.g., brake pedal), and/or clutch of a car or truck, or to simulate the rudder pedals of an airplane or aircraft (e.g., emulate an aircraft rudder pedal).

[0430] Referring once more to FIG. **20**A, the actuated degrees of freedom **2158** and the actuator assemblies **2092***a/b* of the lower-body exoskeleton **2046** (e.g., as shown in FIG. **15**) may be coupled to each platform **2128** (e.g., foot platform **2128**). In example embodiments, each platform **2128** may include a bottom portion **2130** and an upper portion **2134** physically separated by insert(s) **2020**, and/or membrane(s) **2022**, **2024** (e.g., as shown in FIG. **16**) during operation and as described in the disclosure. The bottom portion **2130** and the upper portion **2134** each may include a mechanical interface to insert(s) **2020**, and/or membrane (s) **2022**, **2024** (e.g., as shown in FIG. **16**) that may permit in-plane motion with minimal friction. For example, first and second portions (e.g., the bottom portion **2130** and the upper portion **2134**) of each platform **2128** may be coupled to the insert(s) **2020** and/or membrane(s) **2022**, **2024** by use of the mechanical interface with a coefficient of friction of less than about 0.05. In example embodiments, the mechanical interface may include a rolling element, such as a caster **2142** (e.g., as shown in FIG. **20**B) or a ball bearing plate. In another example embodiment, the mechanical interface may include an air bearing. In example embodiments, as shown in FIG. **20**A, the cover **2136** may be an upper cover for the upper portion **2134** and another cover **2132** may be a bottom cover for the bottom portion **2130**, respectively. The bottom cover **2132** may be the counterpart to the upper cover **2136**.

[0431] Referring now to FIG. **20**B, in example embodiments, the bottom portion **2130** and the upper portion **2134** of the platform **2128** may be coupled through insert(s) **2020**, and/or membrane(s) **2022**, **2024** (e.g., as shown in FIG. **16**) by use of the magnetic coupling **2146**. In example embodiments, the magnetic coupling **2146** may include magnetic elements **2150** that may be arranged in a checkerboard-like pattern of alternating polarity (as shown in FIG. **20**B) to maximize the shear force required for decoupling relative to the strength of the magnetic elements **2150**. In example embodiments, the magnetic elements **2150** may include through holes through which they may be coupled to a threaded rod **2152**, which may in turn be coupled to a steel pot **2148** by a nut **2154** in order to constrain and focus the magnetic field of the magnetic elements **2150** in the direction of the coupling. The casters **2142** and the magnetic coupling **2146** may be coupled to chassis **2138** by use of fasteners **2144**, **2156** to form a complete functional unit. In example embodiments, the chassis **2138** may include a cutout **2140** to facilitate easy coupling to the cover **2136** (e.g., as shown in FIG. **16**) without the need for additional fasteners. The elasticity of the cover **2136** (e.g., as shown in FIG. **16**) may keep it in place once stretched over the chassis **2138**.

Torso Exoskeleton

[0432] FIG. **15** shows the torso exoskeleton **2160** in accordance with an example embodiment. The torso exoskeleton **2160** may include another actuated degree of freedom **2162** that may be configured to permit translation along the longitudinal axis of a user's body. As shown in FIG. **15**, in example embodiments, the torso exoskeleton **2160** may include the first and the second coupled degree of freedom **2162***a* and **2162***b* to increase force output and rigidity of the actuated degree of freedom **2162**.

[0433] FIG. **21**A shows a perspective view of the actuated degree of freedom **2162** of the torso exoskeleton **2160** configured to permit translation along the longitudinal axis of a user's body in accordance with an example embodiment. FIG. **21**B is a perspective cutaway view of the actuated degree of freedom **2162** of the torso exoskeleton **2160** of the example embodiment of FIG. **21**A omitting structural frames **2168** and **2176** (e.g., lower and upper structural frames **2168**, **2176**, respectively) for increased clarity.

[0434] Referring now to FIGS. **21**A and **21**B, a servomotor **2188** may be coupled to a gear reducer **2190**, which may in turn be coupled to an input flange **2192** of a cantilever axis **2184**. A belt-drive unit **2186** may translate rotation of the servomotor **2188** into controlled linear displacement of a cantilever **2194**. At least one structural frame **2168** may be a lower structural frame **2168**. The lower structural frame **2168** may include structural extrusions **2170**, gussets **2172**, and mounting tabs **2174**. The lower structural frame **2168** may couple the cantilever axis **2184** to the structural frame **2010** (e.g., as shown in FIG. **15**). The other structural frame **2176** may be an upper structural frame **2176**. The upper structural frame **2176** may include structural extrusions **2178** and a diagonal brace **2180** that may be coupled to the cantilever **2194** by use of a mounting plate **2196**. Mounting plates **2182** may be configured to couple to the upper-body exoskeleton **2200** (e.g., as shown in FIG. **22**A). In example embodiments, as shown in FIGS. **15**, **21**A, and **21**B, the servomotor **2188** may be coupled to the gear reducer **2190**, which may in turn be coupled to the input flange **2192** of the cantilever axis **2184** as described in the disclosure. The belt-drive unit **2186** may translate rotation of the servomotor **2188** into controlled linear displacement of the cantilever **2194** which may provide the actuated degree of freedom **2162**. In some example embodiments, a pneumatic actuator (e.g., pneumatic cylinder **2164** as shown in FIGS. **21**A-**21**B) may also or alternatively be utilized to provide the actuated degree of freedom **2162**.

[0435] In example embodiments, a pneumatic cylinder **2164** (e.g., pneumatic actuator) may be coupled to the lower structural frame **2168** by use of a mounting plate **2166** and the upper structural frame **2176** by use of the mounting plate **2196**, in parallel with the cantilever axis **2184**. The pneumatic cylinder **2164** may be coupled to a pressure regulator, which may control its force output. In example embodiments, the pressure regulator may be configured to output a first pressure generating a force of the pneumatic cylinder **2164** substantially equal and opposite to the sum of the weight of components (e.g., most if not all components) of a whole-body haptic device distal to the pneumatic cylinder **2164** such that the distal components may be substantially maintained in an energetic equilibrium relative to the force of gravity acting on them. There may be a second pressure generating a force of the pneumatic cylinder **2164** substantially equal and opposite to the sum of the weight of components (e.g., most if not all components) of a whole-body haptic device distal to the pneumatic cylinder **2164** plus the weight of a user, such that both the distal components and the user's body may be substantially maintained in an energetic equilibrium relative to the force of gravity acting on them. In another example embodiment, the pressure regulator may be proportional, being configured to further output pressure states between about zero and a maximum rated pressure of the pneumatic cylinder **2164**.

[0436] In other example embodiments, the actuated degree of freedom **2162** may include solely a fluidic actuator, such as a pneumatic or hydraulic cylinder or solely an electromechanical actuator such as a cantilever axis or other belt-driven actuator, a ball or lead screw actuator, or a crank-based actuator. However, a combination of both a fluidic and electromechanical actuator in parallel, as shown in the example embodiment of FIGS. **21**A/B may present significant advantages. The pneumatic cylinder **2164** may supply a constant gravity offset force calibrated to support the weight of distal elements of a whole-body haptic device and/or the weight of a user, as described in the disclosure, with minimal power consumption, while the cantilever axis **2184** may provide higher precision force or position control than may be possible with a fluidic actuator alone. The pressure of the pneumatic cylinder **2164** may additionally be calibrated to partially offset a user's body weight to permit greater ease of ambulation, particularly for disabled users, or even to simulate buoyancy (e.g., in a simulated underwater environment) or gravitational constants different from standard Earth surface gravity.

[0437] Referring now to FIG. **15**, in an example embodiment, the torso exoskeleton **2160** may include an actuated degree of freedom configured to permit translation along the sagittal axis of a user's body. The actuated degree of freedom may include a cantilever axis substantially similar to the cantilever axis **2184** (e.g., as shown in FIG. **21**A/B). This additional degree of freedom may be preferably employed to help keep the upper-body exoskeleton **2200** and the user **2002** substantially centered relative to the lower-body exoskeleton **2046** during movement of the motion platform **2030**.

[0438] In addition to the gravity compensation functionality described in the disclosure, the torso exoskeleton **2160** may be further employed to simulate sitting on a virtual surface in a computer-mediated environment. In this example embodiment, the torso exoskeleton **2160** may be locked at a height corresponding to the height of a virtual sitting surface to constrain the user **2002** to the virtual surface by use of the interface garment **2006** and then unlocked as the user **2002** stands up.

Upper-Body Exoskeleton

[0439] FIG. **22**A shows the upper-body exoskeleton **2200** in accordance with an example embodiment. The upper-body exoskeleton **2200** may include the first manipulator **2240***a* and the second manipulator **2240***b* with substantially similar construction. In example embodiments, a manipulator **2240** (e.g., the first manipulator **2240***a* or the second manipulator **2240***b*) may include at least three actuated degrees of freedom. In other example embodiments, as shown in the example embodiment of FIG. **22**A, the manipulator **2240** (e.g., the first manipulator **2240***a* or the second manipulator **2240***b*) may include six or more actuated degrees of freedom.

[0440] The base of the manipulator **2240** may include a first actuated degree of freedom **2240** and a second actuated degree of freedom **2240** coupled to a structural member **2250**. For example, each manipulator **2240***a*, **2240***b* may provide the first actuated degree of freedom **2240** and the second actuated degree of freedom **2240**. The structural member **2250** may be further coupled to a third actuated degree of freedom **2250** and a fourth actuated degree of freedom **2240**. For example, each manipulator **2240***a*, **2240***b* may provide the third actuated degree of freedom **2250** and the fourth actuated degree of freedom **2240** from the coupling to the structural member **2250**. A fifth actuated degree of freedom **2250** may be remotely controlled by an actuator assembly **2240**, which may be coupled to the fifth actuated degree of freedom **2250** by use of a drive rod **2240** and a structural member **2250**. For example, the actuator assembly **2240** may provide the fifth actuated degree of freedom **2250** (e.g., via the drive rod **2240** and the structural member **2250**). A sixth actuated degree of freedom **2250** may be coupled to the haptic glove **2008** by use of a mounting bracket **2260**. For example, the sixth actuated degree of freedom **2250** may be provided due to the interaction of the haptic glove **2008** with the mounting bracket **2260**. The manipulators **2240***a* and **2240***b* may produce grounded force feedback on a user's hands by use of haptic gloves **2008**. In example embodiments, the components in FIG. **22**A may function together to provide the various actuated degrees of freedom **2240**, **2240**, **2250**, **2240**, **2250**, **2250** as shown in FIG. **22**A and described in the disclosure.

[0441] The manipulators **2240***a* and **2240***b* may be coupled to a back support assembly **2232** that may include a structural frame **2234** and a padded backrest **2236**. Referring now to FIG. **22**B, in example embodiments, the structural frame **2234** may be coupled to a fluidic drive system **2228**, which may control the haptic gloves **2008**. In example embodiments, the upper-body exoskeleton **2200** may include at least two additional actuated degrees of freedom, including: a first actuated degree of freedom **2206** configured to permit rotation about the sagittal axis of a user's body, and a second actuated degree of freedom **2210** configured to permit rotation about the frontal axis of a user's body.

[0442] The actuated degree of freedom **2206** (e.g., first actuated degree of freedom **2206**) may include or be provided by a servomotor **2202** coupled to a gear reducer **2204**, which may in turn be coupled to a crossed roller bearing **2208** that permits the rotation about the sagittal axis of the user's body. The actuated degree of freedom **2210** (e.g., the second actuated degree of freedom **2210**) may include or be provided by a servomotor **2214** coupled to a gear reducer **2216**, which may in turn be coupled to a crossed roller bearing **2210** and a structural member **2212** having a geometry that may permit the rotation about the frontal axis of a user's body. In example embodiments, this second actuated degree of freedom **2210** may permit the rotation of distal elements of the upper-body exoskeleton **2200** about the axis of rotation of the second actuated degree of freedom **2210** without mechanical interference. In example embodiments, the structural frame **2234** may include structural extrusions **2228** and gussets **2230** that may support the upper-body exoskeleton **2200** and may couple it to the torso exoskeleton **2160** (e.g., as shown in FIG. **15**).

[0443] In example embodiments, the actuated degrees of freedom **2206**, **2210** (e.g., the first actuated degree of freedom **2206** and/or the second actuated degree of freedom **2210**) may be actuated to substantially match the orientation of the user's upper torso, as reported by a position sensor (e.g., a passive optical type of sensor, an active optical type of sensor, and/or an electromagnetic type of sensor), during operation of a whole-body haptic device. This may minimize the required workspace of manipulators **2240***a* and **2240***b*. In example embodiments, the actuated degrees of freedom **2206**, **2210** may be further actuated to match the orientation of a surface in a computer-mediated environment supporting a user's back, while the user may be sitting. For example, the actuated degrees of freedom **2206**, **2210** may be configured to match the angle of the seatback of a simulated reclining chair. In example embodiments, the orientation of another actuated degree of freedom of the upper-body exoskeleton **2200** may be configured to substantially match an orientation of a user's upper body (e.g., user's upper torso).

Interface Garment

[0444] FIG. **15** shows the interface garment **2006**, **2008** of the whole-body haptic device **2000**, in accordance with an example embodiment. In example embodiments, the interface garment may include at least a pair of haptic gloves **2008**, such as HaptX® Gloves. In an example embodiment, the interface garment **2006** may include a harness-like element coupled to the torso of the user **2002**, capable of substantially supporting the weight of the user **2002**. In another example embodiment, the interface garment **2006** may also include a load-bearing element coupled to the legs of the user **2002** to permit comfortable support of the body weight of the user **2002** in a sitting position. The interface garment **2006** may help ensure the safety of the user **2002**, supporting the user **2002** against accidental falls in the typical manner of a harness.

[0445] In an example embodiment, the interface garment **2006** may further include an interface laminate configured to stimulate the torso of the user **2002** with tactile or thermal feedback. The interface laminate may also be extended to the user's extremities, particularly the upper arms and upper legs.

Example Haptic Implementations with Actuator Assemblies

[0446] In example embodiments, FIGS. **23**-**33** show similar but slightly different implementations of the whole-body haptic device or system and its related components. The implementations of FIGS. **23**-**33** may utilize the same or similar components in order to provide the same or similar degrees of freedom to each of the user's feet, to each of the user's hands, and to the user's torso as shown in FIGS. **15**-**22**B and described in the disclosure. In general, the same or similar degrees of freedom to each of the user's feet, to each of the user's hands, and to the user's torso are being provided between the example embodiments in FIGS. **15**-**22**B and the example embodiments of FIGS. **23**-**33** with main differences relating to the approaches being used in terms of design configurations to provide the same or similar degrees of freedom to each of the user's feet, to each of the user's hands, and to the user's torso.

[0447] FIG. **23** shows another whole-body haptic device **2000** (may also be referred to as a whole-body haptic system **2000**) in accordance with an example embodiment that is similar to the whole-body haptic device **2000** of FIG. **15**. In example embodiments, the whole-body haptic device or system **2000** may be a holodeck type of system that includes a motion platform and a lower body exoskeleton that may provide various degrees of freedom. FIG. **23** shows the same or similar components in FIG. **15** that may provide the same functionality as described in the disclosure which may be implemented the same, in a similar manner, or slightly different from the implementations in FIG. **15**. For example, the user **2002** is shown using the whole-body haptic device **2000** which includes interface garments **2006**, **2008** (e.g., haptic glove **2008**), a structural frame **2010**, a motion platform **2030** having a base **2032**, a lower-body exoskeleton, and gantries **2048** (including first gantry **2050***a* and second gantry **2050***b*). The whole-body haptic device **2000** may also include footplates **2074***a*, **2074***b* and foot platforms **2128***a*, **2128***b* for each foot, respectively. The whole-body haptic device **2000** may also include a torso exoskeleton **2160** and an upper-body exoskeleton **2200** that may include a first manipulator **2240***a* and a second manipulator **2240***b*. The whole-body haptic device **2000** may also include pneumatic cylinder(s) **2164** and an upper structural frame **2176**. In example embodiments, the whole-body haptic device **2000** may include actuated degrees of freedom **2162** including a first coupled degree of freedom **2162***a* and a second coupled degree of freedom **2162***b*. As shown in FIG. **23**, the whole-body haptic device **2000** may include motion platform motors **2280** that may be attached to the base **2032**. In example embodiments, there may be only one motion platform motor **2280** with the other motion platform motors **2280** (e.g., as shown in FIGS. **23** and **24**) being replaced with wheels. In another example embodiment, there may be only one motion platform motor **2280** positioned towards the center of the motion platform **2030**. In example embodiments, the whole-body haptic device **2000** may include a foot platform actuator assembly **2310** that has foot platform actuator(s) **2314**. There may be a foot platform actuator assembly **2310** for each foot as shown in FIG. **23**. In example embodiments, the foot platform actuator assembly **2310** may be a Stewart platform type of actuator assembly (e.g., six foot platform actuators **2314**). These components may be configured as shown in FIG. **23** and may provide the same or similar functionality as described in the disclosure. Together these components provide degrees of freedom to each of the user's feet, to each of the user's hands, and to the user's torso as shown in FIG. **23** similar to the degrees of freedom to each of the user's feet, to each of the user's hands, and to the user's torso in FIG. **15** as described in the disclosure. It will be appreciated in light of the disclosure that the whole-body haptic system **2000** of FIGS. **15** and **23** and various portions contained therein and combinations thereof contain ornamental features individually and in combination that are separate and apart from the many functional aspects disclosed herein.

[0448] FIG. **24** shows the motion platform **2030** of the whole-body haptic device **2000** of FIG. **23** in accordance with an example embodiment. The motion platform **2030** may be configured to rotate similarly to the rotation of the motion platform **2030** in FIG. **15** but with a slightly different approach with the motion platform **2030** in FIG. **24** being circular in shape providing some ease in providing the rotation. As shown, the motion platform **2030** may include the base **2032**. In example embodiments, FIG. **17** also shows motion platform motors **2280** that may be attached to the base **2032** of the motion platform **2030**. These motion platform motors may be used in providing movement (e.g., rotational movement around a user's longitudinal axis) for the motion platform **2030**. In example embodiments, a closer detailed view of the motion platform motor is shown in FIG. **25**.

[0449] FIG. **26** shows a base assembly of the whole-body haptic device **2000** of FIG. **23** in accordance with an example embodiment. In this example, the base assembly may include the lower-body exoskeleton **2046** and the motion platform **2030** that has the base **2032**. As described in the disclosure, there may also be motion platform motors that may be attached to the base **2032** of the base assembly. The base assembly may also include gantries **2050***a*, **2050***b* as well as a foot platform actuator assembly **2310** (having foot platform actuator(s) **2314**) for providing movement for the user's feet and specifically degrees of freedom as shown in FIGS. **23**-**33** and described in the disclosure and similar degrees of freedom at each of the user's feet as shown and described for the example embodiment of FIG. **15**. The base assembly may also include footplates **2074***a*, **2074***b* and foot platforms **2128***a*, **2128***b* (e.g., first and second foot platforms) for each foot, respectively as described in the disclosure.

[0450] FIG. **27** shows at least one gantry **2048** and the footplate **2074** in accordance with an example embodiment. Each footplate **2074** may include the foot platform actuator assembly **2310** (having the foot platform actuator(s) **2314**) for providing movement of the foot platform **2128** in terms of degrees of freedom as described in the disclosure. In example embodiments, as shown in FIG. **28**, the foot platform actuator assembly **2310** may utilize the Stewart type of actuator assembly. Each of the platform actuators may include the force sensor **2120** (e.g., single or multi-axis) to assist with providing the degrees of freedom as described in the disclosure. FIG. **28** shows the foot platform actuator **2314** of the foot platform actuator assembly **2310** in accordance with an example embodiment.

[0451] FIG. **29** shows a hybrid example embodiment in relation to the example embodiment of FIG. **15** specifically the example embodiment of FIG. **20**A (e.g., showing a footplate **2074**). This hybrid example embodiment at least partially combines the example embodiment of FIG. **20**A with the example embodiment of FIGS. **26** and **28** in terms of using similar actuator assemblies **2312** as part of the footplates **2074***a*, **2074***b* and under the foot platforms **2128***a*, **2128***b*. In this example embodiment, there may be several actuated degree(s) of freedom **2158** provided such as an actuated degree of freedom configured to permit rotation about the vertical axis such as the longitudinal axis of a user's foot, an actuated degree of freedom configured to permit rotation about a sagittal axis of the user's foot, an actuated degree of freedom configured to permit rotation about the frontal axis of the user's foot, and an actuated degree of freedom configured to permit translation along an axis extending longitudinally through the user's foot of the lower-body exoskeleton **2046** of the example embodiment of FIG. **26**. As shown in FIG. **29**, this example embodiment may also include a servomotor **2076**, a gear reducer **2078**, and a crossed roller bearing **2090** which may assist in providing the various actuated degrees of freedom as similarly described in the disclosure for FIG. **20**A. For example, the crossed roller bearing provides for rotation of the foot platforms **2128***a*, **2128***b*. As described in the disclosure, there may also be the foot platform actuator assembly **2312** with foot platform actuator(s) **2316** to provide degrees of freedom in relation to the foot platforms **2128**. There may be only two foot platform actuator assemblies **2312** for each foot. The foot platform actuator assembly **2312** may be similar to the other foot platform actuator assembly **2310**, however, each foot platform actuator assembly **2312** includes three foot platform actuators (instead of six foot platform actuators) providing three degrees of freedom for each actuator assembly and allowing various locations of a user's foot to be positioned independently.

[0452] FIG. **30** shows a core assembly of FIG. **23** in accordance with an example embodiment. The core assembly may include the torso exoskeleton **2160** and the upper-body exoskeleton **2200** that may include the first manipulator **2240***a* and the second manipulator **2240***b*. The core assembly may also include the upper structural frame **2176**. As described in the disclosure, the core assembly may also include interface garments **2006**, **2008** (e.g., haptic glove **2008**), the pneumatic cylinder(s) **2164** (e.g., two pneumatic cylinders), and the back support assembly **2232** (having the structural frame **2234** and the padded backrest **2236**). These components may provide various degrees of freedom (e.g., actuated degree of freedom **2162**) for the core assembly such as a first coupled degree of freedom **2162***a* and a second coupled degree of freedom **2162***b* may together provide the actuated degree of freedom **2162** for the torso exoskeleton **2160**. As described in the disclosure, this actuated degree of freedom **2162** may be configured to permit translation along the longitudinal axis of the user's body.

[0453] FIG. **31** shows an interface portion of the core assembly of FIG. **30** in accordance with an example embodiment. This interface portion is similar to the example embodiment of FIG. **22**A. The interface portion may include the haptic glove **2008** and the upper-body exoskeleton **2200**. The upper-body exoskeleton **2200** may include the back support assembly **2232**, the structural frame **2234**, the padded backrest **2236**, and the manipulators (e.g., the first manipulator **2240***a* and the second manipulator **2240***b*). The upper-body exoskeleton **2200** may also include the actuator assembly **2240**, the drive rod **2240**, the structural members **2250**, **2250**, and the mounting bracket **2260**. These components of the interface portion including the components of the upper-body exoskeleton **2200** together provide for the various actuated degrees of freedom as shown in FIG. **31** and described in the disclosure. For example, the actuated degrees of freedom may include the first actuated degree of freedom **2240**, the second actuated degree of freedom **2240**, the third actuated degree of freedom **2250**, the fourth actuated degree of freedom **2240**, the fifth actuated degree of freedom **2250**, and the sixth actuated degree of freedom **2250**. In this example embodiment, there may also be a torso actuator assembly **2300** attached below the upper-body exoskeleton **2200** (e.g., attached to the padded backrest **2236** via its own torso backrest). As shown in FIG. **31**, the torso actuator assembly **2300** may include several actuators providing one or more degrees of freedom towards moving the user's torso. In this example embodiment, the torso actuator assembly **2300** may be the Stewart type of actuator assembly (e.g., six actuators) that may provide degrees of freedom such as six degrees of freedom. In example embodiments, the components in FIG. **31** may function together to provide the various actuated degrees of freedom **2240**, **2240**, **2250**, **2240**, **2250**, **2250** similar to the components in FIG. **22**A as shown in FIG. **31** and described in the disclosure. Each of these degrees of freedom may be achieved and provided in FIG. **31** using the same or similar approaches described in the disclosure with respect to FIG. **22**A.

[0454] FIG. **32** shows a torso portion of the core assembly of FIG. **30** in accordance with an example embodiment. As shown in previous figures and described in the disclosure, the torso portion includes the torso exoskeleton **2160**, the pneumatic cylinders **2164** (e.g., two pneumatic cylinders), the upper structural frame **2176**, the upper-body exoskeleton **2200**, and the torso actuator assembly **2300** (e.g., including several actuators such as six actuators for a Stewart type of actuator assembly). As shown in FIG. **32** and described in the disclosure, the torso exoskeleton **2160** may include the actuated degrees of freedom **2162** having the first coupled degree of freedom **2162***a* and the second coupled degree of freedom **2162***b* that may be provided at least partially by the use of the pneumatic cylinders. In example embodiments, these coupled degrees of freedom **2162***a*, **2162***b* may provide increased force output and rigidity for the actuated degree of freedom **2162**.

[0455] FIG. **33** shows a pneumatic actuator such as the pneumatic cylinder **2164** of the core assembly of FIGS. **30** and **32**. In example embodiments, the pneumatic actuator (e.g., pneumatic cylinder **2164**) may be a guided linear servo-pneumatic actuator for providing movement along the core Z-axis. Each pneumatic cylinder **2164** may provide for each coupled degree of freedom **2162***a* relating to the actuated degree of freedom **2162** for the torso exoskeleton **2160**.

Process Examples for Haptic Device Implementations

[0456] In example embodiments, there may be a motion platform toward the bottom of the whole-body haptic device **2000** such that its base **2032** may be secured to the ground while the motion platform may operate to rotate a portion of the whole-body haptic device **2000**. By way of these examples and with reference to FIGS. **15**, **17**, **23** and **24**, the motion platform may operate to rotate a portion of the whole-body haptic device **2000** to provide a single degree of freedom. By way of these examples, the single degree of freedom may be the bottom rotation such that the user **2002** of the whole-body haptic device **2000** may experience a yawing movement (or a spinning, rotational, etc. movement) about an axis that may be generally orthogonal to the ground on which the base **2032** of the whole-body haptic device **2000** may be secured. For example, rotation of the user **2002** and vast majority of the components of the whole-body haptic device **2000** may rotate (yaw, spin, etc.) in the same reference plan or reference frame.

[0457] In example embodiments, the whole-body haptic device **2000** may be configured such that the motion platform may operate to define actual degrees of freedom generally aligned for rotation around the longitudinal axis to the body. In these examples, the whole-body haptic device **2000** may be configured such that the motion platform may generally align the longitudinal axis of the body of the user **2002** with an axis that may be generally orthogonal to the ground on which the base **2032** of the whole-body haptic device **2000** may be secured. In further examples, the lower-body exoskeleton **400** may rotate around a central bearing and slip ring assembly (including, for example, the crossed roller bearing **2090** in FIG. **20**A) that may route electrical power and signal through the rotating junction of the setup.

[0458] In example embodiments, the whole-body haptic device **2000** may be configured so that such rotation about the motion platform may provide the whole-body haptic device **2000** with its sixth degree of freedom or in some configurations, rotation about the motion platform may provide the seventh degree of freedom. In example embodiments, the motion platform may have six degrees of freedom and those degrees of freedom may align with the actuator axes as shown in FIG. **17**, such that each of the six actuator axes provides a degree of freedom and adding rotation (or yawing, spinning, etc.) about the user may then be a seventh degree of freedom.

[0459] In example embodiments, the motion platform **2030** of the whole-body haptic device **2000**, as depicted in FIGS. **15**, **23** and **24**, may be configured and later adjusted to provide three degrees of freedom but as described in the disclosure, further functionality may additionally provide four, six or seven degrees of freedom. As described in the disclosure, multiple degrees of freedom may be experienced by each limb of the user, and independently by the torso of the user. It may be appreciated in light of the disclosure that there may be challenges moving such large amounts of mass while still providing the agility and response time required to seamlessly coordinate each of the degrees of freedom. In doing so, there may be significant mass (e.g., hundreds of kilos of mass) spread across the structures of the whole-body haptic device **2000**. With this reality in mind, the configuration of the motion platform **2030** may incorporate a specific mass distribution to increase the agility of the whole-body haptic device **2000**. By way of these examples, the rigidity may increase the responsiveness of the movement because of the reduced flexibility (e.g., increased rigidity) may make movements crisper when the frame flexes less through the movement. Further specifics with respect to allocating mass throughout the structure of the motion platform **2030** to increase its agility and to increase safety by reducing the power needed to move that mass are further described in the disclosure. In example embodiments, by proper arrangement of the degrees of freedom in a kinematic chain, it may result in a lowering of the mass being distributed across the structures of the whole-body haptic device **2000**.

[0460] As shown in FIG. **24**, in example embodiments, there may be motion platform motors **2280** to facilitate rotation of the platform but not all of the motion platform motors **2280** need to contain a motor. By way of these examples, some of the assemblies can contain passive guide wheels **2282** and then one or more motion platform motors **2280** can contain a side motor **2284** with a friction wheel **2286**, as shown in FIG. **25**. Returning to FIG. **24**, a central motor **2288** may also be used in lieu of one or more side motors **2284** (FIG. **25**) to also facilitate central rotation. In example embodiments, there may be a powered rotational degree of freedom about the axis either facilitated by one or more of the side motors **2284** or the central motor **2288**. In the various arrangements, the motion platform can include a central slip assembly **2290** through electrical and data connections can pass while the connections continue to rotate.

[0461] In example embodiments, as shown in FIG. **15** and described in the disclosure, there may be a six degree of freedom motion platform **2030**. For example, one of the degrees of freedom being the rotational (e.g., rotation about the longitudinal axis of the user's body). In example embodiments, the one degree of rotation around the longitudinal axis of the user's body may rotate the whole-body haptic device **2000**. In example embodiments, the whole-body haptic device **2000** may be on a three degree of freedom, a four degree of freedom, a six degree of freedom, or a seven degree of freedom motion platform **2030** such that three, four, six, or seven degrees of freedom may be configured. For example, in the four and seven degrees of freedom implementations, the whole-body haptic device **2000** may include a rotary actuator motor **2270** or a motion platform motor **2280** that may be positioned in the middle of the six degree of freedom motion platform **2030** (e.g., Stewart platform) or in the middle of the three degree of freedom Stewart motion platform **2030**. For example, the rotary actuator motor **2270** or the motion platform motor **2280** may spin causing the rotation of the platform. This rotary actuator motor **2270** or the motion platform motor **2280** may be positioned under the gantries (e.g., gantries **2048**, **2050***a*, **2050***b*) of the motion platform **2030**. For example, the entire upper assembly may rotate around the long axis of platform. The motor (e.g., the rotary actuator motor **2270** or the motion platform motor **2280**) may need to have sufficient power to accommodate the moving mass. For example, the motion platform **2030** may be designed to move significant masses (e.g., about hundreds or thousands of kilos) at relatively high accelerations such that the motor (e.g., rotary actuator motor **2270** or motion platform motor **2280**) may be configured to accommodate this type of movement of mass.

[0462] In example embodiments, the whole-body haptic device **2000** may be configured to provide six independent degrees of freedom which may be deployed at and bound by the movement of each of the feet and each of the hands of the user **2002**, as disclosed herein. By way of these examples, the combination of the six degrees of freedom of the motion platform **2030** may functionally be associated with each of the footplates **2074***a*, **2074***b*. It will be appreciated in light of the disclosure that the six degrees of freedom provided to each foot of the user **2002** may, in example embodiments, induce the feeling to the user **2002** that they are climbing a set of stairs. In doing so, the footplates **2074***a*, **2074***b* may be titled to induce the feeling of incline. In addition or in the alternative, the footplates **2074***a*, **2074***b* may independently but cooperatively adjust an angle of the foot plates **2074***a*, **2074***b* to simulate a motion up and down a rise and run of traditional stairs rather than only increasing ramp angle. It may be appreciated in light of the disclosure that the number of degrees of freedom may be deployed by combinations with other components of the whole-body haptic device **2000**. By way of these examples, the whole-body haptic device **2000** of FIG. **26** depicts, in example embodiments, six degrees of freedom provided at each of the footplate assemblies (e.g., footplates **2074***a*, **2074***b*) but also depicts further degree of freedom in that the portions of the lower-body exoskeleton **2046** may rotate by way of the motion platform motors **2280** or the rotary actuator motor **2270**. As such, components of the whole-body haptic device **2000** may cooperatively provide multiple degrees of freedom in the recreation of accurate motion. As such, these example embodiments may vary in design in terms of how these degrees of freedom may be combined and mixed together (e.g., example embodiments may distribute degrees of freedom in various ways).

[0463] In example embodiments, the lower body exoskeleton with its two actuated foot platforms **2128***a*, **2128***b* may be configured for substantially supporting a user's weight during the user's ambulation. In some example embodiments, each of the actuated foot platforms **2128***a*, **2128***b* in FIG. **15** may have five degrees of freedom that may share another (e.g., incorporate an additional) six degrees of freedom provided by further components of the whole-body haptic device **2000** depicted in FIG. **15**. In further example embodiments, each of the actuated foot platforms **2128***a*, **2128***b* in FIG. **23** may have seven degrees of freedom with six from each of the foot platforms **2128***a*, **2128***b* and a further degree of freedom provided by the rotary actuator motor **2270** or the motion platform motor **2280**. It will be appreciated in light of the disclosure that the configuration of the whole-body haptic device **2000**, which is depicted in FIGS. **23** and **26**, may further minimize the effects of moving mass as much as possible relative to the configuration of the whole-body haptic device **2000** depicted in FIGS. **15** and **18**. By way of these examples, the whole-body haptic device **2000** may be configured with a stack up of all the different degrees of freedom or in some examples, a single global rotation may provide the degree of freedom that was otherwise provided by the stack up or put another way, the coordinated movement across multiple degrees of freedom may simulate the single simulated degree of freedom. In example embodiments, the whole-body haptic device **2000** may move all of its system mass and, in addition, the mass of the user **2002**. In one example embodiment, the whole-body haptic device **2000**, which is depicted in FIGS. **23** and **26**, may be shown to represent a reduction of about 250 kilos relative to the configuration of the whole-body haptic device **2000** depicted in FIGS. **15** and **18**.

[0464] In example embodiments where the reduction of mass may be sought, each foot of the user **2002** may connect to something else besides two degrees-of-freedom gantries (e.g., gantries **2048**, **2050***a*, **2050***b*). By way of these examples, XY gantries may be deployed, e.g., a serial or parallel gantry (e.g., gantry **2048**, **2050***a*, **2050***b*), that may require relatively limited mass but may supply sufficient degrees of freedom alone or in combination with other systems. In several example embodiments, there may be a single degree of freedom such that each of the footplates **2074***a*, **2074***b* may have six degrees of freedom in the form of a pneumatic Stewart platform (as shown in FIG. **23**) for each foot platform **2128***a*, **2128***b*. In example embodiments, each platform (e.g., foot platform **2128***a*, **2128***b*) may further ride on a set pair of vertical rails such as pneumatic cylinder **2164** that move in and out, as shown generally at the coupled degrees of freedom **2162***a*, **2162***b* together providing the actuated degree of freedom **2162**. This may provide a redundant seventh degree of freedom hybrid manipulator for both of the actuated foot platforms **2128***a*, **2128***b*. For example, if a sixth-degree-of-freedom parallel system may be used, then there may be an additional seventh-degree-of-freedom that may move in the primary axis of motion, along the frontal axis of the user. This may be shown to provide additional improvements over other example embodiments in addressing relatively high forces, relatively low moving mass, relatively high speeds, and relatively high precision. In some examples, a non-redundant system may be altered to accommodate the full range of motion to users' legs to obtain six-degrees-of-freedom. By way of these examples, the seventh-degree-of-freedom may be added to provide improvements to the user's range of motion (e.g., side stepping, angular deviation such as stepping up on a rock, etc.). In example embodiments, the six-degrees-of-freedom may be provided by the Stewart type of platform with the redundant degrees along the frontal axis of the user in the direction the user may be moving (e.g., walking).

[0465] In the various example embodiments, the whole-body haptic device **2000** may include the lower-body exoskeleton **2046**, actuated foot platforms (e.g., two actuated foot platforms **2128***a*, **2128***b*), and key degrees of freedom of the platforms that are described herein. There may be several differences between some of the example embodiments as depicted in FIGS. **15** and **23**. By way of these examples, the degrees-of-freedom may be arranged differently in the two depicted versions. In some example embodiments, there may be five degrees of freedom per actuated foot platform **2128***a*, **2128***b* and in other example embodiments there may be six-degrees-of-freedom with the footplates **2074** (e.g., footplates **2074***a*, **2074***b*). There may be other examples providing degrees-of-freedom such that most example embodiments may present a six-degree-of-freedom controllable platform for each foot and that may be implemented in different ways. With some example embodiments, there may be implementations with the two exterior platforms and the seventh redundant linear degree of freedom, as depicted in FIG. **23**, which may be different than other example embodiments such as what is depicted in FIG. **15**. In example embodiments, the lower body exoskeleton **2046** may include the footplates **2074** such as footplates **2074***a*, **2074***b* (e.g., including the foot platforms **2128** such as **2128***a*, **2128***b*) for each foot that may include one or more torque sensors. The footplates **2074***a*, **2074***b* may control their position and orientation space using their gantry component in combination with the footplate component such as the foot platforms **2128***a*, **2128***b*.

[0466] In example embodiments, there may be similar modules such that the degree of freedom assemblies may include a six-degree-of-freedom motion platform **2030** supplied by pneumatics (e.g., a Stewart type of platform) and a linear pneumatic actuator (e.g., two Stewart actuation systems for each foot and one Stewart actuation system for the torso as shown in FIG. **23**). Such a configuration may be shown to provide a virtual motion platform between all of these degrees of freedom. For example, in a vehicle configuration, the z-axis may be lowered when the user may be in a sitting position. By way of these examples, pedaling may be simulated with the foot platforms **2128***a*, **2128***b* of the footplates **2074***a*, **2074***b*. In example embodiments, acceleration queuing may occur for vehicle simulations such that the whole-body haptic device **2000** may utilize acceleration simulation by substituting the gravity vector with some degree of tilt for the forward acceleration to use the experienced g-force to provide the needed feel. In example embodiments, there may be a set of ways to essentially trick the vestibular system by combining either tilts or tilts in a relatively small amount of forward acceleration or upper acceleration with washout to create that relative motion. These types of motions may be similar to what may be found in aerospace simulators to induce the feelings (sometimes incorrect feelings) of changing airplane attitude or altitude in the sky. For example, these degrees of freedom may be ranged (or constrained) such as to provide all the same capabilities of that motion platform but no longer have to move around the whole mass (in some examples over 350 kilos) in real time. There may be moving mass at the end effector to minimize the mass of the footplates, the back plate, and the relatively limited moving mass of the actuators. By way of these examples, such an arrangement may eliminate safety concerns that may be necessitated by some of the more complex parts of other example embodiments such as the magnetic floor coupling, which may not be required in other example embodiments. In example embodiments, this may provide at least one linear redundant degree of freedom as described in the disclosure.

[0467] In the various example embodiments, there may be the motion platform **2030**, the lower and upper body exoskeletons **2046**, **2200**, and at least two platforms (e.g., foot platforms **2128**) configured to substantially support each of the user's feet and their total body weight. By way of these examples, the upper body exoskeleton **2200** with two manipulators **2240***a*, **2240***b* may provide at least three actuated degrees of freedom. Example embodiments may be implemented in different ways such that some example embodiments may be relatively more decoupled and decrease the mass that may be moved. In some example embodiments, the motion platform **2030** may have one degree of freedom instead of seven degrees of freedom. There also may be footplates **2074***a*, **2074***b* which may each provide seven degrees of freedom that may be implemented in a hybrid design. In example embodiments, there may be six degrees of freedom provided in parallel rather than the five degrees of freedom in a serial hybrid rear design. Across example embodiments, there may be relatively similar motions. For example, same or similar force torque sensors may be provided along with similar controls. By way of these examples, a similar Z-axis may be implemented pneumatically providing a hybrid pneumatic, electro-mechanical approach. In example embodiments, a full six degrees of freedom may be provided in these examples.

[0468] In example embodiments, reducing the number of degrees of freedom may be a goal in some designs. In some example embodiments, extra motion platform degrees of freedom may be used. To that end, there may be additional mechanisms on top of the motion platform (e.g., motion platform **2030**) when extra motion may be preferred or needed. For example, the rotary electrical coupling may be configured to permit extra degrees of freedom such as the ability for continuous rotation to at least at about 720 degrees. In some examples, the orientation of the center of mass of the user may be adjusted. In some example embodiments, there may be mapping of the motion platform **2030** to the average angle of a ground plane and then the footplates **2074***a*, **2074***b* may be used to map some plates to deviations. For example, if a user is walking up the slope, the system may match the motion platform **2030** to the overall slope and the footplates **2074***a*, **2074***b* (including foot platforms **2128***a*, **2128***b*) that may be configured to provide (that is simulate) the small differences of rough terrain or stairs.

[0469] In example embodiments, the lower body exoskeleton **2200** may have similar or the same degrees of freedom. In example embodiments, a belt driven actuator (e.g., wheel robot such as a robot belt drive actuator) may be used and/or a pneumatic actuator may be used for the drive system. In another example, a cable driven actuator may be used for the drive system.

[0470] In example embodiments, the motion platform **2030** may have seven degrees of freedom. The motion platform **2030** may move to any position or orientation (e.g., within about a few hundred millimeters displacement at about 30 to about 40 degrees such as 35 degrees). In example embodiments, the motion platform **2030** including the user **2002** may be rotated. This may provide the seventh degree of freedom as combined degrees of freedom. For example, in example embodiments, the whole-body haptic device **2000** may rotate or spin around. In example embodiments, there may be six additional degrees of freedom that move the whole-body haptic device **2000**, including forward, backward, any other orientation, etc. Together this may result in seven degrees of freedom. For example, the Stewart type of motion platform **2030** at the base may provide six degrees of freedom in a relatively small area then a full rotation of that frame on top to result in seven degrees of freedom. Thus, it may be considered six degrees of freedom plus one degree of freedom as there may be some redundancy. The Stewart platform may rotate about 30 to about 40 degrees which may provide the six degrees of freedom. The seventh degree of freedom may be provided by the motor that provides the rotation as described in the disclosure such as the motion platform motor **2280** (e.g., motion platform yaw motor) or the rotary actuator motor **2270**. In some example embodiments, this motor may spin around the components on top of the Stewart platform. Thus, the entire system (e.g., whole-body haptic device **2000**) may rotate in addition to the six degrees of freedom being provided. In these example embodiments, the goal may be to provide each foot with six degrees of freedom independently at least, and then the center of gravity of the user (e.g., user's waist) may also provide six degrees of freedom independently of each foot. In these example embodiments, these degrees of freedom may be all combined and mixed together. In other example embodiments (e.g., as shown in FIGS. **23**-**33**), there may be less combinations between the degrees of freedom at each foot and at the center of gravity. Thus, in example embodiments, there may be six plus one degrees of freedom for the entire system (e.g., whole-body haptic device **2000**) plus several degrees of freedom for the feet and the center of gravity. For example, the degrees of freedom may relate to the feet and may relate to moving of the center of gravity around.

[0471] As described in the disclosure, FIGS. **15**, **18**, and **20**A show an example embodiment. For example, FIG. **18** shows the that footplates **2074***a*, **2074***b* may be moved by assembly to provide three degrees of freedom. In other example embodiments, as shown in FIG. **23**, this may occur in parallel with a two degree of freedom parallel mechanism. This may provide tilting in any orientation. In some example embodiments, there may be motion relating to tilt, side to side tilt, and/or front to back movements. Then, the rotation of each footplate **2074** around may result from the bottom rotation (e.g., by motion platform motor **2280** or the rotary actuator motor **2270**) as described in the disclosure. Thus, combined together the footplate **2074** on its own may have zero linear degrees of freedom but has all three of the rotational degrees of freedom. In example embodiments, the footplate **2074** may be oriented any which way and/or rotated around. The linear degrees of freedom in this example implementation may be provided by the gantry **2048** (e.g., as shown in FIG. **18**). In some example embodiments (e.g., as shown in FIG. **23**), the gantry **2048** may provide one degree of freedom per foot. In other example embodiments (e.g., as shown in FIG. **18**), the gantry may have two degrees of freedom per foot. These may be linear Cartesian degrees of freedom such that each footplate **2074** may be moved by these gantries **2048** forward, backward, left, right, etc. as described in the disclosure. Each foot may move forward, backward, left, or right such that there may be three rotational degrees of freedom. This may provide five degrees of freedom per foot. The degree of freedom that may not be present in some assemblies may be the Z-X axes (e.g., feet up and down).

[0472] In example embodiments, these movements may be based on a combination of the motion platform moving and the footplates moving. In example embodiments (e.g., as shown in FIG. **23**), simulating stairs may occur by moving one footplate **2074***a* forward and up and moving the other footplate **2074***b* backwards and down. In other example embodiments (e.g., as shown in FIG. **15**), the motion platform **2030** may be angled down, which may then drop the back foot down and may pull the front foot up. Then, a counter angle of the footplates **2074***a*, **2074***b* may occur such that the motion platform **2030** may be moved up and down as needed. Thus, a combination of the planted foot, the angling, and the counter angling may allow for the plant foot to stay in place while everything else moves. The entire system may be moving around the user's body and the front may provide counter angling such that when the user steps up, they may be able to move up from the user's perspective. The footplates **2074** may be able to move up and down by leveraging these combined degrees of freedom. The seven degrees of freedom in the motion platform **2030** plus the five degrees of freedom per foot may all add up to six degrees of freedom per foot.

[0473] With these example embodiments, the motion of the base may affect the center of gravity. These degrees of freedom may be conflated in some example embodiments (e.g., as shown in FIG. **15**). In other example embodiments (e.g., as shown in FIG. **23**), the degrees of freedom may be separated more distinctly. In example embodiments, some of these degrees of freedom may essentially compensate for the fact that the motion platform **2030** may be moving everything and the degrees of freedom may be moving relative to the motion platform **2030**. For example, when taking a step onto a stair, the entire system may have to reorient itself to keep up with the user **2002**. In example embodiments, each foot platform **2128** may bend forward or backward or side to side when in use which may be similar to movement of bike pedals. This may occur from the user's body being locomotive and from the influence of gravity (e.g., the user may be falling forward with the user being fixed at the waist). In example embodiments, there may be a plane that intersects the center of rotation of each foot that may be no greater in angle than the motion platform.

[0474] As described in the disclosure, FIGS. **21**A-**21**B, **22**A-**22**B, and **23** show example embodiments. In example embodiments, there may be a motion providing movement along the longitudinal axis of a user's body (e.g., a combined electromechanical and pneumatic actuation may be used to provide this motion). In other example embodiments, a pneumatic system only may be used such as a single pneumatic cylinder server. With this longitudinal axis motion, there may be movement raising the entire torso up and down. For example, there may be two degrees of freedom that may angle the back plate (e.g., back support assembly **2232** in FIG. **22**A). This may provide angular motion around the frontal and sagittal axes. The two degrees of freedom may be provided and then each of the arms of the haptic glove **2008** may provide six degrees of freedom (e.g., force feedback to the user's arms). FIG. **22**A shows an upper-body exoskeleton **2200** that may include two arms that may connect the haptic glove **2008** and the back support assembly to a pneumatic controller such as the fluidic drive system **2228** (e.g., as shown in FIG. **22**B). The upper-body exoskeleton **2200** may provide a minimum viable fully immersive motion and force feedback experience. There may be six degrees of freedom provided per arm. For example, the user may virtually ride a motorcycle or a horse. For example, the vehicle simulation may be provided with the motion platform. In example embodiments, this implementation may use a configured robotic type of arm that may provide two degrees of freedom in the shoulder assembly. There may be one degree of freedom in the elbow and there may be three degrees of freedom in the wrist. In other example embodiments, the kinematics may be designed to provide three full degrees of freedom with respect to one elbow, two elbows, and one wrist. Even so, with this example embodiment, six degrees of freedom for the back arm may still be provided. There may be different ways to provide these degrees of freedom, but the specific degree combinations may be generally the same or similar. In most example embodiments, the whole-body haptic device **2000** may provide a full six degrees of freedom, independent control for both feet, both arms, and the torso providing the ability to reproduce most (if not all) human motions or human activities.

[0475] In example embodiments, there may be a tracker on the foot to identify where each foot may be located. Each footplate **2074** may be used to project the position of that tracker onto a virtual ground where the footplate **2074** may be located. Thus, each footplate **2074** may not be following each foot but each footplate **2074** may be following a shadow that each foot projects onto the ground surface. This may be an angle that may be essentially a two-dimensional or a two and a half dimensional representation of the shadow.

[0476] In example embodiments (e.g., as shown in FIGS. **15**, **23** and **27**), extra redundant degrees of freedom may be provided in the Stewart type of motion platform to provide improved mobility. For example, nine degrees of freedom may be provided for example embodiments. For example, this may utilize a hybrid system providing extra mobility. In other example embodiments, there may be three-linear degrees of freedom and three rotational degrees of freedom. In other example embodiments, there may be two coupled three degree of freedom platforms. In example embodiments, as shown in FIG. **27**, there may be an intermediate hybrid type design that instead of having one six degree of freedom Stewart motion platform, there may be two three degrees of freedom Stewart platforms in parallel, both of which Stewart platforms are in series with one rotational degree of freedom. This example embodiment may provide some flexibility in relation to and between the ball and heel of a user's foot which do have some semi-separate movement with respect to each other and the entire foot.

[0477] The example embodiments of FIGS. **15-22**B may be compared to the example embodiments of FIGS. **23-33** in terms of degrees of freedom. For example, the motion platform of FIGS. **15-22**B may provide six plus one degrees of freedom that may include redundant rotation around a longitudinal axis of a user's body whereas the motion platform of FIGS. **23-33** may provide one degree of freedom that may include rotation around the longitudinal axis of a user's body. For the lower body exoskeleton, FIGS. **15-22**B may provide five degrees of freedom per foot including no independent motion along the user's longitudinal axis, whereas FIGS. **23-33** may provide six plus one degrees of freedom per foot including redundant movement along the user's frontal axis. For the actuated motion at the user's torso (e.g., related to the upper body exoskeleton), FIGS. **15-22**B may provide two degrees of freedom whereas FIGS. **23-33** may provide six plus one degrees of freedom including redundant movement along the user's longitudinal axis. In example embodiments, FIGS. **15-22**B include the motion platform **2030** that may match a ground plane and actuated motion at the user's torso may accommodate motion required by a gravity vector, and the lower body exoskeleton may accommodate deviations required by topology (e.g., stairs or pedaling simulation). In example embodiments, FIGS. **23-33** may provide independent control of movement of the user's torso and each of user's feet in six plus one degrees of freedom each.

Haptic Glove and Interface System

[0478] A haptic feedback glove, which may be an example embodiment of a wearable haptic glove **236**, is described below. An interface system **3100** for the haptic feedback glove, which may be an example embodiment of at least some of the components of a haptic interface system **230**, is further described below.

[0479] Haptic feedback gloves have broad commercial applications, including in entertainment, medical and industrial training, and computer-aided design and manufacturing. The human-computer interfaces described herein have generality, realism, and practicality.

[0480] Generality refers to human-computer interfaces with general applicability and have improved flexibility, adaptability, and economy of scale over conventional human-computer interfaces.

[0481] Realism refers to realistic touch sensation with multiple sensory modalities for natural interaction, such as cutaneous feedback (mechanical stimulation of the skin) and kinesthetic feedback (net forces applied to the musculoskeletal system). This realism is due, in part, to the resolution, displacement, frequency response, force output, and other performance characteristics to realistically stimulate a particular sensory modality as described herein.

[0482] Practicality refers to a light and low-profile design that is comfortable enough to be worn on the hand, and robust enough to survive repeated use in a real-world environment. Features described herein lower the cost of the human-computer interface to promote commercial practicality. Even more features described herein increase the speed at which gloves of the human-computer interface may be donned and doffed.

[0483] As used herein, the term "haptic feedback glove" means: a hand portion of a human-computer interface garment. As used herein, the term "finger" means: a digit of the hand, including the thumb. "Digit" and "finger" are used interchangeably throughout the present application. As used herein, the term "mechanical ground" means: a point that is substantially fixed and immovable with respect to a finger of the user. As used herein, the term "position sensor" means: a sensor configured to detect at least one of position and orientation.

Overview of Human-Computer Interface System

[0484] Referring now to FIG. **34**, a schematic diagram illustrates a human-computer interface system **3100** in accordance with the teachings of the present disclosure. Human-computer interface system **3100** includes a drive system **3110**, gloves **3112** for the left and right hands, a power supply **3114**, a headset **3120**, transmission sources **3122**, a workstation **3126**, and an external air supply **3128**.

[0485] Human-computer interface system **3100** may be operated in a wireless configuration or in a wired configuration. The wireless configuration uses wireless data communication and a battery to provide power. The wired configuration uses an optional power connection for indefinite operation, an optional USB connection for data communication, and an optional external pneumatic connection for providing additional compressed air.

[0486] Drive system **3110** includes an air supply, a battery, valves, and electronics to support operation of human-computer interface system **3100**, as will be described below. The electronics selectively actuate the valves to control the flow of air to each of gloves **3112**.

[0487] Drive system **3110** is wearable on the body of the user and the components of drive system **3110** include features that reduce the weight, size, cost, and noise of drive system **3110** relative to conventional systems, as described below.

[0488] Gloves **3112** provide haptic feedback to the left and right hands of the user while permitting finger and wrist movement. Gloves **3112** include features such as soft thimbles and manipulation actuators that are improvements over conventional haptic feedback gloves, as discussed below.

[0489] In some embodiments, gloves **3112** include an inner glove or interface layer and an outer glove or veneer layer, as illustrated below. An undersuit glove may be donned by the user before donning gloves **3112** to prevent direct skin contact between the user and the inside of the haptic feedback glove. The use of an undersuit glove reduces the need to clean the haptic feedback glove and offers improved hygiene, particularly in cases where a single haptic feedback glove is shared between multiple users.

[0490] Power supply **3114** may be any suitable electrical power conversion component to couple available electrical power and electronics of drive system **3110**. For example, power supply **3114** may convert 240 volt split phase or 120 volt single phase alternating current electrical power into 24 volt direct current electrical power for use by drive system **3110**.

[0491] Headset **3120** may be any visual interface or display capable of presenting visual images to the user of human-computer interface system **3100**. In the example provided, headset **3120** is an immersive reality headset, which may be commercially described as a virtual reality headset, augmented reality headset, or mixed reality headset.

[0492] Transmission sources **3122** may be used to communicate external data to gloves **3112**.

[0493] Workstation **3126** may be any computer capable of running software to simulate a virtual environment. In the example provided, workstation **3126** connects to drive system **3110** through a wireless data network.

[0494] External air supply **3128** may be any source of pressurized air or other inert gas. For example, external air supply **3128** may be a separate compressor or pressurized gas supply.

[0495] In the example provided, drive system **3110**, gloves **3112**, and power supply **3114** are packaged together to be acquired by users as a single product. In such an example, the headset **3120**, transmission sources **3122**, workstation **3126** hardware, and external air supply **3128** are supplied by the user using other commercially available products. In some embodiments, fewer or more of the components are provided as the single product.

[0496] Various communication paths connect the components of human-computer interface system **3100**. In the example provided, the communication paths include wireless data paths **3130**, wireless data path **3132**, wired data path **3134**, wired data path **3136**, wired data path **3138**, electrical power line **3140**, glove pneumatic conduits **3142**, external pneumatic supply conduit **3144**, and external pneumatic exhaust conduit **3146**.

[0497] It should be appreciated that any of the data paths may be wireless or wired without departing from the scope of the present disclosure. Wireless data paths **3130** couples transmission sources **3122** with gloves **3112** for communication. Wireless data path **3132** couples drive system **3110** with workstation **3126** for communication. Wired data path **3134** couples drive system **3110** with gloves **3112** for communication. Wired data path **3136** couples headset **3120** with workstation **3126** for communication. In the example provided, wired data path **3138** is a USB connection that couples drive system **3110** with workstation **3126** for communication.

[0498] Power communication line **3140** may be any suitable power cable compatible with drive system **3110** and power supply **3114**. Glove pneumatic conduits **3142** may be any suitable pneumatic conduits (e.g., flexible tubing) capable of communicating pressurized air between drive system **3110** and gloves **3112**. External pneumatic supply conduit **3144** and external pneumatic exhaust conduit **3146** may be any suitable pneumatic conduits (e.g., flexible tubing) capable of communicating pressurized air between drive system **3110** and external air supply **3128**.

Electronics of Human-Computer Interface System

[0499] Referring now to FIG. **35**, and with continued reference to FIG. **34**, a schematic diagram illustrates an electronics system **3200** suitable for use with the haptic glove system of FIG. **34** in accordance with the teachings of the present disclosure. Electronics system **3200** includes various circuitry implemented, for example, with printed circuit boards (PCBs) and flexible printed circuit boards (FPCBs or FCBs).

[0500] Electronics system **3200** includes a peripheral assembly **3210**, a main control board **3212**, battery management circuitry **3214**, a wireless antenna **3216**, a pressure sensor board **3218**, valve boards **3220**, workstation **3126**, and a wireless hub **3222**.

[0501] Peripheral assembly **3210** includes a position sensor system **3224**, peripheral circuitry **3234**, and a motion tracker **3238**.

[0502] Position sensor system **3224** includes position sensors **3232**, magnetic emitter **3230**, and position sensor circuitry **3236**. Position sensors **3232** are receiving coil sensors that receive a magnetic field emitted by magnetic emitter **3230** to provide finger position tracking with six degrees of freedom. In some embodiments, magnetic emitter **3230** is located remotely from the user.

[0503] In some embodiments, position sensor system **3224** includes force sensors configured to transduce a point force on the user's skin, or a net force/torque on a digit of the user's hand to enable closed loop force control. In some embodiments, position sensors **3232** and magnetic emitter **3230** are replaced with an optical sensor.

[0504] Position sensor circuitry **3236** includes digital signal processing and USB communications, ADC/Analog input circuitry, sensor drive circuitry, and power management.

[0505] Peripheral circuitry **3234** provides manipulation actuator control, position sensor interconnects, a USB hub, and power management.

[0506] Motion tracker **3238** provides absolute position of the hand. For example, motion tracker **3238** may provide the absolute position of the hand in space, while the magnetic emitter **3230** and position sensor **3232** provide finger positioning relative to the absolute position. Motion tracker **3238** may be any commercially available motion tracker, such as the VIVE tracker commercially available from HTC.

[0507] Main control board **3212** is disposed in drive system **3110**. Main control board **3212** includes a primary processor, a high voltage power supply, an air controller, power management, a US hub, and a wireless transceiver.

[0508] Battery management circuitry **3214** controls charging and discharging monitoring and control of a battery of drive system **3110**. In the example provided, battery management circuitry **3214** is enclosed in a fire enclosure with the battery. In some embodiments, battery management circuitry **3214** and the battery are configured to manage power consumption of about 20 W to about 40 W for greater than four hours of use of human-computer interface system **3100**.

[0509] Wireless antenna **3216** receives wireless signals for data communication and passes the wireless signals to main control board **3212**, such as by a coaxial cable.

[0510] Pressure sensor board **3218** measures the pressure at various locations in human-computer interface system **3100**. Valve boards **3220** provide actuation control of valves within drive system **3110**. In the example provided, valve boards **3220** manage 118 Air Channels with one microcontroller used for every two channels.

Production Test Flow for Human-Computer Interface System

[0511] Referring now to FIG. **36**, and with continued reference to FIGS. **34-35**, a simplified flow diagram illustrates a production test flow **3300** in accordance with the teachings of the present disclosure. Production test flow **3300** includes a peripheral test flow **3302** and a drive system test flow **3304**.

[0512] Peripheral test flow **3302** includes umbilical process **3310**, glove process **3312**, position sensor process **3314**, and peripheral assembly process **3316**. Umbilical process **3310** includes an umbilical build task **3320** and a blockage test **3322**.

[0513] Umbilical build task **3320** utilizes an automated robot to assemble tubing within connectors of a multi-channel pneumatic connector in an umbilical that will connect drive system **3110** with gloves **3112**. For example, the robot may assemble the tubing as discussed below with reference to FIGS. **55-56**. In some embodiments, the automated robot is a six-axis automated robot. In the example provided, in a first task of umbilical build task **3320**, a three-axis robot places the multi-channel pneumatic connector and a top manifold of a wrist mount assembly in a holding device or nest. In a second task of umbilical build task **3320** of the example provided, a six-axis robot pulls tubing from a spool to the designed length and pinches the tubing with fixturing to hold firmly at the measured length. In a third task of umbilical build task **3320** of the example provided, a pivoting and retractable blade cuts the tubing. In a fourth task of umbilical build task **3320** of the example provided, a six-axis robot places a first end of the tube vertically down into the multi-channel pneumatic connector and other end of the tube into the wrist mount top manifold. In a fifth task of **3320** of the example provided, a two-axis robot applies glue to both end pieces. In a sixth task of umbilical build task **3320** of the example provided, a three-axis robot removes the assembly from the holding device or nest. In the example provided, umbilical build task **3320** makes **60** connections.

[0514] Blockage test **3322** measures air movement through pneumatic conduits between end points of an umbilical that will connect drive system **3110** with gloves **3112**.

[0515] Glove process **3312** includes an inflation test **3324** for identifying tactor cross-talk and inflation problems. For example, glove process **3312** may utilize the computer vision testing further discussed with reference to FIG. **61** below.

[0516] Position sensor process **3314** includes electrical testing and calibration **3326**. For example, the electrical testing may include analog and/or digital printed circuit board assembly electrical testing, break out board testing, and calibration of the position sensor.

[0517] Peripheral assembly process **3316** includes a glove assembly task **3330** and a peripheral completion task **3332**. Glove assembly task **3330** assembles gloves **3112** as will become apparent with reference to FIGS. **71-79** below. Peripheral completion task **3332** may include peripheral definition, position sensor visualizer, and glove leak-down testing.

[0518] Drive system test flow **3304** includes controller process **3340**, core plastics process **3342**, and drive system assembly process **3344**.

[0519] Controller process **3340** includes electrical testing **3350**. For example, electrical testing **3350** may include main PCBA electrical tests, channel sensor PCBA tests, and valve PCBA electrical tests. In some embodiments, a main PCBA electrical test includes program channel control and motor and logic control tests.

[0520] In the example provided, electrical testing **3350** includes electronic design verification tests for drive system **3110**. Electrical testing **3350** includes at least: a WiFi Functional Test, pairing, range, interference immunity, USB communication, communication functionality, and bootload functionality. In the example provided, electrical testing **3350** includes drive channel functional and performance verification for frequency response, stability, and power on/off.

[0521] In the example provided, electrical testing **3350** for the compressor includes functional and performance verification for frequency response, stability, and power consumption. In the example provided, electrical testing **3350** for the display functional and performance verification includes display tests and wake/sleep verification.

[0522] In the example provided, electrical testing **3350** for power management includes voltage supply verification, current limiting, and supply on/off timing. In the example provided, electrical testing **3350** for primary processor functionality includes log file read/write, reset, and peripherals/communication buses. In the example provided, electrical testing **3350** for safety and compliance includes high voltage cutoff (case open), current Limiting/in-rush current, EMC/ESD pre-scan of prototypes, and thermal cutoff/shutdown.

[0523] Core plastics process **3342** includes an assembly task **3352** to assemble and glue manifold plastics, a pressure test **3354** for the assembled manifold plastics, an assembly task **3356** to assemble and glue the compressor module, and a pressure flow test **3358** for the compressor module.

[0524] Drive system assembly process **3344** includes an assembly task **3352** and a drive system completion task **3362**. Assembly task **3352** includes assembly of the drive system, as will become apparent with reference to FIG. **37** below. Drive system completion task **3362** may include software revision verification, adjustment of pressure and flow rate, lead-down and leak-up testing, cross-talk testing, blockage testing, channel mapping, and USB port enumeration.

Drive System

[0525] Referring now to FIG. **37**, and with continued reference to FIGS. **34-36**, components of drive system **3110** are illustrated in an exploded view in accordance with the teachings of the present disclosure.

[0526] Drive system **3110** includes a bottom enclosure **3410**, a top enclosure **3412**, an input/output board **3414**, a fan **3416**, a compressor **3418**, a battery **3420**, a strap system

3422, a valve core **3424**, an LCD display **3426**, a power switch **3428**, and a glove connector **3430**.

[0527] Bottom enclosure **3410** defines a cavity into which other components of drive system **3110** may be placed. For example, bottom enclosure **3410** may integrate valve core **3424**, compressor **3418**, battery **3420**, multi-channel pneumatic connectors (not shown in FIGS. **37-41**), fan **3416**, and strap system **3422**. In the example provided, valve core **3424**, compressor **3418**, and fan **3416** are captured in features defined by bottom enclosure **3410**. Bottom enclosure **3410** further defines slots **3432**, as can best be seen in FIG. **41** and are described below.

[0528] Top enclosure **3412** secures to bottom enclosure **3410** to enclose and protect the components in a housing or case. For example, top enclosure **3412** may secure to bottom enclosure **3410** with four to eight short screws to promote serviceability. Top enclosure **3412** further defines a window **3434**.

[0529] Input/output board **3414** includes connections for data, power, and control. In the example provided, output board **3414** has a USB-Type B port, power supply port, and on/off buttons. Fan **3416** moves air through drive system **3110** to cool components within the housing.

[0530] Strap system **3422** includes an attachable and detachable set of straps, as discussed below with reference to FIG. **41**. The straps include a padded shoulder strap **3440** and padded hip strap **3442**. Padded shoulder strap **3440** includes padded arm straps that extend from an upper padded section to padded hip strap **3442**. Padded hip strap **3442** interfaces with bottom enclosure **3410** using metal plates **3444** that slip through slots in the Bottom Enclosure, as will be discussed below with reference to FIG. **41**.

[0531] Referring now to FIG. **38**, and with continued reference to FIGS. **34-37**, a block diagram illustrates additional wiring and electronics of drive system **3110** in accordance with the teachings of the present disclosure. Drive system **3110** further includes a main control board **3510**, a pressure transducer PCBA **3512**, and a valve distribution PCBA **3514**. Main control board **3510** includes a wireless networking module **3520** (e.g., Wi-Fi).

[0532] Main control board **3510** includes wireless networking module **3520** for wireless data communication, such as by Wi-Fi. Pressure transducer PCBA **3512** measures pressure within various parts of drive system **3110**. Valve distribution PCBA **3514** controls pneumatic valves of drive system **3110**, as shown below. The electronics of drive system **3110** communicate across various data communication paths **3530** internal to drive system **3110** and are powered by various low voltage DC power paths **3532** internal to drive system **3110**.

Valve Core-Compressor Assembly

[0533] Referring now to FIGS. **39-40**, and with continued reference to FIGS. **34-38**, a valve core-compressor assembly **3600** is illustrated in exploded views with compressor **3418** in relation to valve core **3424**.

[0534] Compressor **3418** includes pneumatic circuit body **3610**, top cover **3612**, bottom cover **3614**, and two diaphragm pumps **3616**. Valve core **3424** includes two manifold assemblies **3712** and is coupled to compressor with gasketed channels **3710** and tie rods **3720**. In the example provided, valve core **3424** defines tie rod through-holes **3722** and pneumatic circuit body **3610** defines tie rod through-holes **3722** through which tie rods **3720** pass to couple valve core **3424** to compressor **3418**.

[0535] Gasketed channels **3710** insert into valve core **3424** on one end and insert into pneumatic circuit body **3610** of compressor **3418** on the other end. Gasketed channels **3710** permit pneumatic communication between compressor **3418** and valve core **3424** while restricting air leaks with O-rings **3718** and attenuating vibrations from compressor **3418**.

[0536] The two manifold assemblies **3712** are mounted together using interlocking features in the plastics further secured with screws for serviceability. Main control board **3510** is mounted to the top of valve core **3424** using snap arms. In the example provided, main control board **3510** interfaces with the valve distribution PCBA **3514** and pressure transducer PCBA **3512** using board edge connectors.

Backpack Strap Assembly

[0537] Referring now to FIG. **41**, and with continued reference to FIGS. **34-40**, components of a strap system are illustrated in accordance with the teachings of the present disclosure. For example, the strap system may be strap system **3422** illustrated in FIG. **37**.

[0538] In the example provided, metal plates **3444** define slots **3810** through which strap loops **3812** pass. Strap loops **3812** may be formed, for example, from padded shoulder strap **3440** and/or padded hip strap **3442**. Metal plates **3444** then slide into slots **3432** of bottom enclosure **3410** and restrict strap loops **3812** from pulling away from bottom enclosure **3410**.

[0539] Metal plates **3444**, slots **3810**, and strap loops **3812** define an attachment feature configured to accommodate removal and re-installation of strap system **3422**. It should be appreciated that other attachment features may be incorporated without departing from the scope of the present disclosure.

Manifold Assemblies

[0540] Referring now to FIGS. **42-44**, exploded views illustrate manifold assemblies of a valve core in accordance with some embodiments. For example, manifold assemblies may be manifold assemblies **3712** of valve core **3424**.

[0541] Manifold assemblies **3712** each include a bottom manifold **3910**, a middle manifold **3912**, and top manifold **3914**. Bottom manifold **3910** defines valve cavities **3920** for each valve to be inserted into bottom manifold **3910**. Middle manifold **3912** is glued to bottom manifold **3910** and three of top manifolds **3914**. Top manifolds **3914** include channel aperture portions **3922** that each defines a plurality of channel apertures selectively coupled with pressurized air within manifold assemblies **3712** by actuation of valves **4110** illustrated in FIG. **44**.

[0542] In the example provided, bottom manifold **3910**, middle manifold **3912**, and top manifolds **3914** are injection molded and glued such that manifold assemblies **3712** hold a working pressure of 30 psi with a safety factor of 6×.

[0543] As seen in FIG. **43**, valve boards **4010**, valve housings **4012** defining valve cavities **4014**, sensor gaskets **4020**, channel sensor boards **4022**, manifold connector gaskets **4024**, and manifold connectors **4026** are assembled with manifold assemblies **3712**.

[0544] Three valve boards **4010** are connected to bottom manifold **3910** with 15 valve housings **4012** collectively defining 60 valve cavities **4014** per manifold assembly **3712**.

[0545] Six sensor gaskets **4020**, three channel sensor boards **4022**, six manifold connector gaskets **4024**, and six manifold connectors **4026** are coupled to manifold assembly **3712**.

[0546] Each of top manifold **3914**, sensor gaskets **4020**, manifold connector gaskets **4024**, and manifold connectors **4026** defines channel apertures **4030**. The channel apertures **4030** collectively couple the plurality of channel apertures of channel aperture portions **3922** to a of tubing harness, where each channel aperture **4030** is coupled with a respective tube of a tubing harness, as best seen in FIG. **81**. Accordingly, manifold assembly **3712** integrates many channels very tightly and enables a very dense assembly capable of high-performance proportional balance.

[0547] As seen in FIG. **44**, valves **4110** each have two electrically actuated benders **4112**. Each bender selectively blocks a very small air channel of channel aperture portions **3922**. This arrangement forms two independent 2-2 valves for fully proportional control of the flow in each individual channel with very high fidelity.

Compressor

Pneumatic Flow Circuit

[0548] Referring now to FIG. **45**, and with continued reference to FIGS. **34-44**, a pneumatic diagram illustrates a pneumatic circuit **4200** for a compressor. For example, pneumatic circuit **4200** may illustrate pneumatic flow for compressor **3418**. Pneumatic circuit **4200** illustrates how compressor **3418** uses pressure swing adsorption for pulling moisture out of the compressed air. Generally, pressurized air is passed through a desiccant where the desiccant beads will absorb water in the pressurized air. A separate airflow at a low pressure is simultaneously regenerating or taking the moisture back out of a second desiccant. In the example provided, a vacuum generated by compressor **3418** creates the low pressure to regenerate the desiccant faster and more fully than would be achieved with atmospheric pressure.

[0549] Pneumatic circuit **4200** includes a compressor and cooler portion **4210**, inlet/outlet portion **4212**, pneumatic conduits **4214**, and desiccant portion **4220**. In the example provided, compressor and cooler portion **4210** includes diaphragm pumps **3616**. Inlet/outlet portion **4212** defines pressurized air supply and exhaust to valve core **3424**.

[0550] Pneumatic conduits **4214** couple the various parts of compressor **3418** to each other for pneumatic communication. In the example provided, pneumatic conduits **4214** are at least partially defined by pneumatic circuit body **3610**, as will be described below with reference to FIG. **47**.

[0551] Desiccant portion **4220** includes a first valve **4230**, a second valve **4232**, a third valve **4234**, a fourth valve **4236**, a first desiccant portion **4240**, and a second desiccant portion **4242**. Desiccant portion **4220** provides dry air to valve core **3424** with use of one of first desiccant portion **4240** or second desiccant portion **4242** while regenerating the other of first desiccant portion **4240** or second desiccant portion **4242** for continuous flow of dry air.

[0552] The valves **4230**, **4232**, **4234**, **4236** operate to selectively provide a usable airflow **4250** and a desiccant regeneration airflow **4252**. For the airflows illustrated with a first valve configuration, first valve **4230** directs usable airflow **4250** from compressor and cooler portion **4210** to first desiccant portion **4240** and fourth valve **4236** directs usable airflow **4250** from first desiccant portion **4240** to valve core **3424**. For desiccant regeneration airflow **4252**, third valve **4234** directs some bypass air from usable airflow **4250** through second desiccant portion **4242** and second valve **4232** directs desiccant regeneration airflow **4252** to a vacuum of compressor and cooler portion **4210**. Accordingly, usable airflow **4250** provides a steady stream of dry air using first desiccant portion **4240** while desiccant regeneration airflow **4252** dries second desiccant portion **4242**.

[0553] When first desiccant portion **4240** becomes saturated, usable airflow **4250** and desiccant regeneration airflow **4252** may be directed along different paths by a second valve configuration (not illustrated). For example, first valve **4230** may direct usable airflow **4250** from compressor and cooler portion **4210** to second desiccant portion **4242** and fourth valve **4236** may direct usable airflow **4250** from second desiccant portion **4242** to valve core **3424**. In this configuration, third valve **4234** may direct desiccant regeneration airflow **4252** as bypass air from usable airflow **4250** to first desiccant portion **4240** and second valve **4232** may direct desiccant regeneration airflow **4252** from first desiccant portion **4240** to the exhaust of compressor and cooler portion **4210**.

[0554] Referring now to FIG. **46**, and with continued reference to FIGS. **34-45**, a pneumatic diagram illustrates a pneumatic circuit **4300** for a compressor. For example, pneumatic circuit **4300** may illustrate a pneumatic flow for compressor **3418**. Pneumatic circuit **4300** is similar to pneumatic circuit **4200**, where like numbers refer to like components. Pneumatic circuit **4300**, however, includes a first desiccant portion **4310** and a second desiccant portion **4312** that create a usable airflow **4314** and a desiccant regeneration airflow **4316**.

[0555] First desiccant portion **4310** includes a first valve **4320**, a second valve **4322**, a third valve **4324**, a fourth valve **4326**, a first desiccant portion **4328**, and a second desiccant portion **4329**. Second desiccant portion **4312** includes a first valve **4330**, a second valve **4332**, a third valve **4334**, a fourth valve **4336**, a first desiccant portion **4338**, and a second desiccant portion **4339**.

[0556] Usable airflow **4314** continuously flows through one of first desiccant portion **4328** or second desiccant portion **4329**. Desiccant regeneration airflow **4316** continuously flows through one of first desiccant portion **4338** or second desiccant portion **4339**. First desiccant portion **4338** and second desiccant portion **4339** may be reduced in size relative to desiccants of pneumatic circuit **4200** and desiccant regeneration airflow **4316** may have a reduced flow.

[0557] First valve **4320** and fourth valve **4326** cooperate to direct usable airflow **4314** through one of first desiccant portion **4328** and second desiccant portion **4329** from compressor and cooler portion **4210** to valve core **3424**. Third valve **4324** and second valve **4322** cooperate to direct desiccant regeneration airflow **4316** through the other of first desiccant portion **4328** and second desiccant portion **4329** from second desiccant portion **4312** to the vacuum of compressor and cooler portion **4210**. First valve **4330** and fourth valve **4336** cooperate to direct desiccant regeneration airflow **4316** from compressor and cooler portion **4210** to first desiccant portion **4310**.

Pneumatic Circuit and Sound Suppression Features

[0558] Referring now to FIGS. **47-48**, a top view and a perspective view illustrate details of compressor **3418** and pneumatic circuit body **3610**. As previously described, features of compressor **3418** enable a compact, lightweight, and quiet drive system **3110** that provides continuous dry air without the need to add new desiccants. As seen in FIG. **47**, compressor **3418** further includes supplemental pneumatic connections **4410**, an input/output board **4420**, and compressor gaskets **4421**.

[0559] Supplemental pneumatic connections **4410** are optional connections for users that need a higher air demand than the compressor can supply. In the example provided, supplemental pneumatic connections **4410** use quick disconnect fittings for rapid connection and disconnection.

[0560] Diaphragm pumps **3616** provide positive pressure and a vacuum. In the example provided, diaphragm pumps **3616** are small pumps conventionally used for medical applications such as in oxygen concentrators. Diaphragm pumps **3616** oppose each other in compressor **3418** and are driven out of phase to cancel out vibrations. For example, the diaphragm pumps **3616** may be cooperatively energized for the out of phase vibration cancelation. In some embodiments, the phase driving and orientation features may be used for about 100 times average vibration energy reduction relative to conventional compressors.

[0561] Pneumatic circuit body **3610** defines pneumatic conduits to route usable airflow **4250** and desiccant regeneration airflow **4252** between the various valves and components of compressor **3418**. In the example provided, pneumatic circuit body **3610** defines apertures **4422**, a first conduit **4430**, a second conduit **4432**, a third conduit **4434**, a fourth conduit **4436**, a fifth conduit **4438**, and a sixth conduit **4440**.

[0562] Apertures **4422** couple the various conduits to the various valves and components of compressor **3418**. Second conduit **4432** includes a sound absorbing portion **4442** that includes a sound absorbing material. Sixth conduit **4440** includes a sound absorbing portion **4444** that includes a sound absorbing material. In the example provided, the conduits of pneumatic circuit body **3610** are integrally formed in a plastic material of pneumatic circuit body **3610** to provide pneumatic routing to air components, pressure sources, and vacuum sources. Accordingly, tubing within compressor **3418** may be reduced relative to conventional compressor assemblies.

[0563] In the example provided, user input/output board **4420** includes a kill switch, a power receptacle, a USB Type-B port, a reset switch, and a WiFi connect button. The kill switch cuts power in the event top enclosure **3412** separates from bottom enclosure **3410**. In the example provided, the kill switch is a board mounted optical flag sensor that is triggered by a rib in top enclosure **3412**. Input/output board **4420** may be mounted by any suitable method, including by fasteners or by heat staking. A power cable and ribbon cable connect input/output board **4420** to main control board **3510**.

[0564] Compressor gaskets **4421** are elastomeric and have a shape and arrangement to permit movement of diaphragm pumps **3616** relative to pneumatic circuit body **3610**. Permitting movement restricts vibration transmission from diaphragm pumps **3616** to pneumatic circuit body **3610**. Additional gaskets at gasketed channels **3710** and connections through to strap system **3422** dampen vibrations to limit vibrations felt by the user. In the example provided, each gasket at each gasketed connection is tuned in thickness, durometer, diameter, shape, and other properties based on the frequency and amplitude of vibrations measured at the gasket location in the absence of the gasket.

[0565] In some embodiments, a compressor assembly includes a housing, a first compressor, a second compressor, and an outlet portion. For example, the housing defining a first end and a second end opposing the first end may be pneumatic circuit body **3610**, top cover **3612**, and bottom cover **3614**. The first and second compressor may be diaphragm pumps **3616**. The outlet portion for providing compressed air from the compressor assembly may be compressor and cooler portion **4210**.

[0566] In some embodiments, the compressor assembly includes a main body that defines a pneumatic circuit with a plurality of integral pneumatic conduits. For example, the main body may be pneumatic circuit body **3610** and the internal pneumatic conduits may include first conduit **4430**, second conduit **4432**, third conduit **4434**, fourth conduit **4436**, fifth conduit **4438**, and/or sixth conduit **4440**.

[0567] In some embodiments, the compressor assembly further includes a plurality of valves each in pneumatic communication with at least one of the plurality of integral pneumatic conduits. For example, the plurality of valves may include first valve **4230**, second valve **4232**, and third valve **4234**. In another example, the plurality of valves includes first valve **4320**, second valve **4322**, third valve **4324**, fourth valve **4326**, first valve **4330**, second valve **4332**, third valve **4334**, and fourth valve **4336**.

[0568] In some embodiments, the main body defines a first aperture and a second aperture in each of the plurality of integral pneumatic conduits. For example, pneumatic circuit body **3610** may define apertures **4422**. In some embodiments, each of the plurality of valves is in direct pneumatic communication with at least one of the first aperture and the second aperture of at least one of the plurality of integral pneumatic conduits.

[0569] In some embodiments, the plurality of integral pneumatic conduits defines a first desiccant conduit and a second desiccant conduit, where the first desiccant conduit defines a first desiccant portion between the first aperture and the second aperture of the first desiccant conduit, and wherein the second desiccant conduit defines a second desiccant portion between the first aperture and the second aperture of the second desiccant portion. For example, pneumatic circuit body **3610** may define first desiccant portion **4240** and second desiccant portion **4242**.

[0570] In some embodiments, the plurality of integral pneumatic conduits define at least one sound absorbing portion configured to muffle sound traveling through the pneumatic circuit. For example, pneumatic circuit body **3610** may define sound absorbing portion **4442** and/or sound absorbing portion **4444**. In some embodiments, a sound absorbing material is disposed in the at least one sound absorbing portion.

[0571] In some embodiments, the compressor assembly further includes a first desiccant portion, a second desiccant portion, and a plurality of valves configured to direct a first airflow and a second airflow. For example, the first desiccant portion may be first desiccant portion **4240** or first desiccant portion **4328**. The second desiccant portion may be, for example, second desiccant portion **4242** or second desiccant portion **4329**.

[0572] In some embodiments, the plurality of valves has a first configuration and a second configuration. The first configuration directs the first airflow from the first compressor and the second compressor through the first desiccant portion to the outlet. The first configuration further directs the second airflow from the first airflow through the second desiccant portion for desiccant drying in the second desiccant portion. For example, the first configuration may be the valve positions illustrated in FIG. **45** or in FIG. **46** resulting in the illustrated usable airflow **4250**, desiccant regeneration airflow **4252**, usable airflow **4314**, and desiccant regeneration airflow **4316**.

[0573] In some embodiments, the second configuration directs the first airflow from the first compressor and the second compressor through the second desiccant portion to the outlet. The second configuration further directs the second airflow from the first airflow through the first desiccant portion for desiccant drying in the first desiccant portion. For example, the second configuration may be where first valve **4230** directs usable airflow **4250** through second desiccant portion **4242** or where first valve **4320** directs usable airflow **4314** through second desiccant portion **4329**. In some embodiments, first desiccant portion **4338** is a third desiccant portion and second desiccant portion **4339** is a fourth desiccant portion.

[0574] In some embodiments, the second airflow begins at a portion of the first airflow. For example, the first configuration may direct desiccant regeneration airflow **4252** beginning at third valve **4234** in pneumatic communication with usable airflow **4250**. In some embodiments, the first and the second configuration may direct the second airflow beginning at the first airflow before the first airflow passes through either of the first desiccant portion or the second desiccant portion. For example, first valve **4330** may direct desiccant regeneration airflow **4316** branching from usable airflow **4314** between compressor and cooler portion **4210** and first valve **4320**.

Testing and Assembly Processes for the Drive System

[0575] Referring now to FIG. **49**, a process diagram illustrates testing and assembly processes **4600** for the drive system **3110** in accordance with the teaching of the present disclosure.

[0576] Testing and assembly processes **4600** describe assembly of a full drive system **4610** from an input/output PCB **4612**, a main PCB **4614**, a channel sensor PCB **4616**, a valve PCB **4618**, plastic components **4620**, valves for plastics **4622**, drive system miscellaneous parts **4624**, plastics and tubing **4626**, and compressor pneumatic circuit **4628**.

[0577] The processes include SMT electrical processes **4630**, EA electrical processes **4632**, and suitability tests **4634**. The processes further include a manifold gluing and assembly process **4650**, a valve module assembly process **4660**, and a tubing harness cut/assembly/glue process **4670**.

[0578] Manifold gluing and assembly process **4650** includes: placing three top manifolds in a holding device, use gluing robot to apply cyanoacrylate glue or similar to the top manifolds, place middle manifold on top manifolds, use gluing robot to apply Cyanoacrylate glue or similar to the middle manifold, place the bottom manifold on the middle manifold, clamp parts together, and pressure test. Clamping continues until curing is complete.

[0579] Valve module assembly process **4660** includes: placing two pressure sensor gaskets on each channel sensor board, securing three channel sensor boards to the manifold plastic assembly using screws, and installing three valve boards to the manifold plastic assembly using snap arms.

[0580] Tubing harness cut/assembly/glue process **4670** is described below with reference to FIG. **55**.

[0581] Through the testing and assembly processes **4600**, intermediate assemblies for a manifold assembly **4680**, a core assembly **4682**, and a compressor assembly **4684** are created.

[0582] Assembly of full drive system **4610** includes a first task to slip backpack strap buckles through slots in the bottom enclosure and flatten the buckles against the ribs in the bottom enclosure. A second task is to place the valve core and compressor assembly into the bottom enclosure. A third task is to secure multi-channel pneumatic connectors in the bottom enclosure. A fourth task is to insert the cooling fan into the bottom enclosure and connect wires to the main board. A fifth task is to insert the input/output board into the bottom enclosure and use screws or heat staking to secure the input/output board. A sixth task is to insert the power switch into the bottom enclosure and connect the cable harness. A seventh task is to connect wires to the switch and input/output board. An eighth task is to insert quick connect pneumatic fittings into the bottom enclosure. A ninth task is to connect tubing to the compressor housing. A tenth task is to attach a peripheral data cable harness to the bottom enclosure and to the Main Board. An eleventh task is to place the top enclosure on the bottom enclosure and fasten with screws. A twelfth task is to insert the battery. A thirteenth task is to test full drive system **4610**.

Glove Peripherals

[0583] Referring now to FIG. **50**, and with continued reference to FIGS. **34-49**, electronics of a glove assembly **4700** are illustrated in a simplified block diagram in accordance with the teachings of the present disclosure. Glove assembly **4700** is a haptic feedback glove that includes an interface laminate with a plurality of tactile actuators coupled to the skin of the user's hand. Glove assembly **4700** includes an opisthenar assembly **4712**, a wrist assembly **4714**, an umbilical assembly **4716**, thimble assemblies **4718**, palm panels **4720**, and a tracker **4722**.

[0584] Opisthenar assembly **4712** is disposed on the back of glove assembly **4700** corresponding to the dorsal side of the user's hand. Opisthenar assembly **4712** includes manipulation actuators **4730** and a magnetic emitter **4732**. Manipulation actuators **4730** control tendons **4734** that each provides force feedback to a finger of the user's hand, as will be explained below with reference to FIGS. **68-69**. Magnetic emitter **4732** may be an example of magnetic emitter **3230**.

[0585] Wrist assembly **4714** is disposed on a back of the glove corresponding to the back of the user's wrist. Wrist assembly **4714** includes solenoid valves **4740**, a position sensor **4742**, and a sensor PCB **4744**. Solenoid valve **4740** selectively permits compressed air to enter force feedback tubes **4746** for actuation of manipulation actuators **4730**. Position sensor **4742** may be an example of position sensors **3232**. Sensor PCB **4744** receives signals from position sensors **4750** and position sensor **4742** to calculate the relative position to sensor source **4732**. In some embodiments, wrist assembly **4714** includes fewer components. For example, solenoid valve **4740** may be omitted in wrist assembly **4714** in favor of valve control from the drive unit.

[0586] Umbilical assembly **4716** connects drive system **3110** to gloves **3112**. In the example provided, umbilical assembly **4716** includes 59 pneumatic tubes for actuators in glove assembly **4700**, a sensor USB cable, a solenoid power cable, and two air and vacuum tubes for air supply to solenoid valve manifold **4740**.

[0587] Thimble assemblies **4718** provide tactile feedback to the user, provide counterpressure for the tactile feedback, provide a kinematic termination for tendons **4734**, and define the location of position sensors for finger portions of glove assembly **4700**. In the example provided, thimble assemblies **4718** include position sensors **4750** and tactile panels **4752**.

[0588] Palm panels **4720** includes tactors for tactile feedback to a palm of the user, as is discussed below with reference to FIG. **74**. Tracker **4722** may be an example of motion tracker **3238**.

[0589] Referring now to FIG. **51**, and with continued reference to FIGS. **34-50**, a process diagram illustrates testing and assembly processes **4800** for a glove assembly **4810** in accordance with the teaching of the present disclosure. For example, testing and assembly processes **4800** may be used for glove assembly **4700** or gloves **3112**.

[0590] Testing and assembly processes **4800** result in glove assembly **4810** with inputs of an elastomeric material **4801**, plastics and hardware **4802**, a position sensor PCBA **4803**, FPC **4804** for wrist sensor and position sensor harness, plastics and soft goods **4805**, plastics and soft goods and tracker **4806**, and tubes and multi-channel pneumatic connector **4807**. In the example provided, elastomeric material **4801** is a sheet formed from a raw material elastomer, such as a high consistence rubber material, and may also be known as gum stock silicone. Plastics and hardware **4802** include components described with reference to FIG. **72** below. Position sensor PCBA **4803** may be an example of sensor PCB **4744**. FPCs **4804** may include various electronic components illustrated in FIG. **52**. Plastics and soft goods **4805** may include various components from FIGS. **62-65** described below. Plastics and soft goods and tracker **4806** may include various components from FIG. **53** described below. Tubes and multi-channel pneumatic connector **4807** may include various components shown and described in FIG. **71** below. The inputs may, for example, be assembled or manufactured by the assembling entity in other processes in the same manufacturing facility or by other entities, such as by suppliers.

[0591] Testing and assembly processes **4800** include an SMT electrical process **4812**, suitability test processes **4814**, an elastomer molding process **4820**, a plasma bonding process **4822**, a die cut to shape process **4824**, a manipulation actuator assembly process **4826**, and an automated tube cut and insertion/glue process **4828**.

[0592] Through testing and assembly processes **4800**, intermediate assemblies for a position sensor assembly **4830**, a replaceable glove subassembly **4832**, an inner glove assembly **4834**, a manual sheath assembly **4836**, and a wrist durable subassembly **4838** are created.

Replaceable-Durable Assembly

[0593] Referring now to FIG. **52**, and with continued reference to FIGS. **34-51**, a simplified block diagram illustrates a replaceable-durable assembly **4900**. Replaceable-durable assembly **4900** includes a durable subassembly **4910** and a replaceable subassembly **4912**. Durable subassembly **4910** and replaceable subassembly **4912** each include components on an interface **4914**. The physical locations of durable subassembly **4910** and replaceable subassembly **4912** may vary and span several assemblies described elsewhere based on physical location. In the example provided, wrist assembly **4714** spans durable subassembly **4910** and replaceable subassembly **4912**.

[0594] Durable subassembly **4910** has items with longer service lives than at least some of those of replaceable subassembly **4912**. Durable subassembly **4910** includes a multi-channel pneumatic connector **4920**, an umbilical **4922**, a durable wrist subassembly **4924**, and a tracker **4926**.

[0595] Umbilical **4922** includes a power and data cable **4930**, manipulation actuator pneumatic tubing **4932**, and tactor tubing **4934**. In the example provided, umbilical **4922** includes a protective sheath to limit damage to power and data cable **4930**, manipulation actuator pneumatic tubing **4932**, and tactor tubing **4934**.

[0596] Multi-channel pneumatic connector **4920** is a connector for power and data cable **4930**, manipulation actuator pneumatic tubing **4932**, and tactor tubing **4934** to connect to a drive system, such as drive system **3110**. Multi-channel pneumatic connector **4920** may be an example of the multi-channel pneumatic connector described with reference to FIG. **55** below.

[0597] Durable wrist subassembly **4924** includes wrist PCBA **4940**, miniature pneumatic valves **4942**, wrist position sensor **4944**, position sensor PCBA **4946**, and top manifold **4948**. Wrist PCBA **4940** includes electronics to control miniature pneumatic valves **4942**, interact with tracker **4926**, and to interact with position sensor PCBA **4946**. Position sensor PCBA **4946** includes electronics to interact with wrist position sensor **4944**.

[0598] Miniature pneumatic valves **4942** may be VOVK valves commercially available from FESTO. Tracker **4926** may be an example of motion tracker **3238**.

[0599] Replaceable subassembly **4912** includes gloves **4950**, replaceable wrist subassembly **4952**, opisthenar subassembly **4954**, finger thimbles **4956**, and palm assembly **4958**.

[0600] Gloves **4950** may include an inner glove and an outer glove, as described below with reference to FIGS. **73-79**. Gloves **4950** may come in various sizes to support different sized hands of users. Gloves **4950** are generally less durable than other components of replaceable-durable assembly **4900** due to wear from use.

[0601] Replaceable wrist subassembly **4952** includes a bottom manifold **4960** and a position sensor wire harness **4962**. Opisthenar subassembly **4954** includes a magnetic emitter PCBA **4970** and manipulation actuators **4972**. Magnetic emitter PCBA **4970** may be an example of magnetic emitter **3230**. Manipulation actuators **4972** may be an example of manipulation actuators **4730**.

[0602] Finger thimbles **4956** each include a position sensor **4980** and a panel **4982**. Position sensor **4980** may be an example of position sensors **3232** that track a position of each finger. Panel **4982** may be an example of tactile panels **4752** that provide tactile feedback.

[0603] Palm assembly **4958** includes a plastic counterpressure portion **4990** and a panel **4992**. Plastic counterpressure portion **4990** supports counterpressure grounded at opisthenar subassembly **4954** as shown below in FIGS. **71-74**. Panel **4992** may be an example of palm panel **4720**.

[0604] In the example provided, interface **4914** is defined in part by connections between top manifold **4948** and bottom manifold **4960**. In the example provided, interface **4914** is further defined in part by connections between position sensor PCBA **4946** and position sensor wire harness **4962**. The electrical connections may be any suitable connections for power and data. In the example provided, the connection between top manifold **4948** and bottom manifold **4960** utilizes face seals, as discussed below with reference to FIGS. **55-57**.

[0605] In some embodiments, replaceable subassembly **4912** includes only gloves **4950**, finger panels **4982**, and palm panel **4992** with the other components of replaceable-durable assembly **4900** disposed in durable subassembly **4910**. For example, durable subassembly **4910** may include finger thimbles **4956** without panel **4982** and palms **4958** without panel **4992**. In such embodiments, interface **4914** includes pneumatic connections at each of panels **4982** and panel **4992** for actuation of the respective panel **4982** or panel **4992**. For example, the pneumatic connections may be similar to the face seals discussed below with reference to FIGS. **55-57**.

[0606] In some embodiments, replaceable subassembly **4912** includes gloves **4950**, finger thimbles **4956**, and palms **4958** with the other components of replaceable-durable assembly **4900** disposed in durable subassembly **4910**.

[0607] In some embodiments, replaceable subassembly **4912** includes gloves **4950**, finger thimbles **4956**, palm assembly **4958**, and sensor signal electronics with the other components of replaceable-durable assembly **4900** disposed in durable subassembly **4910**. For example, the sensor signal electronics may include analog to digital conversion electronics to limit the travel and degradation of analog sensor signals. In some embodiments, the sensor signal electronics are integrated with position sensors **4980**.

[0608] In some embodiments, durable subassembly **4910** includes a top housing that is coupled to a bottom housing of replaceable subassembly **4912** using clips or screws.

[0609] Referring now to FIG. **53**, and with continued reference to FIGS. **34-52**, a perspective view illustrates an example of a replaceable-durable assembly **5000**. In the example provided, reusable assembly **5000** is an implementation of replaceable-durable assembly **4900**.

[0610] Peripheral assembly **3210** includes a durable subassembly **5002** and a replaceable subassembly **5004**.

[0611] Durable subassembly **5002** includes a wrist reusable assembly **5010**, a tracker **5012**, an umbilical **5014**, and a glove connector **5016**. Wrist reusable assembly **5010** may be an example of durable wrist subassembly **4924**. Tracker **5012** may be an example of motion tracker **3238**. Umbilical **5014** may be an example of umbilical **4922**. Glove connector **5016** may have electronic connections and pneumatic connections as described below with reference to FIG. **55**.

[0612] Replaceable subassembly **5004** includes a wrist replaceable subassembly **5020**, soft goods **5022**, an opisthenar assembly **5024**, tendon guides **5026**, and thimble assemblies **5028**. Wrist replaceable subassembly **5020** may be an example of replaceable wrist subassembly **4952**.

[0613] Soft goods **5022** may include gloves, fabrics, flexible rubbers, and other materials and components consistent with the "soft goods" engineering term of art. Opisthenar assembly **5024** may be an example of opisthenar subassembly **4954**. Tendon guides **5026** provide guides for force feedback tendons, as will become apparent with reference to FIGS. **63-65** discussed below.

[0614] Thimble assemblies **5028** may be examples of finger thimbles **4956** for providing tactile feedback and terminating force feedback to a user's fingers, as will become apparent with reference to FIGS. **62-63**, as discussed below.

[0615] By including opisthenar assembly **5024** and wrist replaceable subassembly **5020** separated by soft goods **5022**, wrist mobility is retained for the user.

Multi-Channel Pneumatic Connector

[0616] Referring now to FIG. **54**, and with continued reference to FIGS. **34-53**, an exploded view illustrates a multi-channel pneumatic connector **5100**. Multi-channel pneumatic connector **5100** is a compact assembly for connecting a large and dense assembly of tubing to other pneumatic conduits. In the example provided, multi-channel pneumatic connector **5100** connects tubes from drive system **3110** to wrist assembly **4714**. In the example provided, multi-channel pneumatic connector **5100** is a straight plug connector. In some embodiments, multi-channel pneumatic connector **5100** is a threaded twist connector.

[0617] Multi-channel pneumatic connector **5100** includes a wrist top shell **5110**, a wrist bottom shell **5112**, a console insert **5114**, a gasket **5116**, a face plate **5118**, a shell **5120**, an umbilical insert **5122**, and an umbilical sheath **5124**.

[0618] Wrist top shell **5110** and a wrist bottom shell **5112** couple together to form a housing. Gasket **5116** seals against console insert **5114** and umbilical insert **5122** to restrict air leakage.

[0619] Sheath **5124** holds tubing and cables and is captured between shell **5120** and umbilical insert **5122**. Shell **5120** and face plate **5118** snap together with umbilical insert **5122**.

[0620] In the example provided, multi-channel pneumatic connector **5100** is injection molded and couples umbilical insert **5122** and umbilical insert **5122** using clips.

[0621] Referring now to FIGS. **55-57**, and with continued reference to FIGS. **34-54**, a multi-channel pneumatic connector **5200** is illustrated in a cutaway view in accordance with the teachings of the present disclosure. In the example provided, multi-channel pneumatic connector **5200** connects tubing to tactile panels.

[0622] Multi-channel pneumatic connector **5200** includes tubing **5210**, tubing connector **5212**, and pneumatic component **5214**. In the example provided, pneumatic component **5214** is a tactile panel. It should be appreciated that multi-channel pneumatic connector **5200** may be used at any tubing termination point without departing from the scope of the present disclosure.

[0623] Tubing connector **5212** defines air channels **5220**, tubing receiving portions **5222**, glue receiving portions **5224**, and face seal apertures **5226**. Air channel **5220** couples tubing **5210** to the respective face seal aperture **5226** for fluid communication.

[0624] In the example provided, tubing receiving portion **5222** is substantially cylindrical with a tubing stop face **5227** against which tubing **5210** may be pressed during assembly. In the example provided, glue receiving portion **5224** is substantially conically shaped with a reducing diameter as glue receiving portion **5224** extends into tubing connector **5212**.

[0625] Pneumatic component **5214** defines face seal apertures **5310** and air channels **5410**, as can best be seen in FIG. **55** and FIG. **57**, respectively. In the example provided, air channels **5410** are each in fluid communication with at least one tactor in a tactile panel.

[0626] In the example provided, pneumatic component **5214** is formed from a channel side silicone layer **5420** plasma bonded onto a molded elastomer layer **5422**. Face seal apertures **5310** are cut into pneumatic component **5214** to make a connection between the upper surface of channel side silicone layer **5420** and air channels **5410** of molded elastomer layer **5422**.

Tactile Panels

[0627] Referring now to FIG. **58**, and with continued reference to FIGS. **34-57**, a tactile panel assembly **5500** is illustrated in an exploded view in accordance with some embodiments. Tactile panel assembly **5500** includes a reinforced sheeting layer **5510**, a channel side silicone layer **5512**, a molded elastomer layer **5514**, a tactor side silicone layer **5516**, and a panel tail silicone layer **5518**. Tactile panel assembly **5500** defines a tactor side **5502** and a connector side **5504**. Tactor side **5502** defines wings or tabs **5506** with through-holes **5508**. In the example provided, tabs **5506** wrap around a finger of a user and through-holes **5508** accommodate securement of tactile panel assembly **5500** to a fingertip assembly of the glove assembly, as will be described below with reference to FIG. **63**.

[0628] Reinforced sheeting layer **5510** provides backpressure to support flexibility in thimble assemblies. Reinforced sheeting layer **5510** is plasma bonded to channel side silicone layer **5512**. In the example provided, reinforced sheeting layer **5510** is formed from a fabric reinforced silicone. Reinforced sheeting layer **5510** has a shape that cooperates with fingertip assembly plastics to provide counter pressure tasks for grounding forces from the tactors and from the force feedback tendon pulling on the user's finger. The fabric reinforced silicone is strong enough for these counterpressure tasks and grounding forces while still permitting tactile panel assembly **5500** to be flexible enough for stretching to accommodate fingertip size variations while still providing sufficient counter pressure for good tactile sensations. The fabric reinforced silicone is further flexible enough for improved realistic interaction with physical props when compared with rigid plastic fingertips.

[0629] Channel side silicone layer **5512**, tactor side silicone layer **5516**, and panel tail silicone layer **5518** are each plasma bonded to molded elastomer layer **5514**. Channel side silicone layer **5512** defines a termination portion **5520** through which face seal apertures are cut. For example, face seal apertures **5310** may be cut into channel side silicone layer **5512**.

[0630] Molded elastomer layer **5514** defines a plurality of channels **5521** and a termination portion **5522** through which face seal apertures are cut. For example, face seal apertures **5310** may be cut through channel side silicone layer **5512** into molded elastomer layer **5514** at termination portion **5522** to channels **3521**.

[0631] Referring now to FIG. **59**, and with continued reference to FIGS. **34-58**, a finger panel fabrication process **5600** is illustrated in a flow diagram view. Finger panel fabrication process **5600** may be used, for example, to fabricate tactile panel assembly **5500**.

[0632] Finger panel fabrication process **5600** includes providing layers **5610**, a first plasma bonding task **5612**, a second plasma bonding task **5614**, a needleless connection cut task **5616**, a third plasma bonding task **5618**, and a cut to shape task **5620**.

[0633] In the example provided, providing layers **5610** includes providing channel side silicone layer **5512** and molded elastomer layer **5514**.

[0634] First plasma bonding task **5612** includes plasma bonding the layers provided in providing layers **5610** task. In the example provided, first plasma bonding task **5612** includes plasma bonding channel side silicone layer **5512** to molded elastomer layer **5514**. As is appreciated by those of ordinary skill in the art, plasma bonding is a method of forming a direct chemical connection between the two layers of silicone.

[0635] Second plasma bonding task **5614** includes plasma bonding a reinforcement layer onto the assembly produced by first plasma bonding task **5612**. In the example provided, reinforced sheeting layer **5510** is plasma bonded onto the already bonded channel side silicone layer **5512** and molded elastomer layer **5514** assembly. In the example provided, second plasma bonding task **5614** further includes an oven heating process after lamination of reinforced sheeting layer **5510** to the assembly.

[0636] Needleless connection cut task **5616** punches the needleless connection holes in the panel. For example, needleless connection cut task **5616** may punch face seal apertures **5310** through channel side silicone layer **5512** and at least partially through molded elastomer layer **5514** to air channels **5410**.

[0637] Third plasma bonding task **5618** includes bonding additional silicone layers to a tactor side and a panel tail side of the assembly. In the example provided, third plasma bonding task **5618** bonds tactor side silicone layer **5516** and panel tail silicone layer **5518** to the bonded and cut assembly of channel side silicone layer **5512** and molded elastomer layer **5514**. In the example provided, third plasma bonding task **5618** further includes an oven heating process after lamination of needleless connection cut task **5616** and panel tail silicone layer **5518** to the assembly.

[0638] Cut to shape task **5620** includes die cutting the assembly to the final shape.

[0639] Referring now to FIG. **60**, and with continued reference to FIGS. **34-59**, a palm panel fabrication process **5700** is illustrated in a flow diagram view. Palm panel fabrication process **5700** may be used, for example, to fabricate palm panel **4720**.

[0640] Palm panel fabrication process **5700** includes providing layers **5712**, a first plasma bonding task **5714**, a needleless connection cut task **5716**, a second plasma bonding task **5718**, and a die cutting task **5720**.

[0641] Providing layers **5712** task includes providing a sheeting layer **5730** and a molded elastomer layer **5732**. In the example provided, channel side silicone layer **5730** is similar to channel side silicone layer **5512** and molded elastomer layer **5732** is similar to molded elastomer layer **5514**. Channel side silicone layer **5730** and molded elastomer layer **5732**, however, have shapes suited to use with palm panel **4720**. In the example provided, the left and right gloves use palm panels with mirrored shapes to fit the left and right hands. Molded elastomer layer **5732** defines air channels **5740**, tactor portion **5742**, and tail connection portion **5744**. The number of air channels **5740** and tactors in tactor portion **5742** is based on the desired haptic feedback to a palm of the user.

[0642] First plasma bonding task **5714** includes plasma bonding channel side silicone layer **5730** to molded elastomer layer **5732**. In an intermediate task (not illustrated), a reinforced sheet similar to reinforced sheeting layer **5510** may be plasma bonded to the bonded assembly of channel side silicone layer **5730** and molded elastomer layer **5732**.

[0643] Needleless connection cut task **5716** includes punching needleless connection holes **5750** in the assembly of bonded channel side silicone layer **5730** and molded elastomer layer **5732**. Needleless connection holes **5750** are similar to face seal apertures **5310** and define a conduit through which a multi-channel pneumatic connector may make a face seal connection to air channels **5740**.

[0644] Die cutting task **5720** cuts the bonded and cut assembly to a final shape for use in a glove assembly.

[0645] Referring now to FIG. **61**, and with continued reference to FIGS. **34-60**, tactile panel testing images **5800** are illustrated in a simplified diagram in accordance with the teachings of the present disclosure.

[0646] Tactile panel testing images **5800** illustrate a testing process using machine vision to confirm tactor operability and channel mapping. In the example provided, a machine vision system is coupled with a pneumatic valve to inflate each tactor and confirm that each tactor inflates in response to actuation of the respective pneumatic valve. In the example provided, an inflated tactor **5810** is illustrated among a plurality of uninflated tactors **5820**.

[0647] Accordingly, the vision system may be used to confirm the inflation properties of each tactile panel after fabrication. In the example provided, tactile panels are tested using tactile panel testing images **5800** after fabrication and before being secured to an inner glove of a glove assembly.

Finger Assembly

[0648] Referring now to FIG. **62**, and with continued reference to FIGS. **34-61**, a fingertip assembly **5900** is illustrated in an exploded view in accordance with the teachings of the present disclosure.

[0649] Fingertip assembly **5900** includes a thimble **5910**, a tendon clip **5912**, a position sensor shell **5914**, a cable strain relief **5916**, a puck **5918**, and a finger portion **5920** of inner glove **6010**.

[0650] Thimble **5910** is a rigid plastic part that secures tendon clip **5912**, position sensor shell **5914**, and cable strain relief **5916** to puck **5918**. Tendon clip **5912** secures a tendon (illustrated as **6024** in FIG. **63**) to fingertip assembly **5900**. Position sensor shell **5914** houses finger position sensors. Cable strain relief **5916** provides cable protection for sensor wires for the finger position sensors in position sensor shell **5914**.

[0651] Puck **5918** is secured to finger portion **5920** over a fingernail portion of the user's expected finger position to ground tactor counterpressure at the user's fingernail. Puck **5918** defines projections **5930** extending away from finger portion **5920** to accommodate a tactile panel assembly, as described below with reference to FIG. **63**. For example, puck **5918** may be heat stake riveted or glued to a fabric of inner glove **6010**.

[0652] Referring now to FIGS. **63-64**, and with continued reference to FIGS. **34-62**, a glove assembly **6000** is illustrated in a simplified cutaway side view in accordance with the teachings of the present disclosure.

[0653] Glove assembly **6000** is illustrated on a user's finger **6002** and includes an inner glove **6010**, a puck **6012**, a thimble **6014**, a silicone panel **6020**, tendon guides **6022**, tendons **6024**, and an outer glove **6026** (not shown in FIG. **64**).

[0654] Inner glove **6010** is a flexible glove that serves as an interface layer between the glove assembly and the user's skin. Inner glove **6010** may be made from LYCRA or another lightweight, elastic fabric.

[0655] Puck **6012** is an example of puck **5918** and is secured to inner glove **6010** with glue **6030** above an expected location of a user's fingernail. In some embodiments, puck **6012** is riveted to inner glove **6010**.

[0656] Puck **6012** grounds reaction forces from actuator of tactors at the fingertip of silicone panel **6020**. For example, as tactors actuate and press against finger **6002**, tensile forces in silicone panel **6020** hold the tactors against finger **6002**. The tensile forces in silicone panel **6020** pull against projections **5930** and also present on puck **6012** to create compressive forces against glue **6030** and the fingernail of finger **6002**. Because puck **6012** and the fingernail are substantially rigid, the compressive forces are distributed across the surface area of the fingernail to limit tactile sensations from the counterpressure for the user.

[0657] Thimble **6014** is an example of thimble **5910** and is secured against puck **6012** to clamp silicone panel **6020** and tendons **6024**.

[0658] Silicone panel **6020** is an example of tactile panel assembly **5500**. Silicone panel **6020** is routed and curved around finger **6002** to puck **6012**. In the example provided, projections **5930** are also present on puck **6012** to hold silicone panel **6020** in place on puck **6012**. Silicone panel **6020** extends along finger and through tendon guides **6022** and ultimately to a multi-channel pneumatic connector. In the example provided, silicone panel **6020** includes 24 tactile actuators capable of producing a displacement of at least 1 mm.

[0659] Tendon guides **6022** are secured to inner glove **6010**, such as by glue **6030** or rivets. Tendon guides **6022** are described below with reference to FIG. **65**.

[0660] Tendons **6024** are secured to thimble **6014** and define a load path between the fingertip of finger **6002** and a manipulation actuator, such as manipulation actuators **4972**.

[0661] In the example provided, tendons **6024** are located on the dorsum of the user's hand and apply forces to the user's finger during grasping motions involving finger flexion while allowing unhindered finger extension. In the example provided, tendons **6024** are 65 lb fishing line. In some embodiments, tendons **6024** are ribbon shaped with a ratio of width to thickness of at least 10.

[0662] Finger motion resisted by the manipulation actuator results in reaction forces that are distributed via the load path to the user's finger. In the example provided, the reaction forces terminate at the distal phalange of the finger. In some embodiments, the reaction forces are distributed approximately evenly across the palmar surface of the phalange by silicone panel **6020**.

[0663] Distributing the net force on the user's fingertip produced by the action of the manipulation actuator promotes approximation of the physical point forces resulting from a particular object interaction. For example, pressing on a simulated pin and a simulated flat surface in a virtual environment might produce identical net forces on the user's fingertip, as rendered by the action of the manipulation actuator. These interactions, however, would produce very different point forces on the skin of the fingertip as rendered by the action of tactile actuators.

[0664] Outer glove **6026** may be an outer glove as described below with reference to FIGS. **73-79**.

Tendon Guide Assembly

[0665] Referring now to FIG. **65**, and with continued reference to FIGS. **34-64**, a tendon guide assembly **6200** is illustrated in an exploded view. Tendon guide assembly **6200** includes tendon guide tops **6210**, tendon guide bottoms **6212**, and FPC **6214** for panel and position sensor.

[0666] Tendon guide tops **6210** define an FPC cavity **6218** and a guide aperture **6220** through which tendon **6222** passes. Tendon guide tops **6210** secure to tendon guide bottoms **6212** with, for example, screws or snap features.

[0667] Tendon guide bottoms **6212** are glued or riveted to an inner glove of a glove assembly. Tendon guide bottoms **6212** receive FPC **6214** and cooperate with FPC cavity **6218** of tendon guide tops **6210** to restrict movement of FPC **6214**.

Wrist Assembly

[0668] Referring now to FIGS. **66-67**, and with continued reference to FIGS. **34-65**, a wrist assembly **6300** is illustrated in perspective views. In the example provided, wrist assembly **6300** is mounted to a glove assembly at the back of the user's wrist.

[0669] Wrist assembly **6300** includes a durable wrist portion **6310**, a replaceable wrist portion **6312**, a tracker **6314**, an enclosure top **6316**, and an enclosure bottom **6318**.

[0670] Durable wrist portion **6310** is an example of durable subassembly **4910** that attaches to umbilical **4922**. Replaceable wrist portion **6312** is an example of replaceable subassembly **4912** connected to the soft goods of the glove assembly. Tracker **6314** is an example of tracker **4926**.

[0671] Enclosure top **6316** is secured to enclosure bottom **6318** with snaps or screws for user disassembly when replaceable wrist portion **6312** has worn out and must be replaced. For example, the soft goods may rip or become frayed and need replacing after extended use.

[0672] Durable wrist portion **6310** includes miniature pneumatic valves and top manifold **6320**, position sensor PCBA **6322**, and wrist position sensor **6324**. Replaceable wrist portion **6312** includes a multi-channel pneumatic connector **6330**, a valve bottom manifold **6332**, a palm needleless connector **6334**, and a thumb needleless connector **6336**, a wiring harness **6410**, and tubes for panels in Opisthenar (not shown).

[0673] Miniature pneumatic valves and top manifold **6320** may be examples of miniature pneumatic valves **4942** and top manifold **4948**. Position sensor PCBA **6322** may be an example of position sensor PCBA **4946**. Wrist position sensor **6324** may be an example of wrist position sensor **4944**.

[0674] Multi-channel pneumatic connector **6330** may be an example of multi-channel pneumatic connector **5100**. Valve bottom manifold **6332** may be an example of bottom manifold **4960**. Palm needleless connector **6334** and a thumb needleless connector **6336** may use face sealing features as shown in FIG. **65**. Wiring harness **6410** may be an example of position sensor wire harness **4962**.

Opisthenar Assembly

[0675] Referring now to FIG. **68**, and with continued reference to FIGS. **34-67**, an opisthenar assembly **6500** is illustrated in a perspective view in accordance with the teachings of the present disclosure. Opisthenar assembly **6500** is an example of opisthenar assembly **4712**.

[0676] Opisthenar assembly **6500** includes an enclosure top **6510**, an enclosure bottom **6512**, a magnetic emitter **6514**, and five manipulation actuators **6520**. In the example provided, enclosure top **6510** and enclosure bottom **6512** are injection molded and form a housing with dimensions of about 87 mm wide, 36 mm tall, 66.6 mm long, and a 21 mm ledge height tall. Accordingly, opisthenar assembly **6500** is well sized to be mounted to the glove assembly at the back of the user's hand. Enclosure top **6510** defines finger tendon apertures **6522** and a thumb tendon aperture **6524**.

[0677] Manipulation actuators **6520** selectively provide force feedback to tendons **6530**. Tendons **6530** each pass through one of the four finger tendon apertures **6522** or thumb tendon aperture **6524**, route through tendon guides, and connect to thimbles on a respective finger assembly, as discussed above.

[0678] In the example provided, manipulation actuators **6520** snap into the housing with snap-fit features. For example, five manipulation actuators **6520** may cooperate with enclosure top **6510** or enclosure bottom **6512** to form an undercut tab and snap feature.

Manipulation Actuator

[0679] Referring now to FIG. **69**, and with continued reference to FIGS. **34-68**, a manipulation actuator **6520** is illustrated in an exploded view. Manipulation actuator **6520** is a component of a force feedback exoskeleton that produces a net force on a body segment of a user, such as a finger.

[0680] Manipulation actuator **6520** includes a cover **6610**, a bladder **6612**, a brake pad **6614**, a wave spring **6616**, a retaining ring **6618**, a spool **6620**, a power spring **6622**, and a housing **6624**.

[0681] Cover **6610** defines clip receiving portions **6630** and housing **6624** defines clip portions **6632**. Covers **6610** assembles to housing **6624** such that clip portions **6632** snap into portions **6630** and secure cover **6610** to housing **6624**.

[0682] Housing **6624** further defines a cavity **6634** and a rotation restriction slot **6636**. Cavity **6634** receives power spring **6622**, spool **6620**, retaining ring **6618**, wave spring **6616**, brake pad **6614**, and bladder **6612**.

[0683] Bladder **6612** inflates and deflates according to properties of air supplied to bladder **6612** by a drive system, such as drive system **3110**. Inflation of bladder **6612** causes brake pad **6614** to contact spool **6620** to apply forces opposing rotation of spool **6620** and extension of tendon **6530**, as described below.

[0684] Brake pad **6614** defines protrusions **6640** and a Hirth joint mating surface **6642**. Protrusions **6640** extend radially out from brake pad **6614** and have shapes and locations corresponding to the shape of rotation restriction slot **6636**. Accordingly, protrusions **6640** are disposed within rotation restriction slot **6636** when cover **6610** is installed on housing **6624** and brake pad **6614** is disposed in cavity **6634**.

[0685] Wave spring **6616** biases brake pad **6614** away from spool **6620** such that brake pad **6614** does not restrict rotation of spool **6620** or extension of tendons **6530** when bladder **6612** is not inflated.

[0686] Retaining ring **6618** restrains movement, but not elongation and compression, of wave spring **6616**. Retaining ring **6618** in turn is restrained from moving by spool **6620**.

[0687] Spool **6620** accommodates tendons **6530** and defines a Hirth joint mating surface **6650** opposing Hirth joint mating surface **6642** of brake pad **6614**. Hirth joint mating surface **6650** and Hirth joint mating surface **6642** each define substantially triangular teeth. The term "substantially triangular" means that the teeth may have curved faces or other variations that maintain the function of the Hirth joint.

[0688] Power spring **6622** increasingly biases spool **6620** to a retracted state in response to rotation of spool **6620** and extension of tendon **6530**.

[0689] To actuate manipulation actuator **6520**, high pressure air inflates bladder **6612**. Bladder **6612** presses against both cover **6610** and brake pad **6614**. Because cover **6610** is secured to housing **6624**, cover **6610** does not substantially move and is able to provide counter pressure as bladder **6612** biases brake pad **6614** toward spool **6620**. When actuation forces from bladder **6612** on brake pad **6614** are sufficient to overcome the biasing forces of wave spring **6616**, brake pad **6614** translates toward spool **6620**.

[0690] When Hirth joint mating surface **6642** of brake pad **6614** contacts Hirth joint mating surface **6650** of spool **6620**, friction and interference between teeth in the Hirth joint restrict movement of spool **6620**. The friction forces include a component that opposes separation of brake pad **6614** and spool **6620**. The friction forces further include a component that opposes rotation forces between brake pad **6614** and spool **6620**, but this component is less than the forces for flat surfaces at any given actuation force to an extent determined by the angle of teeth in the Hirth joint. The forces required to overcome interference between teeth of the Hirth joint, however, are significantly different from flat surface brake pad assemblies. For example, in the absence of deformation of materials or separation of brake pad **6614** from spool **6620** (such as by movement of bladder **6612**), there will be no movement of spool **6620** relative to brake pad **6614**. Accordingly, materials and actuation pressures are selected based on these considerations for an intentionally slip-permitting Hirth joint. In the example provided, the Hirth joint is actuated in an on/off control scheme and has a predetermined brake design holding force that may be overcome for limiting component breakage in the human-computer interface system. For example, the brake design holding force may be a threshold force on tendon **6530**. Above the threshold force, bladder **6612** may compress enough to permit a yielding translation and relative rotation between brake pad **6614** and spool **6620** even when manipulation actuator **6520** is actuated.

Glove PCB

[0691] Referring now to FIG. **70**, glove electronics **6700** are illustrated in a simplified block diagram. Glove electronics **6700** are implemented as a combination of flexible printed circuits (FPCs) and a rigid circuit board. The shape and angles of the FPCs are based on average hand sizes, and the shapes and angles may be reliably fabricated with an automated FPC process.

[0692] In the example provided, glove electronics **6700** includes durable wrist electronics **6710**, replaceable electronics **6712**, and board connector **6714**.

[0693] Durable wrist electronics **6710** include a wrist sensor **6720**, a wrist sensor FPC **6722**, an FPC to PCBA connector **6724**, a position sensor PCBA **6726**, and a stiffener **6728**. Wrist sensor **6720** is an example of wrist position sensor **6324**. Wrist sensor FPC **6722** is a flexible printed circuit connecting wrist sensor **6720** to position sensor PCBA **6726** through PCBA connector **6724**. FPC to PCBA connector **6724** may be any suitable electronic connector, such as a ZIF connector. Position sensor PCBA **6726** is an example of position sensor PCBA **4946**. Stiffener **6728** restricts flexing of the FCB near wrist sensor **6720**.

[0694] Replaceable electronics **6712** include a position sensor harness FPC **6730**, a magnetic emitter **6732**, position sensors **6734**, and stiffeners **6736**.

[0695] Position sensor harness FPC **6730** is a flexible printed circuit that connects magnetic emitter **6732** and position sensors **6734** to position sensor PCBA **6726** through board connector **6714**. Magnetic emitter **6732** may be an example of magnetic emitter **3230**. Position sensors **6734** may be examples of position sensors **3232**. Stiffeners **6736** restrict flexing of the FCB near magnetic emitter **6732** and position sensors **6734**.

[0696] In the example provided, position sensor PCBA **6726** mounts to a durable assembly with screws or heat staking, wrist Sensor FPC **6722** mounts to the durable assembly with screws or heat staking, and position sensor harness FPC **6730** mounts to a replaceable wrist assembly with board connector **6714** exposed to interface with the durable assembly.

Integration of Hard and Soft Goods

[0697] Referring now to FIGS. **71-72**, and with continued reference to FIGS. **34-70**, a palm assembly **6800** is illustrated in perspective views. Palm assembly **6800** includes substrates **6810**, rivets **6812**, and lacer guides **6814**. Rivets **6812** are secured to substrates **6810**, and lacer guides **6814** receive and guide lacers, which are discussed below with reference to FIG. **45**. Substrates **6810** are counter pressure features formed into a thin and semi-flexible piece of plastic that is connected to the soft goods by rivets **6812**.

[0698] Referring now to FIG. **73**, and with continued reference to FIGS. **34-72**, an outer glove **7000** is illustrated in a perspective view according to the teachings of the present disclosure. Outer glove **7000** includes a cuff portion **7010**, a palm portion **7012**, and reinforcements **7020**. Reinforcements **7020** may be flexible plastic, leather, or other materials. Reinforcements **7020** define rivet holes **7022** through which rivets pass through on assembly of the glove.

[0699] Referring now to FIG. **74**, and with continued reference to FIGS. **34-73**, a palm tactile panel layout **7100** is illustrated in a front view in accordance with some embodiments. Tactile panel layout **7100** includes a first tactile panel **7110**, a second tactile panel **7112**, and a third tactile panel **7114**. With reference back to FIG. **73**, it can be seen that the locations of the tactile panels correspond to the locations of reinforcements **7020**. Accordingly, counterpressure to the tactile panels may be spread across reinforcements **7020** and into outer glove **7000**.

[0700] Individual tactors within tactile panels may be grouped into zones, as indicated by zones **7120** of first tactile panel **7110**. In the example provided, each tactor within a zone is actuated by the same pneumatic channel as the other tactors in the same zone.

[0701] Referring now to FIG. **75**, and with continued reference to FIGS. **34-74**, a glove assembly **7200** is illustrated in a simplified diagram in accordance with the teachings of the present disclosure. Glove assembly **7200** includes an outer glove **7210**, tendon guides **7212**, and pucks **7214**. Outer glove **7210** may be an example of outer glove **6026**. Tendon guides **7212** may be an example of tendon guides **6022**. Pucks **7214** may be examples of puck **6012**.

[0702] Outer glove **7210** defines a service slot **7220** through which inner glove **7211** may be inserted or removed.

[0703] Referring now to FIG. **76**, and with continued reference to FIGS. **34-75**, an opisthenar and glove assembly **7300** is illustrated in a perspective view in accordance with some embodiments. Opisthenar and glove assembly **7300** includes opisthenar assembly **7310** and outer glove **7312**. Outer glove **7312** includes a mounting component **7320** that defines locating features **7322** and has a hook and loop surface. Opisthenar assembly **7310** includes a hook and loop portion **7330** and locating features **7332**. In the example provided, mounting component **7320** is sewn into outer glove **7312**.

[0704] During assembly of opisthenar assembly **7310** to outer glove **7312**, features **7322** align with features **7332** to ensure a correct alignment of opisthenar assembly **7310** on outer glove **7312**. It should be appreciated that similar alignment features may be used on other complementary connectors having complementary connector components without departing from the scope of the present disclosure. The hook and loop surface of mounting component **7320** couples with hook and loop portion **7330** to secure opisthenar assembly **7310** to outer glove **7312**. Opisthenar assembly **7310** installs onto outer glove **7312** as indicated by alignment indicator lines **7340**.

[0705] Referring now to FIG. **77**, and with continued reference to FIGS. **34-76**, an opisthenar and glove assembly **7400** is illustrated in a perspective view in accordance with the teachings of the present disclosure. Opisthenar and glove assembly **7400** includes a glove **7410**, an opisthenar assembly **7412**, and a mesh layer **7414**.

[0706] Glove **7410** includes a rigid plate **7420** sewn into mesh layer **7414** and the fabric of glove **7410**. Rigid plate **7420** defines locating bosses **7422** configured to interact with and align opisthenar assembly **7412**. During assembly of opisthenar and glove assembly **7400**, opisthenar assembly **7412** may be secured to rigid plate **7420** with screws interacting with threaded bores formed in plate **7420**.

[0707] Referring now to FIGS. **78-79**, and with continued reference to FIGS. **34-77**, a glove assembly **7500** is illustrated in a perspective view according to the teachings of the present disclosure. Glove assembly **7500** includes a glove **7510**, an opisthenar assembly **7512**, and a lacer knob **7514**. Glove **7510** includes a cuff portion **7516**. Lacer knob **7514** includes lacers **7520**. Lacer knob **7514** is configured to tighten lacers **7520**, such as by turning lacer knob **7514**. Tightening lacers **7520** tightens the opisthenar assembly by way of lacer guides **6814**.

[0708] Glove assembly **7500** further includes a cuff lacer **7610**. Cuff lacer **7610** is disposed on cuff portion **7516** and includes lacer **7612**. Cuff lacer **7610** tightens lacer **7612** to tighten cuff portion **7516**, as can be seen in FIG. **79**. The use of two separate lacers improves the grounding of forces in the glove assembly to the body of the user.

Pneumatic Routing

[0709] Referring now to FIG. **80**, a pneumatic routing assembly **7700** is illustrated in a simplified block diagram. Pneumatic routing assembly **7700** illustrates the route taken by supply and exhaust air between a drive system and tactile panels in a glove assembly.

[0710] Pneumatic routing assembly **7700** includes an umbilical **7712** and a wrist assembly **7714**. Umbilical **7712** may be an example of umbilical assembly **7716**. In the example provided, umbilical assembly **7716** includes a first multi-channel pneumatic connector **7710** and a second multi-channel pneumatic connector **7710**. Wrist assembly **7714** is an example of wrist assembly **6300** and includes a top manifold and multi-channel pneumatic connector, such as multi-channel pneumatic connector **5100**.

[0711] Referring now to FIG. **81**, a tubing harness **7800** is illustrated in a perspective view. Tubing harness **7800** illustrates how the supply and exhaust air is routed between a drive system manifold **3712** and pneumatic routing assembly **7700**.

[0712] Tubing harness **7800** includes a multi-channel pneumatic connector **7810**, tubing **7812**, tube management features **7814**, manifold connectors **7816**, and manifold connector gaskets **7818**. Manifold connectors **7816** interface with valve core manifolds of a drive system, such as at manifold connectors **4026** of manifold assemblies **3712**. In the example provided, multi-channel pneumatic connector **7810** is an example of first multi-channel pneumatic connector **7710**. In the example provided, tubing **7812** includes 60 lengths of Tygon® brand tubing, manifold connectors **7816** include six connectors, and manifold connector gaskets **7818** include six gaskets.

[0713] Referring now to FIGS. **82-85**, and with continued reference to FIGS. **1-81**, a compressor assembly **7900** is illustrated according to some embodiments of the disclosure. In some embodiments, compressor assembly **7900** includes a motor assembly **7910**, a second motor assembly **7912**, a motor mounting plate **7914**, a pneumatic circuit **7916**, an assembly mounting plate **7918**, first dampers **7920**, second dampers **7922**, third dampers **7924**, and fourth dampers **7926**.

[0714] The motor assembly **7910** and the second motor assembly **7912** include rotors (not illustrated) that rotate about the axis of rotation **7930**. The first motor assembly **7910** includes a first motor **7950** and a first compression portion **7952**. The second motor assembly **7912** includes a second motor **7954** and a second compression portion **7956**.

[0715] The motor mounting plate **7914** defines damper receiving bore **7932**.

[0716] The pneumatic circuit **7916** defines a damper receiving bore **7934** and includes a bottom portion **7940** and a top portion **7942**.

[0717] The assembly mounting plate **7918** defines a damper receiving bore **7936**.

[0718] The first dampers **7920**, second dampers **7922**, third dampers **7924**, and fourth dampers **7926** damp vibrations between the coupled components.

[0719] Referring now to FIG. **85**, opposing forces during operation of compressor assembly **7900** are illustrated in accordance with some embodiments.

[0720] Rotation motion **8210** results in a first force **8212** and a second force **8214** from momentum during a cam stroke of the compression portion. The first force **8212** and the second force **8214** oppose each other and occur during different portions of the motor operation.

[0721] The first force **8212** and the second force **8214** transfer through motor mounting plate **7914** to cause swinging motion **8220** of the motor mounting plate **7914**. The first motor **7910** and the second motor **7912** are controlled as described herein to reduce swinging motion **8220**.

Motor Phase Shifting

[0722] Phase shifting of the motors and compressors uses a principle of constructive and destructive interference. The two motors are physically aligned on the same axis of rotation **7930** and on a vibration damping swing **8220**, this allows motion in the direction of motor momentum **8212** and **8214**. The rotation of the motor causes the cam shaft and "pistons" within the motor head to create oscillating forces. This shifts momentum back and forth during each stroke. During these events, the swinging mount can also pivot about the center axis to dissipate this excess motion without transmitting it further into the system.

[0723] When one motor spins up and oscillates back and forth the other motor can spin in the opposite direction and oppose the prior motions. When timed correctly the two motors can rotate at the same speed in sync but with opposing direction of momentum. This mirroring of motion allows a majority of the vibration, produced individually by each motor, to be partially canceled out by the combined motion of the whole swing assembly.

[0724] In embodiments, to achieve this effect one motor will be run to meet the system requirements (compressor for pressurized air needs). The second motor will be spun up to match the same rotation speed via the internal rotational sensors. Once they are rotating together in this physically mirrored orientation the vibration and the acoustic noise begin to interfere periodically. This is identified by a pulsing intensity of both noise and vibration.

[0725] Once they are synchronized together via the rotation speed they need to be placed in opposite phase. This will be achieved by slowly throttling the speed of the vacuum motor such that the rotational position of the motor drifts relative to the compressor motor position. The goal is to place them 180 degrees out of phase, or in other words, the hardware is physically at opposite orientations (opposing stroke positions) during each rotation. While the motors are rotating in tandem and with the correct offset, the internal mechanisms produce opposing forces that help to cancel out a large portion of the vibration load.

[0726] In the example provided, the motors have equally spaced pulses to identify the speed of rotations, but no internal sensors to identify piston position or rotation position. Vibration intensity is monitored to track the overall magnitude of acceleration produced by the motors. This is done by measuring the amplitude with board level accelerometers on the system PCBAs.

[0727] In some embodiments, a vibration source is located nearby each compressor motor to specifically cancel out the vibrations caused by the respective motor using accelerometer data. For example, in an embodiment: 1) The accelerometer is mounted directly to the top of the compressor; 2) Data from the accelerometer is processed by a local microcontroller; 3) The microcontroller computes a canceling waveform that outputs to a Cymbal piezo actuator, and 4) The piezo actuator that is sandwiched between the bottom of the compressor and mounting pad/frame counteracts the vibration of the compressor.

[0728] In some embodiments, the system includes: 1) Using analog circuitry between the accelerometer and piezo actuator instead of the microprocessor to reduce computation and communication delay from the microprocessor, and/or 2) using the piezo actuator as both a vibration measurement device and as an actuator to eliminate the accelerometer. An advantage of this method may be that each of the two motors in the system will have its own active canceling device, potentially making the vibration isolation more effective. These potential benefits may be weighed against: 1) mechanical complexity associated with assembling the 3-layers of mount/frame, piezo actuator, and compressor, 2) additional cost of the piezo actuator, and 3) finding a suitable piezo actuator if the existing units in the market are not adequate.

CONCLUSION

[0729] Clause 1. A method comprising: receiving, by a haptic interface module that is configured to interact with one or more haptic interface devices and an application that generates a computer-mediated environment comprising an avatar corresponding to a user wearing the one or more haptic interface devices, respective sensor data for each respective haptic interface device, wherein the respective sensor data for a respective haptic interface device indicates respective positioning of respective sensors of the respective haptic interface device; processing, by the haptic interface module, the respective sensor data to generate respective relative location data for each respective haptic interface device, wherein the relative location data is relative to a reference location defined with respect to the corresponding haptic interface device; receiving, by the haptic interface module, tracked location data from one or more motion tracking sensors, wherein the tracked location data indicates respective locations of the one or more haptic interface devices relative to a spatial environment of the user; generating, by the haptic interface module, a series of motion capture frames based on the tracked location data and the respective relative location data for each respective haptic interface device, wherein each respective motion capture frame indicates a set of locations and orientations for each respective haptic interface device at a given time; generating, by the haptic interface module, a series of kinematic frames based on the series of motion capture frames and one or more mediation processes that collectively convert, for each of the motion capture frames, the set of locations and orientations of the one or more respective haptic interface devices into a set of intended locations and intended orientations for configuring the avatar in the computer-mediated environment; and outputting the series of kinematic frames to the application, wherein the kinematic frames are provided to the application as user input.

[0730] Clause 2. The method of clause 1, wherein the avatar is a computer-mediated representation of a telerobot, wherein the set of intended locations and intended orientations are usable by the application to configure the telerobot.

[0731] Clause 3. The method of clause 1, wherein the computer-mediated environment is a game or simulation.

[0732] Clause 4. The method of clause 1, wherein the one or more mediation processes comprise performing touch compensation to simulate a detected touch intent of the user.

[0733] Clause 5. The method of clause 4, wherein performing touch compensation comprises: determining that the user has made a self-touch gesture based on the relative location data corresponding to two or more body parts of the user, wherein the two or more body parts of the user are not physically touching; updating the kinematic frame to indicate that the two or more body parts are touching in response to the determining that the user has made the self-touch gesture.

[0734] Clause 6. The method of clause 5, wherein at least one of the one or more haptic interface devices is a haptic glove, wherein the two or more body parts include a first finger and a second finger of the user, and wherein updating the kinematic frame comprises generating respective updated locations of at least one of the first finger and the second finger such that the kinematic frame indicates that the first finger is in contact with the second finger.

[0735] Clause 7. The method of clause 6, wherein the series of kinematic frames cause a first avatar component corresponding to the first finger of the user to touch a second avatar component corresponding to the second finger in the computer-mediated environment.

[0736] Clause 8. The method of clause 6, wherein determining the self-touch gesture comprises: detecting that a distance between a first relative location associated with the first finger and a second relative location associated with the second finger is less than a threshold distance.

[0737] Clause 9. The method of clause 8, wherein the threshold distance is defined based on a respective thickness of at least one of a first finger thimble of the haptic glove that receives the first finger and a second finger thimble that receives the second finger.

[0738] Clause 10. The method of clause 1, wherein the avatar is associated with different dimensions compared to the user, wherein generating the set of intended locations and orientations comprises one or more of: adjusting for a scale difference between the user and the avatar; or adjusting for a morphology difference between the user and the avatar.

[0739] Clause 11. The method of clause 1, further comprising: detecting a predefined gesture based on the series of motion capture frames; and outputting the detected gesture to the application.

[0740] Clause 12. A method of providing haptic feedback to one or more haptic interface devices, the method comprising: receiving, by a haptic feedback controller: first data indicating properties of one or more objects within a computer-mediated environment comprising an avatar corresponding to a user wearing the one or more haptic interface devices; and second data indicating a haptic effect associated with a current state of the computer-mediated environment; processing, by the haptic feedback controller, the first data and the second data to determine haptic feedback for the user based on an amount and type of contact of the avatar with the one or more objects within the computer-mediated environment and the haptic effect associated with the current state of the computer-mediated environment; generating, by the haptic feedback controller a series of haptic frames based on the determined haptic feedback, each haptic frame specifying a plurality of displacement distances for simulating the amount and type of contact and the haptic effect at a particular point in time; generating, by the haptic feedback controller, a series of actuator frames based on the series of haptic frames, each actuator frame specifying respective amounts of pressure to be provided to tactile actuators of the one or more haptic interface devices to provide a high precision simulation of the amount and type of contact and the haptic effect; and transmitting a plurality of instructions to respective actuator controls of the haptic interface devices, wherein the plurality of instructions are configured to cause actuation of the tactile actuators using the specified respective amounts of pressure to provide the high precision simulation of the amount and type of contact and the haptic effect to the user.

[0741] Clause 13. The method of clause 12, wherein the avatar is a digital twin of a telerobot, wherein the one or more objects are one or more digital twins of real objects in the environment of the telerobot.

[0742] Clause 14. The method of clause 13, wherein the digital twin is a haptic digital twin that has one or more of a different scale, a different morphology, or a different location than the telerobot.

[0743] Clause 15. The method of clause 12, wherein the computer-mediated environment is a game or simulation.

[0744] Clause 16. The method of clause 12, wherein the series of haptic frames simulate a soft-body deformation generated by a soft-body physics engine implemented by the haptic feedback controller, wherein the soft-body physics engine is not implemented by an application that generates the computer-mediated environment.

[0745] Clause 17. The method of clause 12, wherein processing the first data comprises: generating a ray trace originating from each of a plurality of locations associated with the avatar; and detecting intersections of each ray trace with the one or more objects within the computer-mediated environment.

[0746] Clause 18. The method of clause 12, wherein the first data indicates a stiffness or compliance of the one or more objects.

[0747] Clause 19. The method of clause 12, wherein the second data indicates one or more of an object effect associated with a haptic wave form that simulates the feel of contact with a moving object, a spatial effect associated with a haptic wave form that simulates the feel of a specific environmental effect, or a direct effect associated with a specific pattern for actuating the tactile actuators.

[0748] Clause 20. The method of clause 12, wherein generating each haptic frame comprises: calculating a first set of displacement distances associated with the amount and type of contact with the one or more objects; calculating a second set of displacement distances associated with the haptic effect, wherein the haptic effect is a time-varying haptic effect; and combining the first set of displacement distances and the second set of displacement distances to yield the plurality of displacement distances for simulating the amount and type of contact at the particular point in time.

[0749] Clause 21. The method of clause 12, wherein the displacement distances for simulating the amount and type of contact at the particular point in time are based on a simulated amount of force applied between the avatar and the one or more objects within the computer-mediated environment.

[0750] Clause 22. The method of clause 12, wherein the properties of the one or more objects comprise a temperature property, wherein the series of haptic frames further specify an amount of thermal flux associated with the contact, wherein the plurality of instructions indicate thermal properties of a fluid used to cause actuation of the tactile actuators.

[0751] Clause 23. The method of clause 12, wherein the series of actuator frames further specify an amount of resistive feedback to be provided to a plurality of brake actuators of the one or more haptic interface devices.

[0752] Clause 24. The method of clause 12, wherein the series of actuator frames further specify an amount of force feedback to be provided to a plurality of magnetorheological actuators of the one or more haptic interface devices.

[0753] Clause 25. A system for interfacing between a haptic glove and a computer-mediated environment generated by an application, the system comprising: a motion capture module configured to perform steps comprising: receiving sensor data from a plurality of sensors arranged on the haptic glove and at least one tracked location determined by a motion tracker arranged on the haptic glove; processing the sensor data to generate relative location data specifying relative locations and orientations of a plurality of components of the haptic glove; generating a series of motion capture frames based on the at least one tracked location and the relative locations and orientations, wherein each motion capture frame indicates a spatial arrangement of the plurality of components of the haptic glove at a specific point in time; generating a series of kinematic frames based on the series of motion capture frames, wherein the series of kinematic frames map the spatial arrangement of the plurality of components of the haptic glove to a hand model corresponding to an avatar within the computer-mediated environment; and outputting the series of kinematic frames to the application, wherein the series of kinematic frames cause the application to reposition a hand of the avatar within the computer-mediated environment; and a haptic feedback module configured to perform steps comprising: receiving data indicating properties of one or more objects within the computer-mediated environment; processing the received data to determine an amount and type of contact of the hand of the avatar with the one or more objects within the computer-mediated environment; generating a series of haptic frames, each haptic frame specifying a plurality of displacement distances for simulating the amount and type of contact at a particular point in time; generating a series of actuator frames based on the series of haptic frames, each actuator frame specifying: respective amounts of pressure to be provided to tactile actuators of the haptic glove; and respective amounts of resistance to be applied to actuate finger brakes of the haptic glove; and transmitting a plurality of instructions to respective actuator controls of the haptic glove based on the series of actuator frames, wherein the plurality of instructions are configured to cause actuation of the tactile actuators using the specified respective amounts of pressure and the finger brakes using the specified amount of resistance to provide a high precision simulation of the amount and type of contact to the user.

[0754] Clause 26. The system of clause 25, wherein the avatar is a haptic digital twin of a telerobotic hand, wherein the series of kinematic frames further cause the application to transmit instructions to the telerobotic hand that cause repositioning of the telerobotic hand.

[0755] Clause 27. The system of clause 25, wherein the computer-mediated environment is a game or simulation.

[0756] Clause 28. The system of clause 25, further comprising adjusting the series of motion capture frames by performing touch compensation to simulate a detected touch intent.

[0757] Clause 29. The system of clause 28, wherein the detected touch intent is a touch involving a first finger and a second finger of the user, wherein adjusting the motion capture frames comprises adjusting a location associated with a first finger so that the first finger touches a second finger of the avatar, the haptic feedback module further configured to perform steps comprising: detecting the touch intent based on the location associated with the first finger and a location associated with the second finger being within a threshold distance, wherein the threshold distance is defined based on a size of a finger thimble of a haptic glove.

[0758] Clause 30. The system of clause 25, wherein mapping the spatial arrangement of the plurality of components of the haptic glove to the hand model corresponding to the avatar within the computer-mediated environment comprises optimizing the position of the hand model to compensate for one or more size differences between the user's hand and the avatar's hand.

[0759] Clause 31. The system of clause 25, further comprising: detecting a gesture being performed by the user's hand based on the series of motion capture frames; and outputting the detected gesture to the application, wherein the application is configured to perform an action based on the detected gesture.

[0760] Clause 32. The system of clause 25, wherein the series of haptic frames simulate a soft-body deformation of the hand.

[0761] Clause 33. The system of clause 25, wherein processing the received data to determine an amount and type of contact of the hand of the avatar with the one or more objects within the computer-mediated environment comprises: generating a ray trace originating from each of a plurality of locations of the hand of the avatar, where each location corresponds to a tactile actuator of the haptic glove; and detecting intersections of each ray trace with the one or more objects within the computer-mediated environment.

[0762] Clause 34. The system of clause 25, wherein the received data indicates a stiffness or compliance of the one or more objects within the computer-mediated environment.

[0763] Clause 35. The system of clause 25, wherein the received data further indicates one or more of an object effect associated with a haptic waveform that simulates the feel of contact with a moving object, a spatial effect associated with a haptic waveform that simulates the feel of a specific environmental effect, or a direct effect associated with a specific pattern for actuating the tactile actuators.

[0764] Clause 36. The system of clause 25, wherein the plurality of displacement distances for simulating the amount and type of contact at the particular point in time are based on a simulated amount of force applied between the hand of the avatar and the one or more objects within the computer-mediated environment.

[0765] Clause 37. The system of clause 25, wherein the properties of the one or more objects comprise a temperature property, wherein the series of haptic frames further specify an amount of thermal flux associated with the contact, wherein the plurality of instructions indicate thermal properties of a fluid used to cause actuation of the tactile actuators.

[0766] Clause 38. The system of clause 25, wherein the received data further indicates a haptic effect associated with a state of the computer-mediated environment, wherein generating the series of haptic frames further comprises simulating the haptic effect.

[0767] Clause 39. The system of clause 38, wherein the haptic effect is one or more of an object effect associated with a haptic wave form that simulates the feel of contact with a moving object, a spatial effect associated with a haptic wave form that simulates the feel of a specific environmental effect, or a direct effect associated with a specific pattern for actuating the tactile actuators.

[0768] Clause 40. The system of clause 38, wherein generating each haptic frame comprises: calculating a first set of displacement distances associated with the amount and type of contact with the one or more objects; calculating a second set of displacement distances associated with the haptic effect, wherein the haptic effect is a time-varying haptic effect; and adding the first set of displacement distances and the second set of displacement distances to yield the plurality of displacement distances for simulating the amount and type of contact at the particular point in time.

[0769] Clause 41. A system for interfacing between a haptic glove and a computer-mediated application for controlling a telerobot, the system comprising: a motion capture module configured to perform steps comprising: receiving sensor data from a plurality of sensors arranged on one or more haptic interface devices and at least one tracked location determined by respective motion trackers arranged on the one or more haptic interface devices; processing the sensor data to generate relative location data specifying relative locations and orientations of a plurality of components of each respective haptic interface device; generating a series of motion capture frames based on the at least one tracked location and the relative locations and orientations, wherein each motion capture frame indicates a spatial arrangement of the plurality of components of each respective haptic interface device at a specific point in time; generating a series of kinematic frames based on the series of motion capture frames, wherein the series of kinematic frames map the spatial arrangement of the plurality of components of the respective haptic interface device to a model of an avatar of the user; and outputting the series of kinematic frames to the application, wherein the series of kinematic frames cause the application to reposition the avatar within a haptic digital twin of the environment of the telerobot based on the series of kinematic frames, wherein the application is further configured to transmit commands to reposition the telerobot based on the avatar; and a haptic feedback module configured to perform steps comprising: receiving data that is descriptive of one or more objects within an environment of the telerobot, wherein the data is generated by the application using the haptic digital twin of the environment of the telerobot; processing the received data to determine an amount and type of simulated contact of the avatar with the one or more objects; generating a series of haptic frames, each haptic frame specifying a plurality of displacement distances for simulating the contact at a particular point in time; generating a series of actuator frames based on the series of haptic frames, each actuator frame specifying respective amounts of pressure to be provided to tactile actuators of the one or more haptic interface devices to provide a high precision simulation of the contact; and transmitting a plurality of instructions to respective actuator controls of the haptic interface devices, wherein the plurality of instructions are configured to cause actuation of the tactile actuators using the specified respective amounts of pressure to provide the high precision simulation of the contact to the user.

[0770] Clause 42. The system of clause 41, wherein the avatar and the telerobot differ in one or more of scale or morphology.

[0771] Clause 43. The system of clause 41, wherein the user and the avatar differ in one or more of scale or morphology.

[0772] Clause 44. The system of clause 41, further comprising adjusting the series of motion capture frames by performing touch compensation to simulate a detected touch intent.

[0773] Clause 45. The system of clause 44, wherein the detected touch intent is a touch involving a first finger and a second finger of the user, wherein adjusting the motion capture frames comprises adjusting a location associated with the first finger so that the first finger touches a second finger of the avatar, the haptic feedback module configured to perform steps comprising: detecting the touch intent based on the location associated with the first finger and a location associated with the second finger being within a threshold distance, wherein the threshold distance is defined based on a size of a finger thimble of a haptic glove, wherein the haptic glove is one of the haptic interface devices.

[0774] Clause 46. The system of clause 41, wherein mapping the spatial arrangement of the plurality of components of the one or more haptic interface devices to the model of the avatar comprises optimizing a position of a hand of the avatar to compensate for one or more size differences between a hand of the user and the hand of the avatar.

[0775] Clause 47. The system of clause 41, further comprising: detecting a gesture being performed by a hand of the user based on the series of motion capture frames; and outputting the detected gesture to the application, wherein the application is configured to perform an action based on the detected gesture.

[0776] Clause 48. The system of clause 41, wherein the series of haptic frames simulate a soft-body deformation of the hand.

[0777] Clause 49. The system of clause 41, wherein processing the received data to determine the amount and type of simulated contact of the avatar with the one or more objects comprises: generating a ray trace originating from each of a plurality of locations of the avatar, where each location corresponds to a tactile actuator of the one or more haptic interface device; and detecting intersections of each ray trace with the one or more objects.

[0778] Clause 50. The system of clause 41, wherein the received data indicates properties of the one or more objects based on sensor data received from the telerobot.

[0779] Clause 51. A compressor assembly, comprising: a first compressor with a first motor having an axis of rotation; a second compressor with a second motor, the second motor opposing and coaxial with the first motor by having the axis of rotation; a pneumatic circuit body in pneumatic communication with the first compressor and the second compressor; an assembly mounting plate coupled to the pneumatic circuit body; at least one compressor damper coupled to damp vibration between the pneumatic circuit body and both the first compressor and the second compressor; and at least one pneumatic body damper coupled to damp vibration between the pneumatic circuit body and the assembly mounting plate.

[0780] Clause 52. The compressor assembly of clause 51, wherein the first compressor has a first compression portion and the second compressor has a second compression portion, and wherein the first compression portion is aligned with the second compression portion to oppose forces of operating the second compression portion.

[0781] Clause 53. The compressor assembly of clause 52, wherein the first compression portion and the second compression portion are cam driven piston compressors positioned to have piston travel directions parallel to and offset from each other.

[0782] Clause 54. The compressor assembly of clause 51, further comprising a compressor plate, and wherein the first compressor and the second compressor are rigidly mounted to the compressor plate.

[0783] Clause 55. The compressor assembly of clause 54, wherein the compressor plate defines at least one damper receiving portion configured to receive a corresponding compression damper of the at least one compressor damper, and wherein the at least one compressor damper is disposed at least partially within the at least one damper receiving portion.

[0784] Clause 56. The compressor assembly of clause 55, wherein the at least one compressor damper is positioned to permit and damp a swinging motion of the compressor plate.

[0785] Clause 57. The compressor assembly of clause 56, wherein the at least one compressor damper has a longitudinal axis that is substantially aligned to intersect the axis of rotation.

[0786] Clause 58. The compressor assembly of clause 51, wherein the pneumatic circuit body defines a plurality of pneumatic circuits.

[0787] Clause 59. The compressor assembly of clause 51, wherein the first compressor and the second compressor collectively provide a pressurized air source and a vacuum source.

[0788] Clause 60. A method of controlling a compressor assembly, the method comprising: commanding a first compressor to operate at a first frequency; commanding a second compressor vibrationally coupled with the first compressor to operate at the first frequency; phase shifting of second compressor components of the second compressor relative to the first compressor by commanding the second compressor to operate at a second frequency that is different from the first frequency; aligning the second compressor components to be substantially opposing momentum from operation of first compressor components of the first compressor using the phase shifting; and operating the first compressor and the second compressor at the first frequency in response to aligning the second compressor components to be substantially opposing momentum from operation of first compressor components.

[0789] Clause 61. The method of clause 60, further comprising measuring a vibration caused by operating the first compressor and the second compressor at the first frequency, and wherein aligning the second compressor components is based on minimizing the vibration.

[0790] Clause 62. The method of clause 60, wherein the first frequency and the second frequency vary with a demand on the compressor assembly.

[0791] Clause 63. A method of performing motion capture on sensor data received from a wearable haptic interface, the method comprising: receiving, by a simulation engine, the sensor data from the wearable haptic interface, wherein the sensor data indicates one or more locations associated with the wearable haptic interface; processing, by the simulation engine, the sensor data to generate a motion capture frame, wherein the processing comprises: mapping the location data to a model corresponding to an avatar that is simulated by the simulation engine; and adjusting the mapped data to compensate for the wearable haptic interface; and modifying the simulated avatar based on the motion capture frame.

[0792] Clause 64. The method of clause 63, wherein adjusting the mapped data to compensate for the wearable haptic interface comprises performing touch compensation to adjust for a detected touch intent.

[0793] Clause 65. The method of clause 64, wherein the detected touch intent is a touch involving a first finger and a second finger, wherein the adjusting comprises adjusting a position of the first finger such that the first finger touches the second finger.

[0794] Clause 66. The method of clause 65, wherein the adjusting is based on a size of a finger thimble of a wearable haptic glove of the wearable haptic interface.

[0795] Clause 67. The method of clause 63, wherein adjusting the mapped data to compensate for the wearable haptic interface comprises adjusting the mapped data to compensate for sensor slip.

[0796] Clause 68. The method of clause 63, wherein the sensor data comprises tracking data from one or more motion tracking sensors and relative location data from one or more magnetic sensors.

[0797] Clause 69. A method of providing haptic feedback to a wearable haptic interface, the method comprising: receiving, from a simulation application, first data indicating a state of an avatar and second data indicating a state of a simulated environment containing the avatar; processing the first data and the second data to detect a contact of the avatar with an entity that exists within the simulated environment; calculating an estimated soft-body deformation resulting from the contact; and actuating a plurality of tactile actuators of the wearable haptic interface based on the estimated soft-body deformation.

[0798] Clause 70. The method of clause 69, wherein calculating the estimated soft-body deformation is performed by an on-board physics engine that is separate from the simulation application.

[0799] Clause 71. The method of clause 69, wherein processing the first data and the second data to detect a contact of the avatar with an entity that exists within the simulated environment comprises: generating a field of ray traces associated with the avatar; and detecting intersections of the field of ray traces with the entity that exists within the simulated environment.

[0800] Clause 72. The method of clause 69, wherein the estimated soft-body deformation is based on a simulated amount of force applied between the avatar and the entity that exists within the simulated environment.

[0801] Clause 73. The method of clause 69, wherein the estimated soft-body deformation is based on a distance between the avatar and the entity that exists within the simulated environment.

[0802] Clause 74. The method of clause 69, wherein calculating the estimated soft-body deformation comprises applying one or more of a low-pass filter or a high-pass filter to body contour data associated with the simulated entity.

[0803] Clause 75. A method of providing interaction assistance within a simulated environment for a user of a wearable haptic interface, the method comprising: determining, based on first data indicating a state of an avatar corresponding to the user and second data indicating a state of a simulated entity within the simulated environment, an intended interaction of the avatar with the simulated entity; adjusting one or more simulated physics properties within a region of the simulated environment that corresponds to the avatar and the simulated entity; determining an end of the intended interaction of the avatar with the simulated entity; and resetting the one or more adjusted simulated physics properties.

[0804] Clause 76. The method of clause 75, wherein the intended interaction is a grasp interaction.

[0805] Clause 77. The method of clause 76, wherein determining the intended interaction comprises determining that a hand or finger of the avatar is applying pressure to the simulated entity within the simulated environment.

[0806] Clause 78. The method of clause 76, wherein determining the intended interaction comprises detecting at least two forces associated with an avatar that are directed towards the simulated entity within the simulated environment.

[0807] Clause 79. The method of clause 78, wherein a first force of the at least two forces is associated with a first finger and a second force of the at least two forces is associated with a second finger.

[0808] Clause 80. The method of clause 75, wherein adjusting the one or more simulated physics properties comprises damping a contact between the avatar and the simulated entity to allow the avatar to more easily hold the simulated entity.

[0809] Clause 81. A method of providing haptic feedback to a wearable haptic interface, the method comprising: receiving, from a simulation application, first data indicating a state of an avatar and second data indicating a state of a simulated environment containing the avatar; processing the first data and the second data to detect a contact of the avatar with an entity that exists within the simulated environment; determining haptic feedback instructions to provide to the wearable haptic interface based on a simulated force associated with the contact; determining, based on one or more governance standards, to modify the haptic feedback instructions in order to reduce an amount of haptic feedback applied to a user of the wearable haptic interface; and providing the modified haptic feedback instructions to the wearable haptic interface.

[0810] Clause 82. The method of clause 81, wherein the governance standards comprise one or more safety limitations.

[0811] Clause 83. The method of clause 82, wherein the safety limitations specify a maximum acceleration associated with force feedback applied to a user.

[0812] Clause 84. The method of clause 82, wherein the safety limitations specify a maximum range of motion associated with a body part of a user.

[0813] Clause 85. The method of clause 82, wherein the safety limitations specify a maximum amount of thermal flux for a user.

[0814] Clause 86. The method of clause 81, wherein the governance standards comprise one or more user preferences for haptic feedback.

[0815] While only a few embodiments of the disclosure have been shown and described, it will be obvious to those skilled in the art that many changes and modifications may be made thereunto without departing from the spirit and scope of the disclosure as described in the following claims. All patent applications and patents, both foreign and domestic, and all other publications referenced herein are incorporated herein in their entireties to the full extent permitted by law.

[0816] The methods and systems described herein may be deployed in part or in whole through machines that execute computer software, program codes, and/or instructions on a processor. The disclosure may be implemented as a method on the machine(s), as a system or apparatus as part of or in relation to the machine(s), or as a computer program product embodied in a computer readable medium executing on one or more of the machines. In embodiments, the processor may be part of a server, cloud server, client, network infrastructure, mobile computing platform, stationary computing platform, or other computing platforms. A processor may be any kind of computational or processing device capable of executing program instructions, codes, binary instructions and the like, including a central processing unit (CPU), a general processing unit (GPU), a logic board, a chip (e.g., a graphics chip, a video processing chip, a data compression chip, or the like), a chipset, a controller, a system-on-chip (e.g., an RF system on chip, an AI system on chip, a video processing system on chip, or others), an integrated circuit, an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), an approximate computing processor, a quantum computing processor, a parallel computing processor, a neural network processor, or other type of processor. The processor may be or may include a signal processor, digital processor, data processor, embedded processor, microprocessor or any variant such as a co-processor (math co-processor, graphic co-processor, communication co-processor, video co-processor, AI co-processor, and the like) and the like that may directly or indirectly facilitate execution of program code or program instructions stored thereon. In addition, the processor may enable execution of multiple programs, threads, and codes. The threads may be executed simultaneously to enhance the performance of the processor and to facilitate simultaneous operations of the application. By way of implementation, methods, program codes, program instructions and the like described herein may be implemented in one or more threads. The thread may spawn other threads that may have assigned priorities associated with them; the processor may execute these threads based on priority or any other order based on instructions provided in the program code. The processor, or any machine utilizing one, may include non-transitory memory that stores methods, codes, instructions and programs as described herein and elsewhere. The processor may access a non-transitory storage medium through an interface that may store methods, codes, and instructions as described herein and elsewhere. The storage medium associated with the processor for storing methods, programs, codes, program instructions or other type of instructions capable of being executed by the computing or processing device may include but may not be limited to one or more of a CD-ROM, DVD, memory, hard disk, flash drive, RAM, ROM, cache, network-attached storage, server-based storage, and the like.

[0817] A processor may include one or more cores that may enhance speed and performance of a multiprocessor. In embodiments, the process may be a dual core processor, quad core processors, other chip-level multiprocessor and the like that combine two or more independent cores (sometimes called a die).

[0818] The methods and systems described herein may be deployed in part or in whole through machines that execute computer software on various devices including a server, client, firewall, gateway, hub, router, switch, infrastructure-as-a-service, platform-as-a-service, or other such computer and/or networking hardware or system. The software may be associated with a server that may include a file server, print server, domain server, internet server, intranet server, cloud server, infrastructure-as-a-service server, platform-as-a-service server, web server, and other variants such as secondary server, host server, distributed server, failover server, backup server, server farm, and the like. The server may include one or more of memories, processors, computer readable media, storage media, ports (physical and virtual), communication devices, and interfaces capable of accessing other servers, clients, machines, and devices through a wired or a wireless medium, and the like. The methods, programs, or codes as described herein and elsewhere may be executed by the server. In addition, other devices required for execution of methods as described in this application may be considered as a part of the infrastructure associated with the server.

[0819] The server may provide an interface to other devices including, without limitation, clients, other servers, printers, database servers, print servers, file servers, communication servers, distributed servers, social networks, and the like. Additionally, this coupling and/or connection may facilitate remote execution of programs across the network. The networking of some or all of these devices may facilitate parallel processing of a program or method at one or more locations without deviating from the scope of the disclosure. In addition, any of the devices attached to the server through an interface may include at least one storage medium capable of storing methods, programs, code and/or instructions. A central repository may provide program instructions to be executed on different devices. In this implementation, the remote repository may act as a storage medium for program code, instructions, and programs.

[0820] The software program may be associated with a client that may include a file client, print client, domain client, internet client, intranet client and other variants such as secondary client, host client, distributed client and the like. The client may include one or more of memories, processors, computer readable media, storage media, ports (physical and virtual), communication devices, and interfaces capable of accessing other clients, servers, machines, and devices through a wired or a wireless medium, and the like. The methods, programs, or codes as described herein and elsewhere may be executed by the client. In addition, other devices required for the execution of methods as described in this application may be considered as a part of the infrastructure associated with the client.

[0821] The client may provide an interface to other devices including, without limitation, servers, other clients, printers, database servers, print servers, file servers, communication servers, distributed servers and the like. Additionally, this coupling and/or connection may facilitate remote execution of programs across the network. The networking of some or all of these devices may facilitate parallel processing of a program or method at one or more locations without deviating from the scope of the disclosure. In addition, any of the devices attached to the client through an interface may include at least one storage medium capable of storing methods, programs, applications, code and/or instructions. A central repository may provide program instructions to be executed on different devices. In this implementation, the remote repository may act as a storage medium for program code, instructions, and programs.

[0822] The methods and systems described herein may be deployed in part or in whole through network infrastructures. The network infrastructure may include elements such as computing devices, servers, routers, hubs, firewalls, clients, personal computers, communication devices, routing devices and other active and passive devices, modules and/or components as known in the art. The computing and/or non-computing device(s) associated with the network infrastructure may include, apart from other components, a storage medium such as flash memory, buffer, stack, RAM, ROM and the like. The processes, methods, program codes, instructions described herein and elsewhere may be executed by one or more of the network infrastructural elements. The methods and systems described herein may be adapted for use with any kind of private, community, or hybrid cloud computing network or cloud computing environment, including those which involve features of software as a service (Saas), platform as a service (PaaS), and/or infrastructure as a service (IaaS).

[0823] The methods, program codes, and instructions described herein and elsewhere may be implemented on a cellular network with multiple cells. The cellular network may either be frequency division multiple access (FDMA) network or code division multiple access (CDMA) network. The cellular network may include mobile devices, cell sites, base stations, repeaters, antennas, towers, and the like. The cell network may be a GSM, GPRS, 3G, 4G, 5G, LTE, EVDO, mesh, or other network types.

[0824] The methods, program codes, and instructions described herein and elsewhere may be implemented on or through mobile devices. The mobile devices may include navigation devices, cell phones, mobile phones, mobile personal digital assistants, laptops, palmtops, netbooks, pagers, electronic book readers, music players and the like. These devices may include, apart from other components, a storage medium such as flash memory, buffer, RAM, ROM and one or more computing devices. The computing devices associated with mobile devices may be enabled to execute program codes, methods, and instructions stored thereon. Alternatively, the mobile devices may be configured to execute instructions in collaboration with other devices. The mobile devices may communicate with base stations interfaced with servers and configured to execute program codes. The mobile devices may communicate on a peer-to-peer network, mesh network, or other communications network. The program code may be stored on the storage medium associated with the server and executed by a computing device embedded within the server. The base station may include a computing device and a storage medium. The storage device may store program codes and instructions executed by the computing devices associated with the base station.

[0825] The computer software, program codes, and/or instructions may be stored and/or accessed on machine readable media that may include: computer components, devices, and recording media that retain digital data used for computing for some interval of time; semiconductor storage known as random access memory (RAM); mass storage typically for more permanent storage, such as optical discs, forms of magnetic storage like hard disks, tapes, drums, cards and other types; processor registers, cache memory, volatile memory, non-volatile memory; optical storage such as CD, DVD; removable media such as flash memory (e.g., USB sticks or keys), floppy disks, magnetic tape, paper tape, punch cards, standalone RAM disks, Zip drives, removable mass storage, off-line, and the like; other computer memory such as dynamic memory, static memory, read/write storage, mutable storage, read only, random access, sequential access, location addressable, file addressable, content addressable, network attached storage, storage area network, bar codes, magnetic ink, network-attached storage, network storage, NVME-accessible storage, PCIE connected storage, distributed storage, and the like.

[0826] The methods and systems described herein may transform physical and/or intangible items from one state to another. The methods and systems described herein may also transform data representing physical and/or intangible items from one state to another.

[0827] The elements described and depicted herein, including in flow charts and block diagrams throughout the figures, imply logical boundaries between the elements. However, according to software or hardware engineering practices, the depicted elements and the functions thereof may be implemented on machines through computer executable code using a processor capable of executing program instructions stored thereon as a monolithic software structure, as standalone software modules, or as modules that employ external routines, code, services, and so forth, or any combination of these, and all such implementations may be within the scope of the disclosure. Examples of such machines may include, but may not be limited to, personal digital assistants, laptops, personal computers, mobile phones, other handheld computing devices, medical equipment, wired or wireless communication devices, transducers, chips, calculators, satellites, tablet PCs, electronic books, gadgets, electronic devices, devices, artificial intelligence, computing devices, networking equipment, servers, routers and the like. Furthermore, the elements depicted in the flow chart and block diagrams or any other logical component may be implemented on a machine capable of executing program instructions. Thus, while the foregoing drawings and descriptions set forth functional aspects of the disclosed systems, no particular arrangement of software for implementing these functional aspects should be inferred from these descriptions unless explicitly stated or otherwise clear from the context. Similarly, it will be appreciated that the various steps identified and described in the disclosure may be varied, and that the order of steps may be adapted to particular applications of the techniques disclosed herein. All such variations and modifications are intended to fall within the scope of this disclosure. As such, the depiction and/or description of an order for various steps should not be understood to require a particular order of execution for those steps, unless required by a particular application, or explicitly stated or otherwise clear from the context.

[0828] The methods and/or processes described in the disclosure, and steps associated therewith, may be realized in hardware, software or any combination of hardware and software suitable for a particular application. The hardware may include a general-purpose computer and/or dedicated computing device or specific computing device or particular aspect or component of a specific computing device. The processes may be realized in one or more microprocessors, microcontrollers, embedded microcontrollers, programmable digital signal processors or other programmable devices, along with internal and/or external memory. The processes may also, or instead, be embodied in an application specific integrated circuit, a programmable gate array, programmable array logic, or any other device or combination of devices that may be configured to process electronic signals. It will further be appreciated that one or more of the processes may be realized as a computer executable code capable of being executed on a machine-readable medium.

[0829] The computer executable code may be created using a structured programming language such as C, an object oriented programming language such as C++, or any other high-level or low-level programming language (including assembly languages, hardware description languages, and database programming languages and technologies) that may be stored, compiled or interpreted to run on one of the devices described in the disclosure, as well as heterogeneous combinations of processors, processor architectures, or combinations of different hardware and software, or any other machine capable of executing program instructions. Computer software may employ virtualization, virtual machines, containers, dock facilities, portainers, and other capabilities.

[0830] Thus, in one aspect, methods described in the disclosure and combinations thereof may be embodied in computer executable code that, when executing on one or more computing devices, performs the steps thereof. In another aspect, the methods may be embodied in systems that perform the steps thereof and may be distributed across devices in a number of ways, or all of the functionality may be integrated into a dedicated, standalone device or other hardware. In another aspect, the means for performing the steps associated with the processes described in the disclosure may include any of the hardware and/or software described in the disclosure. All such permutations and combinations are intended to fall within the scope of the disclosure.

[0831] While the disclosure has been disclosed in connection with the preferred embodiments shown and described in detail, various modifications and improvements thereon will become readily apparent to those skilled in the art. Accordingly, the spirit and scope of the disclosure is not to be limited by the foregoing examples, but is to be understood in the broadest sense allowable by law.

[0832] The use of the terms “a” and “an” and “the” and similar referents in the context of describing the disclosure (especially in the context of the following claims) is to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The terms “comprising,” “with,” “including,” and “containing” are to be construed as open-ended terms (i.e., meaning “including, but not limited to,”) unless otherwise noted. Recitations of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., "such as") provided herein, is intended merely to better illuminate the disclosure and does not pose a limitation on the scope of the disclosure unless otherwise claimed. The term "set" may include a set with a single member. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the disclosure.

[0833] While the foregoing written description enables one skilled to make and use what is considered presently to be the best mode thereof, those skilled in the art will understand and appreciate the existence of variations, combinations, and equivalents of the specific embodiment, method, and examples herein. The disclosure should therefore not be limited by the above-described embodiment, method, and examples, but by all embodiments and methods within the scope and spirit of the disclosure.

[0834] All documents referenced herein are hereby incorporated by reference as if fully set forth herein.

1-11. (canceled)

12. A method of providing haptic feedback to one or more haptic interface devices, the method comprising:

receiving, by a haptic feedback controller:

first data indicating properties of one or more objects within a computer-mediated environment comprising an avatar corresponding to a user wearing the one or more haptic interface devices; and

second data indicating a haptic effect associated with a current state of the computer-mediated environment;

processing, by the haptic feedback controller, the first data and the second data to determine haptic feedback for the user based on an amount and type of contact of the avatar with the one or more objects within the computer-mediated environment and the haptic effect associated with the current state of the computer-mediated environment;

generating, by the haptic feedback controller a series of haptic frames based on the determined haptic feedback, each haptic frame specifying a plurality of displacement distances for simulating the amount and type of contact and the haptic effect at a particular point in time;

generating, by the haptic feedback controller, a series of actuator frames based on the series of haptic frames, each actuator frame specifying respective amounts of pressure to be provided to tactile actuators of the one or more haptic interface devices to provide a high precision simulation of the amount and type of contact and the haptic effect; and

transmitting a plurality of instructions to respective actuator controls of the haptic interface devices, wherein the plurality of instructions are configured to cause actuation of the tactile actuators using the specified respective amounts of pressure to provide the high precision simulation of the amount and type of contact and the haptic effect to the user.

13. The method of claim **12**, wherein the avatar is a digital twin of a telerobot, wherein the one or more objects are one or more digital twins of real objects in the environment of the telerobot.

14. The method of claim **13**, wherein the digital twin is a haptic digital twin that has one or more of a different scale, a different morphology, or a different location than the telerobot.

15. The method of claim **12**, wherein the computer-mediated environment is a game or simulation.

16. The method of claim **12**, wherein the series of haptic frames simulate a soft-body deformation generated by a soft-body physics engine implemented by the haptic feedback controller, wherein the soft-body physics engine is not implemented by an application that generates the computer-mediated environment.

17. The method of claim **12**, wherein processing the first data comprises:

generating a ray trace originating from each of a plurality of locations associated with the avatar; and

detecting intersections of each ray trace with the one or more objects within the computer-mediated environment.

18. The method of claim **12**, wherein the first data indicates a stiffness or compliance of the one or more objects.

19. The method of claim **12**, wherein the second data indicates one or more of an object effect associated with a haptic wave form that simulates a feel of contact with a moving object, a spatial effect associated with a haptic wave form that simulates a feel of a specific environmental effect, or a direct effect associated with a specific pattern for actuating the tactile actuators.

20. The method of claim **12**, wherein generating each haptic frame comprises:

calculating a first set of displacement distances associated with the amount and type of contact with the one or more objects;

calculating a second set of displacement distances associated with the haptic effect, wherein the haptic effect is a time-varying haptic effect; and

combining the first set of displacement distances and the second set of displacement distances to yield the plurality of displacement distances for simulating the amount and type of contact at the particular point in time.

21. The method of claim **12**, wherein the displacement distances for simulating the amount and type of contact at the particular point in time are based on a simulated amount of force applied between the avatar and the one or more objects within the computer-mediated environment.

22. The method of claim **12**, wherein the properties of the one or more objects comprise a temperature property, wherein the series of haptic frames further specify an amount of thermal flux associated with the contact, wherein the plurality of instructions indicate thermal properties of a fluid used to cause actuation of the tactile actuators.

23. The method of claim **12**, wherein the series of actuator frames further specify an amount of resistive feedback to be provided to a plurality of brake actuators of the one or more haptic interface devices.

24. The method of claim **12**, wherein the series of actuator frames further specify an amount of force feedback to be provided to a plurality of magnetorheological actuators of the one or more haptic interface devices.

25-40. (canceled)

41. A system for interfacing between a haptic glove and a computer-mediated application for controlling a telerobot, the system comprising:

a motion capture module configured to perform steps comprising:

receiving sensor data from a plurality of sensors arranged on one or more haptic interface devices and at least one tracked location determined by respective motion trackers arranged on the one or more haptic interface devices;

processing the sensor data to generate relative location data specifying relative locations and orientations of a plurality of components of each respective haptic interface device;

generating a series of motion capture frames based on the at least one tracked location and the relative locations and orientations, wherein each motion capture frame indicates a spatial arrangement of the plurality of components of each respective haptic interface device at a specific point in time;

generating a series of kinematic frames based on the series of motion capture frames, wherein the series of kinematic frames map the spatial arrangement of the plurality of components of the respective haptic interface device to a model of an avatar of a user of the one or more haptic interface devices; and

outputting the series of kinematic frames to the application, wherein the series of kinematic frames cause the application to reposition the avatar within a haptic digital twin of an environment of the telerobot based on the series of kinematic frames, wherein the application is further configured to transmit commands to reposition the telerobot based on the avatar; and

a haptic feedback module configured to perform steps comprising:

receiving data that is descriptive of one or more objects within an environment of the telerobot, wherein the data is generated by the application using the haptic digital twin of the environment of the telerobot;

processing the received data to determine an amount and type of simulated contact of the avatar with the one or more objects;

generating a series of haptic frames, each haptic frame specifying a plurality of displacement distances for simulating the contact at a particular point in time;

generating a series of actuator frames based on the series of haptic frames, each actuator frame specifying respective amounts of pressure to be provided to tactile actuators of the one or more haptic interface devices to provide a high precision simulation of the contact; and

transmitting a plurality of instructions to respective actuator controls of the haptic interface devices, wherein the plurality of instructions are configured to cause actuation of the tactile actuators using the specified respective amounts of pressure to provide the high precision simulation of the contact to the user.

42. The system of claim **41**, wherein the avatar and the telerobot differ in one or more of scale or morphology.

43. The system of claim **41**, wherein the user and the avatar differ in one or more of scale or morphology.

44. The system of claim **41**, wherein the motion capture module is further configured to adjust the series of motion capture frames by performing touch compensation to simulate a detected touch intent.

45. The system of claim **44**, wherein the detected touch intent is a touch involving a first finger and a second finger of the user, wherein adjusting the motion capture frames comprises adjusting a location associated with the first finger so that the first finger touches a second finger of the avatar, the haptic feedback module configured to perform steps comprising:

detecting the touch intent based on the location associated with the first finger and a location associated with the second finger being within a threshold distance, wherein the threshold distance is defined based on a size of a finger thimble of a haptic glove, wherein the haptic glove is one of the haptic interface devices.

46. The system of claim **41**, wherein mapping the spatial arrangement of the plurality of components of the one or more haptic interface devices to the model of the avatar comprises optimizing a position of a hand of the avatar to compensate for one or more size differences between a hand of the user and the hand of the avatar.

47. The system of claim **41**, further comprising:

detecting a gesture being performed by a hand of the user based on the series of motion capture frames; and

outputting the detected gesture to the application, wherein the application is configured to perform an action based on the detected gesture.

48. The system of claim **41**, wherein the series of haptic frames simulate a soft-body deformation of a hand.

49. The system of claim **41**, wherein processing the received data to determine the amount and type of simulated contact of the avatar with the one or more objects comprises:

generating a ray trace originating from each of a plurality of locations of the avatar, where each location corresponds to a tactile actuator of the one or more haptic interface device; and

detecting intersections of each ray trace with the one or more objects.

50. The system of claim **41**, wherein the received data indicates properties of the one or more objects based on sensor data received from the telerobot.

51-86. (canceled)

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